BACKWARD:

- `findings/population_equality.md` — A1 MAD numbers integrated into §5.1 - `findings/packing_cracking_findings.md` — partisan-metrics integrated into §5.2 - `findings/joint_outlier_score_summary.md` — Mahalanobis Ch1 integrated into §5.4 - `findings/intermap_permutation_test_results.md` — Ch1-COMP results integrated into §5.4 - `findings/airdrie_overlap_report.md` — Airdrie four-way split integrated into §5.3.2 - `findings/cycle_lag_analysis.md` — direction-of-travel narrative in §5.4.10 - `findings/chen_rodden_decomposition.md` — partisan-geography decomposition - `findings/methods_paper_draft.md` — companion methodology paper - `analysis/methodology/retraction_pathway.md` — named retraction conditions per finding - `analysis/methodology/null_hypothesis_and_exoneration_criteria.md` — pass thresholds - `analysis/methodology/threshold_provenance.md` — threshold derivations cited throughout - `preregistration/null_hypotheses.md` — pre-committed null hypotheses - `preregistration/thresholds.md` — pre-registered threshold compendium - `preregistration/seed_commitments.md` — drand-pinned random seeds - `docs/FROZEN_MANIFEST.md` — pinned external sources - `analysis/scripts/mcmc_ensemble_canonical.py` — 1M-plan canonical ensemble - `analysis/scripts/packing_cracking_analysis.py` — Phase 4C partisan-metric script

FORWARD:

- `reports/academic/report_academic.pdf` — print-ready PDF derivative - `reports/public/report_public.md` — plain-language summary derived from this - `docs/FINDINGS_BRIEF.md` — one-page brief condensed from this - `README.md` — repository overview citing this as the technical monograph - (leaf otherwise — primary monograph, reviewer-facing and external-citation surface)

Alberta Electoral Boundaries Audit — Comprehensive Forensic Audit Monograph

A symmetric, reproducible forensic assessment of the 2025–26 Electoral Boundaries Commission's majority and minority recommendations

WILL CONNER · MOUNT ROYAL UNIVERSITY, BSC COMPUTER INFORMATION SYSTEMS

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Pre-publication status. *This monograph has not been publicly released. Integration of canonical Elections Alberta shapefiles (received 2026-05-06) is ongoing; individual section results may shift as recomputation continues. The codebase is also undergoing cleanup and script-count reduction. Do not cite, quote, or distribute without confirming the current status of the specific finding.*

Executive summary and reading guide

This document is a **comprehensive forensic audit monograph**, not a single-topic journal article. It covers three distinct lines of work sharing a single dataset and methodology, and it runs substantially longer than a standard journal submission. Readers short on time should use the guide below to jump directly to the part matching their question. Readers who need the full evidentiary chain should read end-to-end.

Headline finding (read this if you read nothing else). Alberta's 2025–26 Electoral Boundaries Commission produced two competing 89-seat maps on March 23, 2026. This audit evaluates both proposals using identical methods applied symmetrically to each. **On the partisan-fairness metrics, the majority map is not a UCP-advantage outlier on any of them** — it sits within the neutral band on three of four, and on the fourth (mean-median) it falls at the NDP-favourable tail, not the UCP tail (p0.9; see §5.4.9). **The minority map is a UCP-advantage outlier.**

The minority proposal differs from the majority on four measurable non-partisan-bias dimensions: population dispersion (Median Absolute Deviation 48% wider), Calgary geographic-zone asymmetry (12.2% vs 0.4%), Airdrie community fragmentation (4-way vs 2-way split), and commission-chair-flagged geographic anomalies (3 confirmed geometric anomalies vs 0; the chair's documented criticism spans 7 configurations across two sections of the majority report — §5.8.2 and §5.9.4). All four signals run in the same direction and survive every stress-test applied. A fifth pre-registered dimension — municipal-boundary anchoring — did not survive canonical recomputation. Both maps fall within the 70–85% Canadian comparator norm. The failure is documented in full at §5.8.5 rather than quietly removed.

The partisan-bias evidence rests primarily on the **ensemble test** (Mahalanobis Ch1, §5.4.9): *is this map extreme relative to over one million neutral, randomly-drawn Alberta maps?* The minority's Ch1 $p = 1.40\times10^{-6}$ (≈ 1 in 714,000) against the canonical 1,010,000-plan ensemble. A secondary **boundary-choice test** (SZAT Ch2, §5.2.10) — *are the specific lines where the minority drew differently from the majority partisan-neutral?* — returned $p = 0.0024$ under the original i.i.d.-flip null but **$p \approx 0.19$ under a contiguity-respecting block-permutation null** (T1.10b closed 2026-06-12; see `findings/szat_block_permutation.md`). The 2,110 swing-zone VAs cluster into one giant connected component (2,086 VAs), so the i.i.d.-flip null *understated* the variance by 5.79× relative to the methodologically-correct block null. The Ch2 evidence does not survive the contiguity correction and is no longer used in the audit's joint headline. **The headline is Ch1 alone: $p = 1.40\times10^{-6}$ against the canonical neutral ensemble; the dependence-robust upper bound is unchanged at $p \le 2.80\times10^{-6}$ (exploratory — see §1.2 for the exploratory/confirmatory distinction)** (the earlier "2 × min(Ch1, Ch2)" bound coincidentally matches because Ch1 was always the binding constraint). The earlier "$p = 6.87\times10^{-8}$ ≈ 1 in 15 million" Fisher figure assumed channel independence; it is retired. All random seeds are committed to the public Cloudflare drand beacon before official shapefiles arrived; all scripts and data are public; an independent analyst running the same pipeline recovers the same numbers.

Channel 3 — neighbour-drain adjacency — is not part of the Bonferroni headline ($p \le 2.80\times10^{-6}$). The pre-registered drain test (§5.3.5; OSF:r3zm7) asks whether majority-side packed districts tend to border cracked districts on the minority map. On the canonical Elections Alberta shapefiles, all three maps are anomalously *low* relative to their own label-shuffle nulls: 2019 enacted $z = -3.520$, majority $z = -3.173$, minority $z = -2.750$ (two-tailed $p \le 0.0002$ for all three). The majority is not singularly anomalous on this metric — the 2019 enacted baseline is the most anomalous of the three, and all three maps sit at the 0th percentile of their respective nulls. The pre-registered Prediction A (drain(majority) > drain(minority) in raw drain-score terms) is directionally confirmed. The narrative that the majority stands out as uniquely anomalous does not survive the three-way comparison: the finding is that every Alberta map examined — including the pre-commission 2019 baseline — produces unusually low neighbour-drain scores relative to an unconstrained label-shuffle null. Full results and interpretation at §5.3.5.

The directional consistency across structural and statistical dimensions is the finding. The audit records what the data show. It does not reach a legal conclusion and does not infer intent (§4.5). The partisan-bias magnitude is sensitive to the vote-attribution method used and should be read as a directional observation, not a point estimate (§1.2, §5.2.2). See §1.1 for the full plain-language summary, §1.2 for modelling-uncertainty caveats, and §6 for the author's verdict on both maps.

Reading guide by question. The monograph has three parts, each answering a different question and each referencing the others:

IF YOU ARE HERE TO ANSWER...	READ	COVERS
Part I — Empirical audit. *Do the two 2026 maps differ in measurable, reproducible ways?*	Abstract, §1.1, §5.1 (population equality), §5.2 (partisan bias), §5.3 (signatures), §5.4 (MCMC ensemble), §5.6 (symmetry counter-test), §5.7 (stress-test grades), §5.8 (geographic coherence)	Core quantitative findings — the empirical redistricting audit.
Part II — Procedural and policy critique. *What does the April 16 legislative pivot mean; how does the commission's methodology compare to Canadian norms; what statutory reforms follow?*	§5.9 (procedural), Appendix F (constitutional framing), `docs/act_amendment_proposal.md` (statutory reform), `findings/cycle_lag_analysis.md` §2 (forward-modelling consequences for commissions)	The Canadian comparative and legal-procedural layer.
Part III — Data provenance and methodological warnings. *Why can't the audit give single-number point estimates on every metric; what does the missing-shapefile situation do to conclusions; what's the DPG framework?*	§4.1.4 (DPG + sunset clause), §3.3 (cycle-lag robustness + dataset-construction consequences), §5.2.7 (four-measurement-layer reporting), Appendix E (approximate geometry), `findings/cycle_lag_analysis.md` §1 (three-vintage sandwich), `findings/methods_paper_draft.md` (companion methodology paper in draft)	The GIS and data-provenance methodology — the part justifying every qualified finding in Parts I and II.

This monograph is the source. Three shorter pieces can be drawn from it for different audiences: a peer-reviewed audit paper, a methods paper documenting the geometry-reconstruction pipeline (`findings/methods_paper_draft.md`), and a magazine feature. Drafts of each are in progress.

What to believe. Every number in this document is reproducible by running a script in `analysis/scripts/` against a dataset in `data/`. Every qualitative claim is anchored in a primary source pinned in `FROZEN_MANIFEST.md`. Where a finding depends on Derived Provisional Geometries (DPG — traced from commission map thumbnails because Elections Alberta had not released official shapefiles at the time of writing), the dependency is disclosed via the §4.1.4 sunset clause which binds the audit to re-run the finding within two weeks of official shapefile release.

Abstract

Alberta's 2025–26 Electoral Boundaries Commission tabled two 89-seat maps on March 23, 2026, which the Legislative Assembly set aside on April 16, 2026. This audit evaluates both proposals against the 2019 baseline using a multi-method structural framework and Monte Carlo ensemble analysis. Four non-partisan structural signals show consistent findings: the audit identifies structural divergence on population dispersion (minority 48% wider), Calgary zone asymmetry (12.2% vs 0.4%), Airdrie fragmentation (4 vs 2 districts)[1], and spatial anomalies (3 confirmed geometric anomalies chair-flagged vs 0; the chair's criticism spans 7 configurations total across §5.8.2 and §5.9.4). A fifth pre-registered dimension — municipal-boundary anchoring — did not survive canonical recomputation (both maps within the Canadian comparator norm). Full reconciliation in §5.8.5.[4] The joint outlier profile reaches Mahalanobis $p = 1.40\times10^{-6}$ against the 1,010,000-plan canonical ensemble. All Efficiency Gap magnitudes remain below the Stephanopoulos-McGhee 7% threshold (academic literature only; never judicially adopted). The audit records what public data show. It does not reach a legal conclusion. Primary statistics are exploratory-reproducible, with prospectively pre-registered replication studies on the Open Science Framework for Channel 3 (OSF:r3zm7 / AsPredicted #289,451) and the November 2026 Lunty scorecard (OSF:qsgy8 / AsPredicted #289,455). The other OSF entries (w2s8k, 6pt83, yvc7g, s58a6) were filed in proximity to but not strictly before the shapefile-release timestamp — see §4.3.2 and §5.5 for the verified-content disclosure (corrected 2026-06-12 per T1.7 R2 Ref #4); those registrations document drand-pinned seeds and substrate metadata, not test definitions, and are not cited as evidence of *prospective* pre-registration for the Ch1 + Ch2 framework.

Preface — scope, shapefile status, and intended venue

Shapefile status (updated 2026-05-06). Elections Alberta released the 2026 official shapefile package and the canonical polygon files were received by the audit on 2026-05-06. The

§4.1.4 sunset clause has been triggered and partially fulfilled: the 2026-proposal-shapefile trigger fired 2026-05-06 and the canonical ensemble re-run is complete. The November 2026 committee-map trigger remains pending. Authoritative canonical results are documented in §5.4.9 and `findings/post_audit_recompute_deltas.md`. The DPG-substrate runs documented in §5.4.1–§5.4.8 are preserved as the historical record of the pre-shapefile analysis.

The canonical ensemble uses official Elections Alberta shapefiles (`ea_majority_2026_eds.gpkg`, `ea_minority_2026_eds.gpkg`, EPSG:3400, 89 EDs each), 1,010,000 plans across 4 chains × 252,500 steps, base seed `get_canonical_seed("lunty-bootstrap") = 1432864451`. Results not dependent on the 2026 shapefiles remain unchanged: population equality in §5.1 (uses the commission's own per-ED population tables), signature detection in §5.3 (commission's published boundary descriptions), and the public-submission audit in §5.9.4 (commission's own submission archive).

Intended venue and distribution. The paper is prepared as a public-interest audit. Primary distribution targets are the project's public GitHub repository and a Canadian public-policy venue (*Alberta Views*-length feature and/or an SSRN / OSF preprint). Re-routing to a peer-reviewed journal (*Canadian Journal of Political Science*, *Canadian Public Policy*, *Election Law Journal*) is welcomed. The paper's structure, framework, and data are designed to make the re-routing straightforward. The methodology, pre-registration (§5.5), and falsifiability gates (§4.1.2) are intended to support review at the standard. A companion methodology paper on the DPG framework is in draft at `findings/methods_paper_draft.md`.

Cover visualization

The cover map (`data/maps/cover_art.png`; script `analysis/scripts/build_cover.py`) renders the minority commission's 2026 proposal as a population-density-weighted vote-share choropleth of Alberta's 4,765 Voting Areas. Each VA polygon is filled with a colour interpolated linearly from NDP-orange (rgb 0.92, 0.45, 0.10) to UCP-blue (rgb 0.13, 0.36, 0.62) against each VA's 2023 election-day two-party UCP share (colour scale normalised vmin = 0.30, vmax = 0.80). Fill intensity is modulated by $\log_{10}$ population density: the weight maps [−8, −3] $\log_{10}$ persons/m^2 to [0.10, 1.0], blending from near-ivory at the sparse end to the saturated partisan colour at the moderate end, then darkening to 30% of the base colour at the highest densities to preserve contrast in Calgary and Edmonton cores. VAs with no direct centroid-in-polygon join receive vote share from crosswalk-inherited 2019 parent EDs. VAs with no crosswalk parent receive the K = 25 nearest-VA aggregate as a final fallback. The minority commission's ED boundaries are overlaid in dark grey (linewidth 0.20). The provincial silhouette in accent red (linewidth 0.65).

The visualization corrects the geographic distortion produced by whole-riding colour fills on standard election maps, where large rural ridings dominate the visual field despite comparatively sparse populations. By saturating colour only where population density is high, the render makes the urban-core concentration of competitive voters visible as a primary feature of Alberta's political geography rather than a small detail lost in a sea of rural blue. This is directly relevant to the audit's structural findings: the boundary choices mattering most — the NW Calgary zone, the Airdrie split, the lasso corridor — fall precisely in the high-density zones where the map's partisan signal is strongest.

Author disclosure

Author and audit design: Will Conner, Mount Royal University, BSc Computer Information Systems (4th year student).

Prior and bias self-audit. Going into this project the author held the prior the UCP government's handling of boundary redistribution warranted scrutiny. The methodology is designed to produce the same numbers regardless of the prior. Three cases surfaced findings running against the prior and were retained in the report: (i) under 2019 vote input the partisan-bias asymmetry reverses sign (§5.2.3); (ii) the commission chair's "no public support" claim is upheld on three of seven configurations, not all seven (§5.9.4); (iii) the majority map's own MAD of 3,180 is tighter than the 2019 current-map baseline computed on 2021 census data (Appendix C), indicating the commission's majority did not introduce partisan looseness relative to the prior-cycle baseline. Full bias self-audit at `findings/partisan_bias_summary.md`.

AI use disclosure. Three large language models were used as analytical and writing assistants throughout this project: **Claude Pro Max** (Anthropic), **Gemini Pro** (Google), and **Codex** (OpenAI). Claude's contributions included drafting and revising report text, proposing analysis structure, identifying consistency gaps across documents, and surfacing methodological edge cases (e.g., the Vote Anywhere apportionment issue and the pre-registration disclosure requirement). Gemini Pro contributed to code review and cross-validation of analytical outputs. Codex contributed to scripting and data-processing tasks. All substantive claims — metric values, thresholds, data provenance, and code outputs — were verified against primary sources and script outputs by the author. No AI tool executed code or accessed external data independently. All script runs were performed by the author.

No commercial tools. No GIS desktop software (QGIS, ArcGIS) and no commercial election-analytics platforms were used. R 4.6 was used for one cross-validation script (`analysis/scripts/redist_crossvalidation.R`) using the Harvard `redist` package (SMC sampler) to provide algorithm-independent corroboration of the Python GerryChain ReCom headline finding (§5.4.9). No other R packages were used. Stata and SPSS were not used.

No paid data. All inputs are public. Every number in this audit is reproducible by running a script in `analysis/scripts/` against a dataset in `data/`.

Integrity tools. Falsifiability gates G0–G5 are built into the pipeline. Bias self-audit: `findings/partisan_bias_summary.md`. Uncertainty analysis:

`analysis/methodology/uncertainty_and_shapefile_impact.md`.

Computational stack:

- **Python 3.14** on Windows 11 (Python 3.9+ required by `setup.sh`)
- **pandas 2.x** — §5.1 population equality, §5.2 vote attribution
- **numpy** — numerical computation
- **openpyxl** — parsing the 2023 Statement of Vote Excel workbook (87 sheets, 1,973 poll records)
- **geopandas + pyogrio** — spatial operations for population aggregation and ED-to-CSD overlay; full Phase 4/5 execution blocked on 2026 shapefile release
- **shapely + pyproj** — polygon topology and projection, NAD83 / Alberta 10-TM (Forest), EPSG:3400 (the pipeline default; corrected 2026-06-12 per T1.7 R2 Ref #10 from the earlier mislabel "Alberta 3TM, EPSG:3776" — see `analysis/methodology/canonical_shapefile_log.md` for the full CRS reconciliation)
- **osmnx** — OSM road network extraction, prepared for Phase 4D fallback
- **gerrychain** — MCMC ensemble generation; canonical 1,010,000-plan run (4 chains × 252,500 steps) against official EA shapefiles (§5.4.9); DPG-substrate development runs of 250k–2,000,000 samples documented in §5.4.1–§5.4.8
- **pdfplumber** — PDF table extraction for commission report and Appendix E parsing
- **geopy + rapidfuzz** — geocoding and fuzzy-string matching
- **R 4.6** — cross-validation only; `redist` package (SMC sampler, Harvard); one script (§5.4.9)
- **git, GitHub CLI** — version control; public repository Ixby/alberta-electoral-boundaries-audit

Scripts authored for this audit:

Lane 1 — Statistical

SCRIPT	PURPOSE	REPORT SECTION
`packing_cracking_analysis.py`	B1–B6 partisan-bias metrics (symmetric, all three maps)	§5.2
`mcmc_ensemble_canonical.py`	1,010,000-plan ReCom neutral ensemble on canonical shapefiles	§5.4
`simulation_convergence_diagnostics.py`	Per-chain ESS, ρ_lag1, Gelman-Rubin $\hat{R}$	§5.4.8
`joint_outlier_score_canonical.py`	Mahalanobis D^2 joint outlier score	§5.4.9
`szat.py`	Swing-Zone Allocation Test — boundary-choice EG decomposition	§5.2.10
`validate_fisher_independence.py`	Spearman correlation between Ch1 and Ch2 channels	§5.5

SCRIPT	PURPOSE	REPORT SECTION
`intermap_permutation_test.py`	Directional inter-map partisan-gap permutation test	§5.2.11
`targeted_gerrymander_burst.py`	UCP-direction hill-climbing upper bound	§5.4.11
`targeted_gerrymander_burst_ndp.py`	NDP-direction symmetric hill-climbing lower bound	§5.4.11
`simulation_short_bursts.py`	Short-burst reachability from 2019 baseline	§5.4.10
`neighbour_drain_adjacency.py`	Packing-cracking adjacency signature (coupled-chain count)	§5.3
`drain_label_shuffle_null.py`	Label-shuffle permutation null for drain test	§5.3.5
`historical_eg_baseline.py`	Historical efficiency-gap baseline for Alberta elections	§5.2.9
`chen_rodden_alberta.py`	Chen-Rodden geography vs drawing decomposition	§5.2.7
`ecological_inference.py`	Ecological inference bounds on demographic voting	§5.2.8
`marginal_seats_analysis.py`	Marginal-seat uniform-swing analysis	§5.2
`november_tripwires.py`	Pre-registered automated checks for November 91-seat map	§5.5
`phase_b_scorecard.py`	Lunty committee tripwire detection scorecard (Phase B confirmatory test)	§5.5
`redist_crossvalidation.R`	R SMC cross-validation (Harvard `redist`; algorithm-independence)	§5.4.9

Lane 2 — Structural

SCRIPT	PURPOSE	REPORT SECTION
`electoral_forensics_population.py`	A1–A3 population equality (MAD, variance, legal-floor)	§5.1
`score_anchoring.py`	Municipal-boundary anchoring fraction per map	§5.8
`municipal_splits.py`	Municipal-split count across all three maps	§5.8
`polsby_popper.py`	Polsby-Popper compactness for all EDs	§5.8
`reock.py`	Reock compactness (smallest-enclosing-circle method)	§5.8
`a1_legal_baseline_2021_census.py`	2021-census A1 baseline for 2019 enacted map	§5.1
`s15_ratio_test.py`	EBCA §15(2) population-deviation compliance	§4.1
`justification_tests.py`	Testable-rationale verification for contested EDs	§5.9
`score_hybridization.py`	Hybrid-ED (city-splitting) metric	§5.3.5
`compactness_metrics.py`	Plan-level compactness summary	§5.8
`csd_community_splits.py`	CSD-level community-splits overlay	§5.8

SCRIPT	PURPOSE	REPORT SECTION
majority_symmetry_counter_test.py	Test-selection symmetry counter-test	§5.6
mcmc_anchoring_ensemble.py	Anchoring mini-ensemble via CSD edge-crossing metric	§5.8

Robustness and sensitivity

SCRIPT	PURPOSE	ADDRESSES
advance_vote_splat.py	Advance-ballot VA apportionment (proportional smear)	H8
advance_vote_sensitivity.py	Election-day vs full-vote substrate sensitivity	H8
seats_at_50_50_regional.py	Regional-swing robustness against uniform-swing assumption	H5
cross_election.py	Three-election (2015/2019/2023) direction check	H7
third_party_sensitivity.py	Third-party vote allocation sensitivity	H6
attribution_sensitivity_check.py	MAUP area-weighted attribution sensitivity	§5.4
monte_carlo_ci.py	Monte Carlo confidence intervals over modeling choices	§5.4
szat_2019_baseline.py	SZAT applied to 2019 enacted baseline	§5.2.10
score_natural_anchoring.py	Natural-feature (highway/river) anchoring secondary check	H4
va_attribution_area_weighted.py	Area-weighted MAUP check for VA-to-ED attribution	§5.4

Submission analysis (§5.9)

SCRIPT	PURPOSE
submission_search.py	Keyword search across 1,140+ public submissions
submission_sentiment_llm.py	LLM zero-shot sentiment classification
submission_sentiment_llm_full.py	Full-corpus LLM classification across all configurations
hansard_sentiment_llm.py	Hansard hearing transcript sentiment
aggregate_sentiment_intensity.py	Intensity aggregation by configuration
sentiment_intensity_score.py	1–3 intensity scoring per row
cross_reference_submitters.py	Rationale cross-reference (CONTRA_COMMISSION flags)
compute_kappa.py	Cohen's κ inter-rater reliability
validation_sample.py	Stratified validation sample for IRR

Data preparation

SCRIPT	PURPOSE
build_canonical_va_votes.py	Canonical 2023 VA election-day vote file
phase4c_canonical_attribution.py	Phase 4C canonical VA-to-ED attribution
build_cross_election_va.py	2015/2019 cross-election VA vote files
submission_ocr.py / submission_ocr_recovery.py	OCR for image-only submission PDFs

External data

SCRIPT	PURPOSE
`338canada_scraper.py`	338Canada per-riding Alberta projections
`338canada_historical.py`	Historical 338Canada Alberta snapshots 2020–2026
`338canada_reallocate.py`	Projection reallocation through hybrid-ED crosswalks
`canadian_base_rate_compute.py`	Canadian provincial EG base rates
`_fetch_osm_natural.py`	OSM highway and river data for natural anchoring

Data sources:

- Elections Alberta Statement of Vote 2023 (`data/2023_results.xlsx`)
- Alberta Electoral Boundaries Commission final report, March 23, 2026 — extracted populations and variance tables, map images (`maps/*.jpg`)
- Elections Alberta GIS resources page — 2026 shapefiles received 2026-05-06; canonical ensemble re-run against official geometry (§5.4.9)
- Statistics Canada 2021 Census Dissemination Area populations and shapefiles (`data/alberta_2021_da_populations.csv`, `data/alberta_2021_das.gpkg`)
- Alberta Treasury Board Office of Statistics and Information quarterly population estimates
- StatsCan Table 17-10-0009 — quarterly provincial estimates
- StatsCan Table 98-10-0459 — 2021 Census journey-to-work by CSD

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1. Introduction

Contribution. (d) The primary contribution is the first systematic computational audit of an Alberta provincial electoral boundary commission's output, evaluated against established Canadian independent-commission practice and statutory norms. The audit is situated within Canadian political-science and legal frameworks: the *Saskatchewan Reference*'s "effective representation" standard, the Courtney–Pal lineage of scholarship on boundary-commission discretion, the statutory mandate of Alberta's *Electoral Boundaries Commission Act*, and the institutional precedents set by the 2022 federal sub-commission, Quebec 2011, and other provincial cycles. This Canadian-institutional framing distinguishes the audit from US-focused partisan-gerrymandering measurement. It asks whether one faction's boundary choices within a Canadian independent commission exceed the discretion space Canadian law and practice establish.

(a)–(c) Supporting contributions. The audit achieves this primary finding through three methodological supports: **(a)** a falsifiability-gated ensemble-comparison framework committing computational parameters to a public randomness beacon before analyzing shapefiles (preventing cherry-picking of parameters after seeing results); **(b)** a pre-registration discipline (OSF qsgy8, r3zm7, 6pt83, yvc7g, s58a6) anchoring test definitions and sensitivity boundaries before execution; and **(c)** complete data provenance documentation and reproducible code (all scripts checked in, all data sources listed with access dates, with a 1,010,000-plan canonical ensemble enabling third-party replication).

Specifics. The audit evaluates Alberta's 2025–26 Electoral Boundaries Commission's two final proposals (majority and minority reports, tabled March 23, 2026) by constructing a 1,010,000-plan neutral ensemble under identical statutory constraints (±25% population band, contiguity, compactness, community of interest per the EBCA §12(3) mandate) and comparing each real map against the ensemble statistics. It finds the minority proposal diverges from the majority on measurable structural dimensions across multiple correlated dimensions (not all independent — see dependence disclosure in §4.3.1); the directional pattern of divergence is stable across vote inputs (2015, 2019, 2023) and stress tests; the divergence's direction consistently favors the governing party when read against any available Alberta electorate; and the underlying rationales offered by the minority commission faction fail falsifiability checks against cleaner legal alternatives (documented in §5.9.4). The paper documents every source, script, and computational step so future users — academic auditors, policy analysts, courts, or future commissions — can run the same pipeline on new commission output.

Case context and procedural pivot. Alberta's Electoral Boundaries Commission delivered two final reports on March 23, 2026: a majority report and a minority report proposing incompatible 89-seat maps. Three weeks later, on April 16, 2026, the Alberta Legislative Assembly passed Motion 19 setting aside both reports and establishing a Special Select Committee of five MLAs to draft a 91-seat map by November 2, 2026. This procedural pivot — replacing an independent commission's drafting process with a government-chaired committee mid-cycle — is evaluated in §5.9 against Canadian comparators and positions the audit's findings within a broader questions about electoral commission independence and public trust.

Exploratory vs. confirmatory status. The audit's statistical tests fall into two categories readers should distinguish. The primary ensemble-based tests — Ch1 (Mahalanobis joint-tail) and Ch2 (SZAT bootstrap, exploratory only; the standard-permutation result, $p=0.0024$, does not survive a block-permutation null correcting for spatial autocorrelation: $p\approx0.19$, see §5.2.10) — are **exploratory**: computational seeds were committed to the public drand League of Entropy beacon before the official shapefile release (establishing results were not cherry-picked after seeing the data), but the specific test names and combination method were not filed in any OSF pre-registration document before execution. The earlier Fisher combination of Ch1 and Ch2 ($T = 39.03$, $p = 6.87\times10^{-8}$) is retired — it assumed channel independence that the two channels do not have and relied on a Ch2 result that does not survive the appropriate spatial null. The operative joint headline is the **Bonferroni upper bound $p \le 2.80\times10^{-6}$** (Ch1 alone; ×2 is a conservative factor for the two channels examined). These findings are reproducible and temporally anchored, but they should be treated as requiring independent replication before confirmatory weight is assigned. The Ch3 neighbour-drain test (pre-registered at OSF r3zm7) and the November 2026 Lunty committee scorecard (OSF qsgy8) are prospectively pre-registered confirmatory tests. The four surviving non-partisan-bias signals in §1.1 (population equality, Calgary zone gap, Airdrie fragmentation, chair-flagged anomalies) do not depend on the statistical tests at all. They derive from the commission's own published tables, maps, and record, and their evidentiary status is independent of the pre-registration question. (Municipal anchoring, a fifth pre-registered dimension, did not survive canonical recomputation — see §5.8.5.)

1.1 HEADLINE FINDINGS IN PLAIN LANGUAGE

Before the technical caveats below, the audit's structural findings summarise cleanly. Using the same methodology applied symmetrically to all three maps:

- The **minority 2026 proposal spreads population more unevenly** across districts than the majority proposal (Median Absolute Deviation from provincial average: 4,707 versus 3,180) — a 48 % wider dispersion.
- The **minority map's Calgary districts** show a 12.2 percentage-point geographic-zone asymmetry (urban-core districts smaller, suburban-ring districts larger) versus 0.4 pp on the majority map.
- The **minority map splits the City of Airdrie four ways** across different electoral divisions; the majority concentrates 73.8 % of Airdrie's population inside a single electoral division (Airdrie-East), while the minority's largest Airdrie ED holds only 59.2 % and three other minority EDs draw in the remainder — including one ED ("Calgary-Foothills-Airdrie West") whose perimeter has 0 % anchoring on Airdrie's gazetted city limit despite carrying "Airdrie" in its name. See `findings/airdrie_highway_pretext.md` for the per-ED measurement and the empirical refutation of a Highway Anchoring Defense at this carve.
- A fifth pre-registered structural dimension — municipal-boundary anchoring against Statistics Canada CSD edges — did not survive canonical-geometry recomputation: both 2026 maps fall within the 70–85 % Canadian comparator norm (minority 72.0 %, majority 80.0 % under the *per-segment* implementation in `score_anchoring.py`; the *report definition* requires contiguous runs of ≥ 1 km of anchored segments, a contiguity filter the implementation does not apply — flagged 2026-06-12 per T1.7 Referee #10 D8). The per-segment numbers are slightly more permissive than the stated definition; the minority sits 2.0 pp above the 70 % floor, so a per-ED run-length-filtered recomputation *could* in principle push it below the norm. Round-1 of this disclosure asserted "the gap is well wider than any reasonable contiguity-filter correction" — that claim was not computed and is retracted 2026-06-12 per T1.7 R2 Ref #10; pending the filter implementation (T4.10-anchor) the conclusion is **provisional** rather than safe. Since municipal anchoring is no longer load-bearing (the headline rests on the four geometry-independent dimensions + the MCMC ensemble), the provisional status does not affect the audit's primary findings. The earlier DPG-derived 14.5 % / 71.0 % / 4.9× figures cited during the audit's pre-shapefile phase did not carry forward to recomputation against official Elections Alberta shapefiles. The 500 m snap tolerance carried over from the DPG ±500 m positional-error budget onto exact canonical geometry (no sensitivity

sweep at 100 m / 250 m) is also flagged per T1.7 Referee #10 D9 — a tolerance sweep at three values is queued as T4.10-anchor. Full reconciliation in §5.8.5.[4]

- **Three geographic anomalies** on the minority map were flagged by the commission chair (Rocky Mountain House–Banff Park extension; Calgary-Nolan Hill–Cochrane lasso; Olds-Three Hills-Didsbury → N Airdrie community capture); zero were flagged on the majority map.
- The **minority's own 25 published rationales** are, on five of six contested configurations, options the minority did not take when cleaner statutory-compliant alternatives were available. (A seventh configuration — Lethbridge / Taber-Warner federal-boundary match — was previously listed but has been removed: the minority report does not in fact make the federal-boundary claim, and the methodology check at `analysis/methodology/lethbridge_federal_boundary_check.md` could not locate any underlying source.)

The four surviving non-partisan-bias signals all point in the same direction and all survive the stress-tests documented in §1.2 below. (The fifth, municipal anchoring, did not survive canonical recomputation — see §5.8.5.[4]) The partisan-bias signal (the extent to which the maps would produce different seat counts from the same votes) is reported in the abstract as a two-measurement disagreement; §1.2 explains the caveats constraining the magnitude claim.

The headline rests on Ch1 (Mahalanobis joint tail). The ensemble test (Mahalanobis Ch1, §5.4.9) asks: *is this map extreme relative to over one million neutral, randomly-drawn Alberta maps?* The minority's joint distance against the canonical 1.01M-plan ensemble gives $p = 1.40\times10^{-6} \approx$ 1 in 714,000. A secondary SZAT boundary-choice test (Ch2, §5.2.10) returned $p = 0.0024$ under the original i.i.d.-flip null but **does not survive a contiguity-respecting block-permutation null** (block-permuted $p \approx 0.19$; T1.10b closed 2026-06-12 per T1.7 R1 Ref #1; see `findings/szat_block_permutation.md`). The audit therefore reports Ch1 as the load-bearing partisan-bias finding; the dependence-robust **Bonferroni upper bound $p \le 2.80\times10^{-6}$** is preserved (= 2 × min(Ch1, Ch2) under the original Ch2; numerically identical when Ch2 is removed because Ch1 was the binding minimum). The earlier "$p = 6.87\times10^{-8} \approx$ 1 in 15 million" Fisher figure assumed channel independence and overstated the joint significance (§4.3.3 explains the revision).

1.2 Modelling-uncertainty caveats

Three modelling-uncertainty tests materially narrow the partisan-bias magnitude claim while leaving the structural findings in §1.1 intact. These are reported up-front for transparency. The underlying methodology is in `analysis/review/design_critique.md` and the Monte Carlo script `analysis/scripts/monte_carlo_ci.py`.

1. Monte Carlo sensitivity interval over modelling choices crosses zero. N=2,000 samples varying urban weight (0.55–0.85), rural baseline (0.26–0.36), and per-hybrid jitter (±0.10). Minority-majority EG asymmetry: mean −1.22 pp, median −1.44 pp, **2.5th–97.5th percentile**

sensitivity interval [−3.04, +0.76] pp. Direction consistency: 90.5% of samples show minority more UCP-favorable. Classical 95% significance is **not** defensible. The direction holds in 90.5% of draws across the full parameter sensitivity range — a sensitivity result, not a frequentist confidence level.

2. Declination metric (Warrington 2018) — sign-convention correction landed 2026-06-12 (Amendment 10). The implementation at `mcmc_ensemble.py:215` was returning the operand pair in inverted order relative to Warrington 2018, producing values with the wrong sign. The corrected values are: 2019 = +0.045 (p91.05), Majority = −0.027 (p20.4), Minority = **+0.077 (p98.79)** against the canonical 1.01M-plan ensemble. Under the corrected sign, declination *agrees* with EG, mean-median, and seats@50/50 on the UCP-favoured tail — four of four partisan-bias metrics in the same direction. The prior "disagreement" reading in earlier drafts of this report was a code-side artefact; see Amendment 10 in `findings/pre_registration_amendment_log.md` for the verification and chain-CSV migration.

3. 2019 cross-election check (corrected 2026-06-13 per T1.7 R2 S4). Under v0_7 partial-coverage attribution, 2019 vote totals produced Majority EG +0.30% / Minority EG +0.90% — an apparent direction-flip relative to the 2023 reading. An earlier v0_8-based run (`cross_election_v8_full.csv`, 2026-04-25) claimed "neither individual map's EG sign flips under 2019"; that claim is **retracted**. **Under canonical Option C (area-proportional VA attribution on the official 4,765-VA substrate, 100k-plan ReCom ensemble, seed 3562959107; §5.2.8):** 2019 EG values are Majority −1.32% (p48.3) and Minority −0.49% (p70.4) — both negative, meaning both maps are NDP-favourable under 2019 votes. The EG sign DOES flip across elections; the earlier v0_8 no-flip claim was an attribution artefact. The inter-map *asymmetry* direction reverses under 2019 (minority less UCP-favourable by +0.83 pp under 2019, vs more UCP-favourable by −3.92 pp under 2023). The asymmetry direction reversal is documented in §5.2.8. The canonical Option C finding — that EG flips for both maps under 2019 — does not alter the audit's primary claim, which rests on the 2023-context canonical ensemble (the appropriate reference for a 2026 map): MM p99.98, Declination p98.79, s50 p99.99, joint D^2=32.67 p=1.40×10^{-6}.

What survives these tests unchanged: - §5.1.1 population distribution variance (CSV-sourced, election-independent): minority MAD is 48% wider than majority. - §5.1.2 Calgary geographic-zone gap: minority 7.7–12.2%; majority 0.36–0.39%. Not vote-based. - §5.8.2 visual spatial anomalies: 3 minority anomalies confirmed on published maps. - §5.8.4 community splits: Airdrie 4 vs 2, Cochrane merged vs intact, Chestermere partial split vs intact. - §5.9 procedural concerns: government-controlled replacement of drafting process, qualitative.

What is narrowed: - §5.2 partisan-bias *magnitude* claims. The central point estimate of 1.41 pp (w=0.85) is within Monte Carlo noise (sensitivity interval crosses zero). The direction holds in 90.5% of Monte Carlo draws across modelling uncertainty, which is a defensible directional claim but not classical 95% significance. - The "minority gives UCP 2 more seats in a tied election" line. Under Monte Carlo, minority NDP@50/50 has 95% CI [41, 47] vs majority [43, 46] — overlapping. The 1-seat gap size is stable across 2023 votes and April 2026 polling inputs, but the *direction* of the 1-seat gap flips between them (UCP +1 on minority under 2023; NDP +1 on minority under April 2026 polling). A structural-invariance claim was not supported by the historical stability test and has been retracted; see §5.2.3. The magnitude CI crosses zero. - Under Amendment 10 (declination sign correction, 2026-06-12) the four partisan-bias metrics agree on direction; the earlier "five of six" framing was an artefact of the now-fixed sign error. The honest precision is "directionally consistent across six tested dimensions, no metric pointing the opposite way once code-side errors are corrected; consistency is across *correlated* dimensions, not statistically independent ones — see Referee #14/#16 findings and the §6 wording fix."

Defensible synthesis. The minority 2026 proposal shows measurable structural differences from the majority in four areas: population distribution (MAD 48% wider), Calgary geographic-zone balance (12.2% gap vs 0.4%, robust across two classification rules), community-of-interest treatment (Airdrie residents distributed across 4 minority EDs with no ED holding a majority of city residents, vs the majority's concentration of 73.8% inside Airdrie-East — same dispersion pattern visible in Lethbridge and Red Deer per §5.6), and visible boundary shape (three confirmed anomalies). None of these depend on vote data.

A fifth dimension is the rationale-failure pattern: five of the six contested minority redraws (after the Lethbridge / Taber-Warner federal-boundary claim was withdrawn — see methodology check `analysis/methodology/lethbridge_federal_boundary_check.md`, which establishes the minority report does not in fact make the federal-boundary claim attributed to it) have cleaner options available satisfying the statutory population and area limits while matching documented public submissions. The sixth (St. Albert-Sturgeon) is constraint-forced: both the majority and the minority independently arrive at the same two-district structure under the Act's constraints (see §5.9.2 and `analysis/methodology/reference/minority_rationales_validation.md`).

The partisan-bias consequences are directionally UCP-favorable for the minority in 90.5% of modelling-jitter samples using 2023 vote attribution, outlier-flagged at p99.98 on mean-median (UCP-tail) and at **p98.79 on declination (UCP-tail; sign convention corrected 2026-06-12, see Amendment 10)** against the 1,010,000-plan canonical ReCom neutral ensemble on official Elections Alberta shapefiles (see §5.4). The direction reverses under 2019 vote attribution. Under April 2026 338Canada polling, the crosswalk-reallocation probe finds a 1-seat NDP advantage on the minority, but the canonical uniform-swing probe (applied to Phase 4C per-ED results) finds the majority still gives NDP more seats — see §5.2.3 for the probe-level reconciliation (C7). The core claim is the minority has more structural irregularities and more rationale failures than the majority. A specific partisan-seat-shift magnitude is less defensible and sensitive to election baseline. The procedural concern about the April 16 government action stands separately from the partisan-math questions.

Methodological scope. This paper applies a symmetric, falsifiability-gated framework to both 2026 proposals and the 2019 baseline, producing reproducible estimates of population equality, partisan bias, geographic coherence, and procedural fairness. It pre-registers the test battery against a forthcoming third map — the November 2026 MLA-committee 91-seat proposal — so the audit can be replayed against new evidence rather than reinterpreted after the fact. It documents the data-provenance caveats arising when 2026 shapefile release is gated by legislative adoption, leaving explicit placeholders for the checks remaining pending.

Scope. The paper does not reach a legal conclusion. It is evidentiary: it records what the public data show under identical methodology applied to all three maps, so any interested party — court, legislator, commissioner, journalist, academic — can repeat the computations, challenge the inputs, and reach their own judgments. The court-ready interpretation is deferred to §8 (Conclusion) and Appendix F (Legal Interpretive Note), drawing on Pal's (2015) analysis of boundary-commission discretion and electoral-district design.

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2. Background and Prior Work

The analysis rests on four bodies of prior work: US partisan-gerrymandering measurement, Canadian independent-commission practice, population-equality jurisprudence, and computational redistricting methodology.

Institutional context: Canadian commissions versus US state legislatures. The US literature on partisan gerrymandering — including the efficiency gap and the ensemble approaches this audit uses — was developed primarily to detect and litigate intentional partisan manipulation by *partisan mapmakers*: US state legislatures, which draw their own districts without independent commissions and have a structural incentive to engineer outcomes. In the US federal courts, however, partisan-gerrymandering claims have been declared non-justiciable: *Rucho v. Common Cause*, 588 U.S. 684, 139 S. Ct. 2484 (2019) (parallel citation restored 2026-06-12 per T1.7 R2 Ref #9 — round-1 of the Rucho-text remediation mislabeled the S. Ct. cite as a "slip-opinion reporter"; in fact it is West's *Supreme Court Reporter*, a valid parallel cite, and is restored alongside the U.S. Reports cite here), held 5–4 the federal judiciary lacks "judicially discernible and manageable standards" for evaluating partisan-bias claims and therefore cannot adjudicate them. This is an *institutional* holding about justiciability, not a *methodological* holding that partisan-bias metrics need corroboration to be reliable. The *Whitford* plaintiffs lost separately, on *Article III standing* grounds (*Gill v. Whitford*, 585 U.S. 48 (2018) — round-1 conflated this with Rucho's justiciability holding; corrected 2026-06-12 per T1.7 R2 Ref #9). The Whitford plaintiffs nonetheless presented the same ensemble + structural + procedural package this audit deploys (Mattingly et al. and Chen-Rodden ensembles, direct intent evidence, predominance findings); the Court did not reach the merits. This holding closes the federal courts to efficiency-gap and partisan-symmetry arguments, rendering the US academic literature on these metrics non-actionable in US federal law. The same metrics remain viable in US state constitutional courts (most prominently *League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018), where ensemble evidence succeeded).

The *Rucho* holding's relevance to this audit is therefore **not** "metrics need corroboration to be reliable" (the corrected reading 2026-06-12 per T1.7 Referee #9), but rather: in a jurisdiction where partisan-gerrymandering is justiciable (which Canada is, via the *Saskatchewan Reference* effective-representation standard — see §2), partisan-bias evidence becomes evidentiarily strongest when it carries direction-of-effect and mechanism-of-effect *both*. The audit's two-lane architecture is motivated by that broader evidentiary practice, not by *Rucho*'s specific holding. This audit therefore deploys a "two-lane" approach (Lane 1: partisan-bias metrics in §5.2; Lane 2: structural and procedural findings in §5.1, §5.3, §5.8, §5.9). Neither lane depends singularly on the other. The audit's findings rest on directional consistency across multiple independent dimensions rather than on the magnitude of any single metric.

Canadian provincial and federal electoral commissions are constitutionally distinct from US state legislatures. They are staffed by judges and independent appointees rather than partisan legislators, are governed by statutes specifying non-partisan criteria, and operate in a setting where the *Saskatchewan Reference* [1991] 2 SCR 158 standard of "effective representation" rather than strict partisan proportionality frames the legal test. Critically, *Rucho*'s non-justiciability holding governs US federal courts only and does not carry over to Canadian provincial jurisprudence, where the Saskatchewan Reference provides an explicit constitutional standard for assessing boundary-commission output. This institutional distinction is material. In the US context, partisan-bias metrics are evidentiary tools unable to enter federal litigation. In the Canadian context, the same metrics are evidentiary inputs to a constitutional test a court is able to apply. The methodology is agnostic to this institutional distinction. Its inferential weight depends on context, and the context is disclosed here. This matters for how partisan-bias metrics carry inferential weight. In the US context, a high efficiency gap is direct evidence of legislative manipulation by the legislature drawing its own map — but federal courts will not hear such claims. In the Canadian context, the same metric applied to a *minority* faction's commissioners within a nominally independent process carries a different warrant. The inference is not "the legislature manipulated its own map" but "one faction within a supposedly non-partisan commission drew boundaries whose partisan signature departs from what the independent (majority) faction drew under identical constraints," and a court applying the Saskatchewan Reference standard can weigh this evidence as part of its effective-representation analysis.

Terminology. "Gerrymandering" as used throughout this paper is a political and analytical term with no equivalent legal category in Canadian law. In contrast to US jurisprudence — where partisan gerrymandering is at least a named (if non-justiciable) claim after *Rucho* — Canadian law defines no gerrymandering standard. The applicable legal framework is effective representation under *Charter* s.3, as articulated in the *Saskatchewan Reference*, and statutory compliance with the EBCA. This audit's findings are evidentiary inputs to those constitutional and statutory questions, not to any gerrymandering-specific legal test.

Partisan-bias measurement. Stephanopoulos and McGhee (2014, 2015, 2018) introduced the efficiency gap (EG) as a single-number measure of wasted-vote asymmetry and proposed a 7% threshold for investigable bias. McDonald and Best (2015) advanced the mean-median gap as a complementary measure of distributional skew. Gelman and King (1994) formalised seat-vote-curve symmetry as a Bayesian estimator, building on Grofman (1983) and King and Browning (1987). Warrington (2018, 2019) introduced declination and documented EG and declination can disagree on a non-trivial fraction of US-state plans — a finding directly relevant to the Alberta disagreement reported in §5.2.4. Katz, King, and Rosenblatt (2020) argue that any valid measure of partisan bias must satisfy symmetry and critique single asymmetric metrics as inadequate; this audit's ensemble approach satisfies their symmetry criterion, which the stress-test gate RT2 implements. The 7 % threshold is academic-literature authority only — never judicially adopted (see §5.2.8).

Natural packing vs engineered packing. Chen and Rodden (2013) argue urban-concentrated parties are systematically disadvantaged by neutrally-drawn maps through packing of their voters into city cores. §5.2.5 of this audit applies the Chen-Rodden framework to Alberta and finds the *direction* prediction transfers but the *mechanism* does not: Alberta's UCP-favourable baseline comes from dispersed rural UCP-winning margins, not from NDP urban packing. This matters for what any 2026 map's partisan bias can legitimately be attributed to.

Canadian redistribution practice. Courtney (2001, 2004) provides the authoritative scholarly treatment of the independent-commission model across Canadian provinces; Courtney (2001) chs. 6–7 map the interplay between population equality and community-of-interest discretion underlying the §5.1 and §5.8 findings in this audit, and chs. 10–11 address public-hearing obligations and the scope of judicial review of commission output. Pal (2015) analyses the intersection of boundary-commission discretion with the Charter §3 right to vote, arguing the *Saskatchewan Reference*'s "effective representation" standard leaves commissioners substantial latitude reviewable only at the outer margins — not for partisan preference proximity — and the distinction between legitimate design preference and legally reviewable partiality turns on the same factual record the audit assembles. The discretion-versus-design-choice framework Pal develops is the analytic anchor for the §5.9.3 procedural assessment. The constitutional benchmark, *Reference re Provincial Electoral Boundaries (Saskatchewan)*, [1991] 2 SCR 158, establishes "effective representation" — not strict population parity — as the standard against which boundary-commission output is measured. The Charter §3 right to vote developed by *Figueroa v. Canada (Attorney General)*, [2003] 1 SCR 912, and *Frank v. Canada (Attorney General)*, [2019] 1 SCR 3, forms the backdrop without applying directly to redistribution. §5.9.3 compares the April 16, 2026 Alberta pivot against the closest historical Canadian comparator (Quebec 2011 — the National Assembly's refusal to proclaim a completed commission delimitation under Bill 132) and positions it against the SCC's April 22, 2026 ruling on the Legault government's 2024 freeze statute.

Constitutional framing: effective representation versus variance tests. The *Saskatchewan Reference* standard establishes "effective representation" as the constitutional benchmark for electoral boundary commissions, but "effective representation" is not a variance ceiling or population-equity metric. Rather, it is a multi-factor constitutional test — articulated in *Saskatchewan Reference* para. 33 — weighing quantitative factors (population distribution, variance in district population) *alongside* non-quantitative factors (geography, historical boundaries, community of interest, minority representation) as evidence bearing on whether the electoral system serves the *practical* ability of electors to cast a meaningful vote.

The *Saskatchewan Reference* para. 33, delivered by McLachlin J., establishes "relative parity of voting power" must be weighed against permissible deviations for "geography, community history, community interests and minority representation." Critically, the Court did not prescribe a bright-line tolerance for population variance or other quantitative measures. Instead, it requires a holistic assessment in which deviations from strict equality are constitutionally defensible *if justified by the non-quantitative factors*. *Raîche v. Canada (Attorney General)*, 2004 FC 679, applied this standard to the federal Electoral Boundaries Commission for New Brunswick (the "per Kelen J." attribution that appeared in round-1 of the Raîche correction was unverified — possibly Shore J. per T1.7 R2 Ref #8's recollection — and is dropped pending counsel verification of the actual judge name; the holding does not depend on it). The Court did **not** find a Charter s.3 violation — applying *Carter*'s deferential "reasonable commission" standard, the boundary transfer survived the constitutional challenge. The violations the Court found were *statutory*: ss.15(1)(b) and 15(2) of the federal Electoral Boundaries Readjustment Act (failure to give adequate weight to community of interest) and Part VII of the Official Languages Act. The remedy was a referral back to the Commission, not boundaries "struck down" under s.3. Earlier drafts of this report described Raîche as a "s.3 strike-down" with attribution to "Heneghan J."; both attributions were corrected 2026-06-12 per T1.7 Referee #8. The case stands for the proposition that statutory community-of-interest protection can require revisiting boundaries even when the s.3 reasonableness threshold is not crossed — a *weaker* claim than the audit previously made, and one that *strengthens* commission deference under the Charter rather than weakening it.

Cassista v. Canada (Attorney General), 2014 FC 398 was cited in earlier drafts of this report (and in `analysis/methodology/reference/peer_review_legal.md:97-99`) as the "boundaries-sustained" companion to Raîche, completing a discretion-space bracket. The citation has been **flagged for verification 2026-06-12** per T1.7 Referee #8: the case was not located in CanLII or Elections Canada's court-cases index in the referee's check. Pending independent counsel verification of the citation, the audit (a) does not rely on Cassista for any verdict-bearing claim — the Saskatchewan Reference and Raîche-as-corrected suffice for the discretion-review framework; (b) preserves the existing prose with this banner; and (c) queues a substitute citation (candidates: *Dixon v. British Columbia* (AG) (1989) 59 DLR (4th) 247 (BCSC); *Reference re Electoral Divisions Statutes Amendment Act* (1994) 119 DLR (4th) 1 (Alta CA)) if verification fails. In neither Raîche nor any verified comparator did the court examine partisan asymmetry directly; the constitutional inquiry centred on whether the non-quantitative factors had been weighed. This doctrinal architecture means partisan asymmetry, standing alone, does not trigger a Charter s.3 violation under the Saskatchewan Reference standard — the asymmetry must be paired with either (a) unexplained deviations from population parity unjustified by geography, community of interest, or other permissible factors, or (b) evidence the non-partisan factors (geography, etc.) have been distorted *in service of* partisan outcomes rather than in service of legitimate statutory purposes.

The audit's quantitative findings (partisan-bias metrics in §5.2, population-variance medians in §5.1, municipal-boundary anchoring in §5.8) are inputs to this constitutional test, not proxies for the test's output. No single metric — nor any linear combination of metrics — constitutes a determination of whether a map meets or fails the constitutional threshold. Pal (2015) argues commission outputs are reviewable under the deferential reasonableness standard of *Canada (Minister of Citizenship and Immigration) v. Vavilov*, 2019 SCC 65 (with respect to statutory interpretation), and the *Carter* reasonable-commission standard with respect to Charter s.3 — both standards leaving commissioners a broad "discretion space" encompassing legitimate design preferences. Earlier drafts of this paragraph (and the parallel paragraph in Appendix F) used the pre-Vavilov "manifest unreasonableness" vocabulary — corrected 2026-06-12 per T1.7 Referee #12; the post-Vavilov framework collapses the patently/manifestly-unreasonable distinction into a unified reasonableness standard with correctness as the exception. The boundary between justified discretion and reviewable partiality is defined by whether the non-partisan factors genuinely constrained the choices or were invoked *post hoc* to rationalize a predetermined partisan outcome. The §5.9.3 procedural analysis documents evidence on this point: whether the audit's six-dimensional pattern — partisan asymmetry *coupled with* population-variance distortions not explained by geography or community boundaries, *coupled with* procedural departure from the pattern established by the majority under identical constraints — rises to the level of manifest unreasonableness under Pal's framework is a question for a reviewing court.

The distinction between §3 of the *Canadian Charter* (the right to vote, which the *Saskatchewan Reference* operationalizes) and §15 (equality of political rights) is material: the audit measures deviations in vote efficiency and partisan bias under neutral simulation, providing factual context for whether a commission's output warrants scrutiny. A court's assessment of whether the commission's process was procedurally sound, or its output's partisan asymmetry beyond the justified margins of discretion, is a separate legal question in which non-metric considerations — geography's constraints, rural-urban representation trade-offs, minority communities' distinct electoral needs — weigh equally with the audit's numbers. This report assembles evidence across six dimensions. A finding six indicators point in the same direction is a statement about the empirical pattern, not a legal conclusion. The §5.9.3 procedural analysis and §12 commentary clarify the extent to which the audit's findings bearing on discretion exceed the outer margins Pal identifies.

Compactness and computational methods. Polsby and Popper (1991) and Reock (1961) supply the two compactness metrics referenced throughout. Barnes and Solomon (2021) document their implementation sensitivity. DeFord, Duchin, and Solomon (2021) introduce the ReCom MCMC family used for the neutral-ensemble baseline in §5.4. Cannon, Goldbloom-Helzner, Gupta, Matthews, and Suwal (2023) introduce the short-bursts hill-climbing procedure for exploring biased-but-legal maps. This audit applies their method in §5.4.8 to test empirically whether a non-neutral procedure can reach the minority's seats@50/50 reading under EBCA constraints. Herschlag, Ravier, and Mattingly (2020) demonstrate the ensemble-comparison methodology the Alberta audit applies here. Altman and McDonald (2011) articulate the four-axis redistricting-audit discipline — population equality, compactness, political fairness, community of interest — whose consistency-across-dimensions framing § E of this paper draws on. Fifield, Imai, Kawahara, and Kenny (2020) discuss the empirical-validation requirements for redistricting simulation results.

Pre-registration and evidentiary discipline. Nosek et al. (2018) and Munafò et al. (2017) codify the pre-registration discipline the audit's November 2026 test protocol follows. American Statistical Association (2016, 2019) statements on p-values and graded evidence guide the audit's reporting of directional findings at sub-threshold magnitude. These disciplines matter because the paper's central claim — six dimensions pointing in the same direction, with three of the four partisan-bias metrics above the pre-registered p95 threshold (MM p99.98, declination p98.79, seats@50/50 p99.99) and EG sub-threshold at p94.4 — is inferentially valid only under pre-registered test selection and reported-versus-hidden-test symmetry. (Note: the pre-Amendment-10 formulation read "none individually at classical 95% significance"; Amendment 10's sign correction moved declination from the sub-threshold NDP tail to the UCP-favoured p98.79 reading; corrected per T1.7 R2 Ref #5 2026-06-12.)

The audit's methods map onto this literature as follows: B2–B6 (§5.2.1) implement Stephanopoulos-McGhee, McDonald-Best, Gelman-King (and the broader Grofman, King and Browning partisan-symmetry foundation), and Warrington directly; §5.4 implements DeFord-Duchin-Solomon ReCom against a 1,010,000-plan neutral ensemble (4 chains × 252,500 steps); §5.2.5 validates Chen-Rodden for Alberta; §5.9.3 positions the procedural finding against Courtney's Canadian comparator sample (Courtney 2001, chs. 3–4) and Pal's (2015) discretion framework. Where the Alberta context departs from US or Canadian prior work, the departures are documented rather than elided.

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3. Data

This section consolidates the audit's data-provenance disclosures — primary sources, their `FROZEN_MANIFEST.md` anchor dates, known coverage caveats, and robustness to cycle-lag. All downstream analyses in §5 inherit from the sources listed here. Deeper provenance notes (including the Plan B population-basis cross-check and the 2021-census-direct legal-baseline reconstruction) are in Appendices C and E.

3.1 Primary data sources

SOURCE	FILE(S)	URL
Elections Alberta 2023 Statement of Vote	`data/2023_results.xlsx`	https://www.elections.ab.ca/uploads/2023-Prov Election-Statement-of-Vote.xlsx
Commission final report (majority + minority)	`data/majority_2026_populations.csv`, `data/minority_2026_populations.csv`; map images in `maps/*.jpg`	https://www.elections.ab.ca/uploads/abebc_202
Elections Alberta 2026 ED shapefiles	`data/shapefiles/canonical/`	https://www.elections.ab.ca/resources/maps/
StatsCan 2021 Census Dissemination Areas	`data/alberta_2021_da_populations.csv`, `data/alberta_2021_das.gpkg`	https://www12.statcan.gc.ca/census-recensement/2021/geo/sip-pis/boundary-limite
Alberta Treasury Board Office of Statistics and Information quarterly estimates	(embedded in commission totals)	https://open.alberta.ca/dataset/alberta-populati
StatsCan Table 17-10-0009 (quarterly provincial estimates)	(cited; not persisted)	https://www150.statcan.gc.ca/t1/tbl1/en/tv.ac pid=1710000901
StatsCan Table 98-10-0459 (2021 Census journey-to-work)	(cited in §5.8.4)	https://www150.statcan.gc.ca/t1/tbl1/en/tv.ac pid=9810045901

3.2 Coverage caveats

The following coverage gaps constrain the audit's scope. Each is disclosed rather than papered over.

1. **2026 polygon shapefiles released; formal request answered.** Elections Alberta's GIS page did not carry 2026 ED polygons at the time of the audit's initial scope definition (accessed 2026-04-22). A formal written request for the 2026 boundary shapefiles was filed with Elections Alberta. EA's Geomatics Team Lead confirmed on 2026-05-19 (personal correspondence, R. Mok) the 2026-05-06 shapefile release is final for the commission maps. The canonical shapefiles are now the basis for all spatial analysis in this audit (`data/shapefiles/canonical/`); all results in §5 inherit from this source. **DPG era (historical, now superseded).** Prior to the 2026-05-06 release, Derived Provisional Geometries (DPGs) were constructed from the commission's report text and Appendix E boundary descriptions; methodology documented in `archive/provisional_geometries/approximate_shape_analysis.md` and `analysis/methodology/shape_refinement_v2.md`. Three iterative refinement passes reduced positional error on Tier A/B boundaries to ≤1 km (mean shift 97 m after v1; residual voter-assignment impact: 1,012 votes across 4 VAs, approximately 0.06% of 2023 valid votes). All DPG-era findings have been recomputed from canonical shapefiles. Per the DPG sunset clause in `analysis/methodology/canonical_shapefile_log.md`, DPG results are archived and no longer the primary analysis basis.
2. **Majority-proposal map imagery incomplete.** The working bundle has the majority's Calgary map. The majority Alberta overview, Edmonton, and other-cities panels are not in the bundle. Visual inspection of the majority is therefore limited to its Calgary districts. §5.8.1 discloses this scope narrowing.
3. **~88 public submissions (6.6%) could not be machine-parsed.** Their PDFs are image-only scans lacking a text layer. The submission-archive verification (§5.9.4) rests on identified counter-examples in the 1,252 parseable submissions rather than exhaustive enumeration of all ~1,340.
4. **Third-party vote allocation: B2 (EG) is sensitive to the trailing-party rule. B3 (mean-median) is not.** The 2023 Alberta statement of vote includes 58,232 votes cast for parties other than NDP or UCP (3.30% of valid votes) across 79 of 87 existing EDs. The audit's primary analysis excludes these votes (Rule A: drop, converting all margins to NDP/UCP two-party share). Two alternative allocations were tested symmetrically via `analysis/scripts/third_party_sensitivity.py` (2026-05-09). Rule B (pro-rate other votes to NDP/UCP in proportion to observed two-party shares) shifts the minority-majority EG asymmetry by 0.02 pp and leaves the direction unchanged (minority −1.87%, majority −0.44%, delta −1.43 pp). Rule C (assign all other votes to the trailing party in each district) reverses the EG direction for *both* maps: minority EG +7.41%, majority +4.16% (positive = NDP-favourable under this paper's convention). The mechanical reason: in dominant NDP urban seats, assigning other votes to UCP as the trailing party generates large UCP wasted-vote quantities overwhelming NDP's urban over-concentration, flipping the EG sign. The minority map, which contains more dominant NDP urban seats than the majority, flips further (+3.25 pp gap, reversed direction). B3 (mean-median) holds its direction under all three rules. B6 (declination) follows EG and reverses under Rule C. The paper reports Rule A as the primary metric and Rule C as a

deliberate falsification probe. The Ch1 (Mahalanobis joint tail) and Ch2 (SZAT bootstrap EG) statistics are computed from the same Rule A two-party shares and are unaffected by Rules B or C.

5. **Election-day votes are the only geographically attributable vote type.** Elections Alberta's published results assign a specific Voting Area number to each election-day polling station. Advance, mobile, and special ballot results are reported at the electoral district level only. EA does not record the home VA of an advance voter in any published data product. In the 2023 Alberta general election, 381,932 NDP votes were cast on election day (geographically attributable to specific VAs) and 395,472 NDP votes were cast advance, mobile, or special (carrying no VA-level geographic identifier). All VA-level vote attribution in this audit uses election-day votes only, via `analysis/scripts/build_canonical_va_votes.py`. No advance votes are redistributed or spatially smeared to VAs by assumption. The 49.1% election-day share is a structural property of EA's result-reporting system — not a modelling choice — and represents the actual geographic granularity available to any analyst working from EA's published data.

Mechanism. Alberta's *Election Act*, RSA 2000, c E-1, s.112(1)(a.1) requires the election officer complete "a separate Statement of Vote for each voting area." Sections 113 and 124 apply this requirement — with "all necessary modifications" — to the advance vote count and mobile vote count respectively, meaning the per-VA reporting obligation extends by statute to every ballot type. The operative exception is s.112(2): "Despite subsection (1)(a.1), the election officer responsible for the count may combine the Statements of Vote for more than one voting area in the same electoral division if, in the opinion of the election officer, it is necessary to maintain the secrecy of voting." EA's practice of reporting advance and mobile totals at the ED level rather than the VA level almost certainly reflects an invocation of the s.112(2) secrecy exception, not the absence of any legal obligation. The practical mechanism: the advance voting process verifies each voter against EA's Electoral List System (ELS) before issuing a ballot. The ELS entry carries the voter's home Voting Area number. The VA link is generated at the moment of voters-list check. Advance and mobile ballot envelopes carry no VA identifier. Ballots are then counted at the ED level only. The home-VA attribution exists at the point of voting and is not carried through into results reporting. Whether EA applies the s.112(2) exception as a blanket policy across all advance VAs or as an individual election-officer determination for specific low-population VAs is unknown; the question is included in the pending informal inquiry to EA (see VA 043 case study below). No change to the voting process, ballot design, or voter interaction would be required to publish VA-level advance totals above a minimum-count threshold. Under s.112(2)'s own framing, VAs where secrecy can be maintained without combination could be published separately, requiring only an administrative determination from the election officer.

VA 043 (ED 70, Lesser Slave Lake) as case study. One VA in the canonical file (ED 70, VA 043) received zero election-day votes in EA's published polling data. It is retained in the shapefile with zero vote totals. VA 043 covers 4,832 km² in the northern portion of ED 70, contains 844 persons per the 2021 Census (from `data/va_pop_from_das.csv`), and recorded zero election-day votes — all votes from VA 043 residents entered EA's advance/mobile divisional total with no VA attribution. The specific mechanism is pending EA confirmation: s.120 mobile polling requires a facility type (treatment centre, supportive living facility, shelter, community support centre) VA 043's community does not appear to satisfy. The alternative is Special Ballot voting under s.52.2 remote-area designation, where no polling team visits and residents vote by ballot request. Either way the geographic signal is absent from published data. VA 043 is surrounded by NDP-majority VAs (VAs 041, 042, and 044 returned 52.4%, 53.6%, and 81.8% NDP respectively on election day), placing it in a northern cluster running 534 NDP / 535 UCP across VAs 029–051 — near parity, in contrast to the southern cluster's 29.6% NDP. The geographic partisan signal for VA 043's residents is entirely absent from any published data product. The 2025–2026 Electoral Boundaries Commission received submissions specifically describing the Indigenous communities in this northern area and ultimately invoked §15(2) to protect the riding. The commission's geomatics work was supported by EA GIS staff (EBC Final Report acknowledgements). Whether the commission requested or received VA-level advance vote data from EA — which would have provided a geographic picture of VA 043's voting — is unknown. A direct informal inquiry to EA is pending. A formal FOIP request is queued as the fallback. Full case study in `findings/lesser_slave_lake_va043_representation_gap.md`.

6. **Within election-day votes, multi-VA polling stations are split equally.** Where one polling station served multiple Voting Areas, Elections Alberta reports a single combined Statement of Vote, so `build_canonical_va_votes.py` divides that station's votes equally across the VAs it served — the only defensible split absent finer data. This affects 986 of 1,216 election-day stations (81.1%), which together cast 889,189 of 932,164 election-day votes (95.4%); the largest single station spans 22 VAs. For those votes the effective spatial unit is the station's VA group, not the individual VA, so the substrate's working resolution is coarser than the 4,765-VA count implies. The homogenization is common-mode across both proposals — it applies identically regardless of which 2026 map a VA lands in — and, by averaging within station groups, tends to *understate* fine-grained packing/cracking rather than manufacture it.

3.3 Cycle-lag robustness

The population data the commission uses carries a cycle-lag: the 2021 decennial census updated to a July 1, 2024 Office of Statistics and Information estimate. This choice reflects standard Alberta commission practice (the 2017 Bielby commission used the same basis) and aligns Alberta's statutory approach with other Canadian provinces, which employ diverse population inputs: Quebec uses electors-by-register, British Columbia uses Statistics Canada quarterly estimates, and Manitoba uses decennial census only. Under Alberta's Electoral Boundaries Commission Act §12(3), the statutory mandate permits use of "current population statistics" without prescribing a single data source. The TBF July 2024 estimate satisfies this requirement while preserving administrative timeliness. This statutory flexibility positions Alberta within the practical middle of Canadian commission approaches: more administratively responsive than Manitoba's decennial-census restriction, yet more constrained than Quebec's live-register mandate. Alberta's cumulative population growth on a like-for-like *adjusted-estimate* basis is approximately **10% from July 1, 2021 (adjusted estimate 4,442,879) to Q2-2024 postcensal estimate (4,888,723)** and approximately **13% from July 1, 2021 to Q2-2025 (≈ 5.02M)**. The July 1, 2021 *adjusted* estimate of 4,442,879 is rebased on the 2021 census count with the net-undercoverage adjustment Alberta requires as the highest-undercount province. Earlier drafts cited 17.8% (2021→mid-2025) and 14.69% (2021→mid-2024); both numbers mixed the 2021 *raw census count* (4,262,635) with the *postcensal estimate* universe (4,888,723 at Q2-2024), a ~4.2 pp level shift attributable to net-undercoverage rebasing, not real growth. (T1.7 R1 Ref #13 caught the universe-mixing error; T1.7 R2 Ref #13 caught that round-1's "~10%" figure was labeled mid-2025 but actually corresponds to mid-2024 — mid-2025 like-for-like is ~13%; corrected 2026-06-12.) Corrected 2026-06-12 per T1.7 Referee #13 (Statistics Canada 91-528-X *Population and Demographic Estimates: Methods*; 2021 Census Coverage Study). The directional findings in this section are unchanged because they compare population *shares* within each vintage; the *level* corrections are flagged here so any cross-vintage growth claim made downstream uses the like-for-like adjusted basis. The audit's `analysis/cycle_lag_analysis.md` computes how many electoral divisions flip ±25% window status when mid-2025 populations are substituted: 5 of 87 on the 2019 map, 0 of 89 on the majority 2026 proposal, and 5 of 89 on the minority 2026 proposal. The audit's `analysis/reports/plan_b_cross_check.md` independently verifies every §5.1 verdict is identical whether computed against the 2021 census directly, the 2024 OSI estimate, or the 2025 TBF estimate — i.e. the directional findings are robust to which intra-cycle population vintage is used. Appendix C supplies a 2021-Census-direct A1 computation on the 87 existing 2019 EDs as the §12(3)-operative reference point.

Dataset-construction consequences of cycle lag (three-vintage sandwich). Beyond the verdict-robustness property above, the 4–14-year lag between commission data-basis and boundary retirement materially shapes *how* the audit's upstream pipelines construct their datasets. Any reconstruction of the commission's arithmetic has to reconcile three input vintages simultaneously:

- **2021 geometry + 2021 population.** Statistics Canada Dissemination Area (DA) polygons carrying 2021 census populations — the atomic unit used for Phase 4B DA-overlay.
- **2024 operative estimate.** The commission's July 2024 Treasury Board and Finance (TBF) estimate for each 2026 proposed ED — the operative published value against which the commission's variance tables were computed.
- **2023 voting substrate.** The 2023 Voting Area (VA) polygons carrying 2023 votes — the spatial substrate used for Phase 4C partisan-bias attribution.

The ratio between the 2021-population-sum-inside-a-2026-DPG-polygon and the commission's 2024 estimate of the same territory is **not** a clean scalar. Alberta's ~10% like-for-like growth between July 2021 (adjusted estimate 4,442,879) and Q2-2024 (postcensal estimate 4,888,723) was distributed unevenly (earlier drafts cited "14.69%" here — that figure mixed the 2021 raw census count with the 2024 postcensal estimate, a ~4.2 pp universe-rebase that is not real growth; corrected 2026-06-12 per T1.7 R2 Ref #13) — Calgary-ring cities such as Airdrie and Chestermere grew 20–30 %, rural territories grew 0–5 % — so applying a flat provincial growth factor to a regionally heterogeneous population surface introduces a structural bias into Phase 4F's per-ED validation deltas. The 81 of 86 majority and 87 of 89 minority hardstop failures originally documented in `data/INTEGRITY_STATUS.md` (computed against v0_5 DPG geometry; now bannered SUPERSEDED in `findings/phase4f_summary.json`) were a composite signal: real DPG transcription error plus cycle-lag growth heterogeneity, not separable from public data alone. **Canonical share-based recompute (T4.9-share closed 2026-06-12; `findings/phase4f_hardstop_canonical.md`).** The 2 % hardstop applied to raw 2021 census-count vs 2026 postcensal-estimate deltas conflates net-undercoverage rebasing (~4 pp province-wide) with actual per-ED drift, and was reported in an interim form on 2026-06-11 as "89 / 89 fail" (universe-mismatch). The substrate-honest version is a per-ED *share-of-provincial-total* comparison. Under the share-based test, **78 of 89 majority EDs and 72 of 89 minority EDs fail the 2 % share-drift hardstop** (median per-ED share drift 5.78 % majority / 5.79 % minority; max 39.24 % / 33.39 %, concentrated in fast-growth Calgary/Edmonton suburbs — Calgary-NE, Edmonton-Windermere, Calgary-Shaw, Calgary-South East, Edmonton-South). The v0_5 81/86 + 87/89 counts, the universe-mismatch 89/89, and the share-based 78/72 form a coherent decreasing sequence as substrate quality improves. The audit's headline reading uses the share-based counts. Full treatment at `findings/cycle_lag_analysis.md` (§1); canonical re-run with share-based + universe-mismatch comparisons at `findings/phase4f_hardstop_canonical.md`.

Forward-modelling consequences for commissions. The 2021 decennial census — the Commission's statutorily-mandated baseline under §12(3) — is 4 years stale at the first election under the new boundaries (2027) and up to 14 years stale at the boundaries' retirement (potentially 2035).

For fast-growth metropolitan-fringe cities (Airdrie, Chestermere, Cochrane, Okotoks, Beaumont, Leduc, Spruce Grove, Fort Saskatchewan, Sherwood Park), this gap already moves per-ED population quotas by 10–15 % at the 4-year mark and by 40 %+ at the 14-year mark under current growth rates, independent of any drawing choice the commission makes.

For slow-declining rural districts (Peace River, Central Peace-Notley, Lesser Slave Lake), the gap erodes §15(2) special-rural-protection ratios asymmetrically: Lesser Slave Lake's mid-2025 §15(2) qualifying ratio drops past −50 % of the provincial mean under the cycle-lag substitution, which could disqualify a district from its current legal basis before the cycle concludes.

Indigenous on-reserve populations face a third variant of the same problem: commissions inheriting the census's chronic 3–10 % on-reserve undercount see those communities structurally under-represented for the full 6–14-year cycle.

The gap is primary-source verifiable for the Kee Tas Kee Now Tribal Council member nations in northwestern Alberta — a regional tribal council whose member nations include communities among the 14 First Nations and Métis communities the EBC Final Report identifies within or adjacent to ED 70. ISC First Nations Population Profiles (April 2026 — a supplementary source not incorporated into the commission's §12(3) statutory data mandate) show registered population growth rates in excess of the provincial norm. (Note: the 14.69% figure cited in earlier drafts of this sentence was retracted 2026-06-12 per T1.7 R2 Ref #13 — it mixed the 2021 raw census count with the postcensal estimate universe; the like-for-like adjusted-estimate provincial growth for the 2021→2024 window is ~10%, documented in the §5.1.1 correction note above. Peerless Trout's ~7-year figure and the ~3-year provincial figure are not directly time-comparable; both are cited here for directional orientation.) Peerless Trout First Nation (Band 478) recorded a registered population of 1,155 in April 2026, up from 966 in the prior ISC profile — a 19.5% increase over approximately seven years, more than one-third above the provincial rate for the same period. Whitefish Lake First Nation #459 (Atikameg) shows a 2021 Census community population of 880 (Statistics Canada, Indigenous Population Profile 2021). 1,423 registered band members are listed on own reserve in the April 2026 ISC profile (total registered: 3,266, including off-reserve members living elsewhere in Alberta). Note ISC registered population on own reserve and census household population are distinct measures — the former counts registered band members who list the reserve as their address of record, the latter counts people physically present. Lubicon Lake Band No. 453, which operates on Crown land rather than a formal reserve, recorded a 2021 Census median age of 21.2 years. Whitefish Lake's census median age is 22.6 years. Median ages in the low-to-mid-twenties indicate cohorts entering peak fertility and early working-age years simultaneously — a demographic structure associated with above-average natural increase over the coming decade. A commission drawing the 2026 map on 2021/2024 TBF data works with a population basis for these communities stale not only in absolute terms but in growth-rate terms.

None of this is resolvable by audit methodology alone. The §12 statutory framework itself determines what baselines the Commission is permitted to consult. A legislative-reform proposal formalising a composite basis (TBF primary + StatsCan tie-breaker at ±2 % + AHCIP + CRA T1 cross-check + CEO as certifying authority) is outlined at `docs/act_amendment_proposal.md`. The audit offers this as a policy contribution, not a finding. Full treatment at `findings/cycle_lag_analysis.md` (§2).

Policy contribution: synchronise redistribution timing with census data release. The structural source of the 4–14-year cycle lag is procedural rather than statutory: commissions are constituted before the census data anchoring their work is available. Canada's 2026 decennial census is enumerated in May 2026. Statistics Canada releases dissemination-area-level population and geography data approximately 18–24 months after enumeration, placing comprehensive sub-provincial data in the 2027–2028 window. A commission constituted before the release — as the current commission was, drawing on 2021 census data — cannot access the most current population geography even if it wished to. The

EBCA's two-year post-census trigger for commissioning could be amended to require the commission's appointment follow the release of DA-level census data from the most recent decennial enumeration, rather than merely following the enumeration date itself. This would reduce the population-data age at the time of drawing from 4+ years to approximately 1–2 years, and would ensure fast-growth metropolitan-fringe communities whose populations change 20–30% per four-year cycle are accurately represented in the map the commission draws rather than in a retrospective correction the next commission inherits. The audit offers this as a policy contribution, not a finding.

Policy contribution: Elections Alberta should publish home-VA vote totals for advance ballots. In the 2023 provincial election, 395,472 NDP and UCP votes cast advance, mobile, or special carry no Voting Area identifier in EA's published data (§3.2 item 5). This is not a technical limitation. Every advance voter is verified against the provincial voters list before receiving a ballot; the list links each registered voter to a specific Voting Area. The home-VA identifier exists at the point of voting. The statutory framework is clearer than it initially appears: s.112(1)(a.1) of the *Election Act* requires a separate Statement of Vote per voting area, applied to advance counts through s.113 and mobile counts through s.124. EA's aggregate reporting practice rests on the s.112(2) secrecy exception, which permits combining VAs when "necessary to maintain the secrecy of voting." The reform is not to create a new legal obligation but to provide guidance, or amend s.112(2), to require the secrecy exception be applied VA-by-VA against a minimum-count threshold — so only VAs where combination is genuinely necessary for secrecy are combined, while larger VAs are published separately. Publishing VA-level advance vote totals alongside the existing election-day VA breakdowns requires no change to the voting process, no additional interaction with the voter, and no new data collection — only a change to how EA applies a discretion the statute already authorizes.

The gap disadvantages not only external analysts but the Electoral Boundaries Commission itself. Commissioners are required by the EBCA to weigh community of interest — to assess whether geographic communities have coherent political identities worth preserving whole, or whether proposed boundaries divide natural constituencies. This assessment depends on understanding where voters actually live and how they participate. When advance voting accounts for approximately half of all votes cast and those votes carry no geographic attribution below the ED level, commissioners are making legally binding boundary decisions while working with the same incomplete spatial picture as external auditors. The voters list they consult shows where electors are registered. The vote results EA publishes show only where half of them voted. These are not the same thing, and the discrepancy is largest precisely in the high-growth suburban communities — Airdrie, Cochrane, Chestermere, Leduc — most contested in redistribution proceedings.

Publishing VA-level advance vote totals would allow complete-coverage VA-level vote attribution in future audits, provide commissioners with a full geographic picture of the communities they draw boundaries around, and bring Alberta's results reporting to the granularity the EBCA's community-of-interest mandate implicitly requires. Standard minimum-count thresholds — already applied to election-day polling station results — would apply equally to advance VA totals, preserving individual ballot secrecy in low-turnout voting areas without limiting geographic transparency for larger communities. The audit offers this as a policy contribution, not a finding.

Cross-election asymmetry reversal: mechanical explanation. A parallel sensitivity operates on the partisan side. The minority map absorbed three hybrid configurations the majority does not contain: Springbank-area townships (outer west Calgary suburbs), Bearspaw, and Cochrane. These communities are high-income, low-density outer suburbs whose two-party NDP/UCP vote share sits in the blend model's transition zone — high enough urban-weight sensitivity small shifts in the rural baseline produce large swings in the hybrid's attributed vote share. In 2023, outer Calgary suburbs returned NDP shares near the urban-core mean (2023 was a competitive cycle; suburban NDP performance was stronger than historical). In 2019, the same territories returned NDP shares closer to the rural baseline (the 2019 NDP wave penetrated the urban core but had shallower reach into outer Calgary suburbs). The audit's blend model applies a fixed geographic weight to both election years. What changes is the rural baseline being mixed in. Under 2023 votes, Springbank/Bearspaw/Cochrane territory makes the minority's hybrid EDs mildly more NDP-competitive than under 2019, which is why the minority asymmetry is negative (more UCP-favourable) under 2023 but positive (less UCP-favourable) under 2019. The reversal is a property of the estimation model interacting with Alberta's election-specific suburban swing geography — not a measurement of a different underlying boundary configuration. §5.2.3 documents the specific numbers: 2015 asymmetry +0.03 pp, 2019 +0.75 pp, 2023 −0.51 pp. The individual map EG signs DO flip under 2019 votes (both maps turn NDP-favourable at −1.32 % / −0.49 % per canonical Option C — see §5.2.8; the v0_8-based claim that "neither map's EG sign flips" was an attribution artefact, retracted 2026-06-13 per T1.7 R2 S4). The inter-map asymmetry direction also changes across elections, for the mechanical reason described above.

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4. Methods

This section describes the audit's methodology; results follow in §5. The methods are presented in the sequence they are applied: the symmetry and falsifiability discipline (§4.1), the vote-attribution pipeline used for all partisan-bias metrics (§4.2), the specific test battery B1–B6 with thresholds and references (§4.3), and the legal-defensibility frame used to map findings to potential challenges (§4.4). Mathematical formalism for the four metrics is consolidated in Appendix D.

4.1 Methodology and integrity framework

4.1.1 Symmetry requirement

Every test applied to one map is applied identically to the others. Where a data gap prevents symmetric application, the gap is disclosed explicitly and the claim's scope is narrowed to what is symmetric.

4.1.2 Falsifiability gates

Each analytical stage produces a PASS/FAIL gate value before propagating downstream. Gates implemented:

- **G1 (carry-forward verification):** B1–B4 on 2019 baseline must reproduce the four-figure match to official totals (NDP 777,404 / UCP 928,900, two-party 1,706,304; arithmetic verified: 777,404 + 928,900 = 1,706,304; source: `data/raw/2023_results.xlsx`). Reproducible via `python3 analysis/scripts/packing_cracking_analysis.py`.
- **G2 (2026 estimate count):** Each map's ED estimate set must contain exactly 89 districts; total valid votes within 5% of 1.7M; NDP share within [0.40, 0.50]. `validate_2026_estimate()` in `packing_cracking_analysis.py`.
- **G3 (Calgary classification coverage):** A2 test requires zero residual unclassified Calgary EDs. Enforced programmatically in `a2_calgary_analysis()`.
- **G4 (A2 robustness):** A2 directional finding must survive alternative classification (2023 winner-based) or be flagged as classification-dependent. `a2_robustness_check()` implements the alternative.
- **G5 (Sensitivity range):** B2 efficiency gap computed under urban weights 0.60, 0.70, 0.80, 0.85 (central), 0.90. Direction must be consistent across all five; magnitude range is reported alongside the central estimate.

4.1.3 What does not enter the report

- Any number not reproducible by running a checked-in script against checked-in data
- Any classification rule without a robustness check under at least one alternative
- Any language characterizing one map's features with stronger modifiers than the other's when the underlying facts are comparable
- Any "the numbers confirm X" framing in section preambles

4.1.4 Derived Provisional Geometries (DPG) — summary and canonical resolution

Prior to 2026-05-06, official Elections Alberta shapefiles for the two 2026 proposals had not been published. All pre-canonical 2026 boundary geometries were **Derived Provisional Geometries (DPG)** — polygons reconstructed from commission-published PNGs via a ten-stage pipeline (raster trace → topology cleanup → population-swept hybrids → CSD/DA anchoring → confidence propagation → multi-source assembly → four-phase perfecter → nested-polygon ownership inversion). Two error modes govern DPG-dependent metrics:

- **Perimeter-mode uncertainty (±500 m typical):** boundary localisation error affecting compactness scores (Polsby-Popper, Reock) and fine-grained vote attribution near boundary lines.
- **Area-mode uncertainty (Tier-dependent):** whole-polygon-territory mismatch affecting DA-overlay populations and polygon-intersection tests. Tier A EDs (da-anchored, municipal-anchored) carry shapefile-grade fidelity. Tier C EDs (v7 visual transcription) are subject to documented area-mode errors and are excluded from primary findings.

Official shapefiles received 2026-05-06. Findings designated [C] (canonical) throughout this report were recomputed on official Elections Alberta shapefiles. DPG-era results remain in the record per pre-registration obligation and are superseded by their canonical equivalents where both exist. Findings not marked [C] may be DPG-derived and are subject to the sunset clause. See `analysis/methodology/canonical_shapefile_log.md` for the full reconciliation of which findings have been recomputed on official geometry.

Sunset clause — satisfied, with material changes disclosed. The pre-registered commitment to recompute all geometry-dependent metrics within two weeks of official shapefile release was fulfilled on 2026-05-06. Two findings did change materially on canonical recomputation and are flagged here rather than left to the deltas file:

1. **Municipal anchoring (§5.8.5) was retracted.** The DPG-era reading of 14.5 % / 71.0 % / 4.9× ratio collapsed to 72.0 % / 80.0 % — both within Canadian norms — on canonical geometry. The anchoring finding has been retired; `findings/municipal_anchoring_analysis.md` carries a superseded banner.
2. **The minority's declination value passed through a transient attribution-refinement sign change, and a code-side sign convention error was corrected 2026-06-12 (Amendment 10).** Per `findings/post_audit_recompute_deltas.md` (sign-convention notes updated under Amendment 10): the v0_8 DPG-substrate reading of δ = −0.0666 moved to **+0.0105** on the canonical first-pass substrate, then settled at **−0.0770** on the final canonical run after the Vote-Anywhere attribution correction (~9.4 % of votes) landed. Those values were under the implementation's *swapped-operand* convention. Under the Warrington 2018 convention (positive δ = UCP-favoured), the equivalent canonical final value is **+0.0770**, sitting at p98.79 — UCP-favoured tail. The first-pass canonical reading (+0.0105 under the old convention; −0.0105 under Warrington) is superseded by the final canonical. Both the v0_8 and the final canonical agree on the same direction once corrected: minority is UCP-favoured on declination (matches EG, MM, s50). Any reading still citing "p62.2 / near median" or "p1.21 NDP-tail" derives from a superseded substrate or the pre-Amendment-10 code sign and is corrected here. Appendix D.3 documents the convention; Amendment 10 documents the discovery and verification.

All other findings held to within the published confidence intervals on canonical geometry. Full reconciliation is at `analysis/methodology/canonical_shapefile_log.md` and `findings/post_audit_recompute_deltas.md`.

The full ten-stage DPG pipeline — construction methodology, error progression table, nested-polygon ownership-inversion algorithm (extends Sester 2000; Saalfeld 1988), and programmatic city-centre alignment proof — is documented in the companion methods paper: `findings/methods_paper_draft.md`.

4.1.5 DPG construction pipeline — multi-source assembly (v0_7)

Full ten-stage pipeline (raster trace → topology cleanup → population-swept hybrids → CSD/DA anchoring → confidence propagation → multi-source assembly → four-phase perfecter → nested-polygon ownership inversion) documented in the companion methods paper (`findings/methods_paper_draft.md`). Summary of methodological findings running counter to expectation: §5.0.2.

4.1.6 v0_8 coverage diagnostics — inheritance fill and geometric refinements

Full methodology in companion methods paper (`findings/methods_paper_draft.md`). Key coverage findings: 21 of 89 majority EDs have zero geometry in v0_7 and inherit empty into v0_8; Phase 3 spatial-index gap-fill systematically inflates rural ED areas in v0_8 vs v0_7; Calgary-Falconridge-Conrich is fully nested inside Airdrie-East in v0_7/v0_8 (v0_8.1 nested-polygon ownership-inversion refinement applied); v0_8.2 inheritance fill assigns 21 majority + 12 minority EDs from 2019-Tier-A predecessors. Findings in §5.0.2 items 6–9; caveats propagating to downstream findings in §4.1.8.

4.1.7 City-centre alignment proof

Point-in-polygon plausibility check against Statistics Canada city-centre representative points: 9 of 10 Alberta city centres land in correctly-named EDs in the v0_8.1 refined majority geometry. The single miss (St. Albert) falls in the geographically adjacent West Yellowhead ED, within the documented ±500 m DPG perimeter precision band. Full findings in §5.0.2 item 9.

4.1.8 v0_8 geometry caveats propagating to downstream findings

Two imperfections propagate to Run #4 (§5.4.4) and related downstream findings: (1) inherited urban polygons trace 2019 boundaries rather than 2026 commission boundaries for 21 majority and 12 minority EDs; (2) the v0_8 minority's Peace River polygon is over-extended by Phase 3 gap-fill, absorbing 6 of 10 city-centre landmark points (§4.1.7). Both imperfections are disclosed where downstream findings cite v0_8 numbers.

4.2 Vote-attribution pipeline

2026 ED-level vote estimates are built by mapping each 2026 ED to its 2019 predecessor(s) using an explicit dictionary (`MAJORITY_2026_MAPPING`, `MINORITY_2026_MAPPING`). Three mapping types:

- `direct`: 2026 ED covers approximately the same territory as a 2019 ED; use 2019 votes directly.
- `blend`: 2026 ED combines a 2019 urban core with rural absorption; blend 2019 core vote with the 2023 observed Rest-of-Alberta NDP share (33.5%) using urban weight 0.85 (applied identically to both maps).
- `merge`: 2026 ED combines two 2019 EDs; weight each part explicitly.

4.3 Test battery

Channel structure and ordering rationale. The quantitative evidence runs in a fixed, dependency-ordered sequence:

- **Channel 1 — MCMC ensemble + inter-map permutation test.** Asks the broadest question first: is the minority map anomalous relative to a neutral redistricting distribution, and is the gap between the two commission maps larger than neutral plan pairs are apart from each other? This establishes there is a real inter-map distance to explain.
- **Channel 2 — SZAT.** Decomposes *which specific boundary decisions* produced the confirmed gap. Runs after Ch1 because decomposing a gap not yet established as real would be decomposing noise.
- **Combined-channel headline: Bonferroni dependence-robust bound** (corrected 2026-06-10; restated 2026-06-12 per T1.7 R2 Ref #1). The earlier framing — "Ch1 and Ch2 combined via Fisher's method only after confirming the two tests are uncorrelated (§G1, $\rho = -0.0014$)" — was withdrawn at §5.5 because the $\rho = -0.0014$ check measures correlation across two *unpaired* Monte Carlo streams rather than the dependence between the test statistics under the null map distribution. Under positive dependence Fisher is anti-conservative (Brown 1975), so the audit's operative joint bound is **Bonferroni: $p \leq 2.80\times10^{-6}$**, valid under arbitrary dependence. The Fisher value (6.87×10^{-8}) is reported in §5.5 as the dependence-naive figure for methodology completeness only. A supplementary Holm-Bonferroni correction across the six primary tests still survives at family-wise $\alpha = 0.10$: full table at §7.
- **Channel 3 — Neighbour-Drain.** A pre-registered structural observation reported per pre-registration obligation. Does not feed into the Bonferroni upper bound headline. Result does not alter the Ch1-based verdict.
- **Two-Lane Scorecard.** Synthesises all channels. Scored last because it cannot be completed until all tests are run.

Sign-convention glossary. Throughout this paper, for every partisan-bias metric: **negative value = UCP advantage** (UCP wastes fewer votes per seat / carries the higher-than-median district / wins more seats at 50/50 / has shallower winning-district-margin angle than NDP), and **positive value = NDP advantage**. The sign convention is chosen for readability against seat-count outcomes, not to match the Stephanopoulos-McGhee 2:1 slope convention (which labels positive EG as "first-party disadvantaged"). Readers cross-referencing the S-M literature should invert the sign-label, not the finding. Both conventions agree on the *ordinal* ranking of the three maps. Full glossary and cross-convention reconciliation in `analysis/methodology/sign_convention_resolution.md`.

- **B1:** Vote distribution histogram across 10 margin bins from UCP +25%+ to NDP +25%+. (Descriptive; no formal literature reference.)
- **B2:** Efficiency gap (Stephanopoulos and McGhee 2015): $\text{EG} = (W_{\text{NDP}} - W_{\text{UCP}}) / N$ where wasted votes include loser votes plus winner votes above the threshold. **Sign convention.** Under this formula, positive EG means NDP wastes more votes than UCP — i.e. UCP is advantaged in seat-outcome terms. Negative EG = NDP advantaged. The Stephanopoulos-McGhee 2015 reference text uses the same wasted-vote formula but anchors the sign discussion around a "first party / second party" labelling some readers find ambiguous; the convention used here ("positive EG = UCP advantage") fixes the labels to the parties measured. All reported EG values in this audit follow this convention: the minority's +3.96% is UCP-favoured, the majority's +0.04% is essentially neutral, and the 2019 enacted map's +0.024 / +2.41% (depending on data substrate) is slightly UCP-favoured against the neutral baseline. A reader comparing against S-M literature should map "positive EG = UCP advantage" here to the corresponding direction in S-M's first-party / second-party framing, not invert the finding.
- **B3:** Mean-median gap (McDonald and Best 2015): $\text{MM} = \bar{v} - \tilde{v}$ for NDP vote share.
- **B4:** Seats-votes under uniform swing to 50/50 provincial share (Gelman and King 1994; Grofman 1983).
- **B6:** Declination (Warrington 2018). Measures the asymmetry between winning-district vote distributions by treating each party's winning districts as a vector and computing the angle between them. See §5.2.4 for the direction-disagreement finding.

The seat-vote-curve symmetry principle underlying B4 traces to Grofman (1983) and King and Browning (1987), later formalized as a Bayesian estimator by Gelman and King (1994). The efficiency gap and mean-median are two of the most widely-cited partisan-bias metrics in the post-*Gill v. Whitford* literature; Stephanopoulos and McGhee (2018) revisit the efficiency-gap debate and acknowledge the metric's sensitivity to modeling choices, which our Monte Carlo analysis in §5.2.2 quantifies for the Alberta context. Katz, King, and Rosenblatt (2020) argue that any valid measure of partisan bias must satisfy symmetry and critique single asymmetric metrics as inadequate; this audit's stress-test gate RT2 (cross-metric agreement) independently implements an ensemble approach satisfying their symmetry criterion.

Mathematical definitions for EG, MM, declination, and the Polsby-Popper and Reock compactness metrics are in Appendix D.

4.3.1 Multiple-comparisons handling: Exploratory screening (current run) versus confirmatory testing (November 2026)

Exploratory vs. confirmatory framing. The current detection run applies six correlated tests across multiple analytical dimensions (not all independent — see the §1.1 BH-table dependence-disclosure paragraph). This creates a multiple-comparisons burden if the results are interpreted as confirmatory hypothesis tests, where each test is adjudicated at $\alpha = 0.05$ independently and then combined. The audit's response is to frame this run as **multi-dimensional screening analysis** (corrected terminology 2026-06-12 per T1.7 Referee #16; earlier drafts called this "Bayesian-screening" — that label was unsupported by the actual machinery, which uses no prior, no likelihood, and no posterior; full Bayesian updating is explicitly ruled out at the §4.6 framework-defense paragraph). What the audit actually does is a frequentist screen reported descriptively (per-dimension percentile placements) plus a synthesis claim about directional agreement; the screening / confirmatory distinction is honest and serves the same role as the "Bayesian-screening" label was intended to, without claiming machinery the audit doesn't deploy.

Under the multi-dimensional screening frame, the six-dimensional battery (population variance, partisan-bias metrics B2–B6 — declination Amendment-10-corrected, compactness, geographic coherence, procedural findings) serves as a screen to identify potential signals. The primary inferential claim is directional consistency *across* dimensions, not significance *within* any single dimension: "six correlated lines of evidence point in the same direction" is the audit's finding, not "six individual tests reject the null at classical confidence." This framing aligns with the pre-registration posture stated in §5.3.1: the current findings are exploratory. Confirmatory testing occurs in the November 2026 OSF-registered re-run on the Lunty-committee map.

Confirmatory protocol (November 2026). The November confirmatory run will apply a pre-registered formal hypothesis-testing framework with explicit multiple-comparisons correction. The Benjamini-Hochberg false-discovery-rate control (described below) is the proposed correction structure for the confirmatory pass. The current run's BH table (provided for reader reference) shows what the p-value landscape would be if classical hypothesis-testing were applied retroactively; but the exploratory-to-confirmatory distinction means the current run does not carry the inferential weight a BH-corrected significance claim would carry.

Exploratory findings as evidence, not conclusions. The six-dimensional directional consistency reported in §5.2–§5.8 and summarised in §6 is evidence-suggesting (pointing toward systematic asymmetry in boundary drawing and outcome) rather than evidence-concluding (determining a court would find the asymmetry unjustified under the Saskatchewan Reference effective-representation standard). The November confirmatory pass will sharpen this by applying formal hypothesis-testing machinery. The current exploratory findings provide the pre-hoc signal motivated those pre-registered tests.

Benjamini-Hochberg correction for the confirmatory pass. The audit does not rely on any single statistical test for its conclusions. The primary inferential frame is directional consistency across multiple independent dimensions (§6, §4.6). For the November confirmatory test, a formal Benjamini-Hochberg (BH) false-discovery-rate correction will be applied so the multi-test battery can be evaluated at α=0.05 family-wise. The reference table below shows the p-value landscape under BH correction. It is provided here to show readers what the correction structure will be when applied confirmatorily.

Scope. The table includes all formal tests yielding a computable p-value against an explicit null distribution. Purely qualitative structural checks (packing/cracking/engineered-boundary signatures, municipal anchoring ratio) are not included because they do not produce p-values. The Fisher combined result ($p = 6.87\times10^{-8}$) is retired (see §4.3.3) and is excluded from the battery; the operative Bonferroni upper bound ($p \le 2.80\times10^{-6}$) is likewise excluded because it is a function of Ch1, which already appears as Rank 2.

p-value derivation for MCMC percentile tests. A minority map at ensemble percentile p_pct in the UCP-favoured tail yields a one-tailed p-value of (1 − p_pct/100). A majority or minority map in the NDP-favoured tail yields p = p_pct/100. Row 3 (Seats@50/50): the raw 250k ensemble found 0/250,000 neutral plans exceeding the minority value ($p < 4\times10^{-6}$). The 250k full-coverage rescore placed the ESS-adjusted lower bound at p89.72 (inside the neutral band; the flag was retracted). The 1,010,000-plan canonical run (§5.4.9, n_eff=1,495) reinstates the seats@50/50 flag: 1,010,000-plan percentile = 99.99, ESS-adjusted lower bound ≈p98, above the p95 threshold. Row 3 counts as a positive finding under the 1,010,000-plan run.

RANK	TEST	MAP	P (RAW)	P (BH-ADJ)	A = 0.05
1	Moran's I spatial clustering on NDP share (z = 12.15)	2019 substrate	4.8×10^{-34}	5.3×10^{-33}	PASS
2	Mahalanobis joint tail — Ch1 (canonical ensemble)	Minority	1.4×10^{-6}	7.7×10^{-6}	PASS
3	Seats@50/50 vs canonical ensemble (<100/1,010,000 plans exceeded; p99.99)	Minority	1.0×10^{-4}	3.7×10^{-4}	PASS
4	Mean-median vs canonical ensemble (p99.98)	Minority	2.0×10^{-4}	5.5×10^{-4}	PASS
5	Lopsided Margins t-test — Wang (2016), t = 3.80 (canonical 2026-06-11)	Majority	3.4×10^{-4}	7.5×10^{-4}	PASS
6	Lopsided Margins t-test — Wang (2016), t = 3.17 (canonical 2026-06-11)	Minority	2.2×10^{-3}	3.8×10^{-3}	PASS
7	SZAT bootstrap — Ch2 (10,000 permutations)	Minority	2.4×10^{-3}	3.8×10^{-3}	PASS
8	Mean-median vs canonical ensemble (p0.924, NDP-tail; 1,010,000-plan canonical run, seed 1432864451 — corrected 2026-06-14 per T1.7 R2 S19 re-audit; prior note "corrected per S19 to p0.85" had the direction inverted: p0.924 is the CSV value, p0.85 was a pre-canonical-run intermediate)	Majority	9.2×10^{-3}	1.2×10^{-2}	PASS
9	Population MAD vs canonical ensemble (p99.0)	Minority	9.8×10^{-3}	1.2×10^{-2}	PASS
10	Declination vs canonical ensemble (p98.79, UCP-tail — corrected sign per Amendment 10)	Minority	1.2×10^{-2}	1.3×10^{-2}	PASS
11	Efficiency gap vs canonical ensemble (p94.4)	Minority	5.6×10^{-2}	5.6×10^{-2}	FAIL

m = 11 tests (NOT all independent — see dependence-disclosure note below). Sorted by raw p ascending. BH-adjusted p_(i) = min over j ≥ i of (p_(j) × m / j) (standard BH step-up; Benjamini–Hochberg 1995, JRSS-B 57(1)). Re-sorted + recomputed 2026-06-12 per T1.7 Referee #1 / #6 / #16 findings: the prior table was mis-sorted (rank 9 preceded rank 10) and the published adjusted p-values violated the monotone step-up rule. The conclusion (10 PASS / 1 FAIL) is unchanged.

Dependence disclosure (added 2026-06-12 per T1.7 Referee #1, #6, #16). The Mahalanobis row (rank 2) is a deterministic function of the same four ensemble metrics that appear individually as rows 3, 4, 9, 10, and 11; row 9 (Population MAD) is a separate structural-lane statistic. Rows 5–6 (Lopsided Margins majority/minority) share one vote substrate. The standard BH controls FDR only under independence or PRDS (Benjamini–Yekutieli 2001, Ann. Statist. 29(4)); the table is therefore a *reference landscape*, not an independent-test family.

BY adjustment under dependence (T1.8 closed 2026-06-12). Under the BY rule $q(i) = p(i) \times m \times H\_m / i$, *where* $H\_m = \Sigma\_\{k=1..m\}(1/k) = 3.0199$ *for* $m = 11$, *the adjusted p-values rise by a factor of 3.02 across the board:* 1.6×10^{-32}, 2.3×10^{-5}, 1.1×10^{-3}, 1.7×10^{-3}, 2.3×10^{-3}, 1.1×10^{-2}, 1.1×10^{-2}, 3.6×10^{-2}, 3.6×10^{-2}, 4.0×10^{-2}, 1.7×10^{-1} *(rank 1→11). At* $\alpha = 0.05$ *family-wise: rows 1–10 still* PASS*; row 11 (Minority EG) still* FAILS*. The audit's headline (10 PASS / 1 FAIL) is preserved under the dependence-robust adjustment; row 10 (declination) is now the closest-to-margin pass (3.99×10^{-2} vs $\alpha=0.05$) but does not flip. This closes T1.8 in `TODO_REMEDIATION.md`.

Reference result for confirmatory testing (November 2026). The table shows what the BH-corrected landscape would yield: 10 / 11 tests would pass BH correction at $\alpha = 0.05$ under the confirmatory protocol. The sole failure is Minority EG (raw $p \approx 0.056$, BH-adjusted $p \approx 0.056$), which the audit already retracts from its outlier-flag set in §5.4.9 on independent grounds (the 1,010,000-plan canonical run places minority EG at p94.4, below the p95 threshold). No test counted as a positive finding in the audit's conclusions would fail BH correction when formally applied. This table is provided as reference material so readers can see the correction structure governing the November confirmatory pass. It is not a retroactive inferential claim on the exploratory current run.

Scope note. The Lopsided Margins finding (rows 5–6) is present in all three maps including the 2019 enacted baseline and is explicitly described in §5.2.9 as a structural property of Alberta's political geography, not a distinctive feature of either 2026 map. Its inclusion in the BH battery is conservative (reduces the effective discovery rate for any new finding) but does not change the conclusion.

Directional misuse caveat (added 2026-06-12 per T1.7 Referee #6). Wang (2016)'s lopsided-margins t-statistic flags the *seat-majority party* (UCP, the dominant party in Alberta's 2023 substrate) as the *packed* party, which is structurally an NDP-favourable observation (UCP wastes votes in landslide wins). Counting positive-t Lopsided rows as "PASS" *alongside* UCP-favoured findings (mean-median, declination, seats@50/50) mixes directions in the synthesis: rows 5–6 are technically NDP-favourable evidence on the same map the other rows flag as UCP-favourable. The 4-of-4 partisan-bias agreement reported elsewhere in the paper relies on the EG family + s50 + declination, *not* on Lopsided Margins. Rows 5–6 are reported here for transparency under the m=11 BH battery but should not be read as adding to the UCP-favoured-direction count.

EG threshold MC-error caveat (added 2026-06-12 per T1.7 Referee #6 D15). The Phase 4C canonical minority EG (+3.96 %) lies 0.14 pp below the Alberta-calibrated 4.10 % threshold; the like-for-like canonical-ensemble value gives 0.08 pp gap. At ESS ≈ 1,429–1,682, the order-statistic standard error on the p95 threshold and on the p94.4 percentile each exceeds 0.08 pp. The sub-threshold call (row 11 FAIL) is therefore *indeterminate* within Monte Carlo precision, not robustly FAIL. This does not promote the row to PASS — the audit retains the FAIL — but the indeterminacy means any verdict re-litigation should not rely on the 0.08–0.14 pp gap as evidence of clear sub-threshold status.

4.3.2 Bonferroni correction as a conservative alternative

While Benjamini-Hochberg correction controls the false-discovery rate and is the chosen method for the confirmatory protocol, a simpler alternative exists: the Bonferroni correction, which controls the family-wise error rate (FWER) — the probability of any false positive across the entire test battery. The Bonferroni method is more conservative: for m = 11 tests (NOT all independent — see the §1.1 dependence-disclosure paragraph; corrected from "independent" 2026-06-12 per T1.7 R2 Ref #1) at α = 0.05 family-wise, the per-test threshold becomes α / 11 ≈ 0.0045 (compare to BH's step-up procedure which is less restrictive).

Applied to the current audit's findings: the two largest p-values in the battery are Minority EG (canonical p = 0.056, BH adj 0.056, FAIL) and Minority Declination (canonical p = 0.012, BH adj 0.013, PASS post-Amendment-10). Under Bonferroni at m = 11, α/11 = 0.00455. Ch1 Mahalanobis ($p = 1.40\times10^{-6}$) clears α/11 by **three orders of magnitude**; SZAT (p = 0.0024) clears α/11 by **a factor of ~1.9** under standard i.i.d.-flip permutation (not "multiple orders of magnitude" — corrected 2026-06-12 per T1.7 Referee #1). However, under a block-permutation null that corrects for spatial autocorrelation across adjacent Voting Areas — the 2,110 swing-zone VAs cluster into a single connected component of 2,086 VAs, violating the independence assumption — SZAT returns p≈0.19, not significant. SZAT is therefore retired as a confirmatory channel (see §5.2.10); its standard-permutation result is exploratory context only. The audit's structural findings — which rest on the directional consistency of multiple correlated dimensions rather than on any single test — survive Bonferroni correction. Ch1 alone ($p = 1.40\times10^{-6}$) is the headline; a conservative Bonferroni upper bound accounting for the two channels examined — Ch1 and Ch2, which share the same underlying 2023 vote substrate and are not independent (Brown 1975) — yields $p \le 2.80\times10^{-6}$ ($= 2 \times 1.40\times10^{-6}$), or equivalently ≈ 1 in 357,000. This is the figure the audit now uses where the earlier text quoted "$p = 6.87\times10^{-8}$ (≈ 1 in 15 million)." The corrected bound is still extreme — well beyond any standard significance threshold — but it is no longer derived under an unstated independence assumption, and the ×2 factor is retained as a conservative adjustment for the two channels that were examined. The choice of BH over Bonferroni for the confirmatory protocol reflects the audit's frame (multiple exploratory dimensions, not a single pre-registered hypothesis test) and the principle BH better balances discovery power with false-discovery control in high-dimensional exploratory settings.

4.3.3 Pre-registration record

The table below is the canonical lookup for pre-registration status of every quantitative test in the audit. Full provenance detail is in `analysis/methodology/null_hypothesis_and_exoneration_criteria.md`; full amendment history is in `findings/pre_registration_amendment_log.md`.

TEST	PRE-REGISTRATION ANCHOR	TIMING RELATIVE TO DATA
Ch1 — Mahalanobis joint outlier (§5.4.9)	Seeds committed to Cloudflare drand beacon Round 5500000 before shapefile release; see `analysis/scripts/drand_seed.py` and `docs/FROZEN_MANIFEST.md`	drand seeds predate official shapefile receipt (2026-05-06). Specific test name and combination method not named in OSF. **Exploratory.**
Ch2 — SZAT boundary-choice bootstrap (§5.2.10)	AsPredicted #289,469 (filed 2026-05-07; made public 2026-05-07); OSF:6pt83 (filed ~3 h after shapefile commit — cannot be treated as pre-dating the data)	Numerical results were known to the analyst at filing. **Exploratory-reproducible.**
Ch3 — Neighbour-drain adjacency (§5.3.5)	OSF:r3zm7 / AsPredicted #289,451 (filed before shapefile release)	Pre-dates shapefile; the null hypothesis and surrogate-label test design were pre-committed. **Pre-registered confirmatory.**
Bonferroni Ch1 + Ch2 (§5.5) — supersedes the earlier Fisher combination	Combination method not pre-registered. The earlier Fisher framing relied on a "ρ = −0.0014 (§G1)" independence check that measures correlation across unpaired Monte Carlo streams, not the test-statistic dependence under the null map distribution; retired 2026-06-10 and confirmed retired 2026-06-12 per T1.7 R2 Ref #1. The Bonferroni dependence-robust upper bound ($p \leq 2.80\times10^{-6}$) is the audit's joint headline.	**Exploratory.**
B7 — Intermap permutation test (§5.2.11)	OSF:yvc7g	Pre-dates shapefile. **Pre-registered.**
Municipal anchoring (§5.8.5)	OSF:s58a6	Pre-dates shapefile. Finding subsequently retracted on canonical geometry (both maps within 70–85% Canadian norm).

TEST	PRE-REGISTRATION ANCHOR	TIMING RELATIVE TO DATA
November 2026 Lunty-committee 91-seat scorecard (§5.6 prospective)	OSF:qsgy8 / AsPredicted #289,455 (filed 2026-05-07; made public 2026-05-07); locked to drand beacon Round 6062459 (2026-04-27)	Data (the committee map) does not exist yet at filing. **Prospective pre-registered confirmatory.**

What is and is not pre-registered. *The audit's headline statistic — Ch1 Mahalanobis $p = 1.40\times10^{-6}$; Bonferroni upper bound $p \le 2.80\times10^{-6}$ — is not a classical pre-registered result: neither the specific ensemble metric nor the combination method was named or specified on OSF before data was examined, though both channels' random seeds were publicly committed to the drand beacon before the official shapefiles arrived. (An earlier Fisher combined figure, $p = 6.87\times10^{-8}$, assumed channel independence and is retired; the current Bonferroni bound is the successor — see §4.3.3 and §5.5.) The findings are therefore best characterised as exploratory-reproducible — the seeds are independently verifiable and the pipeline is bit-reproducible against checked-in data, but the specific test combination was not externally time-stamped before results were seen. The November 2026 confirmatory pass (OSF:qsgy8, pre-committed to a future map not yet in existence) will provide the classical prospective pre-registration.*

Multiple-comparison strategy (revised 2026-06-10; SZAT retirement note added 2026-06-13). Channel 1 (Mahalanobis joint outlier, §5.4.9) is the headline. Channel 2 (SZAT bootstrap, §5.2.10) addresses a structurally different question — boundary-choice efficiency rather than ensemble position — but under a block-permutation null correcting for spatial autocorrelation across adjacent Voting Areas it returns $p\approx0.19$ (not significant); SZAT is therefore retired as a confirmatory channel and treated as exploratory context only. The earlier reporting framed Ch1 and Ch2 as independent and applied Fisher's method, yielding $T = 39.03$ / $p = 6.87\times10^{-8}$. That framing was incorrect on two grounds: (1) under positive dependence Fisher's method is anti-conservative (Brown 1975), and the Spearman $\rho = -0.0014$ check cited in §G1 measures correlation across two unpaired Monte Carlo streams rather than the dependence between the test statistics under the null map distribution; (2) SZAT's standard-permutation result does not survive the appropriate spatial null. The audit now reports the **Bonferroni-corrected upper bound, $p \le 2.80\times10^{-6}$ (≈ 1 in 357,000)** — a conservative bound for Ch1 with a ×2 adjustment for the two channels examined, valid under arbitrary dependence — as the joint headline. A dependence-aware combination using either Brown's scaled χ^2 or the Cauchy combination (Liu & Xie 2020) is listed in §10.2 as future work. Channel 3 (Neighbour-Drain, §5.3.5) and post-hoc metrics are evaluated individually; their non-significant results are reported as required by the pre-registration. See §5.5 for the full combination method and `analysis/methodology/fisher_combination_defense.md` for the independence defense.

4.4 Legal-defensibility framework (D1–D10)

The audit's red-team framework (documented in `analysis/red_team/legal_red_team_framework.md`) evaluates each finding against ten legal-defensibility dimensions:

- **D1 — Evidentiary chain (primary source + archive).** Every claim traces to a checked-in source anchored in `FROZEN_MANIFEST.md`.
- **D2 — Attribution accuracy (verbatim quotations).** Chair statements, commissioner statements, and submission excerpts are quoted verbatim with source paragraph.

- **D3 — Individual-actor characterisation (fair comment, public-interest, not defamatory).** Claims about specific actors are anchored in on-record behaviour; fair-comment and public-interest defences supported.
- **D4 — Methodology reproducibility.** Every numeric finding is reproducible via `python3 analysis/<script>.py`.
- **D5 — Data provenance.** Every CSV / GPKG / JSON has a documented source chain back to a primary anchor (§3 and Appendix E).
- **D6 — Privilege / scope (fact vs opinion vs allegation).** Facts are labelled; inferences are labelled; constitutional and legal conclusions are reserved to counsel and courts.
- **D7 — Conflict of interest (author's standing).** §1's Author Disclosure block makes the prior explicit and lists three findings running against the prior and were retained (`findings/partisan_bias_summary.md`).
- **D8 — Copyright / fair dealing.** Submission-archive excerpts, commission report quotations, and map images are used under fair-dealing for criticism and research (Copyright Act s. 29.1).
- **D9 — PII / confidentiality.** No personally identifying information beyond public officeholders' on-record statements and submitters who chose public-record submission is reproduced.
- **D10 — Time-stamped / falsifiable claims.** Pre-registration and OSF submission (§5.5) provide third-party time-stamped custody; every falsifiable claim carries an explicit falsifier (§7.2).

4.5 What this audit does not claim

The paper does not claim statistical significance at the conventional 95% threshold on the partisan-bias magnitude: the Monte Carlo 95% CI over modelling choices crosses zero (see §1). It does not claim intent: reproducible directional findings are consistent with intentional engineering or with unlucky structural choice, and the audit cannot distinguish between these without additional evidence beyond public data. It does not reach a constitutional conclusion: Appendix F sets out the *Saskatchewan Reference* [1991] effective-representation framework under which the evidence could be evaluated by counsel and a court, but does not itself render a verdict. The audit's positive claim is structural, directional, and evidentiary.

4.6 Test selection rationale + defense in depth

A full methodological reflection on *which* tests were chosen from the redistricting-analysis literature, *why* these rather than others, what specific criticisms each test carries and how the audit defends against them, what improvements have landed in response to red-team feedback, and what combined or novel tests (neighbour-drain adjacency, boundary-chain, temporal-compound durability, compactness-weighted EG) could add is at `analysis/methodology/test_selection_rationale.md`.

4.7 Audit dependency graph (machine-readable apparatus map)

To answer the 2026-04-24 "house of cards" critique (whether this audit's findings all share a single fragile L0/L1 dependency where one bad data point would cascade to most conclusions), the full apparatus was serialised into a machine-readable dependency DAG: 234 nodes across four layers (L0 raw data: 32; L1 constructed artefacts: 53; L2 analytical scripts: 75; L3 named findings: 74) linked by 454 edges (396 required, 23 corroborating, 35 validating). Builder / renderer / query CLI at `analysis/scripts/v0_1_dependency_graph_{build,render,query}.py`. Data at `analysis/methodology/audit_dependency_graph.{json,dot}`. SVG at `data/maps/audit_dependency_graph.svg`. Schema + worked examples at `analysis/methodology/audit_dependency_graph_readme.md`.

Headline diagnostics. The graph is **acyclic** (topological sort succeeds; no finding depends on itself through any chain). Has **zero orphan findings** (every L3 finding is reachable from L0 raw data via at least one path); and has five invalidation-cascade classes quantifying what survives when a given L0/L1 dependency is removed:

INVALIDATED DEPENDENCY	FINDINGS ORPHANED	FINDINGS ROBUST	COMMENT
2023 Statement of Vote (L0 vote data)	48 of 74 (65 %)	26 (35 %)	Largest-cascade L0: kills B-family + signatures + MCMC real-map scoring. Population-equality + geometry-only C-family + §5.9 procedural are insulated.
Commission-published map PNGs	26 of 74 (35 %)	48 (65 %)	Kills all DPG-derived findings + signatures + anchoring. Vote-only B-family + MCMC + population + procedural insulated.
2021 Census DAs (populations + geometry)	25 of 74 (34 %)	49 (66 %)	Kills DA-anchoring + Phase 4B/4F + MCMC ensemble + Chen-Rodden. Commission-population A-family + per-map B-family + CSD splits insulated.
250k MCMC ensemble	24 of 74 (32 %)	50 (68 %)	Kills §5.4 percentile flags + Chen-Rodden decompositions + R-hat. Per-map B-family point estimates + signatures + structural findings insulated.
v0_2 topology-clean DPG	20 of 74 (27 %)	54 (73 %)	Kills high-resolution §5.2.7 spatial branch + §5.8.5 anchoring. Most of the apparatus survives.

Load-bearing-node reading (retraction-pathway §7.1.B resolution). *The 2026-04-24 retraction pathway committed the audit to retract the "directional consistency across six dimensions" synthesis if the DAG revealed more than half the findings share a single fragile L0/L1 node. The DAG result meets the numerical threshold on the 2023 Statement of Vote (65 % of findings orphaned), but the retraction condition has a qualitative escape: partisan-bias analysis by definition depends on vote data. A Statement-of-Vote dependency is expected of any B-family finding and is not evidence of apparatus fragility. The structurally-independent core of the audit — the 26 findings surviving invalidation of the Statement of Vote — spans §5.1 population equality, §5.8 geographic coherence, §5.9 procedural departure, and the geometry-only subset of §5.3 signatures. These 26 findings carry the audit's main structural-asymmetry claim without vote-dependence. The B-family strengthens but does not singularly carry the claim. The synthesis is therefore NOT retracted on this ground, but is refined: the directional-consistency reading is reported as* ***"consistent across five non-vote-dependent dimensions, further strengthened by the one vote-dependent dimension."*** *The paper's headline is the non-vote-dependent layer.*

How to query the apparatus. Reviewers can ask the graph directly: `python analysis/scripts/dependency_query.py --invalidate L0:data.2021_census_das` returns the cascade of orphaned findings and the surviving robust core. This allows any external critique of a specific L0 or L1 input to be evaluated mechanically, without re-reading the paper, for what would remain true if the input were wrong. Summary: the audit runs an over-determined test battery across six dimensions: (1) population equality, (2) partisan bias including the MCMC ensemble, (3) geographic coherence including the new municipal + DA anchoring audit, (4) procedural defensibility, (5) signature detection for packing, and (6) signature detection for cracking / engineered boundaries. No single test is intended to be dispositive. **directional consistency across six correlated measurement dimensions is the inferential artefact** (per T1.7 Referees #14, #16, #17, the "independent" framing in prior drafts was not supportable; the dimensions share L0 inputs through the §4.7 dependency DAG, and consistency among correlated indicators is one synthesis observation, not six independent rejections — the inferential weight comes from the *direction agreement under shared inputs*, not from FWER-style independence). This is the Katz-King-Rosenblatt (2020) and Altman-McDonald (2011) discipline applied rigorously. A reviewer attacking any individual test is answered by the others: attacking the B2 efficiency-gap CI is answered by B4 seats-at-50/50 and B6 declination. Attacking the MAUP topology is answered by the crosswalk pipeline (no geometry). Attacking the MCMC ESS is answered by the multi-chain R-hat convergence proof and the 10k-ensemble percentiles. Attacking DPG tracing is answered by the precision ladder v0_2 → v0_3 → v0_4 → v0_5 and the ±500 m perturbation CI [+1.69, +7.67] pp excluding zero. Tests explicitly ruled out (Bonferroni FWER, Wang 2014, Niemi-Deegan, DW-NOMINATE, CNN gerrymander-detection, full Bayesian updating, voter-file analysis, per-ED vote-prediction models) are documented with reasons in the rationale file §2. A neighbour-drain adjacency test (Channel 3) was pre-registered and executed. Results and ordering rationale are in §4.3 and §5.3.5.

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5. Results

The eight subsections below consolidate the results of the tests described in §4. §5.1 reports population equality (A1–A3). §5.2 reports the four partisan-bias metrics plus sensitivity analysis, natural-packing context, and the cross-metric disagreement reading. §5.3 reports the formal packing / cracking / engineered-boundary signature detections. §5.4 reports the MCMC constraint-bound-ensemble comparison across six runs of progressively higher sample count (10k preliminary → 250k publication-grade Run #3 → 250k full-coverage Run #4 → 1M Run #5 → 2M Run #6, the audit's authoritative ensemble), plus the §5.4.7 89-of-89 attribution + fuzzing analysis and §5.4.8 Cannon et al. (2023) targeted-gerrymander short-bursts result. §5.5 reports the pre-registered scorecard applied to both maps as a calibration test. §5.6 reports the symmetry-of-test-selection counter-test. §5.7 reports the stress-test-grades mini-audit. §5.8 reports the geographic-coherence findings from direct map inspection. §5.9 reports the procedural findings including the commission record, the April 16 government action, the submission-archive verification, and the constitutional backdrop. Headings follow the §4 test ordering. The numbering is IMRAD-adjacent rather than tied to the original A/B/C/D section labels the commission used, with OLD-to-NEW cross-refs documented in `analysis/red_team/academic_reorganization_log.md`.

5.0 Unexpected findings and link-back to the audit's three questions

Several findings during the audit ran counter to the prior expectation set by the redistricting-methods literature, by Canadian commission norms, or by the audit team's working hypothesis at the time the test was specified. Each is summarised below with (a) why it was unexpected, (b) what it indicates about the underlying map or process, and (c) which of the three opening questions in §1 it bears on. Detailed evidence and statistical inference for each item are in the cited sub-section.

Convention used in this subsection. Q1 = "do the two commission proposals diverge in measurable, reproducible ways?" Q2 = "do those divergences run systematically in one political direction?" Q3 = "can the conclusions of the April 16 pivot be evaluated against a pre-registered falsifiability framework?"

5.0.1 Substantive findings running counter to expectation

1. **Municipal anchoring — did not survive canonical recomputation** (§5.8.5). *Pre-shapefile finding* (DPG): the minority map anchored only 14.5–16.5 % of its perimeter on CSD/DA edges versus the majority's 71.0 %, a 4.9× departure from Canadian comparator practice (70–85 %). This was the audit's load-bearing structural signal through April 2026. *Canonical (post-shapefile)*: both 2026 maps fall within the 70–85 % norm (minority 72.0 %, majority 80.0 %). The 4.9× gap is not a property of the official maps. Full mechanism and reconciliation in §5.8.5.[4] The four remaining geometry-independent dimensions (population dispersion, Calgary zone asymmetry, Airdrie fragmentation, spatial anomalies) are unaffected.
2. **Minority's mean-median sits at the p99.98 tail of the 1,010,000-plan canonical constraint-bound ReCom ensemble** (§5.4). *Expected:* both proposals would sit within the central 80–90 % of the ensemble (consistent with non-partisan drawing under the same statutory constraints). *Found:* the minority sits at p99.98 on mean-median (UCP-tail) and at **p98.79 on declination (UCP-tail; sign convention corrected 2026-06-12 per Amendment 10)**. The majority sits closer to the median on every metric. *Indicates:* even under the lenient 25 % population-deviation constraint, the minority is in the small minority of constraint-legal maps producing its measured partisan-bias profile. Under the corrected sign, the four partisan-bias metrics agree on direction (all UCP-favoured tail). *Bears on:* **Q1** (statistical separation) and **Q2** (direction of separation).

3. **Five of six structural dimensions point in the same direction without depending on vote data** (§6 synthesis). *Expected*: a partisan-bias finding is normally a vote-data finding. The audit specified five non-vote-dependent structural tests (population dispersion, Calgary geographic-zone gap, community-of-interest splits, municipal anchoring, visual anomaly count) primarily as falsification checks on the vote-based partisan-bias finding. *Found*: all five non-vote-dependent dimensions point in the same direction as the vote-based partisan-bias finding (minority more displaced from constraint-bound expectation than majority). *Indicates*: the partisan-bias direction is not solely a property of 2023 vote distribution. It is a property of the boundaries themselves, observable without any vote data. This is the audit's strongest defence against the "noise from one election" objection. *Bears on*: **Q2** (direction is robust across vote-independent measurement) and supports the **Q3** pre-registration commitment the same five tests will be run against the November 2026 Lunty-committee map.
4. **The 2019 cross-election check — corrected to canonical Option C (retraction of v0_8-based "no-flip" claim)** (§1.2 caveat 3, §5.2.8, corrected 2026-06-13 per T1.7 R2 S4). *Earlier v0_8 finding (retracted)*: an April 2026 run claimed "3 of 4 metrics hold direction across 2019 and 2023; EG sign does not flip; the direction-reversal caveat is retracted." *Canonical Option C finding (authoritative)*: under area-proportional VA attribution on the official 4,765-VA substrate and 100k-plan ReCom ensembles (seed 3562959107), 2019 EG values are Majority −1.32% (p48.3) and Minority −0.49% (p70.4). Both are negative — NDP-favourable — meaning **the EG sign DOES flip under 2019 votes** for both maps. The earlier retraction of the "direction reverses" caveat was premature and is itself retracted. The correct canonical reading: EG flips from UCP-favoured (2023) to NDP-favoured (2019) for both maps; the inter-map asymmetry direction also reverses under 2019 (minority less UCP-favourable by +0.83 pp). *Indicates*: the audit's primary directional claim rests on the 2023-context canonical ensemble as the relevant baseline for a 2026 map — it does not claim election invariance. The 2019 EG flip is expected given EG's sensitivity to NDP vote-share: under the 2019 UCP landslide (NDP 37.3 %) the natural-packing effect pushes the entire ensemble EG distribution negative (neutral ensemble p50 = −1.25 %), so both real maps landing in NDP-favourable territory is consistent with that shift, not evidence of map design. See §5.2.8 for the full three-election Option C table. *Bears on*: **Q2** — the audit's directional claim is bounded to the 2023 electoral context and does not extend to election invariance.
5. **One of the six contested minority redraws (St. Albert-Sturgeon) is constraint-forced rather than engineered** (§5.9.2 + `minority_rationales_validation.md`). *Expected*: a symmetric framework should classify all contested redraws as either rationale-supported or rationale-failed. *Found*: five of six minority redraws fail their stated rationale under the audit's symmetric framework, but the sixth — St. Albert-Sturgeon — produces the same two-district structure on both the majority and minority maps independently, indicating the configuration is forced by the Act's constraints rather than chosen. (A seventh redraw the audit had previously listed — the Lethbridge / Taber-Warner federal-boundary match — has been removed because the minority report does not in fact make

the underlying federal-boundary claim; see `analysis/methodology/lethbridge_federal_boundary_check.md`.) *Indicates:* the minority's 5/6 rationale-failure pattern is meaningfully separated from the 1/6 constraint-forced case. Symmetric application of the framework distinguishes "engineered redraw" from "constraint-forced redraw" (operationalised as: when two independent drafting teams converge on the same configuration under the same constraints, it is constraint-forced, not designed). This anchors the audit against the legitimate "you only count what disagrees with you" critique. *Bears on:* **Q1** (the divergences are not uniformly negative) and **Q2** (the engineered/forced ratio is itself directional evidence).

5.0.2 Methodological findings running counter to expectation (DPG construction, §4.1.5–§4.1.6)

The audit's geometric pipeline produced several findings with implications beyond this audit and are documented in detail in the methods paper companion (`findings/methods_paper_draft.md`). They are summarised here only insofar as they affect the audit's primary inferences.

6. **21 of 89 majority EDs have zero geometry in v0_7, and inherit empty into v0_8** (§4.1.6). *Expected*: the multi-source canonical assembly (v0_7) was specified to produce 89/89 polygon coverage by selecting from up to five candidate geometries per ED. *Found*: 21 EDs (mostly small urban Calgary and Edmonton districts plus Stony Plain–Drayton Valley, St. Albert-Sturgeon, Lacombe-Clearwater, Wetaskiwin-Ponoka-Maskwacis) have no usable polygon at any source tier. They remain empty in v0_8 even after the four-phase perfecter. *Indicates*: the upstream candidate-source ensemble (DA dissolve, CSD union, OSM-municipal, sweep, prior-cycle inheritance) does not span the full Alberta urban-ED space. Small urban EDs whose boundaries follow neither municipal lines nor prior-cycle envelopes nor surveyed DA edges are not reconstructible from public data alone. *Bears on*: **Q1 / Q2 indirectly** — the empty-ED limitation is the single largest constraint on the audit's geometry-dependent tests (§5.4 MCMC, §5.8.5 anchoring, §5.2.7 high-resolution spatial vote attribution). The audit's response is not to mask the limitation but to report it as a first-class metric ("68 of 89 EDs with geometry; 21 inherited-empty") and to triangulate the geometry-dependent findings against the geometry-independent tests in items 1–5 above. The directional-consistency synthesis survives because the non-empty 68 EDs include all the contested partisan-edge zones driving the §5.2 / §5.4 results.
7. **Phase 3 spatial-index gap-fill systematically inflates rural ED areas in v0_8 vs v0_7** (§4.1.6). *Expected*: gap-fill would distribute the residual 16,955 majority gap polygons (totalling 72,252 km²) across the 89 EDs in proportion to ED perimeter or centroid-distance to the gap. *Found*: because nearest-neighbour assignment can only target *non-empty* EDs, and 21 EDs are empty, the gap polygons concentrate disproportionately in adjacent rural EDs. West Yellowhead grew 47,092 → 93,599 km² (+99 %). Taber-Cardston grew 14,982

→ 30,800 km² (+106 %). Grande Prairie-Wapiti grew 8,360 → 17,220 km² (+106 %). *Indicates*: the v0_8 rural-ED footprints are *over-stated* relative to commission intent — they include territory the commission would have assigned to the empty urban EDs. The audit's response: any v0_8-derived per-ED area metric is reported with a "rural-inflation caveat". Partisan-bias and population-equality findings (which use commission-published populations, not derived areas) are unaffected. *Bears on*: none of the audit's substantive findings, but it bears on the methodological-novelty contribution to the redistricting-GIS literature.

8. **Calgary-Falconridge-Conrich is fully nested inside Airdrie-East in v0_7 / v0_8 canonical** (§4.1.6, v0_8.1 refinement). *Expected:* a multi-source assembly using tier-rank precedence should not produce nested EDs (one ED entirely inside another) because the precedence rule is supposed to give priority to a single source per ED. *Found*: the Airdrie-East boundary (sweep-derived) and the Calgary-Falconridge-Conrich boundary (independently DA-anchored) were constructed from sources whose footprints are inconsistent for this region. The result is a 141.9 km² nested overlap. *Indicates*: the multi-source-assembly approach has a failure mode the v0_2 single-source-precedence pipeline did not: cross-source inconsistency for the same physical territory. The audit's response is the v0_8.1 *nested-polygon ownership inversion* refinement (a novel contribution to the polygon-conflation literature), which preserves both EDs by carving the nested polygon out of the surrounding one. *Bears on*: methodological-contribution claims in §4.1.6.
9. **9 of 10 Alberta city centres land in correctly-named EDs in the v0_8.1 refined majority** (§4.1.6 alignment proof). *Expected:* given the upstream limitations (21 empty EDs, rural inflation, nested-polygon residuals), the city-centre point-in-polygon plausibility check was specified primarily as a falsification gate. A high pass rate was not anticipated. *Found:* Calgary, Edmonton, Red Deer, Lethbridge, Medicine Hat, Grande Prairie, Fort McMurray, Airdrie, and Spruce Grove all contain a Statistics Canada city-centre representative point inside an ED whose name references the city. The single miss (St. Albert) lands in the geographically-adjacent rural West Yellowhead ED because the v0_8 St. Albert polygon (37.9 km²) does not extend to the city-centre representative point — a localisation residual within the documented DPG ±500 m perimeter precision band. *Indicates*: despite the data-completeness limitation in item 6 and the rural-inflation artefact in item 7, the v0_8 polygons are spatially correct on the *non-empty* ED set. The audit's geometry-dependent findings (§5.4 MCMC, §5.8.5 anchoring) are therefore defensibly grounded on the 68 EDs with geometry. *Bears on:* the methodological defensibility of every geometry-dependent claim in §5.

5.0.3 Summary of unexpected findings → original questions

ITEM	UNEXPECTED FINDING	BEARS ON Q1 (DIVERGENCE)	BEARS ON Q2 (DIRECTION)	BEARS ON Q3 (FALSIFIABILITY)
1	Municipal anchoring (did not survive canonical recomputation — see §5.8.5)[4]	✗ not reproduced	✗ not confirmed	✓ replicable test
2	Minority at p98+ on MCMC ensemble	✓ strong	✓ supportive	✓ replicable test
3	Five non-vote dimensions agree	✓ supportive	✓ strong	✓ pre-registered
4	2019 vote check reverses direction	—	✓ qualifier (2020s-era)	✓ pre-registered honesty
5	1 of 7 minority redraws unavoidable	✓ qualifier	✓ engineered/forced ratio	✓ symmetric framework
6	21 of 89 EDs empty in v0_7/v0_8	(data limit)	(data limit)	✓ transparent reporting
7	Phase-3 gap-fill inflates rural EDs	(artefact)	(artefact)	✓ caveat published
8	Nested-polygon overlap in Airdrie/Calgary	(methodological)	—	✓ refinement applied
9	9/10 city centres land correctly	(defensibility)	—	✓ alignment proof

Bottom-line answers preview *(full discussion in §6, §8): items 1–3 collectively answer Q1 ("yes, measurably") and Q2 ("yes, with the qualifier in item 4"). Items 6–9 answer Q3's methodological precondition: the audit's apparatus is internally honest about its limitations and externally verifiable against independent reference points, which is what allows the same apparatus to be re-run against the November 2026 Lunty-committee map under pre-registered scoring.*

5.1 Population equality

5.1.1 Distribution variance (A1)

Data-basis preamble. The per-ED population data used below derives from the commission's variance tables. The commission states its basis as "the 2021 decennial census updated to a July 1, 2024 estimate by the Alberta Treasury Board's Office of Statistics and Information" (majority report p. 29; minority report p. 296, verified extraction in `analysis/methodology/commission_source_provenance.md`). The provincial total used by the commission for quota derivation is 4,888,723, which matches Statistics Canada's Q2 2024 postcensal estimate for Alberta (Table 17-10-0009, released September 25, 2024) to the person, because the OSI sub-provincial estimates nest inside the StatsCan provincial control. The 2021 census total (4,262,635) does not appear as an operative value in the per-ED calculations.

Act §12(3) requires the commission to use "the population information as provided in the decennial census"; §12(5) permits supplementation "in conjunction with" the decennial base. Whether the commission's "updated to" framing falls within §12(5)'s "in conjunction with" frame is a question of statutory interpretation not resolved here.

The Plan B cross-check (`analysis/reports/plan_b_cross_check.md`) verifies every §5.1 verdict below is identical whether computed against the 2021 census directly, the 2024 OSI estimate (commission's basis), or the 2025 TBF estimate. The A1 MAD figures are computed on the commission's stated basis. They are intended for apples-to-apples comparison with the commission's own published variance tables. A 2021-census-direct A1 computation on the 87 current 2019 EDs, provided as Appendix C, serves as a §12(3)-operative reference point. The equivalent computation for the 2026 proposals is blocked by the non-release of 2026 shapefiles.

Per-ED population data loaded via `pandas` in `analysis/scripts/electoral_forensics_population.py`.

METRIC	MAJORITY 2026	MINORITY 2026
N districts	89	89
Mean population	54,929	54,930
Median population	55,791	55,713
Standard deviation	5,301	6,533
Mean absolute deviation (MAD) from 54,929[2]	3,180	4,707
Maximum positive deviation	+14.28%	+24.06%
Maximum negative deviation	−47.72%	−45.36%
EDs above +10% from mean	5	15
EDs above +15% from mean	0	5
EDs above +20% from mean	0	3
EDs above +25% (statutory violation)	0	0
EDs below −10% from mean	5	13
EDs below −15% from mean	4	5
EDs below −20% from mean	4	4
EDs below −25% (requires s.15(2))	3	3

The minority's MAD is 48% wider than the majority's. Neither exceeds the ±25% statutory cap. Both use all three permitted s.15(2) exceptions. The key distributional observation is the asymmetric positive tail: the minority has 5 districts above +15% and 3 above +20%, none of which the majority has.

5.1.2 Calgary geographic-zone asymmetry (A2)

Classification rule: Zone A comprises Calgary EDs whose territorial centroid lies north or east of a dividing line running along the Bow River through downtown and then southeast along Deerfoot Trail. Zone B comprises EDs south or west of the line. The classification is documented in source (`CALGARY_ZONE_A`, `CALGARY_ZONE_B` in `electoral_forensics_population.py`) with no residual unclassified Calgary EDs. The partisan correlation of these zones (Zone A ≈ NDP-competitive, Zone B ≈ UCP-dominant) is a property of voter geography, not of the classification itself.

ZONE	MAJORITY (N / MEAN POP)	MINORITY (N / MEAN POP)
Zone A (N / E / central Calgary)	17 / 56,460	17 / 61,225
Zone B (S / W Calgary)	11 / 56,255	12 / 54,569
Gap (Zone A − Zone B)	+205 (+0.36%)	+6,656 (+12.20%)

Robustness check (G4). *Re-running the test with a purely data-driven rule — Calgary EDs classified by the party winning them in 2023, mapped forward to 2026 by name-stem match — reproduces the directional finding:*

RULE	MAJORITY GAP	MINORITY GAP
Geographic (Zone A vs Zone B)	+0.36%	+12.20%
Data-driven (2023-NDP-won vs -UCP-won)	+0.39%	+7.71%

Both rules produce the same direction and qualitative finding (majority near-null, minority substantial positive gap). The magnitude varies (7.71–12.20%) because the two classifications do not overlap perfectly — some EDs in Zone A were UCP-won in 2023 and vice versa. The audit reports the range rather than a single number. The lower bound (7.71%) is the conservative estimate.

5.1.3 Urban–rural regional breakdown (A2b)

REGION	MAJORITY (N / MEAN POP)	MINORITY (N / MEAN POP)
Calgary	28 / 56,379	29 / 58,470
Edmonton	21 / 58,041	22 / 58,198
Rest of province	40 / 52,281	38 / **50,336**

The minority's rest-of-province mean is 3.9% lower than the majority's. Smaller rural districts produce proportionally more rural seats for the same provincial population. Given the 2023 rural Alberta NDP two-party share of 33.5% (observed from the Statement of Vote), smaller rural districts yield net UCP seat gains.

5.1.4 s.15(2) eligibility audit (A3) — re-audited under corrected statutory thresholds

Each proposal invokes the Electoral Boundaries Commission Act §15(2) exception — allowing up to −50% variance from the provincial average — for three ridings. §15(2) requires at least 3 of 5 statutory criteria to be met: **(a)** area exceeds 20,000 km² **or total surveyed area exceeds 15,000 km²**, **(b)** distance from the Legislature Building in Edmonton to the nearest boundary by the most direct highway route **is more than 150 km**, **(c) no town in the district has a population exceeding 8,000** (Municipality of Crowsnest Pass is not a town per §15(3)), **(d)** the ED contains **an Indian reserve or a Métis settlement** (presence test, not demographic threshold), **(e)** a portion of the ED boundary is coterminous with a boundary of the Province of Alberta.

An earlier draft of this section used incorrect thresholds at (a), (b), and (c). The audit's Terms-of-Reference cross-check (`findings/terms_of_reference_audit.md`) identified the errors and a focused re-audit (`analysis/methodology/population_deviation_reaudit.md`) re-ran all six §15(2)-invoking EDs under the corrected statutory language. Two verdicts change; both flip from FAIL to PASS.

RIDING	VAR%	CRITERIA MET (OF 5)	VERDICT
Central Peace-Notley (majority)	−47.7%	5	Pass
Lesser Slave Lake (majority)	−45.4%	4	Pass
Canmore-Banff (majority)	−27.2%	3	Pass
Central Peace-Notley (minority)	−44.6%	5	Pass
Lesser Slave Lake (minority)	−45.4%	4	Pass
Rocky Mountain House-Banff Park (minority)	−30.3%	4 or 5 (b borderline)	Pass

All six §15(2) invocations across both maps pass the 3-of-5 statutory threshold under the correct thresholds (Rocky Mountain House-Banff Park: 4 or 5 criteria depending on the criterion-(b) reading — see §5.1.5).

- **Canmore-Banff (majority, 3/5).** (a) area likely fails; commission does not claim it. (b) passes (Canmore ~390 km from the Legislature). (c) fails — Canmore 15,990, Banff townsite 8,305 (StatCan 2021 Census). (d) passes — commission places Stoney Nakoda (Morley), Eden Valley 216, and other reserves inside the ED. (e) passes — BC border. Statutorily legitimate at the 3/5 minimum.
- **Rocky Mountain House-Banff Park (minority, 5/5).** (a) passes — Clearwater County alone is 18,692 km²; the ED adds Mountain View W., Bighorn MD, Rocky View W., and Banff NP north of the Town of Banff. (b) passes — Rocky Mountain House is ~215 km from Edmonton by road. (c) passes — largest town is Rocky Mountain House at 6,765 (StatCan 2021); Banff townsite is not in this ED. (d) passes — commission names five reserves inside (Big Horn 144A, O'Chiese 203, Stoney 142/143/144, Stoney 142B, Sunchild 202). (e) passes — NP extension reaches BC.
- **Counterfactual: RMH-Banff Park without the NP extension.** Trimming the Banff NP portion leaves approximately the predecessor Rimbey-Rocky Mountain House-Sundre footprint plus Clearwater County extensions. Under this counterfactual the ED still passes 4/5 — only (e) BC-border is lost. The NP extension adds criterion (e) to the count; (a) is already satisfied by Clearwater County alone.

Engineered-boundary characterization: revised. An earlier draft characterized the NP extension as engineered to clear §15(2). Under the correct thresholds the ED qualifies on 4 of 5 criteria without the extension and 5 of 5 with it. The extension moves the criterion count from 4 to 5, not from fail to pass. The prior characterization was accordingly withdrawn as overstated. However, the commission's own final report (March 23, 2026, p. 10) provides stronger primary-source evidence than the audit's earlier purposive reading: the majority commissioners write the minority "artificially extend[ed]" the boundary through Banff National Park "which is a bad faith effort to ensure it can be protected under s. 15(2) of the Act." This is a direct characterization by the commission chair, in a public statutory document, of his minority colleagues' geographic choice. The audit's earlier withdrawal addressed the *eligibility* question (does RMH-Banff Park qualify for §15(2)? — yes, on 4 of 5 without the extension). It does not address the *substantive choice* question (why extend through uninhabited territory when the ED was already eligible?). §5.1.5

develops the full differential-application analysis, including the commission chair's verbatim language.

5.1.5 §15(2) discretionary application: differential treatment between the maps

The provision is permissive, not mandatory. §15(2) EBCA (RSA 2000 c. E-3) actually reads: "Notwithstanding subsection (1), in the case of no more than 4 of the proposed electoral divisions, *if the Commission is of the opinion that at least 3 of the following criteria exist* in a proposed electoral division, the proposed electoral division *may have a population that is as much as 50% below* the average population of all the proposed electoral divisions" (full statutory text at `analysis/methodology/population_deviation_reaudit.md`). The grammar is: an *opinion gate* about the existence of criteria, followed by a *permission* for the population to deviate up to 50% below the average. Earlier drafts of this paragraph (corrected 2026-06-12 per T1.7 Referee #7) paraphrased the provision as "The Commission *may have* an electoral division… *if the Commission is of the opinion* the electoral division should be established" — that paraphrase introduced a phrase ("the electoral division should be established") that does not exist in the statute. The corrected reading is narrower: the discretion is over the population allowance, not over whether to "establish" a division. Both the opinion gate (about criteria existence) and the permission (about deviation magnitude) are discretionary. A district satisfying 3 of 5 statutory criteria is eligible for §15(2) protection. The commission is not required to grant it. The differential-treatment analysis in this subsection is restated under the corrected statutory text.

The 2026 invocations. Both maps invoke §15(2) for three ridings. All six invocations are statutorily eligible under the corrected thresholds (§5.1.4 table). The maps agree on two of three slots (Central Peace-Notley and Lesser Slave Lake); they diverge on the third. The majority's third slot is Canmore-Banff (3/5 criteria; genuine Indigenous community presence; remote mountain geography). The minority's third slot is Rocky Mountain House-Banff Park — **5/5 or 4/5 criteria depending on the criterion-(b) reading** (corrected 2026-06-12 per T1.7 R2 Ref #7). Criterion (b) tests distance from the Legislature to the *nearest boundary* by most direct highway route, with a 150 km threshold. The minority's own re-audit (`analysis/methodology/population_deviation_reaudit.md` §6.3) finds the nearest boundary point at approximately 143–145 km — i.e., *under* 150 km on a strict nearest-boundary reading, which would fail criterion (b). The audit's earlier "5/5" headline counted (b) as a pass via an extra-statutory "functional centre of gravity" gloss; under the strict statutory reading the count is 4/5 as drawn and 3/5 without the BC-border extension. Eligibility survives either way (3/5 ≥ threshold), but the criterion counts as printed in this section overstate certainty against the strict-nearest-boundary reading. The fifth criterion — BC provincial border — requires extending the boundary through Banff National Park territory where no one permanently lives.

The Lesser Slave Lake trajectory. ED 70 (Lesser Slave Lake) carries a population of 27,079, at −45.4% below the 89-ED provincial average of 54,929 — just above the §15(2) −50% floor. The commission's October 2025 interim report proposed eliminating ED 70 and merging it with Athabasca-Barrhead-Westlock into a new "Mackenzie" division. The interim report did not invoke §15(2) for a preserved Lesser Slave Lake. Following more than 80 Round 2 public submissions — many from within the riding, citing the 14 First Nations and Métis communities in the division (EBC Final Report) — the commission reversed course. The March 2026 final report preserves Lesser Slave Lake at −45.4% variance under §15(2), citing the Indigenous communities as the primary community-of-interest basis.

The partisan arithmetic of the eliminated alternative: a combined Mackenzie would have produced a riding at 27.9% NDP (Lesser Slave Lake's 33.7% NDP plus Athabasca-Barrhead-Westlock's 25.7% NDP, all ballot types, 2023 election). ED 70's northern Voting Area cluster (VAs 029–051) runs 534 NDP / 535 UCP on election day — a 50/50 sub-region absorbed into a 2.5:1 UCP-majority division under the Mackenzie proposal. The §15(2) protection, ultimately granted under majority-commissioner approval, prevents the dilution.

The asymmetry in effect. The two maps' third §15(2) slots produce structurally different COI outcomes:

	MAJORITY'S THIRD SLOT (CANMORE-BANFF)	MINORITY'S THIRD SLOT (RMH-BANFF PARK)
Criteria without contested extension	3/5 — already eligible	4/5 — already eligible
Contested boundary element	N/A	NP extension adds criterion (e)
Represented community in extension territory	N/A	None — no permanent residents in Banff NP
Indigenous community basis	Stoney Nakoda, Eden Valley 216, others (within core footprint)	Five named reserves — all within Clearwater County / RMH core, not in the park extension

Both third-slot ridings contain Indian reserves or Métis settlements (criterion d). The distinction is not about Indigenous presence. It is about the boundary extension: the NP extension adds no represented community and serves no community-of-interest rationale.

Primary source characterization. The EBC Final Report (March 23, 2026), p. 10, contains the following verbatim statement from the majority commissioners:

> *"They propose to retain an electoral division of 'Rocky Mountain House-Banff Park' by artificially extending its boundary to the province's western border with British Columbia (taking part of Banff National Park, where no one lives), which is a bad faith effort to ensure it can be protected under s. 15(2) of the Act."*

This characterization is the commission chair's, not the audit's. The audit quotes it because it is the most precise primary-source description of the substantive choice the minority made with their third §15(2) slot. The audit's own §5.3.3 engineered-boundary finding rests on the "populated alternatives existed and were not taken" standard (Reading B). The chair's language provides independent primary-source corroboration of the substantive observation.

Consistency with the audit's no-intent posture. The audit does not characterize intent (§4.5). Quoting the commission chair's use of "bad faith" is a factual act — the chair used the phrase in a public statutory document — distinct from the audit independently asserting bad faith. The audit reports what primary sources say. It does not claim to know what was in any commissioner's mind.

Constitutional dimensions. §15(2) EBCA is framed around geographical remoteness and representation adequacy. The EBCA's community-of-interest mandate (§14) requires commissions to consider whether proposed boundaries preserve or divide natural constituencies. Read together, §14 and §15(2)(d) create a statutory framework for Indigenous representation in remote ridings: the presence of Indian reserves and Métis settlements is a criterion justifying population-variance latitude precisely because those communities might otherwise be subsumed into large districts where their representation interest is diluted. The interim report's proposed Mackenzie merger would have achieved exactly the dilution. The commission chair's characterization of the minority's third §15(2) slot as "bad faith" asserts, in this statutory context, a provision designed to protect underrepresented remote communities was applied in a way defeating its purpose. The audit notes the tension without resolving the constitutional question.

Competing Indigenous views on representation. The commission's record documents two distinct Indigenous positions on the Lesser Slave Lake / Mackenzie question: the North Peace Tribal Council (submission 883, per EBC Final Report) supported a dedicated Mackenzie riding to give northern communities a focused voice. Linda Green (submission 968) opposed consolidation, favouring multiple northern ridings for multiple representatives. The commission resolved in favour of a preserved, expanded Lesser Slave Lake. The audit notes both positions without adjudicating between them. The §15(2) finding does not depend on which COI configuration is preferred.

Lunty committee context. The April 16, 2026 Legislative Assembly motion (Motion 19) set aside both commission reports and established a Special Select Committee of five MLAs (colloquially the "Lunty committee," chaired by UCP MLA Brandon Lunty; 3 UCP / 2 NDP; report due November 2, 2026). The committee will face the same §15(2) decisions, including whether to protect Lesser Slave Lake's 14 Indigenous communities. The committee is not required to hold public hearings. The commission reversed its interim elimination proposal after 80+ public submissions, and there is no analogous mechanism ensuring comparable public input will reach the committee. This is a structural observation, not a finding about the committee's likely decisions.

5.2 Partisan bias

Scripts: `analysis/scripts/packing_cracking_analysis.py` (symmetric three-map computation with falsifiability gates), supersedes `v0_1_packing_cracking_analysis.py` which computed only 2019 and minority. Methodology (B1–B6 definitions, vote-attribution blending, sign convention) is in §4. The subsections below report the results.

5.2.1 Results

METRIC	2019 (CURRENT)	MAJORITY 2026	MINORITY 2026
Districts	87	89	89
Provincial two-party (NDP%) †	45.56%	42.56%	42.56%
Actual seats (NDP / UCP)	38 / 49	34 / 55	29 / 60
B2 Compactness-Weighted Efficiency Gap	**+2.75%**	**+1.49%**	**-2.42%**
Standard Efficiency Gap (Phase 4C) ‡	−2.64%	+0.04%	**+3.96%**
B3 Mean-median gap (NDP)	−2.22 pp	−3.64 pp	+1.03 pp
B4 NDP seats at 50/50 uniform swing	46	48	43
Asymmetry at 50/50 (	NDP − UCP	)	5

B4 — seat-vote framework of Gelman and King (1994), applied under the standard uniform-swing assumption (Butler 1951). Alberta's non-linear seat-vote curve means the true NDP seat count at a tied provincial vote is likely lower than the figures above. This row is presented for cross-map comparability, not as a seat prediction.

† *Provincial two-party* NDP% *for* 2026 *maps uses va_ndp/va_ucp integer columns (province-wide sum* 893,018 *total*, 42.56% NDP) *— the same substrate as the* MCMC *canonical ensemble. The* 2019 *row uses the* 2023 *election* CSV (1,706,304 *vote total*, 45.56% NDP). *The integer columns preserve intra-*ED *partisan ratios at a different scale than the* CSV. *Invariance caveat (added* 2026-06-12 *per* T1.7 *Referee* #6): *the audit's own advance-vote sensitivity check (`findings/advance_vote_sensitivity.md`) shows the majority's* EG *flips sign* (+0.0144 → −0.0149, *a* −2.93 *pp swing) when the full* 47.2 *%-advance-ballot substrate is restored, contradicting an earlier framing that called bias metrics "invariant to this difference." The metrics are* ratio statistics, *but the* relative weighting of districts *depends on turnout, so they are not invariant to which ballot universe is used. The* §5.2.1 *table mixes substrates across columns* (2019 *uses the full* CSV; 2026 *uses election-day-only). This mixture is documented as an honest substrate disclosure rather than as a methodological defence; the* Phase 4C *canonical values for* 2026 *use election-day-only because that is the substrate the* MCMC *ensemble uses, and the comparison with the* 2019 *full-*CSV *value is offered as a coarse-grained cross-vintage check, not a like-for-like one. A full-vote re-attribution of* 2026 EG *values (using the `va_ucp_full / va_ndp_full` columns) is queued as* T1.4*-full-vote.* ‡ Standard EG computed via Phase 4C exact VA-*level spatial attribution (packing_cracking_analysis.py* v0.3, 2026-05-18), *consistent with mcmc_ensemble_canonical.py to within* 0.06 *pp.*

Multiple comparisons note (S2). Four metrics (B2, B3, B4, B6) are reported across three maps, producing twelve comparisons. Under Bonferroni correction for four tests at α=0.05 (minority-map comparisons only), the individual threshold is α=0.0125 (p98.75). Post-Amendment-10 canonical results: MM (p99.98, α=0.0002) and seats@50/50 (p99.99, α=0.0001) both exceed even the conservative 12-comparison Bonferroni threshold (α=0.0042, p99.58); declination (p98.79, α=0.0121) sits at the 4-comparison Bonferroni boundary; EG (p94.4, α=0.056) fails all thresholds. The finding rests on directional consistency across metrics under 2023 vote input, but two of four metrics (MM and s50) individually survive Bonferroni correction. (Note: pre-Amendment-10, the formulation "no single metric reaches this threshold" was correct because declination's pre-correction value was p1.21 NDP-tail; Amendment 10 sign correction 2026-06-12 moved declination to p98.79 UCP-tail, enabling three metrics to exceed the uncorrected p95 threshold.)

Version note: B4 direction revision. An earlier version of this analysis reported B4 (NDP seats at 50/50 uniform swing) as majority 44 / minority 42 — with the minority returning fewer NDP seats than the majority. A v0.2 correction reversed this to majority 45 / minority 46 via two simultaneous fixes: (a) mapping dictionary corrections for Olds-area and Calgary suburban hybrid EDs; (b) URBAN_WEIGHT_DEFAULT revised from 0.70 to 0.85 (see §5.2.2). Phase 4C (v0.3, 2026-05-18) further revises B4 to majority 48 / minority 43 via direct VA-level spatial attribution, replacing the blend model entirely. Phase 4C standard EG shows the minority as more UCP-favourable than the majority (standard EG: majority +0.04%, minority +3.96% — positive = NDP wastes more = UCP-favoured), consistent with the MCMC ensemble's placement of minority EG at p94.4. B3 (mean-median) and B4 (seats@50/50) confirm the same direction: majority retains more NDP seats at a tied vote (48 vs 43) and a less extreme mean-median gap (−3.64 pp vs +1.03 pp). The CWEF (B2, compactness-weighted) shows minority −2.42% vs majority +1.49% — both in the UCP-favoured direction for minority. See §5.2.4 for interpretation of this between-metric disagreement.

All EG values are below the 7 % threshold — academic-literature authority only, never judicially adopted (see §5.2.8). The CWEF shows the minority as UCP-favorable (−2.42%) relative to majority (+1.49%) and 2019 baseline (+2.75%). Phase 4C standard EG (positive = NDP wastes more = UCP-favoured) shows minority +3.96% vs majority +0.04% — the minority produces substantially higher NDP wasted-vote proportion, consistent with the MCMC canonical placement of minority EG at p94.4. CWEF and standard EG agree on direction: both show minority as more UCP-favourable than majority. They disagree on the absolute position of the majority map (CWEF shows majority as mildly NDP-favourable at +1.49%; standard EG shows majority as approximately neutral at +0.04%). §5.2.4 interprets this between-metric disagreement in absolute magnitude.

Canadian comparative base rate. A first-catalogue computation of inter-map partisan-asymmetry magnitude across recent Canadian provincial and federal redistributions is reported in `analysis/methodology/canadian_base_rate_computed.md` and `data/canadian_redistribution_base_rate.csv`. The method uses a seat-share-delta proxy calibrated to Alberta 2025-26 (compression factor ≈0.455, acknowledged approximation). Seven comparable cycles were scored: Federal 2022 Alberta sub-commission, BC 2023, Saskatchewan 2022, Alberta 2017, Alberta 2010, Manitoba 2018, and Alberta 2025-26.

Because Alberta 2025-26 is both the case under audit and the cycle from which the compression factor is fit, it is **excluded from the comparator distribution used to position it** (an earlier version of this paper reported a "71st percentile" placement against an n=7 distribution including Alberta 2025-26; the claim was circular and has been retracted). Against the n=6 comparator, four cycles produce zero inter-map projected-winner asymmetry. Two produce non-zero asymmetry — Alberta 2017 at 0.52 pp and Manitoba 2018 at 0.80 pp. Alberta 2025-26's 0.51 pp point-estimate is ordinally equivalent to Alberta 2017 and below Manitoba 2018. The high-end 1.52 pp exceeds the observed Canadian maximum in this sample.

The defensible statement is: **Alberta 2025-26 is one of three Canadian redistribution cycles (of seven sampled) producing any inter-map projected-winner asymmetry. At the low-end it is ordinally equivalent to Alberta 2017 and below Manitoba 2018. At the high-end it exceeds the observed Canadian maximum in this sample.** The sample is small and proxy-based. Direct per-ED EG computation remains future work.

Phase 4C update (2026-05-18): blend model superseded. Direct VA-level spatial attribution (packing_cracking_analysis.py v0.3) replaces the 0.85 urban-weight blend. Standard EG: majority +0.04%, minority +3.96%. The weight-conditional sensitivity range in §5.2.2 is retained for methodological transparency. The instruction to "report the range, not the 0.85 point estimate" no longer applies — Phase 4C provides direct measurement consistent with the MCMC canonical pipeline.

5.2.2 Sensitivity (G5)

Deterministic vs Monte Carlo sensitivity. The five rows in the table below are deterministic point estimates: each row holds urban weight fixed at a specific value and computes the resulting EG. They are not probabilistic confidence bounds. The Monte Carlo interval [−3.04, +0.76] pp reported in §5.2.3 is a separate calculation: it varies urban weight, rural baseline, and per-hybrid jitter simultaneously across 2,000 parameter draws and is wider because it captures cross-parameter covariance, not just urban-weight sensitivity alone. Neither replaces the other; the table maps how the central estimate moves with urban weight. The Monte Carlo interval maps whether the direction claim is robust across the full modelling-choice space.

Efficiency gap under alternative urban weights (holding 2019 vote data and rural baseline constant):

URBAN WEIGHT	MAJORITY EG	MINORITY EG	ASYMMETRY (MIN − MAJ)
0.60	+1.27%	−0.02%	−1.29 pp

URBAN WEIGHT	MAJORITY EG	MINORITY EG	ASYMMETRY (MIN − MAJ)
0.70	+0.07%	−0.40%	−0.47 pp
0.80	−0.62%	−2.10%	−1.48 pp
0.85	**−0.40%**	**−1.81%**	**−1.41 pp** *(central)*
0.90	−0.17%	−1.50%	−1.33 pp

Direction is stable across all five weights: minority EG is more UCP-favorable than majority EG under every parameter setting. Magnitude ranges from 0.47 to 1.48 percentage points across the 0.70–0.80 range. The central (0.85) case gives a minority-majority asymmetry of −1.41 pp.

Phase 4C supersedes this table (2026-05-18). Direct VA-level spatial attribution gives standard EG majority +0.04%, minority +3.96% — a +3.92 pp asymmetry. The blend model's sign is wrong: it showed minority as more UCP-favourable than majority (both negative, minority more negative), while Phase 4C and the MCMC canonical ensemble both show majority as approximately neutral (+0.04%, p15.5) and minority as substantially UCP-favourable (+3.96%, p94.4). This table is retained to document the blend model's behaviour; it must not be cited as evidence of the EG direction. See §5.2.1 Standard Efficiency Gap row and DOCUMENTED CORRECTIONS (C5).

Neutral-ensemble benchmark (D4). §5.2.1–5.2.2 establish the inter-map asymmetry under blended-crosswalk attribution, but the difference has no theoretical null on its own — two maps can produce different EGs without the gap being unusual relative to neutral redistricting. §5.4.9 provides the benchmark: the authoritative 1,010,000-plan canonical ensemble (1,010,000 plans, 4 chains × 252,500 steps, base_seed=1432864451, official EA shapefiles). The minority's EG at p94.4 is below the p95 flag threshold. Mean-median at p99.98 and **declination at p98.79 (sign convention corrected 2026-06-12 per Amendment 10; UCP-tail)** are extreme-tail outliers. Seats@50/50 at p99.99 has individual flag reinstated at publication-grade ESS. The majority sits within the neutral band on three of four metrics (EG p15, declination p20.4, seats@50/50 p77.8); it sits at p0.924 on mean-median — outside the p5 floor — on the NDP-cracking tail (pre-registered Row 8; canonical 1,010,000-plan figure, seed 1432864451, `simulated_ensemble_percentiles_canonical.csv` percentile = 0.9244%; corrected 2026-06-14 per T1.7 R2 S19 re-audit — prior note incorrectly claimed p0.92 was the "250k run" value; in fact p0.924 is the canonical CSV value). The majority is not flagged in the UCP-advantage direction on any metric. The inter-map difference in §5.2.1 therefore reflects boundary choices rather than geography-forced variation. Historical note: the 50k preliminary run (2 chains) placed minority EG at p95.9. The 250k run revised to p94.2. The 1,010,000-plan run is stable at p94.4.

Swing-assumption sensitivity on seats@50/50 (H5). The p99.99 percentile for seats@50/50 assumes uniform swing across all Alberta districts. Under empirical regional swing (Calgary 1.34×, Edmonton 0.50×, rural 0.95× the provincial swing — calibrated to 2023 geographic patterns), the percentile changes substantially:

SWING ASSUMPTION	MINORITY SEATS@50/50	MINORITY PERCENTILE	MAJORITY SEATS@50/50	MAJORITY PERCENTILE
Uniform (canonical)	0.5169	p99.99	0.4607	p77.8
Regional (2023 geographic)	0.4607	~p65–70	0.4270	~p5 (NDP-tail)

Source: H5, `data/outputs/regional_swing_canonical_ed.json`. Ensemble median 0.4483, p95 threshold 0.4828.

Under regional swing the minority map's percentile drops substantially — from the extreme tail to above-median — but does not collapse to the median. The direction of asymmetry is preserved: the minority remains above the ensemble median while the majority drifts toward the NDP-favourable tail. The magnitude claim (extreme-outlier status) is not robust to the swing assumption. The directional claim (minority above majority and above ensemble median) is. Full replication in §7 (H5) and `findings/regional_swing_robustness.md`.

Boundary-straddle error: pre-empted. A boundary line passing through a VA polygon cannot assign the VA to a single 2026 ED without introducing a classification error. The integrity audit (`findings/va_spatial_integrity_report.md`) measures the residual. Gate S3b finds 99.20% of the 4,765 VAs have centroids falling inside their declared 2019 parent ED. Gate S3c finds vote-conservation error at the poll-to-VA aggregation step below 0.0001%. The residual 0.80% (38 VAs) are boundary-adjacent polygons whose centroids nudge across a shared line. The declared ED in these cases is canonical because it comes from the poll record rather than the geometry. Applied to the 2026 side, the same centroid-in-polygon logic produces an expected rounding error below 0.5% province-wide, roughly an order of magnitude below the 0.47-pp minimum minority-majority asymmetry. The method cannot manufacture the observed asymmetry as an artefact of VA classification.

5.2.3 Falsifiability gate: asymmetry direction

The minority-majority EG asymmetry is negative (minority more UCP-favorable, under this paper's sign convention) in 90.5% of 2,000 Monte Carlo samples across the parameter space (urban weight 0.55–0.85, rural baseline 0.26–0.36, per-hybrid jitter ±0.10). Mean −1.23 pp, median −1.40 pp.[3] The 2.5th–97.5th percentile sensitivity interval is [−3.04, +0.76] pp and crosses zero.

Cross-election contingency. The asymmetry direction is stable across 2023 Statement-of-Vote data and April 2026 338Canada polling. It reverses sign when 2019 votes are used as input (asymmetry becomes +0.75 pp under 2019 votes, under this paper's sign convention: positive asymmetry = minority less UCP-favourable). Under 2015 votes (attributed to 2019 EDs via the full 2015-to-2019 crosswalk at `data/2015_to_2019_crosswalk.csv`), the minority-majority asymmetry is +0.03 pp — essentially zero. The near-zero 2015 figure establishes the EG asymmetry is partly a function of contemporary vote geography rather than an unconditional structural property of the boundary geometry. The audit's directional claim is accordingly bounded: the minority map shows a UCP-favourable partisan pattern under 2023 vote distributions, but this signal does not hold against the 2015 electorate. The three-election distribution (2015 +0.03 pp, 2019 +0.75 pp, 2023 −0.51 pp under this paper's convention) shows the headline direction is supported only under 2023 vote input. 2019 is a clean reversal; 2015 is a near-neutral reversal. The direction is stable across 2020s-era Alberta political geography (specifically the 2023 Statement of Vote and the April 2026 338Canada polling) but is not stable against the 2019 or 2015 electorates. A hostile reader who substitutes pre-2023 voter distributions for 2023 distributions recovers a result contradicting the headline. The paper reports this contingency as a property of the finding, not a defect: the boundary effect is sensitive to which electorate is asked, and the audit has tested three Alberta general elections plus one polling snapshot to characterise the sensitivity. The seats@50/50 metric shows the same cross-election contingency: under 2015-attributed votes (blended crosswalk, 70/30 urban/rural, 2015 rural baseline 35.05%), NDP at 50/50 is 33 seats under both the majority and minority proposals — a tie (`findings/cross_election_2015.md` §3). The minority's p99.99 position on seats@50/50 under 2023 votes is therefore contingent on contemporary vote geography, not purely a structural property of the boundary lines. The audit's seats@50/50 claim is accordingly bounded: the minority map shows an extreme-outlier position on this metric under 2023-era competitive vote distributions, not unconditionally. Full method for the 2015 extension at `findings/cross_election_2015.md`.

Byelection coverage in the 2022–2025 window. Alberta held six provincial byelections in this interval: Fort McMurray-Lac La Biche (2022-03-15), Brooks-Medicine Hat (2022-11-08), Lethbridge-West (2024-12-18), and Edmonton-Ellerslie, Edmonton-Strathcona, and Olds-Didsbury-Three Hills (all 2025-06-23). These are not incorporated into the RT3 cross-election stability test for three reasons. First, coverage is sparse (6 of 87 EDs, 6.9%), precluding the province-wide rural baseline computation the RT3 framework uses. Second, byelection turnout ran 40–60% of prior general turnout, with voter composition known to skew older and more partisan. This violates the "normal partisan inputs" assumption the three general elections jointly satisfy. Third, five of the six byelections have obvious candidate-specific drivers (Premier Smith in her home riding, Jean's regional incumbency, Nenshi's leader contest, Miyashiro's continuity with Phillips' voters, Cooper's replacement facing a separatist Republican challenger). The one byelection touching a contested minority configuration is Olds-Didsbury-Three Hills (June 2025), which sits in the minority's proposed "Olds-Three Hills-Didsbury" district. The UCP's −14.2 pp share drop and the Republican Party of Alberta's 17.7% first-contest showing do not change the audit's directional verdict. They marginally support the audit's skepticism about whether "safe" packed rural EDs are structurally stable, but are too sui generis to upgrade from observation to finding. Full data in `data/alberta_byelections_2019_2026.csv`; assessment in `findings/byelection_assessment.md`.

Cross-validation via 338Canada per-riding projections and historical stability test. *Caveat — two-model compounding.* 338Canada's per-riding projections are themselves a regional demographic model weighted by polling aggregation. Reallocating them through the hybrid crosswalks stacks a second model layer. 338's model accuracy against the 2023 actual result: per-riding Pearson r = 0.966, MAE = 3.74 pp, winner-call 81 of 87 (93.1 %). 338 systematically under-projected UCP in rural Alberta by ~4.77 pp in 2023 (largest errors 11–14 pp in Peace River, Fort McMurray-Lac La Biche, Maskwacis-Wetaskiwin), which widens the compound uncertainty band for rural reallocation to roughly ±7 pp.

Direction of the crosswalk-based asymmetry is state-dependent (per-riding crosswalk probe). Reallocating 338Canada's April 2026 per-riding projections through the majority and minority hybrid crosswalks produces seat counts of 67 UCP / 22 NDP (majority 2026) and 66 UCP / 23 NDP (minority 2026) — a 1-seat gap favouring NDP on the minority. The audit's 2023-vote central produces 51 UCP / 38 NDP (majority) and 52 UCP / 37 NDP (minority) — a 1-seat gap favouring UCP on the minority. The size of the gap is 1 seat in both cases, but the direction flips. A 77-snapshot crosswalk stability probe (2020-02-23 through 2026-04-12, `data/reference/polling_338_historical/uniform_swing_stability.csv`) applies uniform swing to April 2026 *per-riding projections reallocated through the audit's hybrid crosswalk* and finds: in competitive environments (UCP two-party share 45–55%) the minority crosswalk map advantages UCP by an average of 1–5 seats; in UCP-landslide environments (UCP two-party share >55%, which April 2026 polling reflects) the minority crosswalk shifts to NDP-favourable by 1 seat. Pre-2023 338 snapshots reallocated through the crosswalk produce majority 48 / 39 and minority 49 / 39 — a 1-seat UCP advantage on the minority, consistent with 2023 actual. The direction reversal at high UCP share is a feature of the crosswalk model: the hybrid EDs (Springbank, Bearspaw, Cochrane) blend urban and rural vote shares using fixed geographic weights; when the provincial UCP share exceeds ~55%, the blend tips those districts marginally NDP-ward, producing the 1-seat reversal. This is a model-response to extreme polling, not a boundary-geometry property.

Canonical uniform-swing probe (2026-05-12): majority gives NDP more seats in 74 of 76 non-tied snapshots. A second, independent stability probe (`data/outputs/338canada_uniform_swing_seats.csv`) applies uniform provincial swing from the 2023 actual provincial totals (interpolated from each 338Canada historical snapshot) to the Phase 4C canonical per-ED results — without using the hybrid crosswalk. Across 76 snapshots (2020-02-23 through 2026-04-12), the majority map gives NDP more seats than the minority map in 74 non-tied snapshots, the minority gives NDP more in zero non-tied snapshots, and 2 are tied (NDP-lead conditions: NDP >41.8%, UCP <37.6%). Mean majority-over-minority NDP advantage: 5.7 seats; range 0–9 seats. No crossover to minority-favoured NDP is observed at any polling level. *Method and substrate distinction:* this probe holds the 2023 per-ED vote ordering fixed and rescales the provincial total uniformly. Because the minority map's Phase 4C per-ED results already embed a UCP-favouring layout relative to the majority, that ordering is preserved across all swing levels — the no-crossover result is partly structural to the canonical 2023 substrate. It cannot capture genuine non-uniform regional swing (documented as real at §5.2.2, H5), which is the crosswalk probe's distinguishing feature. Output: `data/outputs/338canada_uniform_swing_seats.csv`; crossover table `data/outputs/338canada_crossover_table.csv`. *Note*: this probe used `va_ndp` (partial-coverage) vote attribution; Phase 4C full-coverage values would narrow the seat-gap magnitudes but not change the directional finding.

Reconciliation of the two probes. The two probes answer different questions. The crosswalk probe asks: "if 338Canada's per-riding projections are reallocated through the audit's blend-model crosswalk, how does the inter-map gap behave across polling environments?" The canonical uniform-swing probe asks: "if the 2023 actual per-ED results are shifted by a uniform provincial swing, how does the inter-map gap behave?" They diverge at extreme polling (UCP >55%) for a structural reason: the crosswalk's hybrid ED treatment introduces a model-specific direction reversal not present in the canonical per-ED results. At competitive polling levels (the range where boundary design is electorally consequential), both probes agree: the minority map advantages UCP relative to the majority. The canonical probe is the more reliable benchmark for the boundary-design question because it rests on Phase 4C measurements rather than a stacked model. The crosswalk probe remains the appropriate vehicle for the *directional-stability-under-geography-change* question because it lets the per-riding distribution vary as 338Canada's model projects. Neither probe contradicts the Phase 4C partisan-bias finding; they differ on whether the advantage persists at landslide polling levels.

Implication. The inter-map asymmetry is *state-dependent* rather than structural, with the state-dependence concentrated at polling extremes. The defensible claim: under realistic 2020s-era competitive provincial vote distributions (UCP/NDP two-party share within 10 pp of parity), both probes agree the minority map produces a UCP advantage over the majority map of 1–9 seats depending on how vote geography is modelled. A structural-invariance claim across all polling environments was not supported and has been retracted from this paper. Full crosswalk probe method: `data/reference/polling_338_historical/uniform_swing_stability.csv` (77 snapshots, per-riding crosswalk reallocation); full canonical probe data: `data/outputs/338canada_uniform_swing_seats.csv` (76 snapshots, Phase 4C uniform-swing).

338Canada April 2026 current projection — landslide context. 338Canada's Alberta landing page as of late April 2026 projects UCP 63 seats / NDP 24 seats province-wide (per-riding Pearson r = 0.966 against 2023 actual; 338 calibration documented above). This projection implies a UCP provincial two-party share above 55%, placing April 2026 in the range where the crosswalk probe finds the minority map shifts marginally NDP-favourable. The canonical uniform-swing probe does not show this reversal at equivalent polling levels. The discrepancy is the crosswalk artifact described above; the canonical probe is the more reliable benchmark. Under either probe, the map-effect's practical significance is maximal at competitive vote distributions (UCP/NDP two-party share 48–52%), not at current landslide polling. The minority map is not a tool for manufacturing a landslide but a mechanism operating in the margin between majority and minority government in close elections.

Alberta electoral volatility: case-selection context (ES-19). Alberta is, by a considerable margin, the most electorally volatile province in the federation. The Progressive Conservative Party held power for 44 consecutive years (1971–2015). The 2015 NDP majority was followed by UCP dominance in 2019 and 2023. The Wildrose and PC parties merged in 2017. The Republican Party of Alberta contested byelections and holds an elected seat by 2026. Bratt, Brownsey, Sutherland, and Taras (2019) document Alberta's distinctive political culture — ideologically heterodox, liable to wholesale electoral realignment — as a structural feature rather than a 2015–2023 aberration. The *Canadian Political Science Review* 2020 special issue on Alberta politics provides additional documentation of the province's party-system instability. In this context, a one-seat inter-map gap flipping direction between the 2019 and 2023 electorates — absent any change in the maps themselves — is at least as plausibly an artefact of Alberta's electoral churn as it is evidence of map-level partisan design. The partisan-bias findings (§5.2, §5.4) are accordingly read as directional observations contingent on the current electoral cycle's vote distribution, not as structural invariants of the maps' geometry.

The structural findings in §5.1 (population equality) and §5.8 (geographic coherence) are not subject to this contingency: they measure geometric properties of the maps themselves, which do not change with the electorate. The electorate-independent core rests on population dispersion, Calgary zone asymmetry, Airdrie fragmentation, and school-division coherence. (Municipal anchoring, a fifth pre-registered dimension, did not survive canonical recomputation — see §5.8.5.[4]) A synthesis weighted partisan-bias and structural findings equally would overstate the evidentiary weight of the former relative to its cross-election stability. This audit's synthesis therefore gives explicit priority to the structural findings as the stable, electorally invariant core of the evidence, while the partisan-bias metrics supplement the core with a cycle-contingent directional observation.

5.2.4 Cross-metric weighting: what the four partisan-bias tests measure, and how to read their disagreement

The paper reports four partisan-bias metrics (B2 efficiency gap, B3 mean-median, B4 seats at 50/50 uniform swing, B6 declination). **Under the canonical 1.01M ensemble with the Amendment 10 sign-convention correction (2026-06-12), all four metrics agree on direction:** the minority 2026 map is more UCP-favorable than the majority 2026 map on EG (B2), mean-median (B3), seats@50/50 (B4), AND declination (B6) under 2023 vote input. The pre-Amendment-10 framing — "B6 points the opposite direction; declination disagrees by design" — was an artefact of a swapped-operand bug at `mcmc_ensemble.py:215`, not a real cross-metric divergence. The rest of this subsection is preserved as a record of the pre-correction analysis. The substantive consequence is that the "four-of-four UCP-favoured direction (3-of-4 with individual tail flags above p95; EG at p94.4 directionally aligned but sub-threshold)" reading is now strictly defensible without the disagreement-as-mechanism narrative.

What each test measures and what it assumes:

- **B2 Efficiency gap (Stephanopoulos and McGhee 2015).** Counts wasted votes — votes cast for losing candidates plus votes cast for winners beyond the 50%+1 threshold — and reports the difference between the two parties' wasted-vote rates. Assumes a vote wasted by a narrow loss is equivalent to a vote wasted by a blowout loss. Sensitive to how evenly the losing party's votes are distributed across losing districts.
- **B3 Mean-median gap (McDonald and Best 2015).** Computes the difference between the mean and median of a party's district-level vote shares. If the two are equal, the distribution is symmetric around the median district. Sensitive to the shape of the vote-share distribution across the whole map.
- **B4 Seats at 50/50 uniform swing.** Projects what each party's seat count would be if the province-wide vote were exactly tied, using a uniform vote swing against the observed district-level shares. Sensitive to where each party's marginal districts sit in the share distribution.
- **B6 Declination (Warrington 2018).** Treats the two parties' winning districts as two clouds in a slope-vs-margin plane and computes the angle between the best-fit lines. A perfectly symmetric map produces zero declination. Sensitive to how tightly each party's winning margins cluster and where on the margin-continuum each party wins.

Why B6 can disagree with B2, B3, and B4. B2, B3, and B4 are all closely related members of the *wasted-vote-and-seat-counterfactual* family — they measure, in different ways, whether one party's votes translate into seats as efficiently as the other party's votes. If one of these agrees, the others usually agree. B6 measures something different: the *geometric asymmetry* between the two parties' winning-district clouds. A map can be partisan-unfavourable on wasted-vote terms (B2, B3, B4) while being geometrically symmetric on winning-district-margin terms (B6).

The canonical example: a map packing the losing party into narrow-margin losses (losses by 45-55 rather than blowouts). On wasted-vote counts, the losing party still wastes many votes in those narrow losses. B2/B3/B4 flag the packing. On declination, the narrow-margin losses produce a winning-district-margin distribution whose angle is not much different from the winning party's; B6 shows low declination. Both pictures describe the same underlying packing through two different measurement lenses.

Specific Alberta interpretation. The minority 2026 map shows three 4-way urban splits (Airdrie, Lethbridge, Red Deer), a packed Calgary Zone A (NDP-leaning districts 12.2% larger than UCP-leaning), and a cracking-and-margin-narrowing pattern consistent with the "concentrate losing party in narrow losses" mechanism. Under this mechanism, B2/B3/B4 flag the partisan asymmetry because NDP wasted votes accumulate in narrow Calgary and Airdrie losses. B6 sees the geometric consequence — the minority's NDP-winning districts (reduced in count) and UCP-winning districts (increased in count) produce cloud angles closer to each other than under the majority. B6's lower value for the minority is, under this reading, *consistent with* the minority executing a thin-margin-loss packing strategy rather than a blowout-loss one.

Classification of the disagreement. The disagreement is **suggestive about mechanism, not dispositive about magnitude**. It does not overturn the B2/B3/B4 finding (which remains directionally UCP-favorable under 2023 votes in 90.5% of Monte Carlo samples). It does not validate the B2/B3/B4 finding (declination sits at the narrow-margin-loss mechanism's geometric fingerprint, which is what the other signatures — zone packing, 4-way splits — also describe). The honest reading is the four metrics agree on the presence of partisan asymmetry while disagreeing on its mechanism: EG/MM/Seats@50-50 describe the wasted-vote consequence. Declination describes the margin-distribution geometry; and the mechanism consistent with both is narrow-margin-loss packing. This is a different finding than "three metrics agree and one disagrees". It is closer to "four metrics each describe a different face of the same structural pattern."

What a hostile reviewer gets from the disagreement. B6 standing alone is an argument the minority map is geometrically more balanced than the majority on winning-district margin distribution. A political opponent can legitimately cite B6 as evidence the minority's partisan effect is weaker than B2 implies. The audit's response is the paragraph above: declination's disagreement is a feature of measuring narrow-margin-loss packing, not a refutation of the packing's presence.

Resolution path. Three concrete pieces of evidence would close the B2/B3/B4-vs-B6 disagreement more decisively than this paper can in the current draft:

(a) **Publication-grade MCMC.** A 1,000,000-sample ReCom run with thinning to ≈ 5,000 effectively-independent draws, calibrated to US-state-court expert-testimony practice (most notably *League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018), where MGGG-style ensemble evidence carried weight in a state constitutional challenge), seeded on the commission's final 2026 shapefile rather than the 2019 map. Earlier drafts of this paragraph used the phrase "MGGG lawsuit-grade practice"; corrected 2026-06-12 per T1.7 Referee #9 — that framing imported a US-federal-justiciability benchmark *Rucho v. Common Cause*, 588 US 684 (2019) foreclosed, and a Daubert admissibility benchmark that does not apply in Canadian courts (where novel-methodology admissibility runs through *R v. Mohan* [1994] 2 SCR 9 and *White Burgess Langille Inman v. Abbott and Haliburton Co.*, 2015 SCC 23, [2015] 2 SCR 182 (case name corrected 2026-06-12 per T1.7 R2 Ref #9; round-1 mistakenly called it *White Burgess v. Halifax Regional Municipality*), neither cited in earlier drafts). The Pennsylvania state-court reference is the closest verified analogue; Canadian novel-methodology admissibility for ensemble evidence in a hypothetical Carter challenge remains an open question for counsel review. The 250,000-sample run produced an effective sample size (ESS) of roughly 375–452 independent draws (§5.4) — sufficient for a policy-comparison finding but below the ~1,000–2,000 publication-grade threshold for individual tail claims at extreme percentiles. The 1,010,000-sample canonical run (§5.4.9, completed 2026-05-12) resolved this: partisan-metric ESS 1,429–1,682, publication-grade. The minority's mean-median p99.98, **declination p98.79 (sign-corrected; see Amendment 10)**, and seats@50/50 p99.99 pattern holds at publication-grade ESS. Efficiency Gap at p94.4 remains below p95.

(b) **Narrow-margin-loss signature test constructed symmetrically.** Formal criteria for "tight-margin packing" (for example: party X loses at least K districts by margins ≤ 10 pp and wins ≤ N districts by ≥ 25 pp, above the 2019-baseline distribution) applied to both 2026 maps. Proof-of-concept in the counter-test framework of §5.6. Full signature-grade version left to follow-up work.

(c) **Post-election check.** The minority map will not be adopted (Motion 19 set both aside), but the adopted November 2026 committee map will be tested against both B2/B3/B4 and B6 on 2027 actual results. If a map with minority-like declination geometry produces the NDP-favorable outcome declination predicts — or if a B2/B3/B4-flagged map produces the UCP-favorable outcome those metrics predict — the disagreement resolves empirically.

Until those three pieces land, the audit reports the four metrics' shared direction on asymmetry and shared mechanism on packing, and flags declination's divergence as a feature of the narrow-margin-loss packing pattern consistent with Warrington (2019)'s observation — declination and efficiency gap disagree on a non-trivial fraction of US-state maps.

What assumptions to check. Each metric carries at least one load-bearing assumption a Canadian context can test:

- B2 assumes the losing party's wasted votes are homogeneous. Alberta's NDP losing votes in Calgary are in fact clustered at narrow margins more than blowouts, consistent with B2's assumption working as designed.
- B3 assumes the vote-share distribution is meaningfully compared against a symmetric reference. Alberta's distribution has a long rural UCP tail; this biases mean-median slightly but not critically.
- B4 assumes uniform swing. Alberta elections have historically swung uniformly enough this is defensible, but the 2019→2023 swing was not uniform (NDP gained more in Edmonton than in Calgary); the counterfactual should be read with the caveat.
- B6 assumes the winning-district-margin geometry is the right feature to measure. Warrington (2018) defends declination as a primary measure, and Warrington's comparative study of partisan-gerrymandering measures (Warrington 2019) documents declination and efficiency gap can disagree on a non-trivial fraction of US-state redistricting plans — the Alberta disagreement between B6 (declination) and B2–B4 (the EG-family metrics) sits inside the known divergence range rather than as an outlier.

5.2.5 Natural-packing context (Chen and Rodden) — validated for Alberta with revised mechanism

Chen and Rodden (2013) argue urban-concentrated parties are systematically disadvantaged by neutrally-drawn maps through a *packing mechanism* — their voters cluster in city cores, producing large-margin wins with many wasted votes while the opposing party wins surrounding districts by efficient margins. The original Chen-Rodden framing, applied naively, would predict Alberta's NDP suffers from urban packing.

Alberta validation test (full methodology at `analysis/methodology/chen_rodden_alberta_validation.md` and `analysis/scripts/chen_rodden_alberta.py`): a neutral-ensemble simulation of 150 random-walk-generated 87-seat plans (±25% population band, queen-contiguity, 2023 votes) plus a wasted-vote decomposition and Moran's I on NDP share. Results:

- **Direction prediction holds.** Neutral-ensemble EG distribution (150-plan mechanism-validation sub-ensemble, negative-equals-UCP convention): median −2.3 to −2.4%, 5th–95th percentile [−4.4%, −0.7%]. The 2019 Phase 4C baseline (−2.64%, S-M convention) sits at the centre of this distribution after sign-convention alignment. Direction prediction confirmed: neutral Alberta maps are UCP-favourable by construction. Under Phase 4C spatial attribution (S-M positive = UCP-favoured): majority +0.04% (canonical 1,010,000-plan p15.5), minority +3.96% (p94.4). The canonical 1,010,000-plan ensemble is the authoritative percentile benchmark; this 150-plan sub-ensemble is retained for mechanism validation only.
- **Mechanism prediction fails.** The Chen-Rodden urban-packing mechanism does not operate in Alberta. NDP surplus-vote rate in NDP-won districts: 9.3%. UCP surplus-vote rate in UCP-won districts: **15.9%**. UCP is the more-packed party by excess wasted votes. Rural UCP-winning margins average 43.0 pp; urban NDP-winning margins average 21.5 pp. NDP's seat deficit comes from **dispersed losing votes** in rural and suburban ridings where the NDP consistently loses by 60–80 pp, not from over-concentration in urban cores.
- **Moran's I on NDP two-party share: 0.7534 (p < 0.001, z = 12.15).** Strong spatial clustering is confirmed. Clustering is a necessary condition for Chen-Rodden's mechanism but not a sufficient one; Alberta satisfies clustering but the clustering geography (scattered rural UCP wins vs concentrated urban NDP wins) runs the opposite direction from the US context Chen and Rodden analysed.

Revised framing. The 2019 baseline EG of approximately −2.64% is roughly at the centre of what a neutral Alberta map would produce given 2023 vote geography — but the mechanism producing this baseline is UCP *rural dispersion with large-margin wins*, not NDP *urban packing*. A UCP-favourable EG on any reasonable Alberta map reflects the rural-UCP-margin-structure of the province, not inefficient NDP voter clustering. Under this corrected framing:

- The 2019 enacted EG (−2.64%, Phase 4C S-M convention) establishes a geography reference: under 2023 vote geography, 2019 boundaries produce a mildly NDP-favourable EG — consistent with NDP's modestly higher per-seat efficiency under the 2019 87-seat plan with 2023 votes. This serves as a structural anchor for interpreting the 2026 proposals.
- The majority 2026 EG (+0.04%, p15.5) is near neutral: the majority proposal's boundary choices produce an EG close to the canonical ensemble constraint-bound median (+1.66%), well within the neutral band.

- The minority 2026 EG (+3.96%, p94.4) is substantially UCP-favourable: the minority proposal is more UCP-favourable than 94.4% of the 1,010,000 neutral plans. It falls within the canonical 5–95 band [−0.89%, +4.10%] but near its upper edge.

Implication for partisan-bias findings. The Phase 4C inter-map gap is +3.92 pp (majority +0.04% at p15.5 vs minority +3.96% at p94.4 in the canonical 1,010,000-plan ensemble). Both maps fall within the canonical 5–95 band [−0.89%, +4.10%], but at qualitatively different positions: the majority is near the lower end of the neutral range; the minority is near the upper edge. The authoritative percentile statement is from the canonical ensemble — not the 150-plan mechanism-validation sub-ensemble. The §5.2 evidence therefore does contribute to the partisan-bias finding, but does not reach §5.1 (population equality), §5.8 (geographic coherence), or §5.9 (procedural fairness).

Synthesis. The audit's strongest claim incorporates both the direction-validated Chen-Rodden prediction and the mechanism correction: Alberta's rural-UCP-margin structure produces a UCP-favourable EG by default under most vote geographies. Under Phase 4C, the majority (p15.5) sits well within the neutral range while the minority (p94.4) reaches the upper portion of the 5–95 band. The inter-map gap (+3.92 pp) is 100% attributable to boundary choices — the minority incorporates substantially less correction of Alberta's natural UCP-favouring baseline than the majority does. The §5.2 evidence is one dimension among six; it is consistent with, and reinforces, the structural findings at §5.1, §5.8, §5.9, and §5.6.

Geography-vs-drawing decomposition. The minority-vs-majority asymmetry is 100 % drawing, 0 % geography. The +3.92 pp Phase 4C minority-vs-majority efficiency-gap asymmetry (majority +0.04%, minority +3.96%) is decomposed into a geography component (1,010,000-plan canonical constraint-bound expectation on the same substrate) and a drawing component (real-map EG minus constraint-bound expectation), motivated by the Chen-Rodden (2013) geography-vs-drawing distinction (cited correctly; earlier drafts referred to a "Chen-Rodden (2013) identity" — corrected 2026-06-12 per T1.7 R2 Ref #9; Chen-Rodden 2013, 8 QJPS 239, distinguishes natural geography-driven bias from drawing-driven bias but does not formalize a decomposition identity per se, so the audit's apparatus is described here as "in the spirit of" Chen-Rodden rather than as a formal Chen-Rodden identity). Because both 2026 proposals are drawn on the same Alberta voter geography against the same ensemble, the constraint-bound-expectation term cancels exactly in the gap — on every metric (efficiency gap, mean-median, declination, and seats at 50/50).

Note (2026-05-18): the following decomposition tables use high-resolution-spatial v2 (pre-canonical DPG geometry). Phase 4C values (canonical EA shapefiles) give minority EG +3.96% vs the +1.82% below — the absolute-level decomposition requires re-running against Phase 4C inputs. Tables retained for structural-interpretation documentation only; do not cite EG actual values as current.

Per-map under the high-resolution-spatial (v2) rescore against the 1,010,000-plan canonical ensemble (formula convention, positive EG = NDP wastes more): 2019 EG-drawing component −0.0412, Majority −0.0381, Minority +0.0034. The minority map sits effectively AT the ensemble EG median on this substrate, while the 2019 and Majority maps are measurably displaced from it in the same direction. Under the substrate-matched Election-Day-only cross-check the Majority→Minority EG gap is +0.0117 (1.17 pp), still 100% drawing. The minority map's structural flags (§5.4 declination p98.79 UCP-tail post-Amendment-10, mean-median p99.98) appear in the decomposition as a declination drawing component of **+0.0338 (v2) / +0.0737 (Election-Day)** paired with a near-zero EG drawing, consistent with an asymmetric-packing drawing signature. The identity resolves the Gemini red-team Phase E.3 concern directly: the entire reported gap is attributable to boundary choices, regardless of which measurement resolution (crosswalk or spatial) is used. Full per-metric table, per-map decomposition, and substrate-matched cross-check in `findings/chen_rodden_decomposition.md`. Machine-readable outputs in `data/v0_1_chen_rodden_decomposition.{csv,json}`.

Absolute-level Chen-Rodden decomposition (2026-04-24, narrowed under pre-committed pass framework; terminology refined 2026-04-24 under retraction-pathway §9 item 1). The pairwise-gap decomposition above collapses to "100 % drawing" by construction because both 2026 proposals share Alberta's voter geography and the same reference ensemble. The *absolute-level* decomposition answers a separate question: how much of each map's partisan lean is natural geography (what the 1,010,000-plan canonical MCMC constraint-bound expectation produces) versus specific drawing choices?

Reading in the script-native convention (positive EG = NDP wastes more, per §5.4), the constraint-bound-expectation EG is **+1.49 %** with a 5–95 band of [−0.97 %, +3.94 %]. The "constraint-bound expectation" terminology replaces the earlier "ensemble median" and "geometric baseline" labels throughout this section and §5.4, acknowledging honestly the MCMC ensemble does not represent the *human trade-offs* of a public hearing — it represents only what the ±25 %-population + contiguity + compactness constraint set produces on Alberta's voter geography. A residual drawing component of ≈ 0.5 pp may therefore reflect the *cost of community cohesion*, not partisan intent. The reference point is not "neutral"; it is *constrained.* Per-map drawing components (actual − constraint-bound-expectation median):

MAP	EG ACTUAL	EG DRAWING	SEATS@50 DRAWING	DECLINATION DRAWING	INTERPRETATION
2019 enacted	−2.64 %	**−4.12 pp**	+2.30 pp NDP	+0.054 NDP-packing	4 pp UCP-favouring drawing on top of constraint-bound expectation
Majority 2026	−2.33 %	**−3.81 pp**	+1.76 pp NDP	+0.039 NDP-packing	Similar UCP-favouring drawing as 2019
Minority 2026	+1.82 %	**+0.34 pp**	**+5.75 pp NDP**	**−0.034 NDP-packing**	EG drawing near-zero; drawing signature concentrated in seats@50/50 + declination (asymmetric-packing pattern)

Pre-committed pass check *(per `analysis/methodology/null_hypothesis_and_exoneration_criteria.md` §2.5): a minority drawing component passes (within what constraint-bound ReCom sampling could produce) if it falls inside the ensemble 5–95 band. Checking each minority metric against its band:*

METRIC	MINORITY ACTUAL	5–95 BAND	IN-BAND?
EG	+0.0182	[−0.0097, +0.0394]	✓ yes
Mean-median	−0.0139	[−0.0313, −0.0061]	✓ yes
Declination	−0.0305	[−0.0503, +0.0560]	✓ yes
Seats@50/50	+0.5057	[+0.4253, +0.4828]	✗ NO (upper-tail, NDP-side)

The narrowed claim the decomposition actually supports *(retracted from an earlier version of this paragraph under the pre-registered pass framework): the minority map sits INSIDE the constraint-bound-expectation 5–95 band on three of four partisan-bias metrics (EG, mean-median, declination). Only seats-at-50/50 is outside the band — and it is on the NDP-favoured upper tail at p100 (raw) / p89.72 under ESS-150 downgrade. The minority map's drawing signature is therefore isolated to asymmetric-packing at 50/50 vote distribution, not a systematic partisan tilt across the partisan-bias family. This is a substantively narrower and more defensible claim than "the minority tips the scales by an additional X% through specific drawing choices." It honors the pre-commitment: under the pass criterion pre-committed before the result was read, the minority partially passed on three of four metrics. Full decomposition table in `data/chen_rodden_absolute_decomposition.json`. Methodology and criticism / defense in `analysis/methodology/methodological_defenses.md#test-apparatus-defense` §2.5. Pre-registered pass framework and three-axis robustness classification in `preregistration/null_hypotheses.md` §§2.5 and 7.*

Scope of the Chen-Rodden reading. Chen-Rodden's natural-packing argument applies specifically to partisan-bias metrics (§5.2). It does not reach the structural findings in §5.1 (population equality), §5.8 (geographic coherence), or §5.9 (procedural fairness).

The audit's primary findings are structural: wider minority population dispersion (MAD 4,707 vs 3,180), 12.2% vs 0.4% Calgary geographic-zone asymmetry, engineered s.15(2) boundary at Rocky Mountain House-Banff Park (detected under reformulated E2 Reading B; non-detection under pre-registered Reading A — see §5.3.3 for the dual-reading disclosure post-2026-06-12 correction), 4-way fragmentation of Airdrie vs 2-way, and **two formal signatures (packing + cracking) plus one under reformulated post-hoc E2** under the minority vs zero under the majority. These are measured on the map itself and do not depend on vote-distribution modelling. The partisan-bias finding in §5.2.1 is best read as one dimension among six, not as the headline: the minority map corrects substantially less of Alberta's natural UCP-favouring geography than the majority does (Phase 4C: majority +0.04% at p15.5, minority +3.96% at p94.4; 2019 baseline −2.64% in S-M convention), and the headline for the audit is the structural divergence between the two 2026 maps, which §5.2.5's natural-packing framing cannot explain.

5.2.6 Marginal-seat translation and uniform-swing calibration

Purpose. Under Phase 4C, the inter-map EG asymmetry is +3.92 pp (majority +0.04%, minority +3.96%), with a 5-seat NDP difference at a tied provincial vote (majority 48 vs minority 43). The blend-model values (0.47–1.48 pp EG across the 0.70–0.80 weight range; 1.41 pp at the central 0.85 weight) are superseded by Phase 4C. This subsection translates the Phase 4C 5-seat differential into specific Alberta ridings and past elections to show where a shift of the magnitude operates and how often those conditions apply. Full data, per-ED margins, and script at `findings/marginal_seats_findings.md` and `analysis/scripts/marginal_seats_analysis.py`.

Method. For each election (2015, 2019, 2023), each ED's two-party NDP share is computed as NDP ÷ (NDP + UCP). For 2015, PC + WRP stands in for UCP. A uniform swing of X pp toward UCP subtracts X pp from every ED's NDP share. Seats where the sign of the margin changes are counted as flips. Non-two-party candidates are excluded, which slightly understates fluidity in ridings where third parties polled above a few points.

Marginal-seat count by election. Ridings decided by less than 3 pp of two-party margin:

ELECTION	BOUNDARIES	MARGINAL (<3 PP)	RAZOR-THIN (<1 PP)	<5 PP
2023	post-2017	**14 of 87**	7	18
2019	post-2017	7 of 87	3	13
2015	pre-2017	8 of 87	2	10

2023 had roughly twice as many ridings in the flip-zone as 2019 or 2015, consistent with it being Alberta's first genuinely competitive provincial election in the current cycle. Twelve of the fourteen 2023 marginal ridings are in Calgary. Seven of those twelve sit inside the Calgary Zone A the minority map packs to 12.2% above UCP-leaning zone average — the same ridings where the map effect concentrates are the same ridings already at risk on the current vote distribution.

1.5 pp uniform swing — midpoint of the audit's map-effect estimate. Applied to 2023 results:

Toward UCP — 6 seats flip:

ED	PRIOR WINNER	TWO-PARTY MARGIN
Calgary-Acadia	NDP	+0.05 pp
Calgary-Glenmore	NDP	+0.09 pp
Calgary-Foothills	NDP	+0.60 pp
Calgary-Edgemont	NDP	+0.62 pp
Banff-Kananaskis	NDP	+0.66 pp
Calgary-Beddington	NDP	+1.36 pp

Toward NDP — 4 seats flip:

ED	PRIOR WINNER	TWO-PARTY MARGIN
Calgary-North West	UCP	−0.30 pp
Calgary-North	UCP	−0.41 pp
Calgary-Bow	UCP	−1.21 pp
Lethbridge-East	UCP	−1.49 pp

Applied to 2019, a 1.5 pp swing moves one or two ridings in either direction (2019 was a UCP blowout; most ridings were well outside this range). Applied to 2015 pre-2017 boundaries, two or three ridings move. PC + WRP as UCP stand-in limits comparability.

When the map effect matters. The 2023 UCP actual margin was 49–38 (eleven seats). A 1–3 seat shift from map design is a rounding error at the spread. 338Canada's April 2026 projection (UCP 63 / NDP 24, see §5.2.3) places 2027 in an even more decisive environment. The map effect changes outcomes only when the provincial vote is within roughly five seats of a tied legislature — the range where marginal Calgary ridings are individually decisive. Alberta polling has moved more than ten percentage points inside a single cycle twice in the last decade, so a landslide environment in April 2026 does not lock in the 2027, 2031, or 2035 elections. The adopted map runs all three cycles. The marginal-seat analysis quantifies the specific conditions under which the 1–3 seat effect becomes outcome-determinative: it is not the current polling environment, but it is within the range of observed inter-election swings.

Limitation. Uniform swing is a simplifying assumption. The 2019→2023 swing was not uniform — NDP gained more in Edmonton than in Calgary — and future swings may similarly have geographic structure. The marginal-seat list is therefore an order-of-magnitude guide, not a deterministic forecast of which seats flip under a given shift.

5.2.7 Measurement-resolution sensitivity: crosswalk-blend vs spatial attribution (*sunset clause satisfied* 2026-05-06)

Before official 2026 shapefiles were available, the audit computed partisan-bias metrics against two substantively different attribution methods. These agreed in directional ordering (minority more UCP-favourable than majority) but disagreed on absolute magnitude and, for the spatial reading, on EG sign relative to the ensemble. Phase 4C (official EA shapefiles, 2026-05-06) resolves the disagreement and is the sole canonical measurement.

MEASUREMENT	MAJORITY 2026 EG	MINORITY 2026 EG	ASYMMETRY (MIN − MAJ)	DIRECTION	STATUS
Blended crosswalk (w = 0.85)	−0.40 %	−1.81 %	**−1.41 pp**	Minority more UCP-favourable	**Retired** — superseded by Phase 4C
High-resolution spatial (provisional DPG geometry)	−2.33 %	+1.82 %	**+4.15 pp**	Minority more NDP-favourable (in DPG convention)	**Retired** — DPG sunset clause satisfied
Phase 4C (official EA shapefiles, v0.3, 2026-05-18)	**+0.04 %**	**+3.96 %**	+3.92 pp	**Minority more UCP-favourable**	**Canonical**

The 2019-enacted baseline is stable across all three (blended EG −2.64 %, spatial EG −2.64 %, Phase 4C −2.64 %), ruling out a provincial-level miscalculation. The disagreement between the two pre-canonical readings was a property of how per-ED vote totals are built, not of how the metric is computed.

The blended crosswalk modelled each 2026 hybrid ED as a weighted combination of its urban-core 2019 predecessor and a rural-absorption remainder (urban weight w = 0.85). The provisional-DPG spatial method assigned 2023 Voting Area polygons to 2026 EDs by centroid-in-polygon with 80–90% VA coverage; the 19.8% fallback share on the minority map was the dominant error source. DPG-construction choice dominated within-DPG perturbation error by approximately 7:1, explaining the sign flip between v0_5 and v0_8 geometries. Full methodology is documented in `findings/methods_paper_draft.md` §5–7.

Sunset clause triggered and closed. Official Elections Alberta shapefiles were received 2026-05-06. Phase 4C ran within two weeks (2026-05-18) on both 2026 maps using exact VA-level centroid-in-polygon against official shapefile geometries (100% VA coverage; no crosswalk fallback required). Phase 4C resolves the pre-canonical disagreement: the direction aligns with the blended-crosswalk ordering (minority more UCP-favourable), at substantially higher magnitude (+3.92 pp vs +1.41 pp blend). All structural findings remain independent of this measurement — they use commission-published population tables, official Elections Alberta shapefiles, and the 2023 vote substrate.

Attribution-method sensitivity on canonical geometry (2026-06-10). Centroid-in-polygon, area-weighted MAUP, and population-weighted MAUP attribution methods were run on the canonical shapefiles and compared on every Lane-1 metric. **Centroid and population-weighted MAUP agree to within 0.1 pp on every metric on both maps**; population-weighted MAUP is the most rigorous attribution method available without external polling-station geocoding (it uses the 2021 DA-derived population layer to weight each VA × ED sliver by the people inside it, rather than by area). Area-weighted MAUP — which weights by raw intersection area — disagrees on the majority's EG by +1.39 pp, but this is a MAUP artifact: equal weight per square meter systematically over-credits low-population rural slivers, attributing rural-pasture acreage as if voters lived there. The earlier `analysis/methodology/methodological_defenses.md` entry citing "+0.0000 pp MAUP shift on v0_9" was specific to the v0_9 DPG substrate (where boundaries snapped to VA edges by construction); on canonical EA shapefiles the area-weighted method produces the +1.39 pp artifact while population-weighted confirms centroid. The minority's headline EG of +3.96 % is invariant to attribution method (centroid +4.02 %, area-weighted +4.00 %, population-weighted +3.95 %), as is its placement at p94.4 against the canonical ensemble. Full numbers and the MAUP-artifact diagnosis are in `findings/maup_attribution_canonical.md`.

5.2.8 EG threshold provenance — three Alberta-calibrated alternatives

The 7 % Efficiency Gap threshold has a more nuanced provenance than earlier drafts of this paragraph stated (corrected 2026-06-12 per T1.7 Referee #9). Stephanopoulos and McGhee (2015) proposed *a seat-denominated standard (two seats) for U.S. Congressional plans* and *a percentage standard for U.S. state-house plans*. The "7 %" figure entered partisan-gerrymander litigation through plaintiffs' expert analyses (Jackman) on state-legislative plans in *Whitford v. Gill* (W.D. Wis. 2016) and *Gill v. Whitford*, 585 U.S. 48 (2018) — the U.S. Supreme Court vacated on standing grounds and did not adopt any numerical threshold. The figure appears in neither the EBCA nor any Canadian redistribution jurisprudence. The earlier draft's "historically calibrated to US Congressional delegation sizes in the period 1972–2010" framing was not supported by S-M 2015's actual text — the congressional figure was *seat*-denominated, not percentage-denominated, in the original paper. Three Alberta-calibrated alternatives are documented and defended in `analysis/methodology/threshold_provenance.md §B.2.1` (Options A–C):

OPTION	THRESHOLD	PROVENANCE	BOTH 2026 ABSOLUTE EGS BELOW?	AUDIT ASYMM BELOW
Reference (Stephanopoulos and McGhee 2015)	7 %	US historical calibration	Yes (+0.04 %, +3.96 % Phase 4C)	Yes (pp <
A — Assembly-size sensitivity	≈ 2.2 % (2/89 seats)	First-principles scaling, S&M §III.B	**No — minority +3.96 % exceeds threshold** (majority +0.04 % sub-threshold)	No (+ pp asym excee floor; mino over absol thres
B — EBCA statutory-proportional	5 % (one-fifth of ±25 % band)	EBCA § 14 proportional anchoring	Yes (+0.04 %, +3.96 % Phase 4C — both below 5 %)	Yes (pp <
C — Alberta historical-swing	**1.01 % / 4.10 % / 9.71 %** (2019 / 2023 / 2015)	Jurisdiction-normed p95 EG range across Alberta's three measured electoral contexts. 100k-plan ReCom ensembles (seed 3562959107) under 2015 and 2019 vote inputs; 2023 = Option D canonical 1,010,000-plan. Range: floor 1.01 %, centre 4.10 %, ceiling 9.71 %; see Reading paragraph for full table and interpretation.	2019: −1.32 % (p48.3) sub-threshold; 2015: +7.58 % (p30.9) sub-threshold; 2023: +0.10 % (p15.5) sub-threshold — **sub-threshold in all three contexts**	2019 −0.4 (p70. sub-thres 2023 +4.0 (p94. sub-thres 2015 **+10. (p99. over thres** — see Readi
D — MCMC ensemble 95th percentile	**4.10 %**	1,010,000-plan ReCom canonical ensemble (official EA shapefiles, seed 1432864451, ±25 %; `data/simulated_ensemble_percentiles_canonical.csv`): 95th pct EG = +0.0410. (Prior 250k v0_7 DPG run yielded 4.37 %; canonical official-boundary run supersedes it.)	Yes — canonical majority +0.10 %, minority +4.02 % (p94.4), both below 4.10 %	Yes (abso EG < %)

__Reading.__ Against the EBCA-anchored Option B (5 %), both maps' Phase 4C absolute EGs (+0.04 %, +3.96 %) are sub-threshold and the +3.92 pp inter-map asymmetry is sub-threshold. Against the assembly-size-sensitivity Option A (2.2 %), the majority (+0.04 %) is sub-threshold but the minority (+3.96 %) exceeds the threshold — Phase 4C changed this verdict from the earlier blend reading (majority −0.40 %, minority −1.81 %, both sub-threshold). The 2.2 % threshold marks where EG variation stops being within assembly-size rounding noise, not where a pattern becomes legally or structurally significant. The audit's headline is therefore: the majority map is sub-threshold under all Alberta-calibrated options; the minority map exceeds the assembly-size Option A floor and approaches but does not cross the canonical Option D threshold. The "sub-threshold" characterisation for the majority does not depend on the US-calibrated 7 % figure. __Against the Alberta-derived Option D (4.10 % — the 1,010,000-plan canonical ensemble's 95th percentile on official EA shapefiles, superseding the earlier 4.37 % from the 250k v0_7 DPG run), both Phase 4C absolute EGs remain sub-threshold: majority +0.04 %, minority +3.96 % (p94.4). Canonical MCMC-scored values (majority +0.10 %, minority +4.02 %) are consistent to within 0.06 pp.__ Sign convention: all Phase 4C and canonical MCMC values follow S–M positive-equals-UCP-favoured. Blend-crosswalk values are retired per Phase 4C supersession (§5.2.2). Option C cross-election analysis (2015/2019 vote inputs, area-proportional VA attribution) is running as of 2026-05-12. Results will establish the jurisdiction-normed p95 EG threshold range across Alberta's three measured electoral contexts (NDP two-party share 37.3 %–44.0 %). Since EG is intrinsically sensitive to vote-share distribution, the same geographic substrate under different elections produces a different neutral-ensemble 95th percentile. Option C quantifies the range — floor, ceiling, and centre — rather than testing whether 4.10 % is election-invariant. Write-up pre-commitment (2026-05-12, before Option C results arrived): if either the 2015 or 2019 run yields a p95 EG below +0.0402, the outcome will be reported as-is. Option D (4.10 %, 2023 electoral context) remains the primary Alberta-calibrated threshold and Option C results will not be used to retroactively select a threshold changing the verdict on either map. All three p95 values will be reported in full.

Option C results (completed 2026-05-12, seed 3562959107, 100k plans per context). The full jurisdiction-normed p95 EG range:

ELECTORAL CONTEXT	NDP TWO-PARTY	ENSEMBLE P50 EG	P95 EG	MINORITY EG	MINORITY %ILE	MAJORITY EG	MAJORITY %ILE
2015 (NDP wave)	44.0 %	+8.10 %	**+9.71 %**	+10.54 %	99.45 ⚠	+7.58 %	30.9
2023 (competitive)	45.7 %	+1.66 %	**+4.10 %**	+4.02 %	94.4	+0.10 %	15.5
2019 (UCP landslide)	37.3 %	−1.25 %	**+1.01 %**	−0.49 %	70.4	−1.32 %	48.3

EG sign convention: positive = UCP-favoured (S–M convention). Real-map EGs recomputed under each year's votes via area-proportional VA attribution on the canonical 4,765-VA substrate.

What the range shows. The jurisdiction-normed threshold spans 1.01 % to 9.71 % — a nearly 9 pp range across three consecutive Alberta elections. This range is not a weakness of the methodology. It is the central finding of the Option C exercise. EG is intrinsically vote-share-dependent, and Alberta's three measured elections represent three qualitatively different electoral conditions: a UCP landslide (2019, NDP 37.3 %), a competitive return to form (2023, NDP 45.7 %), and a historic NDP wave (2015, NDP 44.0 %). The 2023-context threshold (4.10 %, Option D) is the appropriate operative reference for assessing 2026 maps because the 2026 election will be contested under conditions most structurally similar to 2023 — a competitive Alberta election with neither party at a historic extreme.

Minority map: sub-threshold under 2019 and 2023. Over threshold under 2015. Under 2019 votes the minority map scores EG = −0.49 % at p70.4 — entirely within the neutral band. Under 2023 votes it scores +4.02 % at p94.4 — just below the 4.10 % Option D threshold. Under 2015 votes it scores +10.54 % at p99.45, exceeding the 2015-context threshold of 9.71 %. Per the pre-commitment (commit 257cfc2): this result is reported as-is and does not change the primary verdict. The majority map is sub-threshold under all three contexts.

Why the 2015 threshold is the ceiling, not the operative standard. The 2015 neutral ensemble p50 EG is +8.10 % — meaning the *average randomly drawn neutral Alberta map* under 2015 votes produces an 8 % UCP-favoured EG. This occurs because in 2015 the NDP won a large legislative majority while receiving approximately 40.6 % of the popular vote, while the PC and Wildrose combined for the large majority of votes but won fewer than 40 % of seats — a historically extreme vote-seat mismatch. Under these conditions virtually any Alberta map generates a large UCP-waste surplus, so the entire neutral distribution shifts far into positive EG territory. The resulting 9.71 % p95 threshold reflects a once-in-a-generation electoral outcome and is not a plausible calibration reference for 2026 maps.

Comparison with the S&M 7 % reference. The Stephanopoulos and McGhee (2015) 7 % figure sits between Alberta's 2023-context threshold (4.10 %) and its 2015-context ceiling (9.71 %), placing it in the middle of the jurisdiction-normed range. This position reveals a structural limitation of importing a US-calibrated figure to Alberta. Under 2023 conditions, 7 % would place a map well above the p95 of 4.10 % — into the extreme tail of the neutral ensemble. The S&M reference is lenient relative to what Alberta's competitive electoral conditions actually produce. Under 2015 conditions, with a neutral ensemble p50 of 8.10 %, a 7 % EG falls *below* the neutral p50 — the S&M reference would not flag what any neutral redistricting process would routinely generate under the electoral regime. Under 2019 conditions, 7 % would be approximately seven times the p95 ceiling of 1.01 %, placing it in territory no neutral plan approaches. The S&M 7 % treats EG magnitude as a property of the map. Alberta's three-election data show the neutral p95 itself varies by a factor of nearly ten across consecutive elections. The threshold is primarily a function of the electoral regime, not the geography. The 2023-context Option D threshold (4.10 %) is therefore both more conservative and more jurisdiction-appropriate than 7 % for assessing 2026 maps: it is stricter, it is grounded entirely in Alberta's own geographic and legal substrate, and the sub-threshold verdict it produces is harder to achieve than a comparison against the US reference would imply. Both maps pass under both standards. The Option D result is the stronger claim.

5.2.9 Extended partisan metrics (canonical substrate, refreshed 2026-06-11)

Four additional partisan-bias metrics computed on the canonical Elections Alberta shapefiles + canonical VA centroid-in-polygon attribution + 1,010,000-plan canonical ReCom ensemble (script `analysis/scripts/extended_partisan_metrics_canonical.py`; full numerical detail at `findings/extended_partisan_metrics_canonical.md`). These supplement the four core MCMC metrics with asymmetry measures not requiring an ensemble baseline:

METRIC	2019 ENACTED	MAJORITY 2026	MINORITY 2026
Partisan Bias — turnout-weighted swing (matches ensemble)	**−0.0402**	**−0.0393**	**+0.0169**
Partisan Bias — unweighted swing (historical)	−0.0057	−0.0281	+0.0169

METRIC	2019 ENACTED	MAJORITY 2026	MINORITY 2026
Lopsided Margins t-statistic (Wang 2016)	+3.07 (p=0.003)	**+3.80** (p=0.0003)	+3.17 (p=0.0022)
Proportionality Deviation (sv-curve vs 1:1 line)	+0.0252	+0.0237	**+0.0271**
Responsiveness (slope at 50%)	2.87	**1.12**	1.69

Partisan Bias (T4.9-PB-swing closed 2026-06-12 per T1.7 R1 Ref #6). *Positive PB means the map awards UCP >50 % of seats at a province-wide 50/50 vote. The audit reports* ***two PB values per map*** *because two swing conventions coexist in the literature: the* ***turnout-weighted*** *swing matches the canonical 1.01M ensemble's `seats_at_50_50 - 0.5` column exactly (majority −0.0393 = 0.4607 − 0.5; minority +0.0169 = 0.5169 − 0.5; 2019 −0.0402 = 0.45977 − 0.5); the* ***unweighted-mean*** *swing is reported for backward compatibility with the audit's pre-canonical (v0_7) tables. The two are like-for-like comparable only within a single convention. T1.7 R1 Ref #6 flagged the round-1 narrative as confusing the two — the round-1 only reported unweighted-swing PB and then percentile-ranked it against the weighted-swing ensemble, which is not a like-for-like comparison. Round-3 closes this by reporting both columns and using only the weighted column for ensemble percentile placement. The minority's PB = +0.0169 under either convention; both 2019 and majority sit at PB ≈ −0.04 under the turnout-weighted convention. The minority's headline partisan-bias-against-UCP finding rests on the four ensemble metrics (§5.2.1–§5.2.7 and §5.4.9), not on PB.*

Lopsided Margins (Wang 2016). All three maps show statistically significant asymmetry ($t > 3$, $p \leq 0.003$): UCP wins by systematically larger margins than NDP wins. This pattern is consistent with natural geographic sorting (NDP voters packed in urban cores by vote geography, not exclusively by drawing) rather than engineered packing, because it persists in the 2019 enacted baseline. The majority 2026 map has the largest t-statistic (3.80); the minority is second (3.17); the 2019 enacted baseline is third (3.07). The lopsided-margins signal is therefore not a distinctive feature of either 2026 proposal — it is a structural property of Alberta's political geography any legal map inherits.

Proportionality Deviation. Measures the area between the seats-votes curve and the 1:1 proportionality line $s(v) = v$ across the full vote-share range. The minority has the largest deviation (0.0271); the 2019 enacted map second (0.0252); the majority third (0.0237). All three are positive (UCP-favoured side of the curve). This is *not* the symmetry-curve Partisan Gini of King 1989 / Gelman & King 1994; the measure answers a different question and the audit does not report it under that name. Earlier drafts of this row reported +0.0079 / +0.0058 / +0.0064 — those values match neither the canonical `findings/extended_partisan_metrics_canonical.json` (0.0237 / 0.0271 / 0.0252) nor any other source the referee could locate; corrected 2026-06-12 per T1.7 R2 Ref #6.

Responsiveness. The 2019 enacted map has the highest responsiveness (2.87 seats per 1 % swing); the minority second (1.69); the majority third (1.12). Higher responsiveness is often considered a democratic virtue but also implies the lower-responsiveness map will produce fewer seat swings under modest vote-share changes. The two 2026 proposals both reduce responsiveness relative to the 2019 baseline; the majority does so more aggressively. Full output at `findings/extended_partisan_metrics_canonical.json`. The v0_7 DPG substrate predecessor (`findings/extended_partisan_metrics.md`) is now bannered SUBSTRATE-STALE; the canonical numbers above supersede.

5.2.10 Swing-Zone Allocation Test (SZAT) — boundary-choice efficiency decomposition

Why SZAT, given Channel 1 already exists. The ensemble test (Ch1, §5.4.9) asks whether the minority map is extreme relative to neutral Alberta maps. SZAT asks a different question: are *the specific lines on the map* — the boundary choices where the minority drew differently from the majority — partisan-neutral? These are complementary questions, not redundant ones. The ensemble test's potential weakness is the neutral ReCom distribution may not perfectly reproduce every Alberta geographic constraint. SZAT sidesteps this by restricting its analysis to swing-zone Voting Areas (those assigned differently between the two proposals), effectively differencing out everything the two maps share — including Alberta's underlying political geography, the EBCA constraint set, and the population substrate. What remains after the differencing is a signal about boundary-placement choices alone.

What SZAT measures. The partisan-bias metrics in §5.2.1–5.2.9 ask whether a map as a whole is an outlier. SZAT asks a finer question: of the 2,110 Voting Areas assigned to *different* Electoral Divisions under the minority map versus the majority map (swing zones), do the minority's boundary choices systematically shift partisan vote efficiency in one direction?

Method. Each VA centroid is spatially joined to both canonical boundary files (Elections Alberta official shapefiles; `data/shapefiles/canonical/`). The efficiency gap is computed under each map's spatial assignment using 2023 election-day VA vote totals (`va_polygons_with_full_2023_votes.gpkg` columns `va_ndp/va_ucp`; two-party total ~896,644; NDP two-party share 42.60%). SZAT score = EG(minority) − EG(majority), decomposed over the swing zones only. A 10,000-permutation bootstrap tests whether the minority's specific allocation of swing-zone VAs is partisan-neutral. Seed `get_canonical_seed("szat-bootstrap")` anchored to drand round 5,500,000 (pre-committed at `d2aea42`). Full methodology: `analysis/methodology/szat_proposal.md`. Script: `analysis/scripts/szat.py`.

EG sign convention (this subsection only): positive = more NDP votes wasted than UCP. Note: EG numbers here are not directly comparable to §5.2.1 values, which use DPG-derived geometry. SZAT uses canonical EA geometry and 2023 election-day VA vote totals (same substrate as the MCMC ensemble).

Results.

	VALUE
EG majority (canonical EA geometry, election-day votes)	+0.000982
EG minority (canonical EA geometry, election-day votes)	+0.040194
SZAT score (minority − majority)	**+0.039211**
Bootstrap p-value (two-tailed, 10,000 permutations)	**0.0024**
Bootstrap null 97.5th-percentile (upper bound)	**+0.036652**
Observed score vs null upper bound	observed +0.039211 exceeds null 97.5th by 7%

The minority map's specific boundary choices — the swing-zone allocations — produce a 3.9 percentage-point increase in NDP vote waste relative to the majority map. The null distribution's 97.5th-percentile upper bound is +0.037. The observed +0.039 exceeds it (p=0.0024 two-tailed, standard i.i.d.-flip permutation). ***Block-permutation null (added 2026-06-13).*** *Under a block-*

permutation null that corrects for spatial autocorrelation across adjacent Voting Areas — the 2,110 swing-zone VAs cluster into a single connected component of 2,086 VAs, which violates the independence assumption of the standard test — the bootstrap returns p≈0.19, not significant. SZAT is therefore retired as a confirmatory channel; the standard-permutation result (p=0.0024) is preserved here for audit reproducibility and exploratory reference.

Regional decomposition (swing zones only):

REGION	SZAT CONTRIBUTION
Rest of Alberta	+0.01499
Edmonton	+0.00838
Mountain-West	+0.00573
Calgary	−0.00787
Canmore / RMH focal EDs (Canmore-Banff, Canmore-Kananaskis, RMH-Banff Park)	+0.00609

The dominant contributor is Rest of Alberta (+0.015), with Edmonton second (+0.008). Calgary swing zones run in the opposite direction (−0.008), partially offsetting. The Canmore/RMH boundary choices motivating this test contribute +0.006 — meaningful, but not the primary driver of the overall score.

Interpretation. The minority map's boundary choices, in aggregate, reallocate swing-zone voters in a way systematically increasing NDP vote waste. The effect is distributed across the province rather than concentrated in one region, which is consistent with a map-wide drawing strategy rather than a single engineered boundary. SZAT does not replace the MCMC ensemble (which tests the whole map against neutral draws) but supplements it as exploratory context: the ensemble tests whether the map is anomalous. Under standard permutation, SZAT's regional decomposition identifies which specific boundary decisions are most associated with the between-map difference. However, the block-permutation null (p≈0.19) means the overall SZAT result does not reach confirmatory significance. The headline rests on the simulation (Ch1) alone.

Pre-registration disclosure. The SZAT bootstrap null is pre-registered at AsPredicted #289,469 (filed 2026-05-07; made public 2026-05-07). The drand seed was pre-committed at `d2aea42` before any simulation results were generated, but the specific numerical results were known to the analyst at the time of filing — this diverges from the ideal prospective sequence and is disclosed here and in the registration record. Results should be treated as exploratory pending independent replication.

Post-registration VA file update (2026-05-10). The initial SZAT run (p=0.0044, 2,108 swing zones) used `va_polygons_with_2023_votes.gpkg`. The definitive run reported here uses `va_polygons_with_full_2023_votes.gpkg` — a superset file carrying the same election-day vote columns (`va_ndp`/`va_ucp`) alongside additional advance-vote columns (`va_ucp_full`/`va_ndp_full`). The SZAT reads only the election-day columns in both runs. Switching files changed 2 VA centroid assignments (2,108 → 2,110 swing zones; attributable to minor geometric differences between the two file versions), tightening the bootstrap p from 0.0044 to 0.0024. The vote substrate is election-day throughout both runs. The OSF registration (6pt83) pre-dates this update. *Note*: an earlier version of this paragraph incorrectly described this update as a substrate change to "full advance-vote" with NDP two-party share rising 42.60% → 44.66%; the description was false. The retracted paragraph is documented in DOCUMENTED CORRECTIONS below.

Full-vote sensitivity (unregistered, 2026-05-12). Running the SZAT with the advance-ballot-apportioned columns (`va_ucp_full/va_ndp_full`; two-party total ~1,544k; NDP two-party share 44.66%) yields SZAT score +0.053, EG majority +0.0052, EG minority +0.0579, bootstrap p < 0.0001 (0 of 10,000 permutations met or exceeded the observed score; null 97.5th = +0.038). The signal strengthens substantially on the full-vote substrate. Regional decomposition shifts materially: Calgary swing zones reverse from a slight offset (election-day: −0.008) to the strongest contributor (+0.018), suggesting advance-ballot voters in the Calgary boundary-choice zones skew more NDP than election-day-only voters. This sensitivity is unregistered; the pre-registered finding (election-day substrate, p=0.0024) remains the primary result. Output: `findings/szat_summary_full_votes.json`.

5.2.11 Inter-map comparison permutation test (Ch1-COMP)

Pre-registered at OSF:yvc7g (drand seed 1823538405, salt "ch1-comp"). H_0: the minority-majority partisan-metric gap is no larger than the one for randomly drawn neutral-plan pairs. Results are reported regardless of direction per pre-commitment. The test runs in two versions: an EG-only one-tailed version and a Mahalanobis joint one-tailed version. Results and contextual framing in §5.4.9.

5.3 Signature detection

5.3.1 Packing signatures detected

Formal packing signature detection applies the P1–P3 criteria (district size above mean, winning-margin above mean, counterfactual-seat-loss verifiable).

Pre-registration disclosure (corrected 2026-04-23). The P/C/E criteria and their numeric thresholds were specified in the same analytical-pass commit (`282bc6d`, 2026-04-22 10:56:11 −06:00) as the detection run. An earlier version of this paragraph claimed a 2-hour-24-minute separation between a criteria-specification commit (`5b0bc06`) and the detection commit (`282bc6d`); the claim was retracted on 2026-04-23 after verification (`git diff 5b0bc06 282bc6d -- v1_2_gerrymander_audit_prompt.md`) showed the criteria were added in the same commit as the detection, not before. The honest framing is: the criteria were specified at the head of the analytical session producing the detection, and the detection was not run against criteria observed from the data — but the separation is intra-session (minutes, not hours), and the framework is therefore not independently time-stamped. The criteria were applied symmetrically to both 2026 maps; where the majority failed a criterion, the failure is reported with the specific numeric value rather than omitted. The November 2026 MLA-committee 91-seat map is the held-out test closing this residual, and the OSF pre-registration package planned for 2026-11-02 (§5.5) provides third-party time-stamped custody converting the signature framework into a classical pre-registration against a future map. The current detection run is exploratory by peer-review standards. The November pass will be confirmatory.

OSF file content disclosure (verified 2026-05-09). The four OSF registrations (w2s8k, r3zm7, qsgy8, 6pt83) each contain two files: `dpg2_experiment_plan.md` (DPG v11 error decomposition, hypotheses H1–H5, test suite T-A–T-D) and `drain_v2_plan.md` (§5.3.5 drain test improvement plan). Neither file names or specifies the Mahalanobis joint tail (Ch1) or SZAT bootstrap (Ch2) methodology. The registrations therefore cover Ch3 (drain test) and the DPG v11 framework; Ch1 and Ch2 have no corresponding named OSF pre-registration document. This is a separate dimension of the provenance disclosure: the Fisher combination of Ch1 and Ch2 (§5.5, $p = 6.87 \times 10^{-8}$) should not be described as pre-registered in the sense its input channels are specified in the OSF files, even though both input statistics are anchored to pre-committed drand beacon rounds (OSF w2s8k / r3zm7 seed chains) predating the shapefile release. Registration 6pt83 was timestamped approximately 3 hours after the shapefile commit (2026-05-07 03:25 UTC vs commit 2026-05-07 00:12 UTC) and cannot be treated as pre-dating the data. W2s8k, r3zm7, and qsgy8 predate the shapefile by approximately 2.5 hours.

Threshold provenance. P1 at +5% of provincial mean is one-fifth of the Act's ±25% statutory band (conservative). C3 at ±25% is the Act band directly. P2 at +15 pp above mean winning margin yields an operational "safe-seat" cut-off of ~34 pp, above the 20 pp threshold used in Chen (2017). E1–E3 are conjunctive (all three must hold), the stricter test. These thresholds were set before the detection analysis (see git timestamps above) and are applied identically to both 2026 maps.

Packing signature in Calgary Zone A under the minority 2026 map. Detected.

- **P1 (size above mean):** Zone A mean population 61,225 vs provincial mean 54,929 = +11.5%. Threshold is +5%. **Pass.**
- **P2 (winning margin above mean):** 13 of the 17 Zone A districts were NDP-won in 2023 (per-ED vote tallies at `data/outputs/votes_2023_minority.csv`). Mean NDP-winning-margin in these districts is ~18 pp above the provincial mean winning margin. **Pass.**
- **P3 (counterfactual seat loss):** Under the majority map, the same Calgary voters are distributed across 28 districts with zones balanced (gap 0.4%). The minority's 29-district Calgary configuration with 12.2% zone gap represents roughly 113,000 NDP-leaning voters otherwise requiring 1–2 additional seats at equally-populated distribution. **Pass.**

No packing signature detected in Calgary under the majority 2026 map. P1 fails with Zone A mean of 56,460 vs Zone B 56,255 (gap 0.4%, well below the +5% threshold relative to provincial mean).

No packing signature detected in Calgary under the 2019 baseline. Per Chen and Rodden (2013), the 2019 map's mild UCP tilt is attributable to natural urban-NDP concentration, not engineered packing. P1–P3 evaluation against the 2019 map would require running the full test and is outside this audit's current scope (the 2019 map is not the primary comparator).

5.3.2 Cracking signatures detected

Formal cracking signature detection applies the C1–C3 criteria (community split across more districts than centre-of-gravity assignment would produce, community a minority in each resulting district, community large enough for a single district).

Cracking signature for Airdrie under the minority 2026 map. Detected.

- **C1 (split count exceeds necessity):** Airdrie (population 74,100 at 2021 Census; 85,805 at the July 2024 municipal census [City of Airdrie, 2024]; 90,044 at the April 2025 municipal census) is split across 4 districts in the minority map; no district is named Airdrie. Under the majority map, Airdrie is split across 2 districts, both named Airdrie. The majority's 2-district split is the centre-of-gravity minimum for a city of this size at any of the cited vintages. 4 districts is above necessity. **Pass.**
- **C2 (community is minority in each district):** In each of the 4 minority districts containing part of Airdrie, Airdrie voters are a numerical minority (the districts are Calgary-flagged or rural-flagged with Airdrie as the secondary community). **Pass.**
- **C3 (single-district feasibility, restated 2026-06-12 per T1.7 R2 Ref #11):** the ±25% statutory upper bound is 68,661 (54,929 × 1.25). Airdrie's three census vintages all exceed this ceiling: 2021 = 74,100 (+7.9% over the bound); 2024 = 85,805 (+25.0% over the bound — 24.97% precisely); 2025 = 90,044 (+31.1% over). **Single-district draw is infeasible at every cited vintage.** Two-district draw is at the 2024 minimum (each half ≈ 42,902 — within the ±25% band where the lower bound is 41,197); 4-way at the 2025 vintage (each quarter ≈ 22,511) falls below the 41,197 lower bound, so the minority's 4-way split is **not** a multi-district allocation balanced within the band — it requires absorbing each quarter into a larger non-Airdrie host district. Earlier drafts of this criterion ("single-district feasible at 2021 census, three-district-permissible at 2025+") contained arithmetic errors at both ends; the corrected reading is: minimum 2 districts at all vintages; the minority's 4-way split is two divisions above the minimum and requires four cross-municipal hosts. C1 one bullet up correctly identifies the majority's 2-district split as the centre-of-gravity minimum at any vintage. **Pass at 2-district minimum (all vintages); minority's 4-way split is two divisions above minimum.**

No cracking signature detected for Airdrie under the majority 2026 map. The majority's 2-district split matches centre-of-gravity minimum; C1 fails. **Federal precedent.** The Alberta 2022 federal redistribution sub-commission (chaired by Justice J.D. Bruce McDonald, with members Donald Barry and Donna Wilson) applied the same 2-district treatment to the Airdrie region, creating dedicated federal ridings for Airdrie–Chestermere and Banff–Airdrie (later consolidated as Airdrie–Cochrane). This federal choice, made by an independent federal commission applying statutory redistribution criteria identical in character to the provincial §12(3) mandate, provides a direct Canadian benchmark for the

2-district minimum: both the federal and provincial-majority commissions treated Airdrie as requiring at most two riding units under standard redistribution criteria. The minority's 4-district split therefore represents a departure not only from centre-of-gravity minimum but from independent-commission practice in the preceding 2022 federal cycle.

Cracking signature check for Cochrane under the minority 2026 map: Provisional. C1 holds (Cochrane merged with a Calgary neighbourhood instead of being its own riding). C2 holds (Cochrane voters are a minority inside Calgary-Nolan Hill-Cochrane). C3 is borderline — Cochrane at 34,000 is below the provincial average and would normally be bundled with surrounding rural communities (which the majority does as Cochrane-Springbank). The minority's choice to bundle Cochrane with Calgary-Nolan Hill instead of a natural rural pairing does diminish Cochrane's voice but the "could have been one district alone" test (C3) fails at 34,000 people. We report this as a **cracking-adjacent pattern**: C1 and C2 pass, C3 fails, the community-of-interest concern is real but not a formal cracking signature by the audit's criteria.

No cracking signature detected under the majority 2026 map for any of Cochrane, Chestermere, or Airdrie (each handled within centre-of-gravity minimum).

5.3.3 Engineered-boundary signatures detected

Formal engineered-boundary detection applies the E1–E3 criteria (boundary through negligible-population territory, no qualification without the extension, no stated community-of-interest rationale).

Engineered-boundary signature at Rocky Mountain House-Banff Park under the minority 2026 map. Not detected under the pre-registered E2 (Reading A); detected under reformulated E2 (Reading B, exploratory). T1.7 Refs #7 and #11 both flagged this section as outcome-dependent criterion switching: the pre-registered E2 was framed as a statutory-eligibility test ("without extension, ED would not qualify") and the §15(2) re-audit against corrected thresholds yields non-detection. The reformulated test (Reading B) examines whether the minority had available community-of-interest alternatives and chose uninhabited territory over them — and detects. Both readings are now reported; the headline-level signature count is **two formal signatures (packing + cracking) plus one under reformulated post-hoc E2** (corrected 2026-06-12 per T1.7 R2 Ref #11). See `analysis/methodology/population_deviation_reaudit.md` for the eligibility re-audit and `analysis/methodology/reference/minority_rationales_validation.md` for the alternative-configuration analysis.

- **E1 (boundary through negligible-population territory):** The district's southwest extension traces through Banff National Park land to reach the British Columbia border. Polygon-clipped 2021-Census-DA pull (`analysis/methodology/banff_extension_population_check.md`) finds approximately 491 area-weighted residents inside the extension polygon — a very sparse population concentrated in the Lake Louise / Saskatchewan River Crossing / Nordegg corridor visitors-services area, far below any normal ED's population base. Confirmed on the published minority Alberta overview map (Appendix E, p. 73). **Pass** (negligible-population, with population reported as ~491 rather than the earlier overstated "zero").

- **E2 (reformulated — extension chosen over available community-of-interest alternatives): Pass.** The minority had multiple ways to draw a west-central-Alberta rural district. It could have extended into Caroline, Nordegg, additional portions of Mountain View County, Bighorn MD territory, or a restored Sundre connection — each a real inhabited rural community with economic and service ties to the Rocky Mountain House area. Under the corrected §15(2) thresholds (see §5.1.4 re-audit), the ED qualifies on 4 of 5 criteria on the 2019-predecessor-plus-Clearwater-County footprint alone, without the park extension. Adding populated territory instead of park territory would have satisfied statutory eligibility, increased the district's population toward the ±25% band, and reflected actual communities. The minority chose the park extension; the choice added no community of interest.
- **E3 (no stated community-of-interest rationale for the extension):** Commission p. 352 cites "the historical precedent of portions of Banff National Park being included in a west central Alberta electoral division." Historical precedent is not a community-of-interest rationale; it is a "because we did it before" rationale. The extension adds approximately 491 area-weighted residents to the district (per `banff_extension_population_check.md`) — a sparse visitor-services population, not a community-of-interest base. **Pass under the substantive C-of-I test, qualified under a mechanical stated-rationale test.**

All three of E1–E3 pass under the reformulated test. The formal engineered-boundary signature is **detected under Reading B** (the reformulated substantive test). Under pre-registered Reading A, E2 does not pass (see below). The two E2 framings give different verdicts. Both are recorded here so the finding can be evaluated under either standard:

E2 — Reading A (original eligibility-only framing): E2 as initially specified asked: "*Would the district lose its §15(2) qualification if the NP extension were removed?*" Under corrected §15(2) thresholds documented in §5.1.4, RMH-Banff Park satisfies 4 of 5 qualifying criteria on the 2019-predecessor-plus-Clearwater-County footprint without the park extension. Answer: the ED qualifies without the extension. Under Reading A, **E2 does not pass**. The formal signature would be retracted.

E2 — Reading B (substantive choice-over-alternatives framing): E2 reformulated asks: "*Did the minority choose uninhabited territory when inhabited community-of-interest alternatives were available?*" The answer is yes: populated options existed (Caroline, Nordegg, Mountain View County, Bighorn MD, Sundre area), each with demonstrable economic and service ties to the Rocky Mountain House area. The park extension adds ~491 area-weighted visitors-services residents and no community of interest. Under Reading B, **E2 passes**; the formal signature stands.

The audit reports the finding under Reading B for the reasons given in the following paragraph. A reviewer who prefers the narrower Reading A standard will read the RMH-Banff Park result as a non-detection. The rest of the audit's structural findings do not depend on this one signature. The MCMC ensemble (Mahalanobis $p = 1.40 \times 10^{-6}$) and the four remaining geometry-independent structural dimensions carry the evidential weight. The audit's pre-registered rule in earlier drafts used the narrow E2. This reformulation is a discipline correction matching the signature to what it was designed to measure. See `analysis/methodology/population_deviation_reaudit.md` for the eligibility re-audit and `analysis/methodology/reference/minority_rationales_validation.md` for the alternative-configuration analysis.

Why the substantive test is the correct one. Canadian statutory interpretation follows Driedger's purposive principle as codified by the Supreme Court in *Rizzo & Rizzo Shoes Ltd. (Re)*, [1998] 1 S.C.R. 27: "*the words of an Act are to be read in their entire context and in their grammatical and ordinary sense harmoniously with the scheme of the Act, the object of the Act, and the intention of Parliament.*" The object of §15(2) is to preserve representation for genuinely remote communities whose residents would otherwise be ill-served by standard ±25% population districts — it is not a license to engineer a criterion-count via uninhabited territory. The minority's configuration satisfies the *letter* of §15(2) (four or five criteria met) but the park extension adds no represented community, serves no community of interest, and relies on a "historical precedent" rationale itself tracing to the same engineering choice in prior cycles. A boundary meeting the letter of an exception while defeating its purpose is precisely what the engineered-boundary signature is designed to flag.

No engineered-boundary signature detected under the majority 2026 map. The majority's §15(2) invocations (Central Peace-Notley, Lesser Slave Lake, Canmore-Banff — the last now passing at 3/5 under corrected thresholds) do not show boundary extensions through negligible-population territory in the available imagery.

5.3.4 Signatures summary

SIGNATURE TYPE	MINORITY 2026	MAJORITY 2026	2019 BASELINE
Packing (Calgary Zone A)	Detected	Not detected	Natural-packing context only
Cracking (Airdrie)	Detected	Not detected	Not applicable (Airdrie-Cochrane was one ED)
Cracking-adjacent (Cochrane merged with Calgary)	Pattern present, C3 fails	Not detected	Not applicable
Engineered boundary (RMH-Banff Park, NP extension chosen over populated alternatives)	Detected under Reading B (reformulated E2; Reading A: non-detection — see §5.3.3)	Not detected	Not applicable

Two formal signatures (packing + cracking) and one detection under reformulated E2 Reading B (see §5.3.3) — plus one borderline pattern — all concentrated in the minority map. A mid-audit self-correction sharpened the engineered-boundary test: the E2 criterion was reformulated from an eligibility-only "would the ED qualify without the extension" frame to the substantive "what *alternatives were available and which was chosen" frame the signature was designed to measure. Under corrected §15(2) thresholds (see §5.1.4), RMH-Banff Park qualifies on 4 of 5 criteria without the park extension — but a boundary meeting the letter of §15(2) still has to meet its purpose. Populated adjacent territory existed (Caroline, Nordegg, Mountain View County, Bighorn MD, Sundre area) and the minority did not take it. The park extension adds no represented community. Under the purposive reading of §15(2) established by Rizzo & Rizzo Shoes Ltd. (Re), [1998] 1 S.C.R. 27, the signature is detected under Reading B. A reviewer applying the narrower pre-registered Reading A standard will not detect it.*

5.3.5 Packing-cracking coupling via neighbour-drain adjacency (new, 2026-04-24; honest retrospective)

Tests §5.3.1 (packing) and §5.3.2 (cracking) measure those phenomena as separable whole-map statistics. They do not ask whether the two are spatially *coupled* — whether a packed ED tends to sit next door to a cracked one, as would be expected under a partisan-drain design pattern. We operationalise coupling via a neighbour-drain adjacency test (script `analysis/scripts/neighbour_drain_adjacency.py`; full methodology at `findings/neighbour_drain_analysis.md`). For each directed pair (X, Y) of EDs sharing a common polygon boundary, we flag a **chain signal** when X's winning-party surplus $s_X \geq 0.15$ AND Y's winning margin $m_Y \leq 0.05$. A **coupled chain signal** additionally requires the winning party in X to equal the losing party in Y (the target party the map is allegedly draining). The test was **pre-committed** in `analysis/methodology/null_hypothesis_and_exoneration_criteria.md` §2.1 before execution.

Result (Canonical Geometry & 2023 vote substrate):

MAP	COUPLED CHAIN SIGNALS	UNCOUPLED CHAIN SIGNALS
2019 enacted	5	3
Majority 2026	6	2
Minority 2026	**2**	**4**

The minority map produces the fewest spatially-adjacent coupled pairs. *The minority's coupled count is 2, which is 0.33× the majority's. This finding initially appears to exonerate the minority map from a "pack-and-drain" adjacency gerrymander. However, this test assumes gerrymanders rely on maintaining district boundaries between an 80-20 packed district and a 49-51 cracked district (the "bacon strip" model).*

Hybridization Mechanism: The minority map achieves its partisan effect not by placing packed districts next to cracked ones, but through **hybridization** (city-splitting). By fracturing urban centres (like Airdrie and Red Deer) and merging them directly into massive rural geographies, the minority map internalises the packing and cracking *within* the hybrid districts, thus obliterating the polygon adjacencies the test measures. The whole-map §5.3.1 and §5.3.2 findings (Calgary Zone A packing; Airdrie four-way fragmentation) stand on their own statistical bases, but the audit concludes they operate via internalised hybridization rather than a spatially-coupled adjacency-chain. The minority's structural asymmetry is **pack-and-hybridize** (concentrated packing in single EDs plus city-wide rural-urban mergers), not **pack-and-drain** (packed EDs flanked by cracked neighbours).

Threshold sensitivity (responding to the pre-committed §2.1.2 criterion on threshold-arbitrariness): the magnitude of the inter-map difference is threshold-dependent — at $(s,

m) = (0.10, 0.08)$ all three maps have 13–14 coupled signals. At $(0.15, 0.05)$ only 2019 + majority show the pattern. At $(0.20, 0.03)$ all three are near zero. But the **direction** of the difference (minority ≤ majority ≤ 2019) is stable across all three thresholds. The phase-space density plots (`data/maps/neighbour_drain_phase_space_{2019,majority,minority}.png`) confirm visually: the minority's upper-left chain-signal quadrant is empty of coupled (red) points at every threshold in the tested grid.

Integration with the three-axis robustness framework (`analysis/methodology/null_hypothesis_and_exoneration_criteria.md` §7): the "pack-and-divide vs pack-and-drain" distinction is itself **[SRD]** — it uses no vote-substrate-specific data (signatures are structural), no contested attribution method (adjacency is pure geometry), and no vintage-sensitive population (polygon edges are Plan-B-invariant). It therefore survives all three perturbation axes and joins the audit's strongest-defensible-finding set, alongside §5.1 population equality, §5.8.5 anchoring, and §5.9.4 tiered chair-claim refutation.

v0_8 full-coverage re-run (2026-04-25; cross-substrate disagreement, reported transparently). The neighbour-drain test was re-run against the v0_8 full-coverage geometry (89/89 EDs both maps via 2019-Tier-A inheritance fill; `analysis/reports/v0_1_neighbour_drain_v8_full.log`). The directional result *changes*:

MAP	ADJACENT PAIRS	TOTAL SIGNALS	COUPLED	UNCOUPLED	RATE
2019 enacted	486	8	3	5	1.65 %
Majority 2026 (v0_8 full)	582	8	3	5	1.37 %
Minority 2026 (v0_8 full)	460	**14**	**4**	10	**3.04 %**

*Under v0_8 full coverage, the minority's overall chain-signal rate is **2.22× the majority's** and **1.84× the 2019 baseline** — the opposite directional result from the v0_2 partial-coverage reading above. The coupled-count ratio (minority 4 / majority 3 = 1.33×) is still under the pre-committed 1.5× pass threshold, so the pre-registered pass criterion technically holds, but the overall rate signal is in the opposite direction from the partial-coverage finding.*

What changed. The v0_2 substrate used the audit's earlier polygon set, which under-counted central-Calgary urban adjacencies (sub-kilometre topology gaps) and over-counted minority isolates via K-nearest fallback. The v0_8 full-coverage substrate has consistent topology (Phase 4 1m precision pass; v0_8.2 inheritance fill closes the urban-empty-ED gap driving the K-nearest fallback). Under the cleaner topology, the minority's *uncoupled* chain signals (party-A packed in X, party-B cracked in Y) jump from 5 to 10 — these were previously masked because the under-counted urban Calgary adjacencies hid the Calgary-South / Calgary-Currie / Chestermere-Strathmore chain pairs the v0_8 substrate now resolves.

The honest reading. The pre-committed pass criterion (coupled count ratio ≤ 1.5×) was met under both substrates. But the v0_2 substrate's "minority is *less* adjacency-coupled" finding does not survive the cleaner topology — the minority is *more* chain-signal-prone than majority or 2019 on the v0_8 substrate, both in coupled count and especially in overall rate.

The audit reports both readings, retracts the strong "pack-and-divide vs pack-and-drain" archetype claim above (which depended on the v0_2 minority's 0-coupled count), and notes on the v0_8 substrate the minority's neighbour-drain pattern is consistent with the §5.3.1 + §5.3.2 packing-cracking findings rather than inverting them. The cross-substrate disagreement is itself a documented finding per the multi-method-reporting discipline.

Justify-or-drop validation (2026-06-11, `findings/drain_metric_validation.md`). A three-pronged validation was run against the Channel 3 metric:

- **V1 — synthetic ground truth (PASS).** A constructed 3×3 ED grid with a known pack-and-crack pattern (P4 packed 90/10 against NDP, eight outer EDs cracked 48/52 against NDP) produces `drain_score = 0.00996` against a neutral 50/50 jittered grid's `drain_score = 0.0`. The metric detects a constructed pack-and-crack signature, satisfying construct validity.
- **V2 — bivariate Local Moran's I numerical anchor (FAIL).** The audit's framing of Channel 3 as a "directional bivariate Local Spatial Autocorrelation / LISA analog" was checked numerically: for every directed adjacent pair compute `I_pair = z_s(X) × (−z_m(Y))` and correlate with drain-intensity at the same pair. Pearson r = 0.114 (majority) and −0.002 (minority), well below the pre-committed 0.30 anchor threshold. **The LISA framing is rhetorical, not numerical, and is hereby retracted.** Channel 3 is a bespoke pack-and-crack adjacency statistic without external statistical heritage.
- **V3 — pre-registered Prediction A on canonical substrate (PASS).** Recomputing the continuous drain_score on the canonical Elections Alberta shapefiles + canonical VA centroid-in-polygon attribution (the substrate every other audit channel uses) gives `majority = 0.00721`, `minority = 0.00059`. Pre-registered Prediction A (drain(majority) > drain(minority)) confirms. The label-shuffle null on the same canonical substrate gives P(majority > minority | random labels) ≈ 0.91, so the direction is the geometry-favored one and is not strong evidence on its own.
- **Substrate-stale numbers retracted; canonical Phase B null now landed (2026-06-11, `findings/drain_label_shuffle_null_canonical.md`).** The original Phase B numbers (`findings/drain_label_shuffle_null.md`: majority drain = 0.000179, minority = 0.006176; majority z = −2.915 "anomalously low") were computed against a DPG-era / blended-vote substrate and have been bannered SUBSTRATE-STALE. The canonical-substrate Phase B re-run (10,000 label-shuffle permutations on official Elections Alberta shapefiles + canonical VA centroid-in-polygon attribution) reports:

MAP	OBSERVED DRAIN_SCORE	NULL MEAN	Z	PERCENTILE	P (TWO-TAILED)
2019 enacted	0.001530	0.052987	−3.520	0.0 %	0.0000
Majority 2026	0.007213	0.049257	−3.173	0.0 %	0.0002
Minority 2026	0.000591	0.028057	−2.750	0.0 %	0.0002

Three substantive readings emerge from the canonical re-run: (i) Pre-registered Prediction A (drain(majority) > drain(minority)) is CONFIRMED on canonical substrate, reversing the stale-substrate failure. (ii) All three maps are anomalously low against their own label-shuffle null — the 2019 enacted map is the most anomalously low (z = −3.52), the majority second (−3.17), the minority third (−2.75); the metric does not discriminate the 2026 commission maps from the 2019 baseline at the directional level. (iii) The *original framing of the majority as singularly "anomalously clean" was a substrate artefact: on canonical substrate every Alberta map is anomalously clean by this null, consistent with the §5.3.5 hybridization mechanism. Channel 3 remains excluded from the Bonferroni headline combination ($p \le 2.80 \times 10^{-6}$), so this finding does not change the audit's headline.*

5.4 MCMC constraint-bound ensemble

Terminology note (2026-04-24, per retraction-pathway §9 item 1). The language of "ensemble median," "neutral map," and "geometric baseline" is replaced throughout this section — and in §5.2.5 — with **"constraint-bound expectation."** The ReCom ensemble samples from the universe of compact, contiguous maps satisfying the ±25 % population-deviation constraint. This universe is not "neutral" in any politically or statutorily loaded sense. It is the expectation the constraint set produces. The audit does NOT claim the ensemble median is what "fair" drawing would yield — it claims real maps outside the 5–95 band of the constraint set are displaced from what those constraints alone produce, and the displacement is a measurable drawing signature (whether partisan, COI-driven, or otherwise). This responds directly to the 2026-04-24 "Geographic Neutrality Myth" critique: there is no apolitical mathematical object called "the neutral map." The constraint-bound expectation is a reference point, not a normative target.

5.4.1 Run #1 — 10k preliminary ensemble

A Markov Chain Monte Carlo (MCMC) ensemble was run to place each of the three real maps against a distribution of legal ReCom-drawn alternatives. Substrate: 4,765 Voting Area polygons (Elections Alberta 2023) carrying 2023 UCP / NDP / Other votes and 2021 dissemination-area-weighted population. Chain: `gerrychain` 0.3.2 Recombination proposal, ±25% population deviation, seed 42. A 10,000-sample preliminary run (~89 s on laptop) was followed by a 250,000-sample publication-grade run (~12 min on laptop) with a full-coverage rescore using 88-row majority and minority full crosswalks (every VA outside a scored 2026 polygon is assigned to its 2026 ED via the crosswalk; coverage now 100% of VAs on both proposals). Sign convention: positive = UCP-favoured. Full method, convergence diagnostics, and per-sample data in `analysis/methodology/mcmc_250k_and_full_coverage.md` (plus `analysis/methodology/mcmc_ensemble.md` for the 10k preliminary), `data/simulated_ensemble_raw_samples_250k.csv`, and `data/simulated_ensemble_percentiles_full_250k.csv`.

§15(2) deviation constraint gap (twofold; widened 2026-06-12 per T1.7 Referee #2). Two constraint asymmetries between the canonical ensemble and the real legal space need explicit disclosure:

1. **Proposal-level ε is ±12.5% asymptotically; ±25% for inherited initial districts.** The ReCom proposal at `analysis/scripts/mcmc_ensemble.py:558` sets `epsilon = pop_deviation / 2.0`, so for the documented `pop_deviation = 0.25` the *proposal step* explores deviations up to ±12.5% from the ideal. The accept-reject filter

`alberta_statutory_population_constraint` (lines 563-573) enforces ±25% but rarely binds because most proposals already satisfy ±12.5%. **Caveat:** the initial partition is built at `epsilon = pop_deviation` (the full ±25% bound at line ~542), so inherited districts between 12.5–25% persist until ReCom redraws them — the ensemble is asymptotically ±12.5%-bounded but not absolutely. The §15(2) carve-out branch of the constraint function (≤4 districts at ±50%) is unreachable from the ±12.5% proposal regime. The published headline percentiles are anchored to a constraint set the proposal step actually halved. Fix: re-run with `epsilon = pop_deviation` (queued as **T1.11** in `TODO_REMEDIATION.md`; ~8h single-core; corrected reference 2026-06-12 per T1.7 R2 Ref #2 — round-1 erroneously cited "T1.4-eps", which doesn't exist).

2. **§15(2) carve-out unreachable from either proposal regime.** The EBCA §15(2) permits up to 4 electoral divisions to have a population "as much as 50% below the average population of all the proposed electoral divisions" when at least 3 of 5 statutory criteria are met (area, distance from Legislature, absence of large towns, Indian reserve or Métis settlement, provincial boundary coterminous). Both commission maps invoke this exception for 3 EDs — achieving deviations of −27.2% to −47.7% below the provincial average (§5.1.4, `analysis/methodology/population_deviation_reaudit.md`). By comparison, the **2017 Bielby commission** drawing the 2019 enacted map (the 2010 Walter commission had set the count to 87 EDs when expanding from 83 — see `analysis/review/editor_synthesis.md` S-5) used only 2 of the 4 permitted slots (Central Peace-Notley and Lesser Slave Lake), explicitly declining to use a third. Earlier drafts of this paragraph attributed the 2019 map to the "2010 EBC" with "Dunvegan-Central Peace-Notley" naming — corrected 2026-06-12 per T1.7 Referee #7; the 2019 ED names are 2017-cycle, not 2010-cycle. Both 2026 proposals add one additional invocation not present in the 2019 baseline (Canmore-Banff for the majority; Rocky Mountain House-Banff Park for the minority). The MCMC ensemble uses a uniform ±25% constraint for all 89 EDs. It cannot generate maps where any ED falls below −25%, and therefore does not sample the full legal possibility space available under §15(2). The 3 s.15(2) EDs in each real map are by construction outside the ensemble's distribution: they exist in a constraint tier the ensemble does not reach. For aggregate partisan metrics (EG, MM, declination, seats-at-50/50), the 3 s.15(2) ridings — small, remote, and heavily UCP-leaning — contribute a modest fraction of total province-wide votes. The effect on ensemble-relative percentile positions is expected to be small but has not been directly quantified. The minority-majority *asymmetry* comparison is less sensitive to this gap than the absolute percentile scores, because both maps invoke comparable s.15(2) configurations.

Constraint-enforcing first run executed (2026-06-13; T1.4 partial + T1.11; `findings/constraint_enforcing_ensemble_100k.md`). Both gaps above were addressed in a 100,000-step adapted run: (a) the proposal ε was set to the full ±25%; (b) the s.15(2) tier was handled by *freezing* the protected districts at their commission-drawn boundaries (seeding from the real majority map bypasses the recursive_tree_part infeasibility that defeated the earlier H6 freeze attempt) — the freeze set self-identified as 5 districts below 75% of ideal on the 2021 substrate (3 statutory s.15(2) + 2 cycle-lag Calgary suburbs); (c) a per-plan municipal-split tally over 170 multi-VA CSDs was recorded. Results (both seeds run same day; full tables at `findings/constraint_enforcing_ensemble_100k.md`): **three of the minority's four partisan metrics are robust ≥p98.7 tail flags under every null tested** (MM ≥p99.8, declination ≥p98.7, s50 ≥p99.98 across unconstrained-1.01M, majority-seeded-constrained, and minority-seeded-constrained). **The minority's EG is threshold-straddling and seed-dependent: p97.24 under the majority-seeded freeze (5 frozen districts — the flag fires) but p94.17 under the minority-seeded freeze (3 statutory districts frozen; Calgary cycle-lag suburbs redrawn) and p94.4 unconstrained.** The honest EG reading is "straddles p95 in the p94–p97 band depending on the s.15(2)/cycle-lag freeze configuration" — neither a clean fire nor a clean miss. The majority becomes *more* normal on every metric under both constrained nulls (s50 p78 → p51–p57; declination p20 → p19–p34). ReCom plans split 55–67 municipalities (median 61) against the real maps' 23/30 — both real maps at ≈p0, empirically validating the commission-convention auxiliary *symmetrically* (the §5.4.9 severity-audit requirement) and supplying the Amendment-9 S2 specificity rate (Pr(splits ≤ 30 | ReCom) ≈ 2×10^{-4}). Caveats: 100k single chains (ESS ≈ 125–200; p9x.x percentiles within resolution, p99.9x values read as "extreme tail"); split-count universes differ slightly between the per-plan tally (CSD-based) and the battery count (CY/T/SM). The publication-grade 1.01M-scale constrained run remains queued (T1.4-full, ≈3.5 h at observed rate).

Election-Day-only vote coverage (ES-13; full-vote sensitivity recomputed 2026-06-12 per T1.7 R1 Ref #2 D28). The 4,765 VA polygons carry polling-station votes cast on Election Day only. Advance polls, mail-in ballots, and special ballots — approximately 47.2% of 2023 two-party (UCP+NDP) votes province-wide. The Election-Day share was approximately 52.8% in 2023 — are not geographically assigned at the VA level by Elections Alberta and are therefore absent from the MCMC substrate. The absolute percentile positions reported throughout this section (e.g., Minority 2026 EG at p94.4 in the 1.01M canonical ensemble) reflect the Election-Day vote geography.

Full-vote sensitivity (T1.16 closed 2026-06-12). Using `va_polygons_with_full_2023_votes.gpkg`'s `va_ucp_full` / `va_ndp_full` columns (advance ballots apportioned to VAs by election-day two-party shares), the per-map partisan-bias values shift but **direction holds on all four metrics on both maps**:

MAP	METRIC	ELECTION-DAY ONLY	FULL-VOTE SUBSTRATE	Δ
Majority	EG	+0.0010	+0.0052	+0.0042
Majority	MM	−0.0362	−0.0306	+0.0056
Majority	δ (Warrington)	−0.0267	−0.0410	−0.0143
Majority	s50	0.4607	0.4719	+0.0112
Minority	EG	**+0.0402**	**+0.0579**	**+0.0177**
Minority	MM	+0.0104	+0.0063	−0.0041

MAP	METRIC	ELECTION-DAY ONLY	FULL-VOTE SUBSTRATE	Δ
Minority	δ (Warrington)	+0.0770	+0.1037	+0.0267
Minority	s50	0.5169	0.5056	−0.0112

Headline implication: the minority's EG at +5.79% under full-vote attribution is above both the audit's pre-registered Alberta-calibrated 4.10% threshold and the academic-literature 5% / 7% S-M thresholds. *The election-day-only reading places minority EG at p94.4 (sub-p95, EG flag withdrawn). Under full-vote attribution the minority EG crosses every threshold the audit cites and would be flagged on the EG axis. The audit's headline retains the election-day reading as primary because that is the substrate the canonical ensemble uses; the full-vote reading is reported here as a sensitivity check with the explicit disclosure that the EG flag would fire under the full-vote attribution. Declination magnitude also strengthens (+0.077 → +0.104), reinforcing the UCP-tail signal.*

The SZAT analysis (§5.2.10) uses the same election-day substrate as the MCMC ensemble (`va_polygons_with_full_2023_votes.gpkg` columns `va_ndp/va_ucp`). The two-party total is ~896,644 and the pre-registered result (p=0.0024) is substrate-consistent with Channel 1. A full-vote SZAT sensitivity using the advance-ballot-apportioned columns yields p < 0.0001 — even tighter. Neither SZAT run uses DPG-derived geometry.

Earlier drafts of this paragraph asserted ES-13 introduced "no partisan asymmetry in the ensemble comparison" — that framing is corrected 2026-06-12 per T1.7 R1 Ref #2 D28: ES-13 changes vote geography, and the magnitude of metric shifts is *not* uniform across maps (the minority's EG shifts by +1.77 pp under full-vote vs +0.42 pp for the majority). Direction is preserved, but the "invariance" framing was an overclaim.

5.4.2 Run #2 — 250k crosswalk full-coverage rescore

METRIC	2019 ENACTED (10K → 250K FULL)	MAJORITY 2026 (10K → 250K FULL)	MINORITY 2026 V6 (10K → 250K FULL)	250K ENSEMBLE 5TH / 50TH / 95TH
Efficiency gap	+0.0241 (p73.6 → p73.4)	+0.0066 → +0.0241 (p24.6 → p73.4)	+0.0170 → +0.0359 (p57.4 → **p92.1**)	−0.0097 / +0.0149 / +0.0394
Mean-median	−0.0077 (p96.1 → p92.7)	−0.0308 → −0.0077 (p6.6 → p92.7)	−0.0028 → −0.0009 (p100 → **p98.8**)	−0.0313 / −0.0191 / −0.0061
Declination	−0.0451 (p7.6 → p7.2)	+0.0049 → −0.0466 (p52.2 → p6.3)	−0.0259 → **−0.0704** (p18.0 → **p1.6**)	−0.0503 / +0.0033 / +0.0560
Seats at 50/50	+0.460 (p79.2 → p80.9)	+0.421 → +0.459 (p1.7 → p57.9)	+0.486 → +0.482 (p100 → p94.3)	+0.425 / +0.448 / +0.483

Structural floor (key finding). *The constraint-bound expectation on mean-median is −0.019 and on seats-at-50/50 is 0.448 — both UCP-favoured before any drawing choice is made. Alberta's vote geography (NDP concentrated in Edmonton and central Calgary, UCP spread across suburban and rural Alberta) produces a structural UCP floor ReCom-legal maps hit by default. Put another way: at a 50/50 province-wide vote, the median of 10,000 legal Alberta boundary maps draws 44.8 % UCP seats, and the ±25 % population rule and contiguity requirement do not permit a map escaping the floor. The question the MCMC frames is not whether a real map tilts UCP (every ReCom-reachable map does) but how unusual it is for a real map to sit further from the floor than the constraint-bound expectation produces.* ***The question the audit poses is not "is this the correct map?" but "given no uniquely-correct map exists under the constraint set, how statistically improbable is this particular map within the set?"***

Under the 250k full-coverage rescore, the preliminary three-flag set resolves to a two-flag pattern concentrated on the minority map:

1. **Minority 2026 v6 at p95.35 on mean-median (UCP-favoured tail).** Mean-median −0.0009 is closer to zero (less NDP-skewed) than 95.35% of 250,000 full-coverage ensemble alternatives. The effective sample size (ESS) for this metric is ≈ 160, downgrading extreme tail certainty.
2. **Minority 2026 v6 at p98.4 on declination (UCP-favoured tail under corrected Warrington sign; Run #3 v0_7 historical reading).** Declination +0.0704 (sign corrected per Amendment 10 from the pre-Amendment-10 −0.0704 / p1.6 under the swapped operand) is more UCP-favoured than 98.4% of ensemble alternatives. Combined with the mean-median UCP flag, this signals asymmetric packing — winning-margin geometry compressed near 50 % on the UCP-favoured side. Note: this is a Run #3 v0_7 historical reading; the canonical 1.01M reading is p98.79.

Two of the 10k-era flags are withdrawn by the full-coverage rescore: **2019 enacted on mean-median** (10k p96.1 → 250k full p92.7); and **majority 2026 on seats-at-50/50** (10k p1.7 → 250k full p57.9). The minority's 10k-era seats-at-50/50 flag is also downgraded under full coverage to **p89.72** (inside the neutral band) due to ESS limits and crosswalk fallback impacts. Efficiency gap on the minority is a near-outlier at p92.1.

Convergence diagnostics. Per-metric effective sample sizes on the 250k chain are 148–160 (integrated autocorrelation time τ ≈ 625–675 — ReCom is a slow mixer on this 4,765-node graph). Running-mean trace plots for all four metrics stabilise within the first 30–40k samples and drift <0.0003 across the latter 50k, well below the bandwidth of the 5–95 percentile interval on each metric. The chain has not reached US-state-court expert-testimony-grade (most prominently *League of Women Voters v. Commonwealth*, 178 A.3d 737 (Pa. 2018)) ESS (~5,000 independent draws typical for litigation claims). The 150-draw effective information content is sufficient for the audit's policy-comparison framing but not for a standalone statistical-significance claim at a tighter tail than p≈0.7. Plots in `data/maps/mcmc/running_mean_250k_*.png` and `data/maps/mcmc/ensemble_distribution_250k_*.png`.

Explicit tail downgrade under ESS = 150. Raw-percentile claims at the distribution tails must be read through the chain's effective sample size, not against the 250,000 nominal samples. With ESS ≈ 375 per metric, the outermost tail percentile the chain can support at 95% credibility is approximately p ∈ [2.5, 97.5]: anything outside the interval is within the Monte-Carlo noise floor of an ESS-150 chain. Under this downgrade, the table's **p100 and p1.6 values are not statistically distinguishable from p95.35 and p2.5 respectively** at the chain's effective precision. The audit reports the downgraded bounds — **Minority mean-median at p95.35 (UCP-favoured tail flag retained)** and **Minority declination at p97.5 ceiling (UCP-favoured tail flag retained, under the corrected Warrington 2018 sign; earlier draft cited p2.5 NDP-tail under the swapped-operand convention now retired)** — and treats the raw p98.8 / p97.5 decimals as point estimates within an ESS-wide credible interval, not as precision-bearing claims about the true percentile. The minority seats-at-50/50 raw percentile of p94.3 drops to **p89.72** under the same ESS-150 downgrade — inside the neutral band; the flag is retracted. The Gemini red-team (Phase B1) and the Geometric-

Precision-Fallacy plan both identified this disclosure as necessary. The T3 multi-chain ReCom run queued for follow-up work aims to raise ESS above 1,000 and tighten these tail bounds.

Session-12 canonical + full-VA rescore — numbers disagree with the 250k table above. The session-12 remediation (§4.1.4, §5.2.7) repointed the MCMC real-map rescore at the canonical Derived Provisional Geometries and the full-vote VA substrate. Against the same 10k neutral ensemble, minority 2026 now places at **p60.3 on efficiency gap, p81.3 on mean-median, p17.5 on declination, and p100.0 on seats-at-50/50** — a materially different tail pattern from the pre-remediation 250k numbers in the table above, mostly tied to the 2.96 pp province-wide NDP-share correction from Vote-Anywhere splatting. The two readings are both reported: §5.2.7 documents why they disagree. The sunset clause (§4.1.4) binds the audit to re-run both against official shapefiles when released. The 250k table retains the pre-remediation numbers for methodological traceability. The session-12 numbers are the higher-resolution current state of the art and should be cited alongside, not in replacement of, the 250k table.

Multi-chain convergence update (2026-04-24, Gemini Phase D.2 response; refreshed at 150k/chain). Cross-chain convergence is confirmed at strict publication-grade precision. A three-chain ReCom run (seeds 42, 101, 2024; 150,000 ReCom proposals per chain; 10 % burn-in; 91 min total runtime; script `analysis/scripts/simulation_multichain_ensemble.py`) was executed to produce a publication-grade Gelman-Rubin $\hat{R}$ convergence diagnostic. Per-metric split-chain $\hat{R}$ values are **efficiency gap 1.0075, mean-median 1.0099, declination 1.0076, seats at 50/50 1.0014** — all within the strict $\hat{R} < 1.01$ criterion (Gelman et al., 2013, *Bayesian Data Analysis*, 3rd ed., ch. 11) on three of the four metrics, with mean-median at the 1.0099 boundary.

Combined ESS across the three chains is 643–783 per metric (per-chain ESS 165–291; integrated autocorrelation time $\tau \approx$ 463–749 per chain) — an approximate $4\times$–$5\times$ improvement over the original single-chain ESS ~150. At 643–783 the chain is still below the ~1,000–2,000-draw publication-grade ESS threshold for individual tail claims at extreme percentiles (the earlier "MGGG lawsuit-grade" framing is retired 2026-06-12 per T1.7 Referee #9 — see §5.4.2(a)), but firmly in policy-comparison-grade territory. The Monte-Carlo standard error on any 5th or 95th percentile estimate is now approximately ±1.5 percentile points. Under the combined ensemble, the tail-downgrade bounds remain valid: **p100 and p1.6 claims are bounded to p95.35 / p2.5 at the chain's combined effective precision**, and the ~p90 Minority seats-at-50/50 flag remains retracted. Reviewers demanding ESS > 1,000 can reproduce with `--steps 230000` (est. 140 min); the paper's conclusions do not change under the extension. Full convergence diagnostic at `data/simulation_multichain_rhat.json` and `data/simulation_multichain_summary.md`. Raw samples at `data/simulation_multichain_samples.csv`.

Multichain real-map scoring confirmation (2026-05-10; declination percentile re-stated under Amendment 10 convention 2026-06-12). The three real maps were scored against the multichain pooled ensemble (5,001 thinned samples; common thinning factor 81; pooled from seeds 42, 101, 2024 after 10 % burn-in). Minority 2026 canonical places at **EG p93.94, MM p99.94, declination p98.90 (UCP-tail under corrected Warrington sign; earlier draft cited p1.10 NDP-tail under the swapped-operand convention), seats-at-50/50 p99.98**. The maximum drift from the canonical 50k single-chain percentile estimates is 0.77 pp (seats-at-50/50 majority). No metric changes sign, crosses the p95/p5 threshold boundary, or shifts by more than 1.1 pp. The canonical 50k headline claims are confirmed at the multichain-pooled precision level. The Methods reviewer's single-chain-convergence objection is addressed by the R-hat and ESS diagnostics, with the real-map scoring providing empirical confirmation the canonical numbers are not an artefact of chain initialisation. Full scoring data at `data/outputs/mcmc/simulation_multichain_real_map_percentiles.csv`.

Canonical ensemble convergence diagnostics (2026-05-10; OSF s58a6 Section B). Gelman-Rubin $\hat{R}$ was computed on the canonical ensemble (4 chains × 62,500 plans = 250,000 total) under two estimators: the classic Gelman & Rubin (1992) statistic and the rank-normalised variant recommended by Vehtari et al. (2021, *Bayesian Analysis*). Results by metric:

METRIC	$\hat{R}$ GR92	$\hat{R}$ VEHTARI (2021)	GR92 < 1.1	V21 < 1.01	WORST-CHAIN ESS
Efficiency gap	1.01843	1.01847	PASS	marginal	76
Mean-median	1.00179	1.00181	PASS	PASS	63
Declination	1.01343	1.01440	PASS	marginal	70
Seats at 50/50	1.00540	1.00527	PASS	PASS	94

All four metrics pass the GR92 < 1.1 publish-grade threshold. Efficiency gap ($\hat{R}$ = 1.018) and declination ($\hat{R}$ = 1.014) sit marginally above the Vehtari (2021) 1.01 recommendation, indicating mild residual between-chain variation on those two metrics. Mean-median and seats-at-50/50 clear the stricter criterion. Worst-chain ESS ranges 63–94 draws (integrated autocorrelation time τ ≈ 660–990). These ESS values are lower than the three-chain run above (165–291 per chain) because the canonical chains are initialised differently and thinning is not applied. The mild non-convergence on EG and declination does not materially affect the Ch1 conclusion: the Mahalanobis joint-tail p = 1.40×10^{-6} has multiple orders of magnitude of headroom. Even a conservative ESS-adjusted uncertainty of ±2 percentile points on EG and declination cannot move the joint p-value past any interpretive threshold. The disclosure obligation from S2-01 is met by this table. The ESS-precision caveat in the paragraph below ("p100 and p1.6 are bounded to p95.35 / p2.5") remains in force. Diagnostic output: `data/outputs/rhat_diagnostic_section_b.json`. Pre-registered: OSF s58a6, drand round 6099592, committed git hash 72f7e01.

Coverage caveats. Full-coverage rescoring uses polygon assignment where v5/v6 / approximate polygons exist (minority v6: 70 of 89 polygons; majority approximate: 57 of 89) and 88-row full-crosswalk fallback for every other VA via `parent_ed_2019 → 2026 ED`. Coverage is 100% of VAs on both proposals, matching the ensemble's own coverage. A small number of pure Tier-C 2026 EDs with no 2019 parent (4 majority, 5 minority) are not populated by either polygon or crosswalk and are enumerated in the JSON output. The missing EDs are inside the cities where polygon-based scoring is authoritative, so their absence from the crosswalk layer does not affect aggregated metrics.

5.4.3 Run #3 — 250k v0_7 DPG geometry (full 89-ED centroid attribution)

Historical record (DPG-era). *Run #3 and the short-burst analysis below use Derived Provisional Geometries (DPG v0_7) and are preserved as the pre-shapefile analytical record. All percentile placements, including `p100` values in tables below, are superseded by the canonical official-shapefile run in §5.4.9 for the purpose of final conclusions.*

Run #3 — full 89-ED v0_7 geometry (completed 2026-04-24/25; log `analysis/reports/v0_1_mcmc_run3_v7_250k.log`). A third production-grade 250,000-plan ReCom run was completed against the v0_7 canonical DPGs (89 EDs per map; `data/shapefiles/derived/v0_7_canonical_{majority,minority}_2026_eds.gpkg`). The ensemble uses the same 4,765-VA substrate (2023 votes, 2021 population) as previous runs. Real-map scoring uses centroid-in-polygon attribution against the v0_7 polygons (majority: 66 of 89 EDs scored, 66.4 % VA coverage; minority: 71 of 89 EDs, 60.8 % VA coverage — partial coverage arises because v0_7 DPGs do not yet fully tessellate the province, leaving some VA centroids in unclaimed territory). Convergence: ESS 123–219, τ 457–810 (consistent with prior runs). Updated Alberta EG threshold: ensemble 95th-percentile EG = **+4.37 %** (previously 3.86 % from the 10k run), replacing the US-derived 7 % figure in all EG threshold discussions.

Run #3 per-metric percentiles against the 250k ensemble:

METRIC	2019 ENACTED	MAJORITY 2026 V7	MINORITY 2026 V7	ENSEMBLE P5 / P50 / P95
Efficiency gap	+0.0241 (p67.7)	−0.0024 (p12.0)	−0.0102 (p5.3)	−0.0107 / +0.0168 / +0.0437
Mean-median	−0.0077 (p93.2)	**+0.0080 (p99.98)**	−0.0108 (p86.7)	−0.0307 / −0.0194 / −0.0065
Declination (pre-Amendment-10 sign; Run #3 v0_7 historical)	−0.0451 (p11.6)	−0.0203 (p29.1)	**−0.0941 (p0.9)**	−0.0663 / −0.0053 / +0.0555
Seats at 50/50	+0.460 (p63.2)	**+0.515 (p100.0)**	**+0.549 (p100.0)**	+0.425 / +0.460 / +0.483

Outlier flags (≥95th or ≤5th percentile; Run #3 v0_7 historical with pre-Amendment-10 sign): majority MM at p99.98 and s50 at p100. Minority declination at p0.9 (= p99.1 UCP-tail under corrected Warrington sign) and s50 at p100. The EG sign reversal (both 2026 maps showing negative EG under v0_7 centroid attribution) is attributed to the coverage artefact: only 66–71 of 89 EDs are scored, biasing the sample toward urban/suburban areas where NDP performs strongly. MM and s50 are more robust to partial coverage and are the primary cited metrics for this run. The §4.1.4 sunset clause treats Run #3 as the authoritative pre-official-shapefile MCMC estimate, pending the v0_8 tessellation run (Run #4, pending DPG perfecter completion).

Short-burst analysis (completed 2026-04-25; `data/simulation_short_bursts.csv`, `findings/simulation_short_bursts.md`). 500 independent 10-step ReCom chains were run from the 2019 enacted baseline, each with a unique seed, to characterise the *reachable neighbourhood* of the 2019 map. This supplements the ensemble by answering: is the 2026 map's score achievable within 10 random redistricting steps from 2019, or does it require a long, directed walk?

Burst-endpoint distribution (500 × 10-step chains from 2019 baseline):

METRIC	BURST P5	BURST P50	BURST P95
Efficiency gap	+0.0009	+0.0044	+0.0189
Mean-median	−0.0226	−0.0125	−0.0086
Declination	+0.0070	+0.0321	+0.0363
Seats at 50/50	+0.4477	+0.4598	+0.4713

Real map percentile ranks within the burst distribution:

MAP	EG	MM	DECLINATION	S50
2019 enacted	p98.0	p98.2	p0.0	p30.8
Majority 2026 v7	p3.0	**p100.0**	p0.0	**p100.0**
Minority 2026 v7	p0.8	p84.2	p0.0	**p100.0**

Key finding: the majority map's MM=+0.0080 and both 2026 maps' s50 values are at ***p100 of the burst distribution*** *— no 10-step random walk from the 2019 baseline reached these values. The minority's declination=−0.0941 is at p0 in both the 250k ensemble and the burst distribution. This confirms the 2026 boundary changes on these metrics represent a directed departure from the 2019 baseline rather than random redistricting drift. The 2019 enacted itself sits at p98 on EG and MM within its own reachable neighbourhood — consistent with its status as a deliberately drawn boundary at the edge of the random-walk space around it.*

Falsifiability hook (resolved and revised). The 10k preliminary hook's retraction rule has fired and been applied — see the retraction paragraph above. Revised hooks for the remaining claims: if a later commission-shapefile-driven re-run moves the minority map's mean-median below p95 on a 2026-seed ensemble, the mean-median flag is retracted and the minority is reclassified as inside-band on the metric. The declination flag's falsifiability hook is a structural-packing counter-test: if the Calgary-zone packing patterns (§5.6) are independently contradicted — e.g., by a census-block-level verification — the declination flag is downgraded to one-of-two corroborating signals rather than a standalone flag. The 2019-seed ensemble used here is conservative against the minority's held flags. A 2026-seed ensemble would more closely match the minority's own geometry and would deepen the minority's percentile tails rather than narrow them.

5.4.4 Run #4 — 250k MCMC against v0_8 full-coverage geometry (2026-04-25)

> ***Historical record (DPG-era).*** *Runs #4–#6 use Derived Provisional Geometries (DPG v0_8) and are preserved as the pre-shapefile analytical record. All percentile placements, including `p100` and `p100.0` values in tables below, are superseded by the canonical official-shapefile run in §5.4.9 for the purpose of final conclusions.*

A 250,000-sample ReCom run was executed across **4 parallel chains × 62,500 steps each** (chain-specific seed = base * 250k + chain * 1k + chunk; chunked checkpointing every 5,000 samples). Wall time 17.8 min on a 13th-gen i7-1360P (12 physical cores, 16 logical threads) — vs an estimated ~50 min for an equivalent single-chain run. Effective sample size (Geyer initial-positive-sequence) is 491–499 across the four primary metrics, with autocorrelation tau = 501–509.

Critically, this run is the first to use full 89/89 ED coverage on both maps. Earlier runs (10k preliminary; Run #3 at 250k) were against partial-coverage v0_7 geometry (67–71 of 89 EDs measurable), which under-sampled votes from rural-UCP regions. The full-coverage geometry is built via the v0_8.2 inheritance fill (§4.1.6, 21 majority + 12 minority EDs filled from 2019-Tier-A).

METRIC	2019 ENACTED	MAJORITY 2026 (V0_8 FULL)	MINORITY 2026 (V0_8 FULL)
Efficiency gap	+2.41 % (p66.3)	**+6.43 % (p99.94)** ⚠	**+9.21 % (p100)** ⚠
Mean-median	−0.77 % (p91.8)	−1.82 % (p51.5)	−0.97 % (p86.9)
Declination	−4.51 % (p9.4)	−1.18 % (p43.0)	**−6.66 % (p2.9)** ⚠
Seats @ 50/50	46.0 % (p82.9)	43.7 % (p22.7)	**54.2 % (p100)** ⚠

Cross-method shifts vs Run #3 (v0_7 partial coverage). *The substantive changes are large enough to warrant explicit comparison — this is the kind of cross-method disagreement the four-measurement-layer reporting (methods paper §5) is built to surface honestly:*

- **EG sign and magnitude flipped.** Run #3 reported majority −0.0024 (p3.0) and minority −0.0102 (p0.8) — both in the NDP-favoured tail with sub-threshold magnitude. Run #4 reports majority +0.0643 (p99.94) and minority +0.0921 (p100) — both in the UCP-favoured tail with magnitude well above the Alberta-derived 4.37 % threshold (at the time; the canonical 1,010,000-plan run later revised this to 4.10 %). The flip is attributable to coverage: rural-UCP votes systematically un-attributed under partial coverage are now correctly captured.
- **Mean-median is no longer an outlier on either 2026 map.** Run #3 reported majority p100 and minority p100 (raw) / p98.8 (ESS-150 downgraded). Run #4 reports majority p51 and minority p87. The earlier outlier reading was a partial-coverage artefact: the 21 majority empty EDs (small Calgary/Edmonton districts) skewed the median when omitted.
- **Minority declination tail position persists across coverage** — the narrow-margin-loss packing signature is robust to coverage. Run #3 (250k v0_7) had minority declination at the low tail of the implementation's swapped-operand convention (p0 / p2.9 under the *pre-Amendment*-10 sign); under the Warrington-2018 convention restored 2026-06-12, those readings are equivalent to p100 / p97.1 — the UCP-*favoured* tail. The canonical 1.01M run gives the minority at p98.79 UCP-tail. The percentile magnitude survives across Run #3 v0_7 → Run #4 → canonical; only the sign label changes.
- **Minority seats@50/50 at p100 persists.** Run #3 had minority s50 at p100; Run #4 confirms at p100. The "extra UCP seat at a tied election" pattern survives the coverage shift.

- **Majority's outlier signature is now isolated to EG.** Run #3 had majority outliers on mean-median, seats@50/50, and EG-near-tail. Run #4 has majority outlier only on EG (declination p43, mean-median p51, s50 p23 — all inside the central band). The majority's structural geometry under full coverage produces a partisan-bias profile otherwise close to the constraint-bound expectation.

Authoritative reading discipline. Per the methods paper §5 four-layer pattern, the audit does not pick one of {Run #3 v0_7 partial, Run #4 v0_8 full} as authoritative. Both are reported. The cross-method disagreement is itself a finding: under partial coverage neither 2026 map is a Lane-1 EG outlier; under full coverage both are. The structural-pattern reading (Lane 2 in §6.2) is unchanged across coverage choices and remains the audit's most defensible single-direction finding. The §6.2 verdict's qualitative conclusion holds. The magnitudes in Lane 1's verdict table need to be read against both Run #3 and Run #4 numbers, not either alone.

Caveats propagating to downstream §6.2 verdict. The v0_8 full-coverage geometry has documented imperfections (§4.1.6 §4.1.8): inherited urban polygons trace 2019 boundaries, not 2026 commission boundaries. The v0_8 minority's "Peace River" polygon was over-extended by Phase 3 gap-fill and absorbs 6 of 10 city-centre landmark points (§4.1.7 alignment proof). These are disclosed where each downstream finding cites Run #4 numbers.

Outputs: `data/v0_1_mcmc_ensemble_samples_250k_v0_8.csv` (250,000 rows, 4-chain), `data/v0_1_mcmc_ensemble_percentiles_250k_v0_8.csv`, `data/v0_1_mcmc_real_map_scores_250k_v0_8.json`, `data/v0_1_mcmc_convergence_diagnostics_250k_v0_8.json`, `data/maps/mcmc/ensemble_distribution_250k_v0_8_*.png` (4 metric histograms), `data/maps/mcmc/running_mean_250k_v0_8_*.png` (4 convergence trace plots).

5.4.5 Run #5 — 1,000,000-sample MCMC against v0_8 full coverage (publication-grade ESS, 2026-04-25)

A 1,000,000-sample ReCom run was executed across **4 parallel chains × 250,000 steps each** (chain-specific seed = 44 * 250k + chain * 1k + chunk; chunked checkpointing every 5,000 samples). Wall time **71.9 min** on the same 13th-gen i7-1360P (~4× the Run #4 time, exactly linear in sample count). The run was launched explicitly to upgrade the ESS reading and tighten the percentile claims previously hedged at "beyond p95 (likely p98+/p99+)" in Run #4 §5.4.4.

Convergence diagnostics — publication-grade ESS achieved.

METRIC	RUN #4 ESS (250K)	RUN #5 ESS (1M)	IMPROVEMENT
Efficiency gap	515	**2,278**	4.4×
Mean-median	453	**1,967**	4.3×

METRIC	RUN #4 ESS (250K)	RUN #5 ESS (1M)	IMPROVEMENT
Declination	474	**2,278**	4.8×
Seats @ 50/50	430	**1,679**	3.9×

ESS scaling is approximately linear in sample count (4× more samples → 4× more independent draws), confirming the Run #4 chains were not stuck in a local mode. Per-metric ESS are: Efficiency gap 2,278. Mean-median 1,967; Declination 2,278; Seats@50/50 1,679. The percentile claims can be reported as ***point estimates within the chains' precision*** *rather than as bounds-only.*

Per-metric percentiles vs 250,000-row pooled ensemble (2023 vote substrate; v0_8 full-coverage geometry):

METRIC	2019 ENACTED	MAJORITY 2026 (V0_8 FULL)	MINORITY 2026 (V0_8 FULL)
Efficiency gap	+2.41 % (p67.4)	**+6.43 % (p99.93) ⚠**	**+9.21 % (p100.0) ⚠**
Mean-median	−0.77 % (p91.7)	−1.82 % (p53.5)	−0.97 % (p87.4)
Declination	−4.51 % (p9.1)	−1.18 % (p41.5)	**−6.66 % (p3.0) ⚠**
Seats @ 50/50	46.0 % (p82.3)	43.7 % (p21.6)	**54.2 % (p100.0) ⚠**

Cross-run stability vs Run #4. *Every percentile placement reported here is within ±0.5 percentile of the Run #4 value at 250k samples. This confirms Run #4 was not Monte-Carlo-noise-driven. The headline outlier flags (majority EG p99.9; minority EG p100; minority declination p3.0; minority seats@50/50 p100) hold under publication-grade ESS.*

Implication for the §6.2 verdict. The "likely p98+/p99+" hedge from §5.4.4 is now retired. Lane 1 in §6.2 can report majority EG at p99.93 and minority EG at p100 as point-estimate percentiles, not bounds. The published academic-redistricting threshold for "outlier" is p > 95. Both 2026 maps clear the threshold by ≥ 5 percentile points on EG, and the minority additionally clears it on declination and seats@50/50. The §6.2 author's verdict ("structurally consistent with engineering, magnitude-undetectable") was retracted in the 2026-04-25 revision. Under Run #5 ESS, the magnitude is **detectable** at publication grade.

Outputs: `data/v0_1_mcmc_ensemble_samples_250k_v0_8.csv` (1,000,000 rows — output filename retained for downstream-tool compatibility; will be renamed to `_1M_v0_8` in a follow-up commit), per-chain CSVs at `data/mcmc_checkpoints_250k_v0_8/chain{0-3}_samples.csv` (250,001 rows each), `data/v0_1_mcmc_ensemble_percentiles_250k_v0_8.csv`, `data/v0_1_mcmc_real_map_scores_250k_v0_8.json`, `data/v0_1_mcmc_convergence_diagnostics_250k_v0_8.json`, plot PNGs at `data/maps/mcmc/`.

5.4.6 Run #6 — 2,000,000-sample MCMC against v0_8 full coverage (converged-ceiling confirmation, 2026-04-25)

A 2,000,000-sample ReCom run was executed across **4 parallel chains × 500,000 steps each** (chunked checkpointing every 5,000 samples). Wall time **3,675 s = 61.25 min** on the same 13th-gen i7-1360P. Run #6 was launched specifically to test whether the 1,000,000-sample

Run #5 had reached the simulation's true ceiling on the seats@50/50 metric, or whether further sampling would push the maximum higher.

Convergence diagnostics — Run #6 ESS roughly doubles Run #5.

METRIC	RUN #5 ESS (1M)	RUN #6 ESS (2M)	IMPROVEMENT
Efficiency gap	2,278	**4,347**	1.91×
Mean-median	1,967	**3,766**	1.91×
Declination	2,278	**4,258**	1.87×
Seats @ 50/50	1,679	**3,138**	1.87×

ESS scaling is again approximately linear in sample count (2× more samples → ~2× more independent draws). Per-metric ESS are: Efficiency gap 4,347. Mean-median 3,766; Declination 4,258; Seats@50/50 3,138. All four metrics now sit at or above the US-state-court expert-testimony-grade (most prominently League of Women Voters v. Commonwealth, 178 A.3d 737 (Pa. 2018)) threshold (~3,000–5,000 typical for litigation claims).

Per-metric percentiles vs 2,000,000-row pooled ensemble (2023 vote substrate; v0_8 full-coverage geometry):

METRIC	2019 ENACTED	MAJORITY 2026 (V0_8 FULL)	MINORITY 2026 (V0_8 FULL)
Efficiency gap	+2.41 % (p67.5)	**+6.43 % (p99.94)** ⚠	**+9.21 % (p100.0)** ⚠
Mean-median	−0.77 % (p91.7)	−1.82 % (p53.6)	−0.97 % (p87.4)
Declination	−4.51 % (p9.0)	−1.18 % (p41.6)	**−6.66 % (p3.0)** ⚠
Seats @ 50/50	46.0 % (p82.3)	43.7 % (p21.6)	**52.8 % (p100.0)** ⚠ (89-of-89 attribution; see §5.4.7)

Cross-run stability vs Run #5. *Every percentile placement is within ±0.5 percentile of the Run #5 value at 1M samples. This confirms Run #5 was already at convergence. Run #6 adds precision but does not move any flag.*

Converged-ceiling finding (key result of Run #6). The simulation's seats@50/50 ceiling held at **exactly 51.72 %** across both the Run #5 1M ensemble and the Run #6 2M ensemble. Doubling the sample size from 1,000,000 to 2,000,000 did not push the maximum higher. Specifically:

- **Only 104 of 2,000,000 maps reach the 51.72 % ceiling** (one map per ~19,200 samples — consistent with rare-event sampling at the tail of a converged distribution).
- **Zero of 2,000,000 maps reach 52 %.**
- **Zero of 2,000,000 maps reach the minority's 52.8 % seats@50/50 reading.**

The 51.72 % ceiling is therefore not a sampling artifact. It is a **real bound on what a neutral, compactness-preferring procedure can produce under EBCA's ±25 % population-deviation, contiguity, and compactness constraints applied to Alberta's actual 2019 substrate carrying 2023 vote data**. The minority map's 52.8 % reading sits 1.08 percentage points above the simulation's converged ceiling — a placement no neutral procedure across two million draws could match, and the central empirical foundation of the audit's "out-of-distribution" framing in §6.2 (which is now strengthened from "p100 of the ensemble" to "above the ensemble's converged ceiling").

Implication for the §6.2 verdict. The "out-of-distribution rather than tail" reading is now empirically robust. A statistical outlier sits in the extreme tail of a distribution. The minority sits *outside* the distribution Run #6 produces. The probability a random legal map matches or exceeds the minority's seats@50/50 reading is bounded by the resolution of the simulation: **less than 1 in 2,000,000**. This is the headline number cited in `report_public.md` §"The 50/50 test."

Authoritative-run discipline. Run #6 (2M) supersedes Run #5 (1M) as the audit's authoritative MCMC ensemble. Run #5 and Run #4 are retained for historical record and methodological traceability — both produced the same percentile placements within ±0.5 — but downstream §6.2 verdict claims now cite Run #6.

Outputs: `data/v0_1_mcmc_ensemble_percentiles_250k_v0_8.csv` (overwrites the 1M run's file; same path reused), `data/v0_1_mcmc_real_map_scores_250k_v0_8.json`, `data/v0_1_mcmc_convergence_diagnostics_250k_v0_8.json`, per-chain CSVs at `data/mcmc_checkpoints_250k_v0_8/chain{0-3}_samples.csv` (4 × 500,001 rows), full run log at `analysis/reports/v0_1_mcmc_2M_v0_8_full.log`.

5.4.7 Defensible 89-of-89 attribution and fuzzing-scenario analysis (2026-04-25)

The seats@50/50 reading on the minority map depends on how 2023 voting-area centroids are attributed to 2026 minority districts. Earlier readings (Run #4, Run #5) reported **54.2 % (45 of 83 measurable)** because 6 of the minority's 89 polygons did not catch any 2023 voting-area centroids — these are **inheritance-fill slivers**, polygons drawn for districts whose 2026 boundaries the audit reconstructed from 2019 inheritance plus commission text (§4.1.6) and producing thin or fragmented footprints under DA-anchoring. The 83/89 measurable denominator was reported transparently but produced a brittle headline number depending on which 6 districts were excluded.

Defensible 89-of-89 attribution. The new attribution (Run #6 and downstream) fills the 6 unattributed minority polygons with each district's inherited 2019-polygon vote distribution. For each unattributed 2026 minority ED, the audit identifies the 2019 ED whose territory the inheritance-fill geometry traces (from the v0_8.2 inheritance fill in §4.1.6) and assigns the 2019 ED's 2023 vote distribution to the 2026 minority ED at the per-party share level. This is a **conservative inheritance assumption**: the 2019 vote distribution is treated as the best available substrate for territory the 2026 minority polygon was inherited from.

Under 89-of-89 attribution, the minority's seats@50/50 reading is **52.8 % (47 of 89)** — moving from 54.2 % (45 of 83) to 52.8 % (47 of 89) as two additional UCP-leaning seats appear in the previously unmeasurable inheritance-fill set. The 52.8 % is the headline number cited in `report_public.md` §"The 50/50 test."

Fuzzing-scenario analysis. The single 89-of-89 attribution is an estimate. The underlying inheritance assumption is one of many defensible choices. The fuzzing-scenario analysis at `analysis/scripts/fuzz_missing_eds.py` brackets the result by varying the per-ED vote attribution for the 6 inheritance-fill polygons across the full plausible range (random sampling from the broader 2019 vote distribution; alternate inheritance pathways; DA-weighted mixtures; provincial-mean fallback). Across the fuzzing trials:

- **Worst case (most NDP-favourable attribution):** minority seats@50/50 = **51.7 %** — at the simulation's converged ceiling but technically not above it.
- **Best case (most UCP-favourable attribution):** minority seats@50/50 = **57.3 %** — well above the converged ceiling.
- **Central 89-of-89 inheritance-fill attribution: 52.8 %** — above the converged ceiling but below the worst-case bound.
- **89 % of random-attribution trials place the minority above the simulation's ceiling.**

The fuzzing analysis confirms the "above the simulation's ceiling" finding is robust: the only attribution placing the minority *at* the ceiling (51.7 %) is a worst-case construction designed to soften the result. 89 % of plausible attributions place the minority *above* the ceiling. The 52.8 % central estimate is the audit's authoritative reading. The 51.7 %–57.3 % bracket is the audit's transparency about residual attribution uncertainty.

Why this matters for the §6.2 verdict. The fuzzing-scenario analysis closes a falsification pathway a hostile reviewer could have raised: "the 52.8 % reading depends on which 6 inheritance-fill polygons you imputed, and a different imputation would put the minority at the ceiling rather than above it." The honest answer is: only a worst-case imputation puts the minority at the ceiling. 89 % of plausible imputations put it above the ceiling, and the central reading is 1.08 pp above. The audit's "out-of-distribution" framing survives the imputation-sensitivity stress test.

Outputs: `analysis/scripts/fuzz_missing_eds.py` (fuzzing pipeline), per-trial CSVs and the 51.7 %–57.3 % distribution histogram in the run log.

5.4.8 Targeted-gerrymander short-bursts hill-climb (Cannon et al. 2023) — empirical proof of the non-neutral pathway

Run #6's converged-ceiling finding establishes a **neutral** procedure cannot reach the minority's seats@50/50 range across 2,000,000 draws under EBCA constraints. A reasonable-but-distinct question is whether a **non-neutral** procedure — one explicitly tasked with maximising UCP seats while staying inside the same statutory constraints — *can* reach the range. If yes, the minority's reading is consistent with the kind of map a partisan optimiser would produce; if no, the reading is anomalous on every procedure the redistricting-statistics literature has formalised.

To answer this directly, the audit ran a **short-bursts hill-climbing procedure** following Cannon, Goldbloom-Helzner, Gupta, Matthews, and Suwal (2023), "Voting Rights, Markov Chains, and Optimization by Short Bursts" (arXiv:2011.02288). Short-bursts is the standard tool in the redistricting-statistics literature for exploring biased-but-legal maps: it runs short MCMC chains (the "bursts") with the best-scoring map from each burst seeding the next, producing a hill-climb through the constraint-legal-map space toward an explicitly-defined objective.

Procedure. 40,000 ReCom steps with hill-climbing acceptance (accept proposed map iff its UCP seats@50/50 ≥ current best). Same substrate as the Run #4–6 ensemble (4,765-VA 2023 votes, ±25 % population deviation, contiguity). Explicit objective: **maximise UCP seats@50/50**. Implementation at `analysis/scripts/targeted_gerrymander_burst.py`. Full run log at `analysis/reports/v0_1_targeted_burst.log`.

Result. After 40,000 hill-climbing steps, the best map produced gave UCP **52.87 %** of seats at 50/50 votes — **within rounding of the minority map's 52.8 %**. The non-neutral procedure reaches the minority's range. The neutral procedure across 2,000,000 draws does not.

What this confirms. The Run #6 converged-ceiling finding plus the Cannon et al. short-bursts result jointly demonstrate the audit's central empirical claim:

- **Neutral procedures cannot produce the minority's seats@50/50 reading.** 2,000,000 ReCom draws under EBCA constraints, ceiling at 51.72 %, zero draws at 52 % or above.
- **Non-neutral procedures can.** A partisan optimiser using short-bursts hill-climbing under the same constraints reaches 52.87 % in 40,000 steps.
- **The minority sits in the non-neutral procedure's reachable space, not the neutral procedure's.** Whether the minority commissioners used a partisan optimiser is not observable from public data — intent cannot be read off a map. What is observable is the minority map's seats@50/50 reading is one a partisan optimiser would produce and one a neutral procedure does not produce, under identical statutory constraints applied to identical Alberta geometry.

Why this matters for the §6.2 "out-of-distribution" framing. A reviewer could object — "out-of-distribution" is meaningless without specifying *which* distribution. The Cannon et al. result specifies it precisely: the minority map sits inside the distribution a non-neutral procedure produces (reachable via 40,000 hill-climbing steps) and outside the distribution a neutral procedure produces (unreachable across 2,000,000 random draws). The audit's framing is therefore: *the minority map is the kind of map a non-neutral procedure produces. It is not the kind of map a neutral procedure produces*. This is the framing a court would actually apply, and it is the framing `report_public.md` adopts in its §"The 50/50 test" methodology caveat.

What this does not show. The short-bursts result does not show the minority commissioners used a partisan optimiser. It shows only — *if* one had been used, the minority's seats@50/50 reading is in the range the procedure would produce. The audit's restraint on intent is unchanged: intent is not observable from boundary geometry. The procedural-class mapping (neutral vs non-neutral) is.

Outputs: `analysis/scripts/targeted_gerrymander_burst.py` (short-bursts implementation), `data/targeted_burst_best.json` (best map's per-ED seat count and metadata), `data/targeted_burst_trace.csv` (per-step hill-climbing trace), `analysis/reports/v0_1_targeted_burst.log` (full run log).

Canonical-substrate confirmation (2026-06-13; `findings/burst_pathways_canonical.md`). Both burst experiments were re-run on the official Elections Alberta canonical shapefiles. The targeted hill-climb (`targeted_gerrymander_burst.py`, 800 bursts × 50 steps = 40,000 steps, seed 137) reproduces the **52.87 %** ceiling exactly — it is the discrete 46-of-87-seat maximum, robust to substrate, which is why it matches the DPG-era figure. The neutral short-bursts experiment (`simulation_short_bursts.py`, 500 independent 10-step ReCom walks from a single tight 2019-seed start) places the minority map more extreme than *every one* of the 500 neutral walks on all four partisan metrics (efficiency gap, mean-median, declination, seats@50/50), with a neutral seats@50/50 ceiling of 0.4713 — well below the minority's 0.5169. The majority map, by contrast, falls inside the neutral burst envelope (seats@50/50 burst-rank p88, efficiency gap p9.6, mean-median p0.2). The DPG-era ceiling figures above therefore survive on the canonical substrate, and the canonical 1,010,000-plan ensemble (§5.4.9) independently places the minority's seats@50/50 at p99.99.

5.4.9 Canonical ensemble — official Elections Alberta shapefiles (1,010,000-plan, 4 chains × 252,500 steps, completed 2026-05-12)

This is the authoritative MCMC run. The §4.1.4 sunset clause bound the audit to re-run the ensemble against official shapefiles within two weeks of release. Official shapefiles were received 2026-05-06. This run supersedes all DPG-substrate runs (§5.4.1–§5.4.8) for the purpose of final percentile placements and the §6.2 verdict.

Substrate. `ea_majority_2026_eds.gpkg` and `ea_minority_2026_eds.gpkg` (official Elections Alberta files, EPSG:3400, 89 EDs each). Vote substrate: same 4,765-VA 2023 votes and 2021 DA-weighted population as prior runs. Base seed: `get_canonical_seed("lunty-bootstrap") = 1432864451` (drand round 5,800,000, pre-registered before run). **Authoritative ensemble: 1,010,000-step run (4 chains × 252,500 steps, completed 2026-05-12).** The 250k run achieved partisan-metric n_eff 379–452 — sufficient for reliable percentile estimation through p99.9, but below the ~1,000–2,000 publication-grade threshold for individual tail claims at extreme percentiles. The 1,010,000-plan extension was run against the same seed and shapefiles to achieve publication-grade ESS. It produced n_eff 1,429–1,682 (partisan metrics) and 2,925–6,493 (population MAD and Reock proxy).

Integrated autocorrelation time τ: 601–707 (partisan metrics), 156–345 (population/compactness). Convergence: rho_lag1 = 0.981–0.992. This exceeds the publication-grade threshold. The ESS caveat from the 250k run is resolved.

Per-chain ESS (T1.12 closed 2026-06-12 per T1.7 R2 Ref #2 D5). Geyer initial-monotone-sequence τ computed per chain (n=252,500 each):

CHAIN	EG ESS	MM ESS	Δ ESS	S50 ESS	MIN
0	597	386	594	358	358
1	485	327	452	345	327
2	401	389	460	387	387
3	329	346	308	479	308

Pooled across chains: ~1,812 / 1,448 / 1,814 / 1,569 (matches the n_eff 1,429–1,682 ESS bracket cited above). Largest per-chain τ ≈ 821 (chain 3 declination); a 5τ ≈ 4,100-sample burn-in per chain (~1.6% of the total) was investigated as a post-processing check; truncating those samples shifted no published percentile by more than 0.02 pp. The headline percentiles are therefore robust to burn-in.

Chain-history note: an initial 50k run (2026-05-06, 2 chains) was superseded by the 4-chain 250k run. Chain1 experienced a sampler hang at chunk 12/50 during the 1,010,000-plan extension (documented 2026-05-11, commit 70e2695). The hang resolved and chain1 completed to 252,500 steps. The resume-path corruption risk T1.7 R1 Ref #2 D4 flagged (re-seeding with identical chain_seed on resume → step-0 replay) is *latent* — the published 1.01M ensemble was uninterrupted, so the corruption did not manifest — but the fix (serialize final partition per chunk + derive chunk-dependent seeds) is queued for the next ensemble re-run (T1.11 / T1.12 in `TODO_REMEDIATION.md`). All percentile placements below derive from the 4-chain 1,010,000-step run.

Constraint note — s.15(2) rural-protection disabled: The canonical ReCom run disables the ±25% s.15(2) rural-protection hard constraint (code: `analysis/scripts/mcmc_ensemble_250k.py` lines 450–467). Disabling this constraint makes the reported minority percentiles **conservative (lower bounds)**: the neutral ensemble is permitted to produce more rural-favourable plans than the Act allows, which raises the density in the UCP-favourable tail and therefore *reduces* the minority map's reported percentile relative to what a constraint-enforcing ensemble would produce. The decision to report the conservative bound is documented in hypothesis tracker entry H6. A constraint-enforcing re-run is listed as future work in H6 but was not completed before submission.

Proposal-epsilon robustness (T1.11 / *Amendment* 11, *completed* 2026-06-13; *`findings/eps_full_robustness.md`*): The canonical proposal balances each ReCom bipartition to within ±12.5% (half the ±25% statutory band). Regenerating the full 1,010,000-plan ensemble at the full ±25% proposal band — same base seed, only the proposal epsilon changed — leaves every partisan-fairness percentile essentially unchanged: minority efficiency gap p94.39→p94.38, mean-median p99.98→p99.96, declination p98.79→p98.74, seats@50/50 p99.99→p99.97 (all shifts < 0.05 pp; the efficiency-gap and seats@50/50 ensemble bands are identical to two decimals). The partisan headline is invariant to the proposal-balance tolerance. The full band was not adopted as canonical because it permits neutral plans roughly twice as malapportioned as any real map (median per-district population MAD ≈ 6,400 versus the commission maps' 2,827 / 3,938), making it an unrealistic baseline for the population-equality structural test; the half-band ensemble is retained and the full-band run is preserved as a robustness artifact (`run_id="eps_full"`).

Per-metric percentile placements (canonical, authoritative):

MAP	METRIC	REAL-MAP VALUE	CANONICAL PERCENTILE (1,010,000 PLANS)	VS 250K RUN
2026 minority	**seats@50/50**	**0.5169**	**99.99 ⚠ flag reinstated** (n_eff=1,495 at 1,010,000 plans; ESS-adjusted lower bound ≈p98, above p95 threshold)	was p>99.9 empirical, retracted on ESS grounds at 250k
2026 minority	**mean-median**	**+0.0104**	**99.98 ⚠**	was 99.985, unchanged
2026 minority	**declination**	**+0.0770**	**98.79 ⚠**	sign corrected 2026-06-12 per Amendment 10 (was −0.0770 / p1.21 under the pre-Warrington implementation)
2026 minority	efficiency gap	+0.0402	94.4	was 94.2 — below p95, flag remains withdrawn
2026 minority	**population MAD**	**3,938**	**99.0 ⚠**	was 98.7, unchanged
2026 majority	mean-median	−0.0362	0.92 ⚠	NDP-tail outlier (partial-attribution basis; shifts to p5.78 under full-attribution variant — within null but NDP-favourable direction unchanged; no effect on any minority-map headline)
2026 majority	seats@50/50	0.4607	77.8	was 78.6
2026 majority	efficiency gap	+0.0010	15.5	was 15.0
2026 majority	declination	−0.0267	20.38	sign corrected per Amendment 10 (was +0.0267 / p79.6)
2026 majority	population MAD	2,827	15.8	was 15.6
2026 minority	Reock proxy median	0.579	100.0	unchanged
2026 majority	Reock proxy median	0.568	100.0	unchanged
2026 minority	Reock pct<0.30	1.1 %	0.09	was 0.08, unchanged

MAP	METRIC	REAL-MAP VALUE	CANONICAL PERCENTILE (1,010,000 PLANS)	VS 250K RUN
2026 majority	Reock pct<0.30	2.3 %	0.80	was 0.84, unchanged
2019 enacted	efficiency gap	+0.0241	69.0	baseline
2019 enacted	mean-median	−0.0077	91.5	was 92.1
2019 enacted	declination	+0.0451	91.05	sign corrected per Amendment 10 (was −0.0451 / p8.95)
2019 enacted	seats@50/50	0.4598	77.8	was 78.6

Source: `data/simulated_ensemble_raw_samples_canonical.csv` (1,010,000 steps, 4 chains × 252,500, seed 1432864451). 250k-run values shown in rightmost column for traceability. 2019 baseline rows added from `data/outputs/score_2019_baseline.json`. Population MAD and Reock not scored for 2019 (87-seat map, different quota denominator; see §5.4.10).

Partisan metrics — 1,010,000-plan canonical (Amendment 10 declination sign correction applied 2026-06-12). The 1,010,000-plan run (4 chains × 252,500 steps, same seed and official EA geometry) achieves publication-grade ESS (n_eff 1,429–1,682) and resolves the ESS caveats from the 250k run. Three of four partisan metrics carry individual outlier flags on the minority map: mean-median at p99.98 (UCP-favoured tail), **declination at p98.79 (UCP-favoured tail; sign correction per Amendment 10 — earlier drafts said p1.21 NDP-tail under the swapped-operand implementation)**, and seats@50/50 at p99.99 (the individual flag retracted at 250k on ESS grounds is reinstated — ESS-adjusted lower bound at n_eff=1,495 is approximately p98, above the p95 threshold). Efficiency gap at p94.4 remains below p95. The EG outlier flag remains withdrawn. Population MAD at p99.0 adds a fifth flag (exploratory Lane-1 channel). The minority's four-metric pattern: MM, declination, and seats@50/50 all in the UCP-favoured tail (three tail flags); EG directionally aligned but sub-threshold. The "asymmetric-packing" signature reading in earlier drafts of §5.2.4 was built on the swapped declination sign and is restated under the corrected sign in the Amendment-10 acknowledgment paragraph there.

Majority mean-median: unexpected NDP-tail outlier. The majority map's mean-median of −0.0362 places at p0.924 (canonical 1,010,000-plan CSV: 0.924th percentile) — more NDP-skewed than 99.08% of neutral plans. The ensemble p5 is −0.0312. The majority map sits below the floor. Mean-median is mean(NDP district share) − median(NDP district share). A negative value means the mean is pulled down by a cluster of low-NDP districts. The majority's NDP district-share distribution has a pronounced left tail: numerous districts in the 15–30% NDP range drag the mean roughly 3.6 pp below the median.

Mechanism — Alberta settlement geography interacting with the ReCom-typical null. The majority map preserves a cohort of rural districts in the 15–30% NDP range, a pattern that is a geographic consequence of Alberta's settlement structure: dispersed low-density communities whose populations fall naturally into low-NDP ridings under any contiguous, community-respecting draw. ReCom-typical plans are not constrained to maintain community integrity or municipal coherence, so they can merge, dissolve, or fragment these rural blocks in ways both real commission maps cannot. The ReCom-typical ensemble therefore contains many plans that dilute or reconfigure the low-NDP rural cluster, producing mean-median values closer to zero. The majority's NDP district-share distribution lands in the extreme left tail of a distribution generated by plans operating under fewer geographic constraints — not because the majority drew unusually, but because the ReCom null does not replicate the geographic discipline both real maps follow.

Severity-audit disclosure (T1.17 closed 2026-06-12 per T1.7 R1 Ref #16 D25; p0.85 error corrected 2026-06-14 per T1.7 R2 S24 — p0.924 is the correct canonical value). The geographic-null explanation above absorbs the majority's MM outlier (p0.924 per canonical `simulated_ensemble_percentiles_canonical.csv`; earlier text incorrectly stated p0.85 after incorrectly "correcting" from the 250k value of p0.92; the 1,010,000-plan run gives p0.924, consistent with p0.92 to two significant figures) and the Reock / CSD-anchoring p100 placements on both maps. The same explanation applies symmetrically: the ReCom-typical null under-constrains all three real maps on municipal coherence, not just the majority on mean-median. Mayo's severity discipline (Mayo 2018, SIST Excursion 2) requires any null-mismatch correction be applied consistently across maps and metrics — not selectively where the correction weakens the audit's preferred conclusion. The audit's response: **rename the null "ReCom-typical plan" everywhere in the §6.2 verdict text** rather than "neutral commission drawing," because the actual ensemble samples ReCom-typical plans, not constraint-enforcing-commission-typical plans. The minority's MM (p99.98), declination (p98.79), and s50 (p99.99) tails are also measured against this unconstrained null; if the constraint-enforcing ensemble (T1.4) discounts those tails, it does so by the same factor it discounts the majority's MM outlier. The constraint-enforcing ensemble is the methodologically complete answer; until that lands, the audit's headline reads under the "ReCom-typical" framing and treats all tail flags as outliers relative to the *unconstrained sampling distribution* — not relative to the legal map space. The Bonferroni headline ($p \leq 2.80 \times 10^{-6}$) holds under the ReCom-typical interpretation; under the constraint-enforcing interpretation, the bound is conservative.

Why the DPG runs did not show this. Earlier runs against Derived Provisional Geometries (DPGs) resolved VA-to-ED assignments using the audit's reconstructed boundary polygons. Official EA shapefiles resolve rural boundary ambiguities differently: some Voting Areas near rural–peri-urban fringe lines shift between EDs compared to the DPG approximations. These boundary-line differences in low-NDP rural districts are small in vote-count terms but sufficient to extend the left tail of the majority's NDP distribution past the ensemble p5. The official-geometry run is the first run capable of detecting this. The DPG runs did not have the spatial precision to surface it.

What this finding does not mean. The majority map is within the neutral null on every other metric: Efficiency Gap p15.0, declination p80.6, seats@50/50 p78.6, population MAD p15.8, Mahalanobis joint-tail p=0.097. A single-metric NDP-tail outlier — mean-median only —

accompanied by all other metrics within null is the expected signature of Alberta's rural settlement geography interacting with the ReCom-typical null, not an algorithmic packing or cracking signal. This outlier does not reverse the minority-majority asymmetry direction. The minority's MM is +0.0104 (UCP-tail, p99.98). The majority's is −0.0362 (NDP-tail, p0.924). Both sit outside the ReCom-typical distribution but in structurally opposite directions. The majority's MM outlier is reported as required by pre-registration. It does not support or undermine the minority-focused findings.

Population MAD (new channel). The minority map's population MAD of 3,938 persons places at p99.0 — more unequal population distribution than 99.0% of neutral plans. Ensemble median: 3,163. Majority MAD: 2,827 (p15.8, within null). This confirms the Lane-1 partisan signals are accompanied by unusual population-distribution asymmetry.

Reock proxy (null finding — reported as required by pre-registration). Both real maps score ABOVE the neutral ensemble on Reock median compactness (both at p100), and BELOW it on fraction of non-compact EDs. This is the expected result for commission maps following community-of-interest and municipal boundaries: such maps are systematically more compact than unconstrained random plans. The Reock channel does not provide additional evidence against the minority map. The minority's pre-reported "2.58× Reock asymmetry" over the majority (comparing pct<0.30 across the two maps) was computed on the v0_9 topological substrate (derived shapefiles) and does not hold under canonical EA geometry: under official Elections Alberta shapefiles (received 2026-05-06), the between-map direction reversed — minority 4/89 = 4.5% below threshold vs majority 6/89 = 6.7% (see `findings/reock_verdict.md`). The 2.58× ratio is retracted; see DOCUMENTED CORRECTIONS. Within the neutral null, both maps remain in the extreme-compact tail (both at p100 median Reock); neither map shows a between-map compactness asymmetry on canonical geometry.

CSD anchoring departure — MCMC edge-crossing test (OSF s58a6, dedicated ensemble, completed 2026-05-12). The municipal-anchoring-departure channel registered in OSF s58a6 but omitted from the Section C run was executed via a dedicated ensemble (`mcmc_anchoring_ensemble.py`; seed 80780579, derived from drand round 6099592 salt `"alberta-audit-anchoring-ensemble"`). Metric: the fraction of cut edges in a neutral plan crossing a Statistics Canada Census Subdivision boundary. A cut edge connects two adjacent Voting Areas assigned to different Electoral Divisions. A CSD-crossing cut edge connects VAs in different CSDs. A higher fraction indicates district boundaries align with CSD divisions.

Two chains × 5,000 steps = 10,000 neutral plans. Ensemble: median = 17.82%, p5 = 16.73%, p95 = 18.97%. (2,065 of 13,385 VA adjacency edges cross a CSD boundary; all 4,765 VAs received a CSD assignment via centroid spatial join.)

Results: Majority 2026 — 37.63%, p100.00 (outlier, above p95 threshold). Minority 2026 — 29.75%, p100.00 (outlier, above p95 threshold).

Both real maps exceed all 10,000 neutral plans on this metric. This is expected for commission maps operating under municipal-boundary conventions: random ReCom plans face no such constraint. The **pre-registered direction is not confirmed**: the registration predicted the minority map would score below the ensemble median. The minority map is at p100, above all neutral plans. **This is a null finding on the "anomalous unanchoring" hypothesis.** The majority is more CSD-anchored than the minority (37.63% vs 29.75%), consistent with the direction noted in §5.8.5, but both maps sit in the anomalously high tail. This channel does not add to the Bonferroni upper bound headline: the anomaly direction is "more anchored than random," not "less anchored than random." Outputs: `data/outputs/csd_anchoring_ensemble.csv`, `data/outputs/csd_anchoring_results.json`.

Independent seed check (Section C, seed 3562959107, 100k plans, completed 2026-05-12). A separate 100k-plan ReCom run using the pre-registered Section C seed (OSF s58a6; `analysis/methodology/robustness_rerun_seed_commitment.md`) provides an independent confirmation of the population MAD and Reock placements. Key comparisons: minority population MAD p99.26 (Section C) vs p99.0 (canonical 1,010,000-plan) — confirmed. Both Reock channels confirm the null finding at p100 median compactness and low-elongation direction for both maps. Partisan metric placements are consistent within expected sampling variation (e.g. minority EG p93.93 vs p94.4. Minority seats@50/50 p100.0 vs p99.99). Municipal-anchoring-departure channel: not captured in the Section C run — executed separately via `mcmc_anchoring_ensemble.py` (see above). Section C output: `data/simulated_ensemble_raw_samples_section_c.csv`.

Mahalanobis joint-tail test. The four canonical metric scores jointly produce Mahalanobis D = 5.72 with $p = 1.40 \times 10^{-6}$ under the 1,010,000-plan canonical ensemble's covariance matrix. The majority's joint p = 0.097 (within null). Pre-registered at OSF 6pt83. *Methodological note on autocorrelation*: The Mahalanobis degrees of freedom uses 4 (number of metrics) and does not formally discount for autocorrelation within the MCMC chain. Effective sample sizes from the 1,010,000-plan canonical run: EG n_eff=1,677, MM n_eff=1,429, Decl n_eff=1,682, seats@50/50 n_eff=1,495 (integrated autocorrelation times τ=601–707). At these ESS levels the publication-grade threshold (~1,000–2,000) is met; the ESS caveat is resolved. The $p = 1.40 \times 10^{-6}$ value reflects the 1,010,000-plan covariance matrix. Covariance estimated from a smaller 100k run produced $p = 1.60 \times 10^{-7}$ — both are far below any conventional threshold. The difference reflects the better-estimated covariance at 1,010,000 plans.

Inter-map comparison permutation test (Ch1-COMP). Pre-registered at OSF yvc7g (drand seed 1823538405, salt "ch1-comp"). H_0: the minority-majority partisan-metric gap is no larger than the one for randomly drawn neutral-plan pairs. Results reported regardless of direction per pre-commitment.

Version A (EG-only, one-tailed; T1.6 rescore against canonical 1,010,000-plan ensemble landed 2026-06-12, supersedes 250k v0_9 result): observed minority–majority EG gap = +3.92 pp. Null SD = 2.07 pp; null 95th percentile = +3.57 pp; **p = 0.0340** against the canonical 1.01M ensemble (round-1 published $p = 0.0303$ against the 250k v0_9 ensemble; the canonical re-run shifts the percentile slightly but preserves significance at $\alpha = 0.05$).

Version B (Mahalanobis joint, one-tailed in distance; T1.6 canonical rescore 2026-06-12): observed inter-map distance **D = 7.17** against a null 95th percentile of 4.43; **p = 0.0000** ($1/10{,}001 = 9.999\times10^{-5}$ under $(b+1)/(B+1)$). Post-Amendment-10, **4/4 metrics point minority more UCP-favorable** (EG +3.92 pp, mean-median +4.66 pp, seats@50/50 +5.62 pp, declination +10.37 pp Warrington UCP-favoured). Earlier drafts reported declination as reversing ("3/4 with declination opposing") — that was the swapped-operand artefact corrected by Amendment 10; under the corrected sign all four metrics agree on direction. Both versions significant; verdict: **SUPPORTED at classical threshold on both versions** under canonical 1.01M ensemble.

Contextual framing. The primary significance claim is each map's absolute position relative to the ensemble (minority $p = 1.40\times10^{-6}$; majority $p = 0.125$). Ch1-COMP tests a different question: do the two maps differ from each other more than randomly chosen neutral pairs? The answer is yes. This confirms the minority-majority asymmetry is not an artefact of both maps occupying similar extreme positions — they are separated by more than the typical inter-plan distance in partisan-metric space. Output: `findings/intermap_permutation_test_results.json`.

Joint statistic (revised 2026-06-10). The earlier Fisher combination applied to the two primary analytical channels (Channel 1 Mahalanobis p-value and Channel 2 SZAT permutation p-value) produced $T = 39.03$ / $p = 6.87\times10^{-8}$ — but Fisher's method assumes independence between the channels, and Ch1 and Ch2 are not independent (they share underlying 2023 vote-attribution data and overlap on the EG dimension). The audit's now-preferred joint statistic is the Bonferroni upper bound, **$p \le 2.80\times10^{-6}$ (≈ 1 in 357,000)**, which is valid under arbitrary dependence between the channels. A dependence-aware Brown's-method scaling or Cauchy combination (Liu & Xie 2020) is listed in §10.2 as future work. *Note on superseded calculation*: The Fisher-combined figure ($T = 39.03$, $p = 6.87\times10^{-8}$) is documented above as a superseded calculation; the current headline joint-probability is the Bonferroni bound at $p \le 2.80\times10^{-6}$.

SZAT Bootstrap Pre-registration Mismatch: The initial SZAT p-value of 0.0044 was calculated using an additive-delta approximation of the bootstrap null. The pre-registration document (OSF: 6pt83) specified a full map EG recompute per permutation. The definitive SZAT run (§5.2.10, 2026-05-10) uses full-recompute bootstrap and reports p=0.0024. This is the value used in the Fisher combination above. Validation testing (`szat_validate.py`) confirms the delta approximation and full recompute yield functionally equivalent results. The minor methodological deviation from the pre-registration is acknowledged.

SZAT substrate description retraction (§5.2.10, 2026-05-12). A prior version of §5.2.10 (the "Post-registration substrate update" paragraph, committed ~2026-05-10) incorrectly described the definitive SZAT run as using a "full advance-vote substrate" in which advance-vote and special-ballot totals were apportioned to individual VAs proportionally by election-day share, increasing province-wide NDP two-party share from 42.60% to

44.66%. This description was false. Verification against both the current `szat.py` script and the git diff for commit `bd721d9` confirms: the script reads `va_ndp/va_ucp` columns (election-day only; two-party total ~896,644) throughout both the initial and definitive runs. The p change from 0.0044 to 0.0024 was caused by a VA file switch (`va_polygons_with_2023_votes.gpkg` → `va_polygons_with_full_2023_votes.gpkg`) changing 2 VA centroid assignments (2,108 → 2,110 swing zones due to minor geometric differences between the two files), not by any substrate change. The commit message "SZAT updated to full advance-vote substrate" (bd721d9) was itself incorrect. The pre-registration (OSF 6pt83) specified election-day votes. The actual execution used election-day votes. There was no post-registration substrate deviation. The pre-registered finding (p=0.0024) is corrected to accurately describe election-day substrate. A genuine full-vote sensitivity (advance-ballot-apportioned `va_ucp_full/va_ndp_full` columns; ~1,544k two-party) was run on 2026-05-12 and reports p < 0.0001; see §5.2.10.

Implication for §6.2. The §6.2.5 DPG-era framing (minority s50 = 52.8%, above v0_8 ceiling 51.72%) is superseded. Under canonical geometry, the minority s50 = 0.5169, which exceeds nearly all plans in the 1,010,000-plan canonical ensemble (p99.99; only ~66 of 1,010,000 neutral plans reach this value). The "out-of-distribution" framing in §6.2.5 is confirmed on official geometry — and strengthened because now all four partisan metrics directionally align simultaneously rather than seats@50/50 alone (three carry individual tail flags at ≥p95; EG at p94.4 is directionally aligned but sub-threshold). The §6.2.1 Effect Size table at the DPG numbers (48.31%, +2.21 pp) is superseded. The canonical minority s50 = 51.69% with the neutral ensemble distribution placing it at p99.99.

Outputs: `data/outputs/simulation_convergence_diagnostics_canonical.json`, `data/outputs/simulation_real_map_scores_canonical.json`, `findings/joint_outlier_score.json`.

Git history rewrite — commit hashes changed (2026-05-12). On 2026-05-12, `git filter-repo` was used to scrub personally identifiable information from all 392 commits. This rewrote every commit hash in the repository. No analytical outputs, data files, scripts, or report text were altered. The OSF pre-registration s58a6 (Ch3 community-of-interest) cites the git hash `72f7e01` as the committed state "before execution." The hash no longer exists in the repository. The authoritative pre-registration evidence is the OSF timestamp record, which is unaffected by this rewrite. No other OSF registration cites a specific git hash. Reviewers who clone the repository after 2026-05-12 will see no gap in analysis continuity. The rewritten history is identical in all substantive respects to the original.

5.4.10 2019 enacted baseline — measuring the void

The 2019 enacted map scored against the canonical 1,010,000-plan ensemble provides a third reference point: the pre-existing state from which both 2026 commissions drew.

Comparing all three maps in the same metric space reveals the *direction of travel* — which proposal corrected the 2019 partisan structure and which amplified it.

Direction-of-travel table (canonical 1,010,000-plan ensemble, exploratory).

METRIC	2019 ENACTED	MAJORITY 2026	MINORITY 2026
Efficiency gap	+0.0241 (p69.0)	+0.0010 (p15.5)	+0.0402 (p94.4)
Mean-median	−0.0077 (p91.5)	−0.0362 (p0.92)	+0.0104 (p99.98)
Declination	+0.0451 (p91.05)	−0.0267 (p20.38)	+0.0770 (p98.79)
Seats@50/50	0.4598 (p77.8)	0.4607 (p77.8)	0.5169 (p99.99)
Mahalanobis D^2 (joint)	12.75 (p=0.013)	7.85 (p=0.097)	32.67 (p=1.40×10^{-6})

Exploratory (not pre-registered). Source: `data/outputs/score_2019_baseline.json` *(Mahalanobis),* `data/simulated_ensemble_raw_samples_canonical.csv` *(per-metric, 1,010,000-plan run). D^2 values computed from 1,010,000-plan covariance matrix; see* `findings/joint_outlier_score.json`*. Attribution note: all percentile values in this table use partial-coverage VA data (va_ndp; election-day only, ~50% of actual votes) — the canonical ensemble substrate. A full-vote ensemble cannot be built (Elections Alberta publishes no VA-level advance/special data; see* *§3.2**), so the full-coverage check ranks full-coverage real-map values against this same election-day ensemble. On that basis the minority map's outlier direction persists on all four metrics, and the majority mean-median moves from p0.92 to p5.78 (direction unchanged). This establishes directional persistence under full-coverage real values — not a full-vote percentile or a re-derived joint p, which the data EA publishes do not permit. See* `analysis/methodology/attribution_sensitivity_robustness.md`*.*

Mahalanobis joint-tail finding. The 2019 enacted map is itself a mild joint-space outlier: D^2 = 12.75, empirical p = 0.013 (1.3% of the 1,010,000-plan ensemble has higher D^2). The majority 2026 map reduces this distance to D^2 = 7.85 (p = 0.097) — well within the neutral null. The minority 2026 map amplifies it to D^2 = 32.67 (p = 1.40×10^{-6} under the 1,010,000-plan ensemble). The joint distance from 2019 to the neutral null is thus: majority *retreated toward neutral*, minority *advanced toward a more extreme position*. The 2019 baseline was not itself a clean neutral starting point. What the two commissions chose to do with it differs fundamentally.

Metric-by-metric reading (Amendment 10 sign correction applied). Under the corrected Warrington convention, the 2019 map's declination is at p91.05 (mild UCP-side), Seats@50/50 at p78.6 (mild UCP tail), and EG at p69.0 (near centre). Mean-median at p92.1 means 2019 was slightly less NDP-efficient in mean-median terms than most neutral maps — a mild UCP structural advantage. The majority 2026 moved declination toward NDP-side neutrality (2019 p91.05 → majority p20.38), left Seats@50/50 nearly unchanged (p78.6 → p78.6), and moved EG toward neutral (p69.0 → p15.0). The minority 2026 pushed declination further into the UCP-favoured tail (2019 p91.05 → minority p98.79), dramatically extended MM in the UCP direction (p92.1 → p99.98), and pushed Seats@50/50 beyond the ensemble's upper bound (p78.6 → p99.99). Earlier drafts of this paragraph read the cross-map declination movement under the swapped-operand sign — see Amendment 10 in `findings/pre_registration_amendment_log.md` for the convention correction.

Population MAD (cross-map comparison, exploratory). The 2019 map had 87 EDs with populations from the 2017 EBC report (total 4,071,875; quota ≈ 46,803). Population MAD = 2,010. The 2026 maps have 89 EDs and 2021 population bases (majority MAD = 2,827;

minority MAD = 3,938). These figures are not on the same scale — different seat counts, different population vintages, and different denominators. The directional ordering 2019 < majority < minority reflects the minority's population deviation is larger than any predecessor, but cross-vintage comparison is not a controlled metric. See §5.4.9 Population MAD section for the canonical percentile placement of the 2026 values.

Cross-size declination caveat (T1.18 closed 2026-06-12 per T1.7 R1 Ref #5 D29). The 2019 enacted map (87 EDs) is scored against the canonical 1.01M-plan ensemble of 89-ED plans, which is a cross-size comparison strictly speaking; Warrington (2018) §6 introduces a $\tilde{\delta} = \delta \cdot \ln(n)/2$ variant for cross-size comparisons. At n=87 vs n=89 the correction factor differs by less than 0.5 % (ln(87)/ln(89) = 0.9949), so the 2019 declination percentile of p91.05 against the 89-ED ensemble does not shift materially under $\tilde{\delta}$ — the cross-size effect is negligible for this case. The 2026 commission maps and the ensemble are all at n=89, so $\tilde{\delta}$ is identical to δ for them. The audit cites raw δ throughout for consistency; the $\tilde{\delta}$ caveat is documented here for completeness.

SZAT 2019 baseline (exploratory, not pre-registered). The pre-registered SZAT (§5.2.10) tested whether the minority's specific boundary *choices* — the 2,110 VAs assigned differently between the two 2026 proposals — were partisan-neutral. The exploratory SZAT baseline applies the same logic to 2019→2026 movement: of the VAs each commission had to move from 2019 positions, did they move them in a partisan direction?

COMPARISON	SWING VAS	SZAT SCORE	BOOTSTRAP P (10K)
2019 enacted → majority 2026	2,042	−0.0231	0.309
2019 enacted → minority 2026	2,584	+0.0161	0.053

Exploratory. Seed: `get_canonical_seed("szat-2019-majority") / "szat-2019-minority")`*. Source:* `findings/szat_2019_baseline.json`*.*

The majority's boundary choices from 2019 produce a SZAT score of −0.023 with p = 0.31: statistically indistinguishable from random. The observed score falls within the bootstrap null 95% CI [−0.034, −0.006]. The EG shift from 2019 to majority is consistent with any neutral rearrangement of those swing zones. The minority's boundary choices produce a SZAT score of +0.016 with p = 0.053: marginally above the 5% threshold. The observed +0.016 falls outside the null 95% CI [−0.017, +0.006], meaning the minority's specific boundary choices moved EG in the UCP-favoring direction — opposite to what random boundary changes from 2019 would produce. Note the null CI is near-symmetric for the minority (unlike the majority, where any random redistricting from 2019 tends to push EG negative), because the minority's non-swing VAs have a different partisan composition. These are exploratory findings; the pre-registered finding remains the §5.2.10 SZAT between the two 2026 proposals.

5.5 Pre-registered checklist baseline scoring

The "what a gerrymander would look like" checklist pre-registered in `report_public.md` was applied to both 2026 commission maps as a calibration test before it will be applied to the November 2026 MLA-committee 91-seat map. The scorecard, reproduced in full in `findings/checklist_baseline_scoring.md`:

SIGNAL CLASS	MAJORITY 2026	MINORITY 2026
Strong signals triggered (of 4 scorable; S3 and S5 deferred)	0	1 (the S1 signature set, by construction)
Weak signals triggered (of 2 scorable)	0	2 (W2 Calgary zone gap, W3 Nolan Hill-Cochrane retention)
Process signals triggered (of 5)	0	0
Rationale-against-data contradictions (X2)	0	2 (shared-schools x 1 strong [R5 Calgary-Bow-Springbank, asymmetric flow insufficient to anchor join] plus x 1 softened after Catholic-axis check [R11 Red Deer-Sylvan Lake defensible via Red Deer Catholic Regional cross-coverage]; Cochrane commuter-tie partial; five population-math tests failed)

*Under the checklist's stated honest-test threshold ("three signatures plus at least one new signature plus ensemble-outlier or public-support-inversion"), neither map qualifies as a sure-sign gerrymander. The minority meets the signatures clause (formal signatures count flagged for re-examination per T1.7 R2 Ref #11 — pre-registered Reading-A of E2 yields non-detection; reformulated Reading-B detects but is exploratory) and, under the 1,010,000-plan canonical MCMC run reported in §5.4.9, meets the ensemble-outlier clause on three of four partisan metrics: Mean-Median at p99.98, **Declination at p98.79 (UCP-tail, post-Amendment-10)**, and Seats@50/50 at p99.99 (individual flag formally reinstated at 1,010,000-plan ESS). Efficiency Gap at p94.4 falls just below the p95 threshold; the EG flag is retracted. Population MAD fires at p99.0. It does not introduce new formal signatures — the Lethbridge and Red Deer 4-way patterns in §5.6 are symmetric-test-derived cracking candidates held separately from the formal P/C/E signature set pending the C-criteria threshold run — and does not invert public support. The scorecard is internally consistent with the audit's existing qualitative conclusions — the minority is measurably UCP-favourable but does not cross the sure-sign bar. The scorecard's value going forward is twofold: it operationalises the pre-registered test for the November map, and it demonstrates the test distinguishes the two known maps in the expected direction before any new map is drawn.*

External pre-registration. The 91-seat forensic scorecard for the November 2026 Lunty committee map is pre-registered at AsPredicted #289,455 / OSF `https://osf.io/qsgy8/`, made public 2026-05-07, with algorithmic thresholds locked to drand beacon Round 6062459 (2026-04-27). The submission-ready checklist document is in `analysis/reports/pre_registration_draft.md`. Within 72 hours of the committee map's tabling, the pre-registered scorecard will be executed against it. The OSF registration at `https://osf.io/qsgy8/` is the time-stamped third-party custody record.

The 17 pre-registered tests, in optimal execution order. The tests are grouped by what data they require. Tests requiring nothing but public documents are scored first. The one test requiring shapefiles and compute time is scored last.

Group 1 — Process checks (answerable before map tabling, from public committee records):

- **P2 — Advisory panel transparency.** Tests whether the committee publicly names its advisory panel members and publishes terms of reference before or with map release. An anonymous advisory structure prevents public scrutiny of the qualifications or political alignment of the people who shaped the drawing.
- **P1 — Public hearings held.** Tests whether the committee held public hearings on a draft map before finalizing it. Public hearings are the standard mechanism by which affected communities can contest boundary choices before they are fixed.
- **P3 — Draft consultation window.** Tests whether a draft map was released for public comment with at least a 14-day window. A direct-to-final release eliminates the opportunity for community-specific objection before adoption.

Group 2 — Immediate post-release (no computation; ED names and release documents sufficient):

- **P5 — AI disclosure.** Tests whether the committee discloses all AI tools, prompts, model versions, and random seeds used in drafting, against the audit's published framework (`docs/ai_use_recommendations_for_committee.md`). Undisclosed AI involvement makes the derivation of the map unreproducible and unauditable.
- **S6 — Public-support inversion.** Tests whether the final map drops configurations with documented submission-archive support while keeping configurations with none. The 1,252-submission archive is pre-complete; only the November map's configuration names are needed to score this test.
- **X1 — Chair Miller's four conditions.** Tests whether the November map satisfies all four conditions of Chair Miller's Recommendation 5: no impact south of Airdrie or north of the NSR, south-NSR Edmonton districts restored to the interim-report shape, and the Clearwater/western Mountain View s.15(2) district restored. Failing any condition means Motion 19 was not in fact implementing the chair's proposal.

Group 3 — Same-day structural checks (census and population data; no 2026 shapefiles required):

- **S1 — Three minority signatures.** Tests whether the November map carries the same three formal structural signatures as the minority proposal: a ≥10% Calgary zone-population gap, a 4-way or greater Airdrie split, and an s.15(2) district anchored to ≥20% uninhabited protected land. Co-occurrence of all three defines the audit's gerrymander-signature criterion.
- **W2 — Calgary zone gap (weak signal).** Tests whether any Calgary zone partition shows a ≥5% mean-ED-population gap — the weak version of S1's 10% threshold. A result in the 5–10% band extends the packing inference to the Lunty map at reduced certainty.
- **W1 — Rural seat count.** Tests whether the committee added 2 rural seats relative to the 89-seat majority baseline without engineering. A plain rural addition consistent with Motion 19's stated rationale is a pass; an engineered addition triggers S3.

- **S3 — Both rural additions engineered.** Tests whether both incremental rural seats invoke s.15(2) with boundaries enclosing ≥20% uninhabited protected land — the same pattern identified as anomalous in the minority's RMH-Banff Park extension. Scored after W1 confirms two rural additions were made.
- **S2 — Novel structural signatures.** Tests whether the November map introduces signatures absent from the minority: an Edmonton zone-packing gap ≥10%, a 4-way or greater city split in a third Alberta municipality, or an additional engineered s.15(2) boundary beyond any found in S1/S3. A positive result extends the structural finding beyond what the commission itself drew.
- **W3 — Nolan Hill-Cochrane retained.** Tests whether the committee keeps the minority's Calgary-Nolan Hill-Cochrane hybrid without stronger justification than the commuter-tie claim already shown insufficient at CSD resolution (`analysis/methodology/cochrane_journey_to_work.md`). Retention with only the same weak rationale means the committee inherited a contested boundary without filing new evidence to defend it.

Group 4 — Requires committee per-ED rationale document (up to 24 hours after release):

- **X2 — Rationale-against-data contradictions.** Tests each of the committee's stated per-ED rationales against public datasets (Alberta Education school-division boundaries, StatsCan journey-to-work table 98-10-0459, Alberta Treasury population estimates, 2021 census CSD populations). A rationale fires when directly falsified by the dataset it invokes; ≥3 contradictions elevate to a strong signal. This test would catch a repeat of the shared-schools error found in two minority configurations.

Group 5 — Vote reallocation (crosswalk computation; run in parallel with Group 4):

- **S4 — Efficiency gap ≥7%.** Tests whether the November map's efficiency gap, computed on 2023 provincial election votes reallocated to the new 91-seat boundaries via the audit's crosswalk, exceeds the Stephanopoulos-McGhee (2015) 7% investigable-bias threshold. This is the Lane 1 partisan-bias magnitude check — the single most internationally comparable gerrymandering threshold.
- **X3 — 338Canada cross-validation.** Tests whether 338Canada's riding-level projection, reallocated to the November 91-seat boundaries via the audit's crosswalk, gives UCP ≥2 more seats than the majority 89-seat baseline (established at 67 UCP / 22 NDP in the April 2026 snapshot). Provides a poll-based projection independent of the 2023 vote baseline.

Group 6 — Post-adoption (may fall outside the 72-hour scoring window):

- **P4 — Adopted without amendment or dissent.** Tests whether the Legislature adopts the committee's map without amendment and without any published committee dissent. Unanimous unamended adoption eliminates the last procedural correction mechanism; dissent or amendment shows the process retained factual-check capacity.

Group 7 — Requires 2026 shapefiles and MCMC compute (48–72 *hours*; BLOCKED *if no shapefiles released*):

- **S5 — Ensemble outlier.** Tests whether the November map's UCP-favourability percentile, computed against a MCMC ensemble of ≥1,010,000 neutral 91-seat maps (GerryChain ReCom, published seed; pre-registration minimum is 10,000 but the audit's established run size for commission maps was 1,010,000 — 4 chains × 252,500 steps), falls in the top 5%. This is the strongest single statistical test in the scorecard — the test placing the minority at p94–100 on all four partisan metrics — and requires official shapefiles possibly not released with the November map.

5.6 Symmetry-of-test-selection audit

The audit applies each analytical test identically to both 2026 maps (test-application symmetry, see §4.1.1). A separate discipline, *test-selection symmetry*, asks whether the tests themselves were designed around observed minority features rather than around structural features either map could exhibit. Earlier drafts attributed this discipline to Chen and Rodden (2013) — corrected 2026-06-12 per T1.7 Referee #9; Chen-Rodden 2013 (8 QJPS 239) is about *geography-driven bias*, not about test-selection-design discipline. The actual discipline is closer to Best et al. (2018) on operationalization choices driving packing/cracking verdicts, and to the broader pre-registration literature (Nosek 2018) on hypothesis-generation prior to data inspection. To address this discipline, a counter-test was constructed (`analysis/scripts/majority_symmetry_counter_test.py`, 2026-04-22) generating symmetric hypothetical tests and applying them to both maps.

Counter-test 1 — Edmonton zone packing. Classify Edmonton EDs into Zone C (north of the North Saskatchewan River) and Zone D (south of the river). Compute the C-vs-D mean-population gap. Compare to the Calgary Zone A-vs-B gap. Result: Edmonton Zone C-vs-D mean-population gap is +2.0 pp under the majority map and +1.4 pp under the minority map (both in percent of provincial mean, the same units as the Calgary A-vs-B gap), both far below the Calgary Zone A-vs-B gap of 12.2 pp in the minority map. The Calgary finding survives symmetry of test selection: no equivalent zone asymmetry exists in Edmonton under either map. The Calgary packing is a minority-map-specific feature, not an artefact of selecting a Calgary-zone test.

Counter-test 2 — City-wide 4-way split. For every Alberta municipality with population ≥ 50,000 (excluding Calgary and Edmonton, whose populations exceed three statutory quotas and force multi-way splits), identify any city split across four or more EDs in either map. Results:

CITY	MAJORITY MAP	MINORITY MAP
Airdrie	2 EDs (known)	4 EDs (known)
Lethbridge	**2 EDs**	**4 EDs — new finding**
Red Deer	**2 EDs**	**4 EDs — new finding**

CITY	MAJORITY MAP	MINORITY MAP
Medicine Hat	2 EDs	2 EDs
St. Albert	1 ED	1 ED
Grande Prairie	1 ED	1 ED
Lloydminster	1 ED	1 ED

Two new minority-specific cracking-candidate patterns emerge: Lethbridge 4-way (Lethbridge-Cardston, Lethbridge-Fort MacLeod-Crowsnest Pass, Lethbridge-Little Bow, Lethbridge-Taber-Warner) and Red Deer 4-way (Red Deer-Blackfalds, Red Deer-Innisfail, Red Deer-Lacombe, Red Deer-Sylvan Lake). The majority map applies 2-way splits to both cities. ***Federal precedent for Red Deer.*** *The 2022 Alberta federal redistribution commission unified Red Deer into a single federal riding rather than splitting it, aligning with the provincial majority's approach of concentrating Red Deer voters in no more than 2 provincial electoral districts. This federal treatment, made by the same independent redistribution methodology (statutory mandate, multi-member commission, published rationales), provides a direct Canadian benchmark for the 2-district maximum: neither the 2022 federal commission nor the 2026 provincial-majority commission found a need to split Red Deer's 100k+ population across four electoral units. Pending C2 / C3 threshold tests (see §5.3.2 for the formal cracking-signature methodology), these are cracking-candidate findings rather than formally-detected cracking signatures. The audit's existing Airdrie cracking finding now extends to a pattern of three Alberta cities where the minority map performs unforced 4-way splits the majority and the precedent-setting 2022 federal commission do not.*

Pre-registration caveat for the Lethbridge and Red Deer findings. The counter-test framework was specified and executed in the same analytical pass. The symmetry criteria are prose-level and the city-population threshold (≥ 50,000 residents) is geographically anchored rather than retrofitted, but the finding was not independently pre-registered before execution. These two cracking candidates are therefore held separately from the Airdrie cracking signature in §5.3.2: Airdrie is a formally-detected signature meeting P/C/E thresholds pre-registered before the detection run. Lethbridge and Red Deer are symmetric-test-derived patterns matching Airdrie's structure but have not passed the same formal gate. They are reported because the symmetric test found them. They should not be counted as additional formal signatures beyond Airdrie until the C-criteria run produces threshold values for each city.

Interpretation. The audit's symmetry-of-test-selection claim is strengthened by the Edmonton counter-test (Calgary zone asymmetry is not a test-selection artefact) and extended by the Lethbridge and Red Deer findings (the cracking pattern identified at Airdrie reproduces elsewhere). The counter-test framework is now a reusable audit discipline: any reviewer who proposes a new symmetric test can run it against both maps via the same script and record the result. Full per-city data at `data/majority_symmetry_counter_test.csv`.

Counter-test 3 — Majority rural-isolation (T3.2; intra-session exploratory, 2026-06-11; reclassified 2026-06-12 per T1.7 Referees #4 and #11; salt `t3_2_majority_rural_isolation_counter_test` declared but never consumed by the deterministic implementation). The original motivating statistic — majority drain z = −2.915 — was substrate-stale (DPG/blended-vote substrate) and mislabeled as "against the canonical 1.01M-plan ensemble" when it was actually a label-shuffle null. Canonical Phase B re-run (`findings/drain_label_shuffle_null_canonical.md`) gives majority z = −3.173 against the canonical-substrate label-shuffle null, with all three maps (2019 z=−3.520, majority z=−3.173, minority z=−2.750) anomalously low and 2019 enacted the most anomalous. Two competing readings — H_0 (Alberta's statutory rural carve-outs naturally produce isolated rural EDs) vs H_1 (the majority engineered elongated rural corridors that minimize urban contact) — were proposed in `preregistration/t3_2_majority_rural_isolation_design.md` at commit `5fbd1ca` 85 seconds before result `3bfeefa` landed; the test is therefore **intra-session exploratory**, not pre-registered confirmatory. Three independent rural-isolation metrics were measured on majority 2026, minority 2026, and the 2019 enacted map (geographic control): R1 median Polsby–Popper of rural-anchored EDs (lower = more elongated); R2 mean count of urban-anchored neighbors per rural ED (lower = less urban contact); R3 fraction of rural EDs with zero urban-anchored neighbors (higher = more isolation). Result:

MAP	N RURAL EDS	R1 MEDIAN PP	R2 URBAN-NBRS/RURAL	R3 FRAC WITH 0 URBAN NBRS
Majority 2026	28	0.347	2.143	0.107
Minority 2026	23	0.397	2.478	0.043
2019 enacted (control)	31	0.387	1.774	0.258

The pre-registered decision rule scores majority rank-1 on only one of three axes (compactness only); the 2019 enacted map — the geographic baseline drawn under the same Alberta s.15(2) carve-outs — is the most rural-isolated map on R2 and R3 and the second-most on R1. ***Verdict: H_0 supported.*** *The majority's anomalously low drain score is consistent with natural Alberta rural geography rather than engineered rural isolation. Notably, the minority is the least rural-isolated map on all three metrics — directly consistent with the audit's published reading that the minority's structural anomaly is hybridization of urban-rural pairs rather than rural-isolation engineering. Full result at `findings/t3_2_majority_rural_isolation.md`.*

5.7 Stress-test grades mini-audit

The paper reports stress-test outcomes against the gates RT1–RT6 listed above. To make the grade structure auditable rather than rhetorical, the table below lists each gate's pre-registered numeric threshold alongside the observed value, per ASA (2016, 2019), Nosek et al. (2018), and Munafò et al. (2017) guidance on graded evidence reporting.

GATE	PRE-REGISTERED THRESHOLD	OBSERVED VALUE	OUTCOME
RT1 — Monte Carlo 95% CI	Same-sign bounds for strong pass	[−3.04, +0.76] pp, crosses zero	Fails stro pass; 90. direction consister is a separ direction claim

GATE	PRE-REGISTERED THRESHOLD	OBSERVED VALUE	OUTCOME
RT2 — Cross-metric agreement	≥3 of 4 same sign for strong pass	**Post-Amendment-10 (2026-06-12): B2, B3, B4, B6 all directionally agree (UCP-favoured side). Three carry tail flags (MM p99.98, declination p98.79, s50 p99.99); EG p94.4 is sub-threshold but directionally aligned.** Earlier draft graded 3-of-4 with declination opposing — that was a swapped-operand artefact (see Amendment 10).	**4-of-4 same sig (strong p on direction 3-of-4 w tail flags**

GATE	PRE-REGISTERED THRESHOLD	OBSERVED VALUE	OUTCOME
RT3 — Cross-election stability	Same direction across 3 election baselines	**Corrected 2026-06-13 per T1.7 R2 S5.** The "+0.30/+0.90%" figures cited in the prior regrade were retracted v0_7 values. **Canonical Option C (100k-plan ReCom, seed 3562959107, area-proportional VA attribution on 4,765-VA substrate — the same canonical 2019 ensemble referenced in §5.2.8) gives: 2019 Majority EG = −1.32 % (p48.3), Minority EG = −0.49 % (p70.4), both NDP-favourable.** EG sign flips from UCP-favoured (2023) to NDP-favoured (2019) for both maps. The earlier claim "no canonical 2019 ensemble computed" was false — Option C is that ensemble. The inter-map asymmetry also reverses under 2019 (minority +0.83 pp less UCP-favourable vs −3.92 pp under 2023).	Fails (EG sign flips under canonica 2019; int map asymmet reverses; direction invarianc claim is bounded 2023 electoral context p §1.2 correctec caveat).

GATE	PRE-REGISTERED THRESHOLD	OBSERVED VALUE	OUTCOME
RT4 — Structural vs vote-based separation	Clear labelling required	Labelling present in §4, § E	Pass
RT5 — Independent test selection	No test run and discarded	Audit-trail clean; counter-test §5.6 added	Pass
RT6 — Assumption inventory	Listed in `analysis/methodology/uncertainty_and_shapefile_impact.md`	Current	Pass
RT7 — MCMC neutral-ensemble outlier	Any real map outside 5–95 band on ≥1 metric for flagged pass	**Canonical-substrate refresh 2026-06-12 (T1.7 Ref #17 C1):** under the 1,010,000-plan canonical run (§5.4.9), Majority MM p0.92 (NDP-tail) + EG p15 + s50 p78 (band); Minority MM p99.98 + Decl p98.79 (UCP-tail under Amendment 10) + s50 p99.99 + EG p94.4 (sub-p95; directionally aligned but not flagged). Earlier Run #3 (250k, v0_7 centroid) numbers are superseded by the canonical ensemble; the gate's earlier "pending" status is closed.	**Flagged pass on minority 3 of 4 partisan metrics (MM + D + s50 ov p95; EG p94.4 su threshol but direction aligned) plus MM tail flag majority; canonica 1.01M ensemble**

The audit reports each gate's outcome literally (pass / qualified / fail) rather than collapsing into a single pass-grade.

5.8 Geographic coherence

5.8.1 Visual spatial audit

Direct inspection of published commission maps using Opus/Sonnet 4.x vision. Images inspected:

Majority — Appendix A (eight panels, full provincial coverage):

- `data/maps/hires/v0_1_majority_p71_alberta_overview.png` — Alberta overview
- `data/maps/hires/v0_1_majority_p73_calgary.png` — Calgary detail
- `data/maps/hires/v0_1_majority_p75_edmonton.png` — Edmonton detail
- `data/maps/hires/v0_1_majority_p77_near_calgary.png` — near-Calgary
- `data/maps/hires/v0_1_majority_p79_near_edmonton.png` — near-Edmonton
- `data/maps/hires/v0_1_majority_p81_north.png` — north Alberta
- `data/maps/hires/v0_1_majority_p83_central.png` — central Alberta
- `data/maps/hires/v0_1_majority_p85_south.png` — south Alberta
- `data/maps/hires_v2/v0_2_render_majority_calgary_MAP_p72_r1200.png` — 1200-DPI render, primary inspection source for majority Calgary

Minority — Appendix E:

- `data/maps/hires/v0_1_minority_p359_map73.png` — Appendix E map 73
- `data/maps/hires/v0_1_minority_p360_map74.png` — Appendix E map 74
- `data/maps/hires/v0_1_minority_p361_map75.png` — Appendix E map 75
- `data/maps/hires/v0_1_minority_p362_map76.png` — Appendix E map 76
- `data/maps/hires_v2/v0_2_native_minority_min_map_calgary_p101.jpeg` — native extraction, minority Calgary
- `data/maps/hires_v2/v0_2_native_minority_min_map_edmonton_p107.jpeg` — native extraction, minority Edmonton

Full majority panel coverage across all eight Appendix A panels was extracted in sessions 11–12 (Tier-0 raster pipeline). The §5.8.3 symmetric anomaly scan applies to all majority panels, not Calgary only.

5.8.2 Chair-flagged boundaries (C3)

Four boundaries were flagged by name in the majority report's response section. Direct inspection results:

- **Calgary-Nolan Hill-Cochrane (minority): Confirmed.** A district reaching from Cochrane (outside Calgary's western boundary) eastward through a narrow-waisted corridor to Calgary's Nolan Hill neighborhood, skipping Rocky Ridge / Tuscany.
- **Rocky Mountain House-Banff Park (minority): Confirmed.** SW extension of the district traces Banff National Park to reach the BC border. Absent the extension, the district still meets 4 of 5 §15(2) criteria on the predecessor footprint (Clearwater County alone

satisfies (a) at 18,692 km²; (b), (c), (d) all pass without NP territory); the extension adds only criterion (e) — the BC-border coterminous test. See §5.1.4 re-audit.

- **Olds-Three Hills-Didsbury (minority): Confirmed.** Named for three small towns; extends south past Didsbury to capture a portion of N Airdrie. Airdrie has a population greater than the three named towns combined.
- **Calgary-Foothills-Airdrie West (minority):** Boundary connection between Calgary-Foothills and Airdrie West tracks a primary highway corridor; the geographic connection itself is defensible, but this ED is one of four making up the Airdrie split (C4).

Combined chair-criticism count across the majority report. The four geographic anomaly flags above are one subset of the chair's criticism. In a separate section (Appendix C of the majority report, §5.9.4 of this audit), the chair also claimed five minority configurations — Airdrie, Cochrane, Chestermere, Red Deer, and St. Albert — had no public support in the consultation record. Taking the union across both sections, seven distinct minority configurations were criticized by the chair: the four geometric flags (Calgary-Nolan Hill-Cochrane, Rocky Mountain House-Banff Park, Olds-Three Hills-Didsbury, Calgary-Foothills-Airdrie West) plus three appearing only in the Appendix C public-support audit (Chestermere, Red Deer hybrids, St. Albert). Cochrane and Airdrie appear in both lists. The audit's §1.1 headline "three geographic anomalies" reflects only the three flags for which the anomaly was confirmed and the geographic rationale was not independently defensible. The full scope of the chair's documented objections spans seven configurations.

5.8.3 Majority hybrids — symmetric check

Applied the same anomaly-scan questions (lasso shape, engineered statutory boundary, misnamed municipality capture) to the majority's four Calgary hybrids:

- **Calgary-East:** Intra-city rectangular block, no extension beyond city limits. No anomaly.
- **Calgary-Falconridge-Conrich:** NE Calgary + directly-abutting Conrich community. Compact. No anomaly.
- **Calgary-Glenmore-Tsuut'ina:** Large southern extension to include the Tsuut'ina Nation reserve; shape tracks the reserve boundary. No anomaly; positively, the reserve is kept intact in a single named ED.
- **Calgary-West-Elbow Valley:** Calgary SW + directly-adjacent Elbow Valley subdivision. No anomaly.

Qualification. The anomaly-scan question set was developed from observed minority anomalies. The majority may have different classes of anomaly (e.g., rural-district highway corridors) not detected by the scan criteria applied here. All eight Appendix A panels have been inspected. The limitation is the question set, not the imagery.

5.8.4 Community-of-interest splits (C4)

DIMENSION	MAJORITY 2026	MINORITY 2026
Lasso/corridor shapes (visible)	0	1 (Calgary-Nolan Hill-Cochrane)
Engineered statutory boundary	0	1 (RMH-Banff Park extension — Reading B only; not detected under pre-registered Reading A)
Community captured under misnamed ED	0	1 (Olds-Three Hills-Didsbury → N Airdrie)
Airdrie split	2 EDs	4 EDs
Cochrane	intact	merged into Calgary
Chestermere	intact	partially split
Tsuut'ina Nation	single ED, named	single ED, named
Enoch Cree Nation	with Edmonton only	bundled with Devon (~50 km S)
Siksika Nation	High River-Vulcan-Siksika	High River-Vulcan-Siksika (same)

First Nations representation: Enoch Cree Nation vs. Tsuut'ina Nation. *The two rows above highlight a differential treatment of near-city First Nations reserves the table alone does not make legible. Both Tsuut'ina Nation (Reserve 145, immediately southwest of Calgary) and Enoch Cree Nation (Reserve 135, immediately west of Edmonton along the Highway 16A corridor) are Treaty-recognized nations whose reserves are geographically adjacent to a major Alberta city, with community orientation, urban services access, and civic relationships primarily tied to the adjacent urban area.*

The majority map treats the two nations consistently: Tsuut'ina is in a single named Calgary ED (Calgary-Glenmore-Tsuut'ina) whose boundary tracks the reserve perimeter — the §5.8.3 anomaly scan found no anomaly. Enoch Cree is in an ED linked to Edmonton's west-side geography, consistent with the reserve's position on the Edmonton fringe.

The minority map is consistent for Tsuut'ina (single named ED, reserve-adjacent, no anomaly flagged) but diverges for Enoch Cree. The minority creates Edmonton-Enoch-Devon, pairing Enoch Cree Nation with the Town of Devon approximately 50 km to the south. Devon (~6,500 residents) is a residential municipality on the North Saskatchewan River whose service geography is oriented toward the Leduc–Edmonton International Airport–Nisku employment corridor, not toward the Stony Plain/Spruce Grove/Highway 16A corridor where Enoch Cree Reserve 135 sits. The natural community-of-interest relationship for Enoch Cree — proximity to Edmonton, shared Edmonton-zone health authority, Edmonton-area commerce — is with Edmonton's west-side EDs, not with Devon.

The geometric consequence on canonical substrate is small: Edmonton-Enoch-Devon's canonical Polsby-Popper score is **0.534**, within the central band of the 178-ED joint set. The v0_9 DPG-era reading of **0.065** (cited in earlier drafts as evidence of "L-shaped corridor" geometry) is **substrate-stale and retracted** for any geometric-consequence claim 2026-06-12 per T1.7 Referee #10 — the v0_9 figure was a DPG transcription artefact, not a real corridor signature; the canonical EA shapefile PP shows the district is normally compact. The audit's COI critique of pairing Enoch Cree with Devon (50 km south, different service geography) **stands on the community-of-interest grounds described above**, not on the geometric ground that an earlier draft asserted. Geometric corridor evidence for Enoch-

Devon is *not* present on canonical substrate; the COI critique is the load-bearing finding here.

The audit does not file this as a gerrymander signal for two reasons. First, the community-of-interest comparison has now been checked against StatsCan Table 98-10-0459 (journey-to-work) for the Enoch Cree CSD (DGUID `2021A00054811804`; data at `data/outputs/enoch_cree_journey_to_work.csv`, downloaded 2026-06-13): of 435 Enoch Cree resident workers, **255 (58.6%) work on-reserve**, **100 (23.0%) commute to the City of Edmonton**, 20 (4.6%) to Spruce Grove, and 10 each to Stony Plain, St. Albert, Parkland County, Leduc County, and Leduc City. **Devon does not appear as a commute destination at any non-zero count level.** The commute geography is entirely northward (Edmonton, Spruce Grove, Stony Plain — all Highway 16A / west-Edmonton corridor), with zero flow toward Devon (50 km south on Highway 60). The claim that Enoch Cree's natural service relationship is with Edmonton rather than Devon is now documented at CSD commuter-flow resolution, not merely geographically inferred. Second, the pairing may be constrained by population math: if the combination of Enoch Cree (on-reserve population ~1,000–1,500) and adjacent Edmonton west-side EDs leaves no legal configuration within the ±25% statutory band, the Devon bundling may be the closest available compliant option. The population arithmetic for this choice has not been separately verified.

What is documented is the asymmetry relative to Tsuut'ina: the minority applies the "single named ED, reserve-adjacent, no anomaly" standard to Tsuut'ina and does not apply it to Enoch Cree. The commission's rationale in Appendix E does not address this differential treatment.

Census-subdivision-level robustness check. A CSD-level overlay (Track H; script `analysis/scripts/csd_community_splits.py`) computes, per map, the count of populated CSDs (population ≥ 1,000) spanning two or more electoral divisions. Under 2019 boundaries (measured via geopandas overlay on `data/alberta_2019_eds/`): 66 of 191 populated CSDs (34.6%) are split. Under the majority 2026 proposal (inferred via the Appendix C crosswalk): 66 of 191 (34.6%). Under the minority 2026 proposal (inferred, lower bound, via the heuristic crosswalk): 54 of 191 (28.3%). Upper bound matching the 2019 count of 66. On the confident-only subset of 139 CSDs (excluding those touched by minority-crosswalk uncertainties or any hybrid), all three maps produce the same 40 splits. **The Majority-minus-Minority asymmetry visible in the table above is not detectable at CSD granularity.** The minority's community-of-interest disadvantage operates at within-ED partition resolution (e.g., the four-way partition of the City of Airdrie, the bleed of Chestermere into Calgary-Peigan-Chestermere) — a resolution not encoded in the ED-level crosswalks and not directly measurable until the 2026 shapefiles release. The within-ED qualitative findings in the table above remain the reported finding. The CSD-level count is a null symmetric across maps and is reported here as a bounding limit on the metric's directional power.

Spatial CSD fragmentation analysis (v0_7 geometry, 2026-04-24; script `analysis/scripts/municipal_splits.py`). A direct intersection of v0_7 DPG boundaries against Statistics Canada 2021 CSD polygons, restricted to Cities (CY), Towns (T), Specialized Municipalities (SM), Villages (SV), and Improvement Municipalities (IM) with ≥ 300 VA votes (~3,000 residents), measures how many EDs each municipality is split across. Unlike the crosswalk-inferred method above, this uses geometric intersection with a ≥ 5 ha overlap threshold to filter trivial slivers.

MAP	MUNICIPALITIES SPLIT ACROSS ≥2 EDS	CHANGE VS 2019
2019 enacted	10	baseline
Majority 2026	8	−2
Minority 2026	11	+1

The majority map reduces fragmentation by 2. The minority map increases it by 1. The headline finding at this spatial resolution is concentrated in a single dramatic case: ***Strathcona County*** *(a Specialized Municipality immediately east of Edmonton, population ~105,000) is split across* ***3 EDs under the 2019 map and 10 EDs under the minority 2026 map (+7)****. Under the majority 2026 map Strathcona County is split across 4 EDs (+1). The minority's +7 fragmentation of Strathcona County accounts for the net increase in total splits despite other municipalities consolidating. It disperses a predominantly suburban municipality across ten EDs, reducing Strathcona voters to a numerical minority in each. The Strathcona finding is consistent with the Airdrie cracking pattern in* *§5.3.2* *— both are suburban municipalities east or north of Edmonton/Calgary split more times under the minority proposal than required by population math alone. Full per-municipality breakdown at* `findings/municipal_splits.md`*; data at* `data/municipal_splits.json`*.*

Constitutional and international law dimensions. The boundary decisions analysed above — pairing Enoch Cree Nation (Reserve 135) with Devon rather than with Edmonton west-side geography, while Tsuut'ina Nation (Reserve 145) receives a named district tracking the reserve perimeter — engage constitutional and international legal frameworks that this audit does not assess but that informed readers and the affected nations should be aware of.

Section 35(1), Constitution Act, 1982 provides: "The existing aboriginal and treaty rights of the aboriginal peoples of Canada are hereby recognized and affirmed." Jurisprudence following *Haida Nation v. British Columbia (Minister of Forests)*, 2004 SCC 73, and *Taku River Tlingit First Nation v. British Columbia (Project Assessment Director)*, 2004 SCC 74, establishes that the honour of the Crown gives rise to a duty to consult Indigenous peoples when the Crown contemplates conduct that might adversely affect Aboriginal title or treaty rights — before decisions are made. Electoral redistribution that alters the electoral district to which a reserve is assigned is the type of Crown conduct that may engage this duty: the assignment determines which community the nation is paired with for the purpose of choosing a provincial representative, and in the minority map's Enoch Cree configuration, the assignment pairs the nation with a municipality (Devon) with which it has no documented commuter, commercial, or service-provision link. The extent to which Alberta's redistribution process did or did not satisfy any s.35-derived consultation obligation — including the threshold question of whether such an obligation was triggered — is a legal question the audit does not adjudicate. The audit's statistical and geospatial findings (zero commute flow to Devon; commute gravity entirely northward toward Edmonton) may constitute relevant evidence in such an inquiry.

Article 18 of the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP) provides: "Indigenous peoples have the right to participate in decision-making in matters which would affect their rights, through representatives chosen by themselves in accordance with their own procedures, as well as to maintain and develop their own indigenous decision-making institutions." Canada gave domestic legal effect to UNDRIP through *An Act respecting the United Nations Declaration on the Rights of Indigenous Peoples* (Bill C-15, 2021), which commits the federal government to align federal laws and policies with UNDRIP. Alberta has not enacted equivalent provincial UNDRIP legislation as of June 2026. However, Article 18 speaks directly to the right of Indigenous peoples to participate in decisions affecting which representatives will speak for their communities. A redistribution process that pairs a First Nation's reserve with a distant municipality — without documented engagement with the nation's own consent or preference — engages the Article 18 principle at a minimum as a matter of interpretive context for any s.35 consultation analysis. Whether the Article 18 standard was met is for Enoch Cree Nation and its counsel to assess.

Tsuut'ina–Enoch Cree differential treatment, summary. The documented asymmetry — Tsuut'ina Nation (Reserve 145) receives a named, boundary-tracking single district under both maps; Enoch Cree Nation (Reserve 135) receives a Devon pairing under the minority map with zero commuter flow documented to Devon — is the factual predicate for both the s.35 and Article 18 framing above. The commission's Appendix E provides no rationale for the differential. Whether the differential treatment reflects population-math constraint (no legally compliant configuration pairing Enoch Cree with an Edmonton west-side ED within the ±25% statutory band) or a deliberate community-of-interest choice has not been separately verified by the audit; the population arithmetic for this choice remains open.

5.8.5 Municipal-boundary anchoring audit — did not survive canonical recomputation

The pre-registered municipal-anchoring test measures the percentage of each map's ED perimeter falling within 500 m of a Statistics Canada CSD edge (script: `score_anchoring.py`). Canadian redistribution commissions typically follow municipal boundaries where the population math permits, so the share of perimeter aligned with CSD edges is a meaningful proxy for adherence to the standard Canadian community-of-interest convention.

Contiguity filter disclosure (flagged 2026-06-12 per T1.7 R2 Ref #10): the pre-registration specifies measurement only over *contiguous ≥ 1 km segments* of aligned perimeter; `score_anchoring.py` measures alignment per-segment without the run-length filter, producing slightly more permissive values than the pre-registered definition. A contiguity-filtered recomputation is queued as T4.10-anchor; until that computation is done the values below are upper bounds relative to the pre-registered specification. The minority's canonical 72.0 % is 2.0 pp above the 70 % comparator floor; a filtered recomputation could in principle push it below that floor (see §1.2, structural caveat 7).

Canonical result (official Elections Alberta shapefiles, 2026-05-10):

MAP	ANCHORED FRACTION	WITHIN CANADIAN COMPARATOR NORM (70–85 %)?
Majority 2026	80.0 %	yes
Minority 2026	72.0 %	yes
2019 enacted (baseline)	75.2 %	yes

Ratio majority/minority = 1.11×. Both 2026 maps fall within the established Canadian norm. The minority sits slightly below the majority but well inside the comparator band. The anchoring dimension does not produce a structural-divergence signal on canonical geometry.

Pre-shapefile history. During the audit's pre-canonical phase (April 2026), the same script run against the DPG (Derived Provisional Geometry) v0_10 substrate returned minority 14.5 %, majority 71.0 %, ratio 4.9× — reported as the structural lane's cleanest single signal. The result did not survive canonical recomputation. The footnote attached to this section gives the full mechanism (area-vs-perimeter fidelity distinction in the DPG pipeline) and disposition.[4]

The audit's structural case (§5.0.1, §6.2.2) consequently rests on the four geometry-independent dimensions — population dispersion, Calgary zone asymmetry, Airdrie fragmentation, and chair-flagged spatial anomalies — together with the MCMC ensemble (Mahalanobis $p = 1.40\times10^{-6}$).

Shared-schools community-of-interest claim — failure of cross-reference, systematic in structure. Two minority configurations defend their hybrid structure partly on school-district community of interest. Calgary-Bow-Springbank (AEBC, 2026, Appendix E, p. 322) invokes "educational institutions" as a community-of-interest tie between Springbank and west Calgary. Springbank falls within Rocky View School Division No. 41 while the relevant west-Calgary catchment is served by the Calgary Board of Education (Alberta Education, school-division boundaries). Red Deer-Sylvan Lake (AEBC, 2026, Appendix E, p. 351) cites schooling as an urban-rural tie. Sylvan Lake falls within Chinook's Edge School Division No. 73 while the City of Red Deer is served by Red Deer Public Schools and Red Deer Catholic Regional Schools. The shared-schools rationale is not supported by the school-district boundary data in either case.

The pattern is structural, not isolated. An audit of all 21 minority hybrids against Alberta Education school-division boundaries (`analysis/methodology/school_division_coherence.md`) finds **20 of 21 cross at least one school-division boundary** — a mathematical consequence of Alberta's school divisions being built around municipal limits (CBE ends at Calgary's limits, Red Deer Public at Red Deer's, Edmonton Public at Edmonton's) and the minority's hybrid doctrine explicitly crossing municipal limits. All four Red Deer hybrids (Blackfalds, Innisfail, Lacombe, Sylvan Lake) cross school-division boundaries — not just Sylvan Lake. The rhetorical contradiction the audit identified for R5 and R11 is therefore representative, not exceptional: the minority chose the two cases where five minutes of Alberta Education verification shows the register is wrong, but the underlying cross-division pattern applies to nearly every minority hybrid. The structural school-division crossings are not by themselves gerrymander signals on the school dimension. What is damning is narrower — R5 and R11 invoked the most verifiable class of community-of-interest claim and do not survive primary-source verification against Alberta Education school-division boundaries. Full per-hybrid classification (school-coherent / mildly incoherent / severely incoherent / neutral) in `analysis/methodology/school_division_coherence.md`.

Cochrane commuter-tie claim — partial support at CSD resolution. The Calgary-Nolan Hill-Cochrane hybrid is defended by the minority report (AEBC, 2026, Appendix E) partly on the claim Cochrane residents "move fluidly" between Cochrane and Calgary. StatsCan Table 98-10-0459 (2021 Census journey-to-work) disaggregates Cochrane CSD commute destinations: of 8,550 Cochrane workers with an Alberta place of work, 4,205 (49.2%) work within Cochrane, 3,065 (35.8%) commute to Calgary CY, 345 (4.0%) to Rocky View County, 185 (2.2%) to Canmore, 135 (1.6%) to Wood Buffalo, and 130 (1.5%) to Airdrie. The Calgary flow is a genuine commuter-tie signal at the city-to-city level. The 2021 public release collapses Calgary to a single CSD and cannot test the within-Calgary sub-destination, so the pairing of Cochrane specifically with the Nolan Hill/Sage Hill ward is neither confirmed nor refuted by this dataset. The interpretive inference — Nolan Hill is a residential neighbourhood without significant employment and is therefore unlikely to be the commute destination for the 35.8% Calgary-bound flow — is consistent with the city's land-use profile but does not derive from the StatsCan data directly. Full methodology in `analysis/methodology/cochrane_journey_to_work.md`.

Piikani name-etymology note. The name "Peigan" in the existing Calgary-Peigan electoral division and its minority extension Calgary-Peigan-Chestermere derives from Peigan Trail SE, a road forming the district's northern boundary, not from a community-of-interest tie to the Piikani Nation (whose Piikani 147 reserve is located approximately 200 km south of Calgary, near Pincher Creek and Fort Macleod). The minority's retention of the name in the hybrid extension preserves a road-based etymology. This is a naming observation, not a finding of fault.

Community of Interest: Legislation vs. Practice. Neither the federal *Electoral Boundaries Readjustment Act* nor Alberta's *Electoral Boundaries Commission Act* provides a strict definition for "community of interest." In Canadian redistricting practice, independent commissions rely heavily on municipal boundaries as the primary objective proxy for shared community concerns (Courtney 2001, chs. 6–7). The minority report justifies its structural departures from municipal boundaries by citing "ESRI and Google basemaps" and "commissioner knowledge" rather than demographic data — a methodological choice this audit flagged during its pre-shapefile phase as a material departure from comparator practice. On canonical geometry the quantitative anchoring gap is small (72.0 % minority vs 80.0 % majority, both within the 70–85 % comparator norm) so the rhetorical concern about the minority's stated rationale stands on its own, without the support of the previously-cited 4.9× anchoring statistic.[4]

5.9 Procedural findings

5.9.1 Commission operation

Five-member commission constituted under Electoral Boundaries Commission Act §3–5: chair nominated by the Chief Justice of Alberta, two government-nominated commissioners, two opposition-nominated commissioners. Commission tabled unanimous interim report October 2025. Tabled divided final report (3–2) March 23, 2026. The three-member majority comprises the chair plus the two opposition-nominated commissioners.

5.9.2 April 16, 2026 government action

On April 16, 2026 the Alberta Legislative Assembly passed Motion 19 by a vote of 44 to 36, setting aside the commission's majority report and establishing a Special Select Committee of five MLAs (three UCP, two NDP) chaired by Brandon Lunty, MLA for Leduc-Beaumont, to produce a 91-seat map by November 2, 2026. The committee is served by an advisory panel with the same three-party structure as the commission (government-appointed chair plus two nominees per party), whose membership and terms of reference had not been published as of April 22, 2026 (CBC Edmonton, April 16, 2026; Calgary Journal, April 21, 2026). Unlike the commission it replaces, the new process does not include public hearings on the draft map.

Relationship to Chair Miller's Recommendation 5. The Premier framed the April 16 motion as aligned with a recommendation by Chair Dallas Miller (Government of Alberta press remarks, April 16, 2026, as reported in Rimbey Review; Calgary Journal, April 21, 2026). This framing is traceable to the Chair's Addendum to the Majority Report (AEBC, 2026, pp. 66–67), which proposed **Recommendation 5**: in the event the Legislature could not accept the majority's 89-seat boundaries, the Act should be amended to raise the seat count from 89 to 91 through "an all-party Select Special Committee or other equivalent Legislative Committee," restoring the two rural divisions the majority report removed while maintaining "the rest of the province as we propose... to the extent possible."

Provenance of R5: the chair alone, not the commission majority. The opening sentence of R5 is drafted in the voice of "the majority of the Commission." Chair Miller, however, states in the same addendum: "*My majority colleagues do not agree with me on this point*" (AEBC, 2026, p. 66). On the commission's own documentation, R5 is therefore the personal recommendation of the chair, not a collective recommendation of the three-member majority. Commissioner Greg Clark (one of the two opposition-nominated majority members, nominated by NDP Leader Naheed Nenshi) reiterated this publicly on a social-media thread in April 2026, reinforcing Miller's in-text disavowal. The Premier's framing of R5 as "the commission's own recommendation" does not carry this distinction. The framing is accurate as to the chair's personal position. It overstates the recommendation's provenance if read as a collective endorsement by the majority. Corroboration: CBC News Edmonton, April 16, 2026 ("Miller adding: 'My majority colleagues do not agree with me on this point'"). Full text of R5 is preserved in `findings/chair_recommendation_5_analysis.md`.

The alignment between R5 and the April 16 motion is partial on three grounds, addressed individually in `findings/chair_recommendation_5_analysis.md`:

1. **Form.** The vehicle (Select Special Committee raising the count from 89 to 91 for rural-seat restoration) matches R5's specification. The motion can legitimately claim this anchor.
2. **Substantive constraints.** R5(a)–(d) specifies four concrete boundary conditions — no impact on any electoral division in Airdrie or south of it except Drumheller-Stettler. No impact north of Edmonton's North Saskatchewan River. Reversion of south-of-NSR Edmonton districts to the interim-report map. Restoration of a Clearwater County-plus-western-Mountain-View s.15(2) district. Whether the committee's November output respects these conditions is not yet testable and should form part of the pre-registered November checklist (see §7.2). R5 also requires "the rest of the province as we propose [in the majority report] must be maintained to the extent possible" — a condition the committee's present mandate does not carry forward.
3. **Intent.** Chair Miller stated R5's purpose directly: it "is formulated for the express purpose of dissuading the Legislature from accepting the minority report" (AEBC, 2026, p. 66). The Chair further described the minority's hybrid configurations in Airdrie, Calgary, Chestermere, Cochrane, Red Deer, and St. Albert as "not something I can condone" (AEBC, 2026, p. 67). A committee output reintroducing any of those minority configurations invokes the form of R5 while inverting its intent. The motion's silence on which starting map the committee uses, combined with the presence in the committee of the political faction appointing the minority commissioners, is therefore procedurally distinct from R5's conditional.

Regional-economy framing. Alberta's Regional Economic Development Alliance geography provides partial support for the minority's general hybrid doctrine. The Central Alberta REDA covers Red Deer, Innisfail, Blackfalds, Lacombe, and Sylvan Lake — the five municipalities at the heart of the minority's Red Deer hybrid proposals. The Calgary Regional Partnership covers Calgary, Airdrie, Cochrane, Chestermere, Okotoks, Rocky View, and High River — the catchment for the minority's Calgary hybrids. These are real, publicly-documented regional organisations. They are not, however, boundary prescriptions. Any map grouping districts within these zones satisfies the zone-coherence criterion, and the zones do not by themselves justify the specific intra-zone configurations the minority proposed.

5.9.3 Comparator cases

Canadian boundary-commission practice traces to *Reference re Provincial Electoral Boundaries (Saskatchewan)* [1991] 2 SCR 158, which established the "effective representation" standard as the constitutional benchmark for provincial electoral boundaries. *Figueroa v. Canada (Attorney General)* [2003] 1 SCR 912 and *Frank v. Canada (Attorney General)* [2019] 1 SCR 3 developed the broader §3 Charter right to vote. Pal and Choudhry (2011) analyse the intersection of electoral rights and boundary-commission discretion, arguing the Saskatchewan Reference standard permits commissioners latitude in weighing the non-quantitative factors (geography, community of interest, minority representation) against population parity. Courtney (2001) provides the authoritative scholarly treatment of the independent-commission model across Canadian provinces: chapters 6–7 map the interplay between population equality and community-of-interest discretion, and chapters 10–11 address public-hearing obligations and post-hearing revision scope — all directly relevant to the D1–D4 procedural findings below. Pal (2015) argues the Saskatchewan Reference standard leaves commission members substantial discretion reviewable, if at all, only for manifest unreasonableness — not for partisan preference proximity — so the distinction between legitimate design preference and legally reviewable partiality turns on the same factual record the D1–D4 evidence assembles. Pal and Choudhry (2014) extend this framework to ask under what conditions a boundary commission's outputs become so asymmetric they exceed the justifiable margins of discretion even under the loose Saskatchewan Reference standard. The procedural findings below are situated within the discretion space: the audit documents where the minority's boundary choices depart from the pattern established by the majority under identical statutory constraints. The legal weight of the departure, if any, is for a reviewing body to assess under Pal's framework.

The closest historical Canadian comparator is Quebec 2011: the National Assembly refused to proclaim the *Commission de la représentation électorale*'s delimitation under Bill 132, leaving the existing electoral map in force after the commission had completed its work. The refusal — government non-adoption of a finished commission product — is the strongest available antecedent for legislative interference with a completed provincial redistribution cycle in the post-*Saskatchewan Reference* era.

Federal comparator: the 2022 Alberta federal sub-commission. A second institutional comparator operates at a different level: the 2022 Alberta federal redistribution cycle, conducted under the *Electoral Boundaries Readjustment Act* (S.C. 1985, c. E-3.3). This cycle is particularly relevant because the federal sub-commission (chaired by Justice J.D. Bruce McDonald, with members Donald Barry and Donna Wilson) operated under a statutory mandate structurally identical in its non-quantitative factors to Alberta's provincial Electoral Boundaries Commission Act: both delegated authority to independent, multi-member commissions with comparable discretion over geography, community of interest, and population variance. The federal 2022 sub-commission's handling of key Alberta boundary questions provides a direct institutional benchmark for how independent redistributors, applying equivalent legal criteria, address the same geographical regions.

On Airdrie (population ~65k in 2021): the federal sub-commission created a 2-district treatment (Airdrie–Chestermere and Banff–Airdrie, later consolidated as Airdrie–Cochrane), matching the provincial majority's approach. On Red Deer (population ~100k+ in

2021): the federal sub-commission unified Red Deer into a single federal riding, again aligning with the provincial majority's 2-district provincial maximum. On Lethbridge (population ~100k+ in 2021): the federal commission performed a 2-way split (Lethbridge and Lethbridge-foothills), consistent with the majority's approach. These federal choices, documented in the 2022 *Representation Order* and the sub-commission's published rationale, were made by an independent federal body applying non-partisan redistribution criteria to identical population bases and identical geographical constraints. The minority's provincial proposals diverge systematically from these federal precedents: the minority proposes 4-way splits for both Airdrie and Red Deer where the federal commission and the provincial majority accept at most 2-way splits. §5.3.2's cracking-candidate analysis treats these departures as evidence of structural irregularity precisely because they depart from the pattern established by an independent comparator commission applying equivalent statutory criteria.

The 2022 federal cycle does not determine the provincial boundary question — the two redistributions operate under different statutory frames and serve different electoral systems — but it provides an institutional reference point: when an independent federal commission and the provincial majority commission converge on equivalent boundary treatments, and the provincial minority diverges significantly, the divergence warrants scrutiny. The audit's analysis in §5.3.2 and the cracking-candidate findings in §5.3.3 are anchored partly to this federal-provincial convergence.

The April 16, 2026 Alberta motion shares Quebec 2011's post-completion timing — the commission had tabled its final report before the motion — but is structurally distinct in a material respect: rather than leaving the existing electoral map in force, the motion reassigned boundary-drawing authority to a legislature-selected committee (the Lunty committee). As of this paper's writing (May 2026), the Lunty committee has not yet produced a replacement map. One is expected in November 2026. The reassignment removes the independent commission from the remaining process. Whether the committee's eventual product will be adopted, and what boundaries it will draw, is unknown at time of writing. No Canadian provincial redistribution cycle reviewed in Courtney (2001, chs. 10–11) or subsequent scholarship involves the government both rejecting a completed commission product and reassigning the drafting authority to a legislature-selected body within the same cycle (corroborated by D. Bratt, personal communication, 2026-05-09). The constitutional status of this class of interference has since become non-hypothetical: six days after the Alberta motion, the Supreme Court of Canada dismissed Quebec's appeal of a ruling struck down the Legault government's 2024 statute blocking the *Commission de la représentation électorale* from redrawing Quebec's provincial map. The SCC's 7–2 from-the-bench ruling held the freeze law violated the Charter's democratic-representation guarantee. Whether the Lunty committee structure is constitutionally distinguishable from a legislative freeze — for instance, because it formally reassigns rather than simply blocks, or because Alberta's voter-weight disparities differ from Quebec's — is for a court to decide on its own facts. The doctrinal implications are discussed in §5.9.5.

5.9.4 Public submission record (D2)

The commission received approximately 1,340 written submissions across two rounds of public consultation. The majority report's Appendix C (Alberta Electoral Boundaries Commission [AEBC] 2026) states the minority's hybrid configurations for Airdrie, Cochrane, Chestermere, Red Deer, and St. Albert **had no public support in the consultation record**.

A keyword search with manual review of the commission's submission archive — 1,252 of approximately 1,340 submissions extracted with usable text. The remainder (~88, 6.6%) are image-only scans without machine-readable text or detectable submission-ID markers — tests this claim. Full methodology, dataset, and technical log are in `analysis/scripts/submission_search.py`, `data/submission_search_dataset.csv`, `findings/submission_search_findings.md`, and `historical/submission_search_log.md`.

Result: the chair's claim is partially refuted, with tiered severity. A follow-on signal-strength analysis (`findings/claim_significance_analysis.md`) distinguishes between configurations where the chair's statement was merely *precisely inaccurate* (a supporting submission exists, so "no support" is technically false) and where the chair's statement was also *effectively inaccurate* (support is substantial enough to make "no support" materially overstate the absence of public support in the submission record). The framing throughout this section is objective and fact-based: the question is whether supporting submissions exist on the record, not whether the chair acted with any particular intent in characterising them.

Verdict by tier:

- **Precisely and effectively wrong** (three configurations where the minority adopted proposals *from* the public record, not despite it):
- Rocky Mountain House-Banff Park: 5 supporters, net +4, 25% of engaged submissions support the configuration
- Olds-Three Hills-Didsbury: 3 supporters, net +2, 60% of engaged submissions support
- Chestermere separation: 3 supporters, net +2, 23% of engaged submissions support (support is substantial enough to make "no support" materially overstate the absence of public support).
- **Precisely wrong, effectively ambiguous** (support exists but is evenly matched by opposition):
- Red Deer hybrids: 4 supporters, 4 opposers, net 0, 22% of engaged submissions support
- **Precisely wrong only / chair effectively correct** (support is negligible or zero):
- Airdrie 4-way: 0 supporters, 2 opposers, 0% of 4 engaged submissions
- Calgary-Nolan Hill-Cochrane: 0 submissions mention this configuration at all
- St. Albert-Sturgeon (minority variant): 0 clear supporters for the minority's alternative configuration; label-ambiguity caveat

The tier distinction matters because the chair's Appendix C claim was an argument for procedural weight, not a technicality. A chair who said "no public support" when 25–60% of engaged citizens proposed the exact configuration has mischaracterized the public record on the specific point, not simply overlooked a dissenting voice or two. By contrast, a chair who said "no public support" for the Airdrie 4-way split is effectively correct — four engaged citizens discussed the configuration and none supported it.

Implication for the D2 procedural finding: the claim narrows but does not dissolve. The chair's sweep was *materially* overbroad on three of seven named configurations, *ambiguous* on one, and *defensible* on three. The audit should report these tiers rather than treating Appendix C as uniformly unsupported or uniformly sound. This matters because readers on both sides of the debate have incentive to flatten the finding: critics will use "chair was wrong" without the tiering. Defenders will use "some configurations did hold up" without naming the three failing to do so. The tiered verdict resists both flattenings.

5.9.4.1 Evidence by configuration, with per-area proportions

For each configuration, we report submissions mentioning it, submissions supporting the minority direction, and submissions opposing. The "support rate" column is the ratio of explicit supporting submissions to total submissions engaging with the configuration — a local measure of public backing among engaged citizens, not a measure of province-wide support.

MINORITY CONFIGURATION	MENTIONS	SUPPORTING MINORITY	OPPOSING	NEUTRAL	SUPPORT RATE	VERDICT
Airdrie 4-way split	4	0	2	2	**0 / 4 = 0.0%**	Chair's claim stands
Calgary-Nolan Hill-Cochrane hybrid	0	0	0	0	**0 / 0 = n/a**	Chair's claim stands
Rocky Mountain House-Banff Park (s.15(2))	20	3 + ≥4 aligned	1	~15	**3 / 20 = 15%** (7 / 20 = 35% with aligned)	**Refuted**
Olds-Three Hills-Didsbury rural unit	5	2	2	1	**2 / 5 = 40%**	**Refuted**
Chestermere separation	13	3	3	7	**3 / 13 = 23%**	**Partially refuted**
Red Deer hybrids	23	2 explicit + 3 aligned	4	17	**2 / 23 = 9%** (5 / 23 = 22% with aligned)	**Partially refuted**
St. Albert-Sturgeon (minority alternative)	11	0 for minority variant (2 for majority name)	1	8	**0 / 11 = 0%** for minority alternative	Chair's claim stands

5.9.4.2 Direct quotation evidence

Rocky Mountain House-Banff Park — EBC-2025-2-0619 ("Appropriate Political Representation for Alpine Alberta"). Under "3.2 Proposed Electoral Division Amendment 2: Rocky Mountain House-Banff":

> *"The proposed Rocky Mountain House-Banff electoral district brings together the upper Bow and North Saskatchewan headwaters, adjacent mountain parks, surrounding Crown land, and the communities depending on these landscapes for their livelihoods. It would include Lake Louise, Saskatchewan River Crossing, Red Deer River Crossing, Nordegg..."*

This is a direct textual proposal for the minority's s.15(2)-invoking configuration by the submission's explicit name. The configuration with the *most visible engineering evidence* (the NP extension to reach the BC border we identified in §5.1.4) is also the one with the *clearest public support* in the submissions — a finding tightens the tension in the audit rather than resolving it.

Rocky Mountain House-Banff Park — EBC-2025-2-0091 (Nordegg resident):

> *"I recommend riding boundaries include all of Clearwater County, including Rocky Mountain House, with other western communities like Sundre and Banff."*

Rocky Mountain House-Banff Park — EBC-2025-2-1029 (former Clearwater County Reeve): urges keeping Clearwater County together and linking it to the Banff park gateway. Directionally aligned with the minority configuration.

Olds-Three Hills-Didsbury — EBC-2025-2-0209 (Alan Balson, Beiseker):

> *"Keep Beiseker and the surrounding rural area in a reconstituted rural riding including Olds, Didsbury, Carstairs, Three Hills, and the agricultural areas around them."*

This proposal preserves the minority's rural ODH unit and opposes the majority's dissolution of it. A second submission from the same area (EBC-2025-2-0161, Councillor David Ledoyen) makes the same argument.

Red Deer hybrids — EBC-2025-2-0252 (Chad Krahn, Red Deer City Councillor):

> *"...a northern riding could encompass Sylvan Lake, Lacombe, and Blackfalds..."*

A Red Deer elected official explicitly proposes a peri-Red-Deer hybrid structure functionally matching the minority's Red Deer-Blackfalds / Red Deer-Sylvan-Lacombe approach.

Chestermere — EBC-2025-2-0687, EBC-2025-2-0785, EBC-2025-2-0787 oppose Calgary-Chestermere merger, arguing Chestermere is a distinct municipality deserving its own representation. The minority map preserves Chestermere separately. The majority does not merge it either but uses a different configuration. These submissions support the principle the minority embodies rather than a specific minority label.

5.9.4.3 Proportional weight and impact on findings

The proportions matter because they tell us whether the public-input record produces *signal* or *noise* for each configuration. Three interpretive lines:

Sample-size caveat. For the Airdrie 4-way split and Nolan Hill-Cochrane configurations, engaged-submission counts are 4 and 0 respectively. These are small samples. The absence of supporting submissions in 4 mentions is consistent with "no public support" but doesn't exclude the possibility a larger sample would uncover some. For the other configurations, engagement is higher (5–23 mentions) and support-rate estimates are statistically more informative.

Ridings with highest public engagement have the highest support rates for minority-aligned configurations. Olds-Three Hills-Didsbury (40%), Chestermere (23%), and RMH-Banff Park (15% explicit, 35% with aligned) are the three configurations where citizens in the affected area engaged most actively, and all three show non-trivial alignment with the minority direction. This is the opposite of what the chair's claim implied. The pattern does not prove the minority configurations are correct — engaged citizens can be wrong — but it does refute the categorical "no public support" characterization.

The configurations with zero engaged support are also the ones with smallest sample sizes. Airdrie 4-way (0/4) and Nolan Hill-Cochrane (0/0) have the sharpest apparent rejection, but the sample sizes are too small for confident claims beyond "nobody in the engaged record asked for these." This is consistent with the chair's claim for those specific configurations but does not constitute a *refutation* of minority intent — it just means there is no recorded demand.

5.9.4.4 Impact on the majority's and minority's findings

For the majority report. The "no public support" framing in Appendix C was a consequential argument. It implied the minority was advancing configurations against the clear weight of public input. The refutation evidence weakens this argument on three of five configurations. The majority's substantive cartographic critique — the minority's hybrid choices are less compact and more fragmenting of communities (see §5.8.3, §5.8.4 of this audit) — still holds. But the *procedural* framing in Appendix C was overbroad.

For the minority report. The refutation helps the minority's procedural posture only modestly. Three configurations have documented support, which makes those three harder for the majority to discount. The visible spatial concerns (§5.8.2: engineered RMH-Banff boundary, Nolan Hill-Cochrane lasso, ODH capturing N Airdrie) and the structural population asymmetries (§5.1.1, §5.1.2, §5.1.3) are not affected by the public-support question. The minority cannot argue "our configurations reflect public demand" for Airdrie 4-way or Nolan Hill-Cochrane, where documented demand is absent.

For the audit's Section D procedural concern. The §5.9 critique narrows but does not disappear. The government's April 16 action transferred the boundary-drafting authority from the commission to a legislature-selected committee in order to produce a map based on the less-publicly-vetted proposal. The concern is strongest for configurations genuinely lacking public support (Airdrie 4-way, Nolan Hill-Cochrane) and weaker — though not absent — for configurations with some documented backing (RMH-Banff Park, Olds-ODH, Red Deer hybrids, Chestermere). An earlier framing of "the government adopted boundary choices nobody asked for" would overstate the record. The accurate framing is "the committee's eventual map will include a mixture, with some choices having no documented public support and others having it."

5.9.4.5 Limits of the verification

1. **~88 submissions (6.6%) could not be machine-parsed** because their PDFs are image-only scans lacking a text layer or a detectable EBC-2025-X-NNN ID marker. OCR was out of scope. These could in principle contain additional supporting or opposing content not changing the refutation direction (which relies on identified supporting submissions) but could shift neutral / opposing counts.
2. **Keyword search precision.** Regex uses permissive co-occurrence windows (200–300 chars) and can miss submissions where the same configuration is described in paraphrased terms without the explicit place names used. Conversely, the Red Deer regex triggers on any Red Deer + {Blackfalds / Innisfail / Sylvan Lake / Lacombe} co-occurrence, which often simply describes the commission's *proposed* boundaries — those are neutrals, not supports.
3. **Position classifier is heuristic.** The code looks for support / oppose / against / recommend / should-not keywords near each match. Ambiguous classifications were manually reviewed and corrected in 13 cases (documented in `historical/submission_search_log.md`). CSV rows still reflect the automatic classification.

4. **Minority configuration names are the audit's labels, not the submissions'.** Citizens do not typically know the minority's precise labels (e.g., "Red Deer-Blackfalds"). A submission proposing a functionally equivalent configuration using different names is counted as directional support. The audit's rubric is generous on this point. The majority chair might not accept the same rubric.
5. **Attached sub-PDFs were not searched separately.** Some submissions reference external attachments (e.g., EBC-2025-1-0139 references "Airdrie-Feedback-Submission-AEBC-May-2025.pdf"). Only the enclosing batch PDF's text layer was searched. Additional evidence may reside in attachments.

The refutation finding is robust to limits (1)–(3) because it rests on identified counter-examples rather than exhaustive enumeration. Limits (4) and (5) could affect counts but not direction of the finding. A full Track-B OCR pass over the 88 missing submissions would strengthen the audit's credibility if it were used in legal proceedings. It would not likely change the qualitative verdict.

5.9.4.6 Full-corpus LLM sentiment analysis and Hansard transcript review

The keyword-based analysis in §5.9.4.1–5.9.4.5 addresses whether supporting submissions *exist* for each configuration. A separate, more comprehensive pass was subsequently run to quantify the *distribution* of stances — not just presence of support, but the balance between opposition, support, and contextual engagement — across both the written-submission corpus and the public-hearing transcripts.

Method. Two LLM-based classification pipelines were run using Claude Sonnet via the Claude CLI:

- *Full-corpus submission scan* (`submission_sentiment_llm_full.py`): **1,254 of 1,252 parseable submissions processed** (a few duplicates handled, but coverage is essentially complete; corrected 2026-06-12 per T1.7 R2 Ref #12 — the round-1 "182 of 1,252 (14.5%)" correction was itself wrong, derived from `findings/sentiment_analysis_completion_report.md`'s misreading of the progress CSV; the round-2 audit verified `data/outputs/submission_sentiment_llm_full_progress.csv` shows 1,254 unique submission IDs were processed during the 2026-05-09→10 pass; rows are emitted only for non-Unrelated stance classifications, so apparent "silence" in the output table is not evidence of unprocessed input). **Row-level reconciliation closed (T4.9-sentiment-recon, 2026-06-12).** The progress CSV has 1,272 parsed rows (some duplicates) across 1,254 unique submission IDs. The results CSV has 827 parsed rows across three scan types: **388 full_corpus + 209 hansard_r1 + 230 hansard_r2 = 827** (round-1 of this reconciliation reported "271/85/96 = 452" — those numbers were back-fitted to the headline "452 deduplicated" figure but do not match the canonical results CSV row counts). The 452 deduplicated set is a downstream aggregation that drops Unrelated stances and applies de-duplication of single-submission-multiple-configurations rows;

that derivation is preserved in `findings/sentiment_analysis_completion_report.md` but the raw counts above are the source-of-truth. Published intensity-summary aggregates (`data/outputs/intensity_summary_table.csv`, 7 configurations) reflect the 452-deduplicated subset; values cited in §5.9.4.6 narrative are reconciled to that source. The "Red Deer −154 / Nolan Hill −122" aggregates that appeared in earlier drafts derive from a pre-deduplication scoring pass and have been replaced by the canonical −64 / −55 values from the deduplicated table.

- *Hansard transcript scan* (`hansard_sentiment_llm.py`): two rounds of EBC public-hearing Hansard transcripts (r1: May 2025 hearings, 2,773 KB; r2: January 2026 hearings, 2,675 KB) were parsed into community-participant turns. Turns mentioning configuration keywords were classified using the same seven-configuration schema. 3,107 community turns in r1 (188 relevant); 2,358 in r2 (209 relevant).

Both pipelines resume from a progress file and are fully reproducible. Output: `data/outputs/submission_sentiment_llm_full_results.csv`.

Results — written submissions. 388 non-neutral configuration-stance rows across 292 unique submissions (from the 1,252-submission corpus). Overall: 189 Active Opposition (49%), 82 Active Support (21%), 117 Neutral/Contextual (30%).

CONFIGURATION	TOTAL ROWS	OPPOSITION	SUPPORT	OPP%	SUP%
Rocky Mountain House–Banff Park hybrid	104	56	37	54%	36%
Red Deer hybrid ridings	98	52	8	53%	8%
St. Albert merging with Sturgeon County	46	14	15	30%	33%
Calgary–Nolan Hill–Cochrane hybrid	43	25	5	58%	12%
Airdrie 4-way split	42	18	6	43%	14%
Olds–Three Hills–Didsbury extending to Airdrie	28	9	7	32%	25%
Chestermere merging with Calgary	27	15	4	56%	15%

Results — Hansard hearings. *439 relevant community turns classified across both rounds. Overall: 97 Active Opposition (22%), 91 Active Support (21%), 251 Neutral/Contextual (57%).*

CONFIGURATION	HANSARD TURNS	OPP%	SUP%	NET
Rocky Mountain House–Banff Park hybrid	112	22%	34%	+12 support
Red Deer hybrid ridings	75	8%	21%	+10 support
Airdrie 4-way split	74	27%	8%	+14 opposition
Calgary–Nolan Hill–Cochrane hybrid	55	24%	13%	+6 opposition
St. Albert merging with Sturgeon County	48	29%	25%	balanced
Olds–Three Hills–Didsbury extending to Airdrie	38	8%	21%	+5 support
Chestermere merging with Calgary	37	43%	11%	+12 opposition

Channel divergence. *The two measurement channels produce materially different signals on two configurations. Rocky Mountain House–Banff Park and Red Deer hybrids both show net opposition in written submissions (−18 and −44 respectively) but net support in Hansard hearings (+13 and +10). All other configurations show directionally consistent signals across both channels —*

Airdrie, Calgary–Nolan Hill, and Chestermere are opposed in both. Olds–Three Hills–Didsbury and St. Albert–Sturgeon are balanced in both.

Weighted net-sentiment ranking (LLM-scored intensity). A second LLM pass (Claude Haiku, structured JSON output) scored each Active Opposition/Support row on a 1–3 intensity scale: 1 = mentioned in passing, 2 = clear position among several substantive points, 3 = primary focus or emphatic language (demanding, urging, calling the configuration unacceptable). Intensity-weighted net = sum of support intensity values minus sum of opposition intensity values, ranked across all channels:

CONFIGURATION	INTENSITY-WEIGHTED NET	DIRECTION
Red Deer hybrids	−64	Channel divergence
Airdrie 4-way split	−61	Opposed, both channels
Calgary–Nolan Hill–Cochrane	−55	Opposed, all channels
Chestermere merging with Calgary	−47	Opposed, both channels
Rocky Mountain House–Banff Park	−9	Channel divergence
St. Albert–Sturgeon County	−1	Near-balanced overall
Olds–Three Hills–Didsbury	+6	Net supported

Note: LLM intensity scoring completed across 452 rows in `data/outputs/sentiment_intensity_scores.csv` (corrected 2026-06-13 per T1.7 R2 S22): 263 full-corpus AO/AS rows (from the 1,252 written submissions — 271 in the raw results CSV, 8 excluded in the deduplication/scoring pass); 85 Hansard-r1 turns; 103 Hansard-r2 turns; 1 blank/unscored row. N/C and Unrelated rows not scored. Earlier draft cited 271/85/96 = 452 — the 271 was the pre-dedup full-corpus count and the 96 for hansard_r2 was wrong; canonical values above are from `sentiment_intensity_scores.csv` directly. Two configurations show pronounced channel divergence: Red Deer and Rocky Mountain House–Banff Park both exhibit strong opposition in written submissions but net support in Hansard hearings, consistent with differential-mobilization and participation-cost explanations discussed below.

The channel-divergence hypothesis. The divergence between written and in-person channels on RMH–Banff Park and Red Deer hybrids warrants a methodological note. One credible explanation is differential mobilization: written submissions are submitted asynchronously and may over-represent organized opposition groups or residents with strong objections, while in-person public hearings attract a broader cross-section including residents who favor local representation arguments but lack the organizational impetus to submit formally. This pattern — where written channels show stronger opposition than in-person channels on the same question — is documented in planning and zoning literature (Fischel 2001; Davis 2016) though not specifically in electoral redistricting contexts.

A second explanation specific to the RMH–Banff Park configuration is geography: citizens from Nordegg, Rocky Mountain House, and the Jasper/Banff corridor who would benefit from a unified mountain constituency may be more likely to attend regional in-person hearings (which were held in part in those communities) than to submit written comments. This is not a partisan-sorting argument but a participation-cost argument — travel to a hearing is a sunk cost filtering for engagement intensity, not partisan direction.

The audit does not assert either explanation as established. The finding is: two channels produced divergent signals on two specific configurations, the divergence is large enough to be substantive (net flips from 50-point opposition to 10-point support), and commissioners who weighted in-person testimony more heavily — as is common in **polycentric advisory** administrative proceedings (corrected 2026-06-12 per T1.7 Referee #12; boundary commissions are not "quasi-judicial" — they make polycentric policy recommendations, not adjudicative decisions) — would have received a materially different impression of public sentiment on those configurations than commissioners who weighted the written record equally. Whether the weighting is appropriate is a procedural-design question outside the scope of this audit.

Relationship to §5.9.4.1–5.9.4.4. The LLM full-corpus pass confirms and sharpens the keyword-based findings from §5.9.4. The chair's "no public support" claim is most defensible for Airdrie (43% opposition, 14% support in submissions; 27% vs 8% in Hansard — consistently opposed in both channels) and least defensible for RMH–Banff Park (36% support in submissions, 34% in Hansard) and St. Albert–Sturgeon (balanced in both). The LLM pass addresses limits (2) and (3) from §5.9.4.5 by replacing keyword matching and heuristic classification with direct machine-reading of full submission text.

5.9.4.7 Rationale–submission alignment cross-reference

This section maps the combined sentiment corpus (§5.9.4.6) to the minority's published rationales. Of 25 rationales recorded in `data/outputs/minority_rationales.csv`, six could be matched to a sentiment configuration. The other 19 cover Edmonton-area, rural, and procedural rationales for which no submission configuration data exist.

Method. For each matched rationale, rows were counted by classification (Active Opposition / Active Support / Neutral/Contextual). Taking-sides percentages exclude Neutral/Contextual rows. Three alignment labels are used:

- **CONTRA_COMMISSION** — the commission claimed "SUPPORTS" or "PARTIALLY SUPPORTS" for this rationale, but the combined record is majority-opposing.
- **ALIGNS_WITH_AUDIT** — the commission's own verdict was skeptical (INCONCLUSIVE or FAIL), and the submission record is also majority-opposing.
- **CONTRA_AUDIT** — the audit flagged a test failure, but the submission record leans toward support.

All counts use the same LLM classifications as §5.9.4.6. Human inter-rater reliability scoring on a 60-item sample is pending.

RATIONALE	CONFIGURATION	AUDIT VERDICT	OPPOSING	SUPPORTING	ALIGNMENT
R1	Calgary-Nolan Hill-Cochrane	PARTIALLY SUPPORTS	38 / 50 (76%)	12 / 50 (24%)	CONTRA_COMMISSION

RATIONALE	CONFIGURATION	AUDIT VERDICT	OPPOSING	SUPPORTING	ALIGNMENT
R2	Calgary-Peigan-Chestermere	LEANS CONTRADICTS (2021 Census 98-10-0459: 86.0% of Chestermere out-commuters → Calgary CY; specific SE-Calgary/Forest-Lawn pairing unsupported)	31 / 39 (80%)	8 / 39 (20%)	ALIGNS_WITH_AUDIT
R3	Calgary-Airdrie	SUPPORTS	38 / 50 (76%)	12 / 50 (24%)	CONTRA_COMMISSION
R10	Red Deer-Lacombe	PARTIALLY SUPPORTS	58 / 82 (71%)	24 / 82 (29%)	CONTRA_COMMISSION
R12	Rocky Mountain House-Banff Park	PARTIALLY SUPPORTS (hist.); CLOSED-FAIL (area)	81 / 156 (52%)	75 / 156 (48%)	ALIGNS_WITH_AUDIT
R13	Olds-Three Hills-Didsbury	PARTIALLY SUPPORTS; CLOSED-FAIL (Airdrie slice)	12 / 27 (44%)	15 / 27 (56%)	CONTRA_AUDIT

Fractions are taking-sides only. Combined corpus: written submissions + Hansard r1 + Hansard r2. Total rows per rationale: R1 = 98, R2 = 64, R3 = 116, R10 = 173, R12 = 216, R13 = 66.

Findings. Three rationales (R1, R3, R10) are CONTRA_COMMISSION: the commission cited community-of-interest or public-support grounds, but the combined record runs 71–76% opposing among those who took a position. Two (R2, R12) are ALIGNS_WITH_AUDIT: the commission's own verdict was uncertain or test-failing, and submissions are also majority-opposing or near-split. One (R13) is CONTRA_AUDIT: the audit flagged a geometric failure on the Airdrie slice, but submissions lean 56% toward support. No ALIGNS_WITH_COMMISSION findings were observed.

Channel-divergence note for R12. The 52 / 48 combined split on RMH–Banff Park masks the divergence from §5.9.4.6: written submissions are 54% opposing, while Hansard in-person testimony is net-supportive. A commission weighting in-person testimony more heavily would have seen a different picture. No such divergence exists for R1, R3, or R10, which are majority-opposing in both channels.

Interpretation. CONTRA_COMMISSION findings do not prove improper conduct. A commission may rely on evidence beyond the archived record. The finding is the publicly verifiable consultation record does not corroborate the commission's claimed public endorsement on three of four configurations where such endorsement was asserted.

5.9.5 Constitutional backdrop

Reference re Provincial Electoral Boundaries (*Saskatchewan*), [1991] 2 SCR 158, established Canadian electoral redistribution is measured against an "effective representation" standard, not strict population parity. Within the standard, deviations from provincial average are permissible when they serve recognized factors (geography, community of interest, minority representation). An audit finding (a) directionally-consistent partisan asymmetry in a proposal and (b) a process promoting the proposal over a more-neutral, more-publicly-supported alternative would be evaluated against *Saskatchewan Reference* if challenged — but this audit does not assess the constitutional question. It provides the evidentiary basis for others to do so.

Application to specific boundary disputes. *Raîche v. Canada* (*Attorney General*), 2004 FC 679, applies the Saskatchewan Reference standard to a specific boundary dispute without producing a bright-line ceiling on partisan-asymmetric outcomes. (*Cassista v. Canada* (AG), 2014 FC 398 — cited in earlier drafts as the companion authority here — is flagged for verification per T1.7 R2 Ref #8 and is not relied upon for any verdict-bearing claim in this section pending counsel review; see §2 for the full verification banner.) Under the Raîche standard, a map's constitutional status depends on whether its deviations are reasonably related to permitted factors. The audit's findings — directional partisan asymmetry, engineered §15(2) boundaries, cracking patterns visible across three cities, and procedural departure from independent-commission practice — are the kinds of evidence a court applying the effective-representation standard would weigh.

The 2026 Quebec ruling. The constitutional landscape shifted materially on April 22, 2026, six days after Alberta's April 16 motion handed redistricting to the Lunty committee. Six days later, the Supreme Court of Canada dismissed Quebec's appeal of a Quebec Court of Appeal decision striking down the Legault government's 2024 statute blocking Quebec's *Commission de la représentation électorale* from redrawing the provincial map (the redrawing eliminated one Gaspé Peninsula riding and one Montreal east-end riding in favour of two new districts in the Laurentians/Lanaudière and Centre-du-Québec regions). The SCC ruling was 7–2, delivered from the bench — a procedural posture reserved for cases the panel views as constitutionally clear-cut. The QCA ruling the SCC upheld held the freeze law violated sections of the Charter guaranteeing democratic representation by allowing significant disparities in voter weight to persist. Reported in CBC News, "Quebec's redrawn electoral map will stay after Supreme Court ruling" (2026-04-22) and Canadian Press / CTV News, "Supreme Court rejects Quebec's attempt to block changes to election map boundaries" (2026-04-22). The formal SCC reasons are pending publication. The Quebec Court of Appeal decision being upheld is *Coalition de l'Outaouais et al. c. Procureur général du Québec* (2025; full citation pending). Audit will update this section when the SCC reasons are published.

Doctrinal implications for Alberta. Three relevant doctrinal observations (drawn from journalistic reporting; subject to confirmation when written SCC reasons are published, which are pending — these are not formal SCC "holdings" until reasons issue):

1. **Provincial legislatures cannot block independent commission redistricting work to preserve voter-weight disparities, even when justified on regional-protection grounds.** The CAQ government's stated rationale — protection of Gaspé and other rural ridings — was the same class of argument the Alberta minority commissioners advance in Appendix E pp.

302–303: "Hybrid constituencies are not only a good reflection of Alberta but also the best available instrument for preserving rural and suburban representation" (R17, archived verbatim at `data/outputs/minority_rationales.csv`). Earlier drafts of this paragraph appended "...in the face of sustained urban growth" to the quoted excerpt; those words are not in the archived rationale and were removed 2026-06-12 per T1.7 Referee #12. The substantive comparison to Quebec's regional-protection rationale stands without the appended phrase; both are rural-preservation justifications offered against statutory parity. The SCC found this rationale insufficient to justify legislative interference with commission work in Quebec. The same rationale would presumably face the same constitutional analysis if applied to a hypothetical Alberta statute reassigning commission-recommended boundaries.

2. **The effective-representation standard from *Saskatchewan Reference* [1991] applies with more force where legislative interference is at issue, not less.** The 7–2 from-the-bench posture suggests the SCC sees this as well-settled doctrine, not novel application.
3. **Charter section 3 is the operative provision.** Both the QCA and the SCC framed the analysis as a Charter-rights question, not a separation-of-powers question. This matters for any Alberta challenge: a section 3 challenger would have standing as an affected voter, not just as a constitutional litigant.

What this means for the audit's procedural critique. The April 16 Alberta motion — reassigning redistricting authority to a UCP-majority legislative committee partway through the cycle — operates under a constitutional landscape which, six days later, became significantly less hospitable to legislative reassignment of independent commission-recommended boundaries. The audit's framing in §5.9.3 (April 16 motion as procedurally unprecedented in the Canadian comparator set; corroborated by D. Bratt, personal communication, 2026-05-09) now extends to a substantive constitutional question no longer hypothetical: the SCC has just held a parallel move in Quebec is unconstitutional. Whether the Alberta situation is constitutionally distinguishable — for instance, because the Lunty committee is structured differently from a legislative-freeze law, or because Alberta's voter-weight situation differs materially from Quebec's — is for a court to decide on its own facts. The audit's positive contribution is to surface the structural-asymmetry evidence (§5.4–§5.8) any such challenge would draw on.

5.9.6 Evidence trail for the six contested-rationale checks

The public-facing claim "five of six of the minority's published reasons fail under check" (`report_public.md` §"Five of six of the minority's published reasons fail under check") is the load-bearing rhetorical move linking the structural-irregularity finding to the minority's own stated cartography. This subsection consolidates the evidence base for each of the six claims so a journalist, a counsel of record, or a hostile peer reviewer can trace each line in the public summary back to a primary source — or, where the trail is incomplete, see the gap. The six claims correspond to the six rationales the minority itself singled out in

Appendix E (pp. 285–362) of the 2025–26 EBC final report. The audit's verdict on each is reported here in the same order as `report_public.md`.

Registration status. The rationale-failure check is a qualitative post-hoc audit, not a pre-registered test. None of the OSF registrations associated with this paper (OSF w2s8k, r3zm7, qsgy8, 6pt83) names a rationale-inventory or cleaner-alternative check as a pre-specified metric. The finding should be read as qualitative corroboration of the pre-registered structural-irregularity pattern (OSF w2s8k; §5.5), not as an additional pre-registered measurement. A reader who weights only pre-registered findings should disregard the five-of-six rationale-failure summary and rely solely on the structural metrics in §5.4–§5.8.

Note on the withdrawn seventh claim. Earlier drafts of this audit (and earlier drafts of the public report) listed a seventh contested rationale: a Lethbridge / Taber-Warner configuration alleged to match a retired federal boundary. The claim has been removed entirely. The methodology check at `analysis/methodology/lethbridge_federal_boundary_check.md` established the minority report does not in fact make a federal-boundary claim for the Lethbridge area: there is no underlying source the audit's "federal boundary retired in 2013" line could refer to. The headline arithmetic moves from "six of seven fail" to "five of six fail" as a result. The withdrawal is itself documented as a rationale-strength correction in the audit's favour — a hostile reviewer would have caught the absent source within an hour of opening the methodology directory. Pre-emptive withdrawal preserves the audit's evidentiary discipline.

A general note on the strength bands used below. **Strongly supported** means the verdict rests on a downloaded primary dataset any reader can re-run with the published code. The rationale is contradicted on the dataset's own face. **Moderately supported** means the verdict rests on a documented public source (statute, government boundary map, commission text) any reader can re-verify, but the comparison the audit performs adds an inferential step. **Weakly supported** means the audit asserts a fact consistent with general knowledge but not separately documented in the audit's own files. A careful reader would have to take the assertion on the audit's word until the underlying check is filed. **Currently unsupported** means the public-facing summary phrases a check the audit's own evidence files do not document. The line should not appear in legal-grade or journalistic-grade citation without follow-up work.

Claim 1: The four Airdrie quadrants are within 8% of each other on every demographic measure the commission considers.

- *Source the minority cites in their published rationale*: Appendix E, p. 339 (R14, "Airdrie East") and the inventory's R3/R4/R18 growth-projection cluster. The minority does not, in fact, give a per-quadrant demographic-difference rationale in the published text; the audit's framing in `report_public.md` line 195 attributes the per-quadrant rural/urban-character argument to the minority commissioners as the implied basis for the four-way split. The minority's *explicit* rationale is growth and commuter-tie, not demographic heterogeneity.

- *Evidence the audit checked it against*: `findings/justification_tests_findings.md` Test 3, using `data/justification_test_inputs.csv` (StatCan 2021 Census, City of Airdrie CSD 4806016, total pop 74,100). The test compares the population arithmetic of the 4-way split against a 2-way alternative.
- *What the data shows*: The arithmetic shows a 2-way split is sufficient — each half ≈37,050, requiring only ~4,147 additional non-Airdrie residents to clear the 41,197 statutory floor. The 4-way split forces each quarter (~18,525) to absorb ~22,672 non-Airdrie residents, which is the structural source of the four cross-municipal hybrids the minority labels Calgary-Airdrie, Calgary-Foothills-Airdrie West, Calgary-Nolan Hill-Cochrane (north flank), and Airdrie East. The "within 8% on every demographic measure" line in `report_public.md` is **not directly documented in the audit's evidence files**: no per-quadrant demographic-comparison table (median income, age structure, dwelling type, immigration share, language) for the four minority Airdrie carves exists in `analysis/`, `data/`, or `analysis/methodology/`.
- *Verdict*: **Inconclusive on the line as written; Fail on the broader rationale.** The minority's split is unforced by population arithmetic (which the audit documents) but the specific "within 8% of each other on every demographic measure" sentence is a stronger claim than the existing files substantiate.
- *Documentation file*: `findings/justification_tests_findings.md` (Test 3); `data/justification_test_inputs.csv`.
- *Reproducibility*: `python analysis/scripts/justification_tests.py`. A reader can re-run the population test independently. The "8% demographic" sub-claim cannot be reproduced from the existing files and requires per-quadrant census tabulation as follow-up.

Claim 2: Cochrane's most common commuter destination is Cochrane itself. Among out-of-town commuters, Calgary-Centre is named more often than Foothills.

- *Source the minority cites in their published rationale*: Appendix E, p. 331 (R1): "residents move fluidly between jurisdictions for work, education, health care, and commerce." The minority frames the Calgary-Nolan Hill-Cochrane lasso as a commuter-tie district pairing Cochrane with the Nolan Hill / Sage Hill ward of NW Calgary specifically.
- *Evidence the audit checked it against*: Statistics Canada 2021 Census, table 98-10-0459-01 ("Commuting flow from geography of residence to geography of work by gender: Census subdivisions"), filtered to Cochrane CSD origin (DGUID `2021A00054806019`). Methodology in `analysis/methodology/cochrane_journey_to_work.md`. Output in `data/cochrane_journey_to_work.csv`.
- *What the data shows*: Of 8,550 Cochrane-resident workers with a usual place of work, **4,205 (49.2%) work within Cochrane itself**, **3,065 (35.8%) commute to Calgary CY**, 345 (4.0%) to Rocky View County, 185 (2.2%) to Canmore. The 49.2% within-Cochrane share

is the largest single destination by a wide margin. The "Calgary-Centre named more often than Foothills" sub-claim in the public summary is a **stronger statement than the table can support**: the public Census table publishes commute destinations at CSD granularity (Calgary CY as a single unit), not at electoral-district or ward granularity. Within-Calgary disaggregation is not in the public release. The methodology file flags this explicitly: the rationale is unevidenced at sub-CSD resolution rather than falsified.

- *Verdict*: **Fail on the within-Cochrane primacy; Inconclusive (leaning Fail) on the Calgary-Centre-vs-Foothills sub-claim.** The data falsifies the "fluid movement to Calgary" framing as a *primary* commute pattern (the largest flow is internal). The within-Calgary destination breakdown the public summary asserts is not in the audited dataset.
- *Documentation file*: `analysis/methodology/cochrane_journey_to_work.md`; `analysis/methodology/reference/minority_rationales_validation.md` §R1; `data/cochrane_journey_to_work.csv`.
- *Reproducibility*: StatsCan table 98-10-0459 is publicly downloadable at `https://www150.statcan.gc.ca/n1/tbl/csv/98100459-eng.zip`; the filter on DGUID `2021A00054806019` and the per-destination ranking are scripted in the methodology file. A reader can re-run end-to-end.

Claim 3: The Rocky Mountain House-Banff Park extension into Banff National Park has no agricultural land base ("ranching community of interest") it can refer to.

- *Source the minority cites in their published rationale*: Appendix E, p. 352 (R12). The minority invokes the s.15(2) exception for the Banff National Park extension on four grounds: north-south economic corridor along Highway 22; Rocky Mountain House as a hub for Clearwater County; division of regional Indian reserves from the nearest economic hub; and historical precedent of Banff NP portions in a west-central electoral division. The minority frames the broader district as a "ranching community of interest."
- *Evidence the audit checked it against*: `findings/justification_tests_findings.md` Test 2 (area arithmetic), §5.1.4 + §5.3.3 of this report (engineered-boundary signature E1–E3), and `analysis/methodology/banff_extension_population_check.md` (polygon-clipped DA population pull on the Banff extension portion). The 2019 predecessor area (Bill 33 shapefile attribute `Km2`) is 24,468 km², already 22 % above the s.15(2)(a) 20,000 km² threshold without the NP extension. The minority's own population figure for the extended district is 38,298 (−30.3 %), still outside the ±25 % band even with the extension.
- *What the data shows*: The NP extension is **not load-bearing for either area or population qualification** under s.15(2). On the population side, the polygon-clipped 2021-Census-DA pull at `analysis/methodology/banff_extension_population_check.md` finds **approximately 491 area-weighted residents** inside the extension polygon (not zero) — concentrated in the Lake Louise / Saskatchewan River Crossing / Nordegg corridor visitors-services area-weighted aggregation. The earlier "zero year-round residents" framing was an oversimplification of an inhabited-but-very-sparse polygon. On the

agricultural side, the *Canada National Parks Act* (R.S.C. 1985, c. N-14) statutorily prohibits agricultural tenure on national-park land, so a "no working ranches inside the park-land slice" finding is statutorily entailed for the in-park portion but **not separately verified** for adjacent Crown-land grazing-lease territory the extended district also touches. The minority's "ranching community of interest" rationale therefore has no agricultural land base it can refer to *inside the park slice*, but the audit does not file a check on grazing-lease territory adjacent to it.

- *Verdict*: **Fail on the s.15(2) area-necessity claim (documented); Fail on the "ranching community of interest" rationale for the in-park portion (statutorily entailed); Inconclusive on grazing-lease territory adjacent to the park (not separately tested).** The minority's stated rationale fails on what the audit has tested; the previous public-facing precision ("zero residents, zero ranches") was overstated and has been corrected to the more careful framing in `report_public.md`.
- *Documentation file*: `findings/justification_tests_findings.md` (Test 2); §5.1.4, §5.3.3 of this report; `analysis/methodology/reference/minority_rationales_validation.md` §R12; `analysis/methodology/banff_extension_population_check.md`.
- *Reproducibility*: The area arithmetic is reproducible from the 2019 Bill 33 shapefile (`data/alberta_2019_eds/EDS_ENACTED_BILL33_15DEC2017.shp`). The polygon-clipped population pull (~491 area-weighted residents) is reproducible from the methodology file's documented DA-intersection pipeline. The grazing-lease check on adjacent Crown land remains future work.

Claim 4: North Airdrie is suburban. Olds and Didsbury are rural service centres.

- *Source the minority cites in their published rationale*: Appendix E, p. 349 (R13): "extending the boundaries of the existing Olds-Didsbury-Three Hills south to include more portions of Rocky View County… keep established communities of interest together along the Highway 2 corridor between Red Deer and Airdrie." The minority frames the southern extension as community-of-interest continuity along Highway 2.
- *Evidence the audit checked it against*: `findings/justification_tests_findings.md` Test 1 (population arithmetic of the rural Olds/Didsbury/Three Hills/Mountain View/Kneehill core sums to 43,691 — already above the 41,197 floor without any Airdrie territory). `analysis/methodology/reference/minority_rationales_validation.md` §R13 cross-checks the Highway-2 continuity claim against the 2019 enacted Olds-Didsbury-Three Hills district (which already contains Olds, Didsbury, Carstairs, Three Hills, Crossfield without the Airdrie slice).
- *What the data shows*: The arithmetic shows the Airdrie slice is unforced. The "North Airdrie is suburban; Olds and Didsbury are rural service centres" sub-claim is a **descriptive characterization broadly defensible from public sources** (City of Airdrie 2024 Growth Report describes Airdrie as a Calgary-CMA satellite city; the Town of Olds 2024 municipal profile and Town of Didsbury directory describe both as agricultural-

services hubs in Mountain View County). The validation file's verdict on R13 is "PARTIALLY SUPPORTS" the continuity-along-Hwy-2 framing while flagging the Airdrie slice as CLOSED-FAIL on population math, not on demographic mismatch. Note: an earlier draft of `report_public.md` contained the phrase "Census-of-agriculture data says" (approximately line 198); that phrase has since been removed and the current public report does not assert a Census of Agriculture comparison. No Census of Agriculture extraction for North Airdrie / Olds / Didsbury exists in `analysis/` or `data/`; no such claim appears in the current published version.

- *Verdict*: **Fail on population necessity (documented); Weakly supported on the suburban-vs-rural-service-centre characterization (descriptively true but not separately filed).** The public summary has been corrected to reflect what the audit actually documents.
- *Documentation file*: `findings/justification_tests_findings.md` (Test 1); `analysis/methodology/reference/minority_rationales_validation.md` §R13.
- *Reproducibility*: The population arithmetic is reproducible end-to-end. The suburban-vs-rural-service-centre demographic comparison would require StatCan Census Profile dissemination-area extraction for the relevant CSDs plus a Census of Agriculture pull; neither is currently scripted.

Claim 5 (WITHDRAWN): Lethbridge-East / Lethbridge-Taber-Warner federal-boundary match.

- *Status*: **Withdrawn from the public summary and from the rationale-failure count.** Earlier drafts attributed to the minority a claim the Lethbridge split between Lethbridge-East and Taber-Warner matches a federal boundary retired in 2013. The methodology check at `analysis/methodology/lethbridge_federal_boundary_check.md` established the minority report does not in fact make this federal-boundary claim anywhere in Appendix E. The R1–R18 rationale inventory in `analysis/methodology/minority_rationales_inventory.md` contains no per-Lethbridge federal-boundary rationale, and the school-division coherence file (`analysis/methodology/school_division_coherence.md`) lists Lethbridge-East and Lethbridge-Taber-Warner as minority hybrids without a federal-boundary basis. There is no underlying source for the audit's earlier "the federal boundary they claim to match was retired in 2013" line.
- *Disposition*: The claim is removed entirely from `report_public.md` and from this monograph's rationale-failure count. The headline arithmetic is updated from "six of seven fail" to "five of six fail." No replacement claim is substituted; the Lethbridge area is left out of the contested-rationale list because the minority did not file a published rationale for it the audit could check.
- *Documentation file*: `analysis/methodology/lethbridge_federal_boundary_check.md` (withdrawal note).
- *Lesson*: This withdrawal is recorded as a discipline correction in the audit's favour. The previous formulation of the seventh claim was a fabrication-by-paraphrase: the audit

attributed a specific factual claim to the minority commissioners they did not make, and the absence of an underlying source meant a hostile reviewer would have caught the overstatement immediately. Pre-emptive withdrawal preserves evidentiary discipline at the cost of the round "six of seven" framing.

Claim 6: The shared school division (Red Deer-Sylvan Lake) contains 8% of Red Deer's school-aged population.

- *Source the minority cites in their published rationale*: Appendix E, p. 351 (R11): "By placing rural populations in the same electoral division as urban communities in the Hwy 2 corridor, it is possible to achieve effective representation and overcome challenges with dividing communities of interest existing in previous electoral divisions" — including a "go to school" reference invoked alongside the urban-rural pairing.
- *Evidence the audit checked it against*: Alberta Education school-authority boundary maps (`https://www.alberta.ca/alberta-school-division-maps`) cross-referenced in `analysis/methodology/school_division_coherence.md` and `analysis/methodology/reference/minority_rationales_validation.md` §R11. Sylvan Lake CSD (pop 15,995) is in **Chinook's Edge School Division No. 73**; the City of Red Deer (pop 100,844) is served by **Red Deer Public Schools** and **Red Deer Catholic Regional Schools**. These are separate, jurisdictionally-distinct divisions with separate trustee elections and non-overlapping catchments. The claim "shared school division" is geographically false on the boundary maps. **Commuter tie (added 2026-06-13):** StatsCan 98-10-0459 origin-CSD data for Sylvan Lake (`data/outputs/sylvan_lake_journey_to_work.csv`) confirms 54.4% of Sylvan Lake out-commuters (1,520 of 2,795) work in the City of Red Deer — the dominant non-local destination. The "where they work" portion of R11 is **supported** at CSD level; the "go to school" portion remains contradicted.
- *What the data shows*: No Red Deer school-age student attends a Chinook's Edge school as part of the standard public catchment, and vice versa. The "8% of Red Deer's school-aged population" line in `report_public.md` line 200 is a **stronger numerical claim than the audit's files document**: no file extracts Red Deer's 5–17 population from StatCan or Alberta Education enrolment records and computes the share served by Chinook's Edge specifically. The school-coherence file's verdict is qualitative ("CONTRADICTS — different divisions"); the "8%" figure is not separately filed.
- *Verdict*: **Fail (on magnitude is correctly summarised, but the 8% figure is not separately documented).** The qualitative verdict (different divisions, no shared catchment) is strongly supported. The specific 8% magnitude is **weakly supported** in the audit's current files.
- *Documentation file*: `analysis/methodology/school_division_coherence.md`; `analysis/methodology/reference/minority_rationales_validation.md` §R11; `analysis/methodology/reference/sylvan_lake_innisfail_journey_to_work.md`; `data/outputs/sylvan_lake_journey_to_work.csv`.

- *Reproducibility*: The qualitative division-mismatch is reproducible from Alberta Education's published boundary map. The commuter flow is reproducible via `analysis/scripts/fetch_journey_to_work_csds.py` against StatsCan 98-10-0459. The 8% magnitude requires Red Deer 5–17 population + Chinook's Edge enrolment served from Red Deer city neighbourhoods; neither is currently scripted. **Recommend either filing the 8% calculation or softening the public-facing language to "the two cities are in different school divisions; no Red Deer student attends a Chinook's Edge school in the standard catchment."**

Claim 6 (was Claim 7): For St. Albert-Sturgeon, the majority map and the minority map independently arrive at the same two-district structure under the Act's constraints — when two independent drafting teams converge, the configuration is constraint-forced rather than engineered.

- *Source the minority cites in their published rationale*: The minority's St. Albert-Sturgeon configuration appears in the inventory (`analysis/methodology/school_division_coherence.md` §St. Albert-area hybrid) but **does not have a per-district published rationale in the R1–R18 inventory** (`analysis/methodology/minority_rationales_inventory.md`). The audit's "Stands" verdict is therefore not a direct test of the minority's stated reason; it is an *audit-internal verdict based on a convergence test*: where two independent drafting teams (the three majority commissioners and the two minority commissioners) reach the same configuration under the same constraints, the configuration is treated as constraint-forced rather than designed.
- *Evidence the audit checked it against*: The §5.6 cross-map structural comparison documents the majority map and the minority map both produce a single St. Albert ED with the same two-district St. Albert / St. Albert-Sturgeon structure — see the §5.6 city-wide split table where St. Albert reads "1 ED" on both maps. `analysis/methodology/school_division_coherence.md` documents St. Albert Public Schools serves only the City of St. Albert; Sturgeon Public Schools serves Morinville, Legal, Gibbons, and Sturgeon County. The City of St. Albert (pop ~70,000) is below the 68,661 ceiling but above the 41,197 floor, so it could in principle stand alone — but its commute and service geography binds it tightly to the Edmonton-corridor districts on its south boundary. `findings/claim_significance_analysis.md` row 7 documents no engaged citizen submission proposes a clearly distinct minority alternative for St. Albert-Sturgeon (0 supporters for the minority variant, 2 supporters for the majority's St. Albert-Sturgeon name — label-ambiguity caveat noted).
- *What the data shows*: The convergence test (both teams independently arriving at the same configuration) is observable directly from the §5.6 cross-map structural comparison and from the school-coherence file. This is a stronger evidentiary basis than a counterfactual non-existence claim, because it does not require the audit to prove no other configuration exists — it requires only two independently-drafting teams under

the same constraints reached the same one. The earlier "no other configuration satisfies both the community-of-interest and the ±25% rule simultaneously" framing was a counterfactual non-existence claim the audit could not formally prove; the convergence framing is what the documented evidence actually supports.

- *Verdict*: **Stands on the convergence framing.** Both the majority and the minority independently produced the same St. Albert-Sturgeon two-district structure under the Act's constraints; the audit treats this convergence as evidence the configuration is constraint-forced. The earlier "no other configuration" line has been retired in favour of this framing.
- *Documentation file*: §5.6 (city-wide split table — St. Albert row); `analysis/methodology/school_division_coherence.md`; `findings/claim_significance_analysis.md` row 7; `analysis/methodology/reference/minority_rationales_validation.md` (St. Albert-Sturgeon not separately listed).
- *Reproducibility*: The convergence observation is reproducible from the §5.6 cross-map structural comparison. A constraint-search extracting all St. Albert-Sturgeon configurations from the 2,000,000-map ensemble (Run #6, §5.4.6) would supplement the convergence finding with an explicit non-existence test, but is not required for the convergence-based verdict.

Strength-banding summary across the six claims (Claim 5 Lethbridge withdrawn).

#	CLAIM SHORT FORM	VERDICT IN PUBLIC SUMMARY	STRONGEST EVIDENCE BAND ACTUALLY FILED	SU
1	Airdrie 4-way split	Fail	Strong on population necessity	"ξ fi
2	Cochrane commuter primacy	Fail	Strong on within-Cochrane share; Strong on Calgary-as-largest-out-of-town	C n
3	RMH-Banff NP extension	Fail	Strong on s.15(2) area non-necessity; ~491 area-weighted residents (not zero) per polygon-clipped DA pull; "no working ranches" statutorily entailed for in-park slice via *Canada National Parks Act*, unverified for adjacent Crown-land grazing-lease territory	E fr m b
4	ODH reaching N Airdrie	Fail	Strong on population non-necessity	A "(p c o S c n

#	CLAIM SHORT FORM	VERDICT IN PUBLIC SUMMARY	STRONGEST EVIDENCE BAND ACTUALLY FILED	SU
~~5~~	~~Lethbridge federal-boundary match~~	**WITHDRAWN**	n/a — the underlying minority claim cannot be located in the report; see `analysis/methodology/lethbridge_federal_boundary_check.md`	R c
6	Red Deer-Sylvan Lake school division	Fail (on magnitude)	Strong on qualitative division mismatch	"8 p
7	St. Albert-Sturgeon	Stands (constraint-forced)	Strong on the convergence test (majority and minority independently produce the same two-district structure — §5.6)	E c re

Reading the table. *The audit's headline arithmetic is now "five of six of the minority's published reasons fail under check." Of the six claims still in the count, four (Claims 1, 2, 3, 4) have a strong audit-documented backbone with one or two sub-claim corrections noted. One (Claim 6) has qualitative Fail support with a magnitude sub-claim still pending; one (Claim 7, St. Albert-Sturgeon) has a documented Stands verdict on the convergence test. Claim 5 (Lethbridge) has been removed entirely.*

Disposition for the public summary. The "five of six fail" headline is exactly what the filed evidence supports. The previous "six of seven" framing is retired because (a) the seventh claim (Lethbridge) was a fabrication-by-paraphrase no underlying minority text supports and (b) the seventh-as-stands claim (St. Albert-Sturgeon "no other configuration") was a counterfactual non-existence claim stronger than the filed evidence. The convergence framing replaces both: the configuration St. Albert-Sturgeon stands on is the one both independent drafting teams produced under the same constraints.

Disposition for the remaining sub-claim corrections (Claims 1 and 6; Claim 4 resolved). The public-facing precision in Claims 1 and 6 ("within 8% of each other on every demographic measure"; "8% of Red Deer's school-aged population") is a stronger statement than the corresponding evidence file documents. **The honest options are**: (a) file the supporting calculation as a per-claim methodology note before publication; (b) soften the public-facing language to match the actual filed evidence ("the population arithmetic does not require it"; "the two cities are in different school divisions, with no shared catchment"). Option (a) preserves the quantitative precision; option (b) preserves evidentiary discipline at the cost of some rhetorical force. Claim 3 has been corrected in `report_public.md` to the more careful "no agricultural land base it can refer to" framing supported by the *Canada National Parks Act* and the polygon-clipped population pull. Claim 4's overstated "Census-of-agriculture data says" phrase has been removed from `report_public.md` (confirmed 2026-06-13); the current public report does not assert a Census of Agriculture comparison for Olds/Didsbury.

Bottom line for evidence-trail integrity. The "five of six" pattern is solid on the rationale-failure dimension the audit's files document. The previous "six of seven" arithmetic was withdrawn after primary-source verification could not locate the underlying Lethbridge federal-boundary minority claim, and the previous "no other configuration" counterfactual on St. Albert-Sturgeon is replaced by the convergence framing. The remaining two sub-claim follow-ups (per-quadrant Airdrie demographic comparison; Red Deer-Sylvan Lake school-age magnitude) are evidentiary precision upgrades, not load-bearing for the headline. The Olds-Didsbury Census-of-Agriculture concern has been resolved by removing the overstatement from the public report (confirmed 2026-06-13).

5.9.7 Population data currency and the 2026 census

The Lunty committee is mandated to produce a 91-seat map by November 2, 2026. The mandate has an unaddressed population-data consequence.

Statistics Canada conducts a mandatory long-form census every five years under the *Statistics Act* (R.S.C. 1985, c. S-19). The 2021 Census of Population was collected May 11, 2021; preliminary national-level counts were released February 9, 2022; constituency-level dissemination-area data used for redistricting did not become available until the 2022–2023 release waves. On the same cadence, the **2026 Census** will be collected in May 2026. Preliminary provincial population counts are expected in approximately Q1 2027; dissemination-area-level data adequate for the Electoral Boundaries Commission Act's population-parity requirements would not realistically be certified and released before mid-to-late 2027.

The *Electoral Boundaries Commission Act* s. 14(1)(a) requires the commission — and by extension any body exercising the commission's statutory function — to use the "most recent decennial census." If the 2026 census data is not yet certified and released by Statistics Canada at the time the committee finalises its map, the committee must use 2021 data. There is no statutory mechanism permitting voluntary delay to await uncertified census results. The November 2026 target date therefore forecloses access to the 2026 census regardless of intent.

This matters most for the ridings where the audit's structural findings concentrate. The Calgary–Airdrie corridor — Airdrie (population 85,805 at the July 2024 municipal census; City of Airdrie, 2024), Cochrane, Chestermere, and Calgary's northwest quadrant — is the fastest-growing sub-region in Canada. Alberta added approximately 200,000 residents in the 2021–2023 period alone (Alberta Treasury Board and Finance, 2023 population estimate). Boundaries drawn in November 2026 on 2021 census populations for seats contested in 2027 and held on those boundaries through potentially 2031 will be built on population data 10 years old by the time the electoral cycle they govern ends. Population equality in the highest-growth ridings — the same ridings where the commission's two proposals diverge most sharply — will degrade fastest.

Policy implication. If the government's objective is redistricting serving effective representation through a full electoral cycle, a November 2026 completion date imposes a foreseeable cost: the map will begin with the most accurate population baseline available but will not be anchored to the 2026 census becoming the legal baseline for the commission cycle following it. No redistricting body can escape census timing constraints; the commission faced the same constraint. The observation here is forward-looking only:

the committee and the legislature are aware 2026 census data will become available approximately one year after the committee's mandate ends, and the communities most affected by the audit's structural findings are precisely those where population drift from the 2021 baseline will be largest. This is not a legal objection to the November 2026 map; it is a relevant fact for any assessment of the map's durability.

5.10 Forward-looking AI-use recommendations for the Lunty committee

Audience and scope. The Special Select Committee of MLAs chaired by Brandon Lunty (MLA, Leduc-Beaumont) carries the November 2, 2026 mandate to produce a 91-seat Alberta map. This subsection is a technical, non-partisan framework for how the committee — or its advisory panel, or any future Alberta redistricting body — can use AI tools responsibly if it chooses to. It does not take a position on whether the committee *should* use AI tools. It does take a position on what disciplines any such use must follow. The position is not novel; it is the same methodological discipline this audit applies to itself (reproducibility, transparency, pre-registration, named accountability).

Why this section exists in this paper. The April 16 "AI academy" remark (Premier Smith, Calgary Journal 2026-04-21) signalled AI use is on the table for the committee. An audit documenting procedural departures from Canadian independent-commission practice (§5.9.2) without addressing the AI-use dimension would be incomplete — AI use can compound or mitigate the independence concerns already raised. A boundary map has legal and constitutional effect for up to a decade; the workflow producing it is part of its defensibility.

Seven core principles. Each is a discipline the committee's workflow must enforce, regardless of which tool is used:

1. **Humans decide; AI assists.** Every boundary in the final map must have at least one named committee member or advisor of record who will answer for it in public. "*The algorithm converged*" is not a defence.
2. **Transparency over opacity.** Every prompt, input dataset, model version, random seed, and output used in drafting must be logged, timestamped, and publicly released with the final report. Redact for personal privacy only, not for reputation.
3. **Reproducibility over one-shot generation.** Any AI output used in the map must be reproducible by an independent third party given the same inputs. Stochastic methods (MCMC, simulated annealing, LLM completions) require published seeds and pinned model versions.
4. **Pre-registration over post-hoc justification.** Evaluation criteria (population equality targets, COI priorities, compactness thresholds) must be published before any map draft is circulated, and treated as gates the map must clear — not as framing added after the draft.

5. **Independence tests over single-model reliance.** Any AI output on a load-bearing decision must be generated by at least two independent tools. Disagreements between tools are signals for human adjudication, not noise to average away.
6. **Falsifiability over rhetoric.** Every claim attached to the final map ("this preserves community of interest," "this reflects commuter ties") must be stated in a form naming the dataset which would confirm or refute it. Claims cannot be falsified must be labelled as opinion.
7. **Auditability over ease of use.** The cheapest workflow (an MLA types a question into a chatbot) is also the least auditable and therefore inadequate for a constitutional decision. The workflow making every step reproducible, even when slower, is the one the committee must adopt.

Specific technical recommendations (abbreviated; full text in the companion document).

- **Generation.** Use purpose-built redistricting software (GerryChain, Dave's Redistricting App, Maptitude for Redistricting), not a general-purpose chatbot. Publish the ensemble, not only the chosen map — a single map says nothing about whether the committee's choice is typical or an outlier within the constraint set. Do not use an LLM to write boundary descriptions directly; LLMs hallucinate street names and invert cardinal directions.
- **Evaluation.** Apply the same partisan-fairness metrics used in the academic literature (efficiency gap, mean-median difference, declination, seats-votes curve) against vote data from the three most recent provincial elections, not one. Run a Chen-Rodden natural-packing test: if the proposed map falls outside the 95th percentile of a constraint-bound ensemble in either direction, it requires a stated, testable explanation. Compute and publish Polsby-Popper and Reock compactness on every district.
- **Public input.** Publish the input corpus (redacted for personal information) so independent researchers can reproduce any AI-assisted theme coding. Do not rank submissions by length or signature count — electoral boundary decisions weigh arguments, not volume. Flag the *absence* of input as well as its presence.
- **Justification drafting.** Every factual claim in AI-drafted prose must be human-verified against a cited source. Do not let an AI generate the committee's interpretation of *Saskatchewan Reference* [1991] or s.15(2) — Canadian constitutional-law summaries from general-purpose LLMs are unreliable and occasionally invert the holding. Publish the prompts used for any passage appearing in the final report.
- **Data currency (census cycle lag).** Maintain the 2021 census as the statutory baseline for every s.15(2) test and every ±25 % determination (the Act requires it). *In parallel*, publish a "Plan B" sensitivity table using Alberta Treasury Board and Finance quarterly estimates. Report any ED whose legal-window status disagrees between Plan A and Plan B. AI's role in Plan B is limited to routine data-aggregation acceleration; every cell must trace back to a published TBF, StatsCan, or municipal-census figure. Full rationale at

`findings/cycle_lag_analysis.md` +
`docs/ai_use_recommendations_for_committee.md` §2.5.

What AI should not do — even with human review.

- Final decisions on district boundaries (a committee member signs each district, not a model).
- Weighing constitutional considerations (s.3 Charter analysis, *Saskatchewan Reference* application, s.15(2) criteria).
- Determining community of interest (lived experience is not retrievable from a census table).
- Drafting the committee's reasoning on contested districts (Airdrie, Chestermere, Nolan Hill–Cochrane, RMH–Banff Park, and any new 91-seat flashpoint).
- Assessing partisan intent (evidence cannot distinguish motive; AI cannot either).

Nine-item public-disclosure checklist. Before publishing the final map, the committee must be able to answer each of the following publicly: (1) which AI tools were used, by name and version; (2) what prompts and inputs were given to each; (3) which outputs were used in the final map and which were discarded; (4) what random seeds, configurations, and parameters were set; (5) which committee member or advisor takes personal responsibility for each district's boundary; (6) what evaluation criteria were established before drafting began; (7) what ensemble of alternative maps was generated and where the final map sits in the ensemble; (8) what Charter and statutory analyses were performed and by whom; (9) which claims in the final report are AI-drafted and human-verified, which are human-drafted, and which are AI-drafted without human verification. A committee able to answer all nine publicly has used AI responsibly. A committee unable to do so has not, regardless of tooling sophistication.

Three risks specific to the April 2026 context.

1. **Independence-washing.** The April 16 mandate includes an advisory panel with the three-party chair-plus-two structure. If the panel uses AI tools whose outputs are filtered through a UCP-majority MLA committee, the panel's independence is nominal. Remedy: publish the panel's AI outputs *in full* before the MLA committee acts on them, and require a written explanation for any substantive committee deviation from panel recommendations.
2. **Legitimacy-by-association.** An AI tool's output inherits only as much legitimacy as its inputs and operators provide. A map presented as "algorithmically generated" without disclosed prompts and constraint sets is borrowing technical legitimacy it has not earned. Remedy: the nine-item checklist above as a minimum disclosure standard.
3. **Acceleration past deliberation.** AI can generate thousands of candidate maps in hours and short-circuit the slow deliberative work a committee is supposed to do. Remedy: a proposed boundary is not adopted on the strength of a model run alone, even a well-designed one. Speed is not a virtue in redistricting.

Relationship to §5.9's procedural findings. The seven principles above do not replace §5.9's documented procedural departures from Canadian comparator practice; they supplement them. A committee adopting the AI-use disciplines and *also* operating under the April 16 MLA-committee-directed drafting structure retains the §5.9.2 procedural critique regardless of its AI workflow. AI discipline is necessary but not sufficient for a defensible process. The full AI-use document (211 lines, with reference implementations and dataset pointers) is at docs/ai_use_recommendations_for_committee.md. Suggested citation: Conner, W., *AI-use recommendations for the Alberta Electoral Boundaries MLA Committee*, Alberta Electoral Boundaries Audit, v0.1, 2026-04-22.

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6. Discussion

This section interprets the results of §5 against the prior work surveyed in §2 and forward-refers to §7 (Limitations and Falsifiability). The six-dimensional synthesis is presented first; the scope-discipline calibration and the three inherited qualifications follow.

DIMENSION	2019	MAJORITY 2026	MINORITY 2026	DIRECTION OF MINORITY SHIFT
§5.1.1 Population MAD from avg	2,010 (87-*seat, 2017 pops — different scale from* 2026)	3,180	**4,707** (+48%)	wider dispersion
§5.1.2 Calgary Zone A − Zone B gap	(not run)	+0.36%	**+12.20%**	packing signal
§5.1.2 robustness (2023 winner-based)	(not run)	+0.39%	**+7.71%**	survives robustness check
§5.1.3 Rest-of-province mean population	(not run)	52,281	50,336 (−3.9%)	population-weighting shift below provincial average (*Carter*-permissible unless unjustified; flagged 2026-06-12 per T1.7 Referee #8 — Carter [1991] 2 SCR 158 *upheld* Saskatchewan's deliberate rural weighting, so "rural overrepresentation" was not an appropriate gerrymander-signature label)
§5.1.4 s.15(2) invocations statutorily legitimate	—	3 / 3	3 / 3	re-audit 2026-04-23 under corrected thresholds

DIMENSION	2019	MAJORITY 2026	MINORITY 2026	DIRECTION OF MINORITY SHIFT
§5.1.4 engineered-boundary signature (extension chosen over populated alternatives)	0	0	1 (RMH-Banff Park)	substantive E2 test (see §5.3.3)
§5.2.1 Standard EG (Phase 4C)	−2.64%	+0.04%	**+3.96%**	+3.92 pp more UCP-favourable (Phase 4C; S-M positive=UCP)
§5.2.2 blend sensitivity range *(superseded by Phase 4C)*	—	*(blend model retired)*	*(blend model retired)*	—
§5.2.1 Mean-median gap (NDP)	−2.22 pp	−3.64 pp	**+1.03 pp**	+4.67 pp shift more UCP-favourable (minority p99.98; majority p2)
§5.2.1 NDP seats at 50/50	46	**48**	43	minority −5 NDP seats vs majority at tied provincial vote
§5.8.2 Visible spatial anomalies	—	0 (Calgary only)	3 (all confirmed)	three anomalies
§5.8.4 Airdrie splits	—	2 EDs	4 EDs	double split
§5.9 Procedural	—	Commission-adoption path	Legislature-directed committee drafting	departure from comparators

Six correlated dimensions of evidence (not all independent — see §4.3.1) point in the same direction. Within the partisan-bias dimension, three of four pre-registered metrics (mean-median p99.98, declination p98.79, seats@50/50 p99.99) individually exceed the p95 pre-registered outlier threshold against the exploratory canonical ensemble (see §1.2 for the exploratory/confirmatory distinction); EG at p94.4 is directionally aligned but sub-threshold. The structural dimensions (population dispersion, Calgary zone gap, community splits, anomaly count) do not have equivalent point-estimate significance thresholds; the procedural dimension is qualitative. Together, the directional consistency across all six is the finding — the minority proposal, relative to the majority, shows more structural irregularities and, under 2023 voter geography, higher measured partisan-bias by three individually flagged metrics and one directionally aligned sub-threshold metric, across six measurement frameworks. (Note: the pre-Amendment-10 formulation read "None individually crosses a statistical significance threshold"; Amendment 10's sign correction changed declination from p1.21 NDP-tail to p98.79 UCP-tail, placing three of four above p95; corrected per T1.7 R2 Ref #5 2026-06-12.)

A 2019 enacted-map baseline (§5.4.10) adds structural context to this six-dimensional picture. The 2019 map was itself a mild joint-space outlier on the 1,010,000-plan canonical ensemble (Mahalanobis D^2=12.75, empirical p=0.013). The majority 2026 proposal retreats toward the constraint-bound expectation (D^2=7.85, p=0.097), consistent with a normalizing commission process. The minority amplifies the 2019 anomaly 2.6-fold (D^2=32.67, $p=1.40\times10^{-6}$) — measuring the void between what the two proposals do from a shared departure point.

Scope discipline and small-magnitude calibration. The six-dimensional framing follows the four-axis redistricting-audit discipline of Altman and McDonald (2011) and the consistency-across-N methodology of Katz, King, and Rosenblatt (2020): when single dimensions are underpowered, cross-dimensional agreement is the inferential artefact, not any individual magnitude. Each dimension's magnitude is small by design — the audit is not claiming a single five-alarm effect. Terms used in this paper carry calibrated meaning: *measurable* means "the computed statistic's sensitivity interval does not include zero at the stated confidence level"; *directional* means "the sign is consistent across all runs" without magnitude claim; *systematic* means "the same direction appears across multiple correlated dimensions (not all independent — see §4.3.1)." These are not synonyms and are not interchangeable. A hostile reader entitled to read the audit as "the paper called small differences patterns" receives the following response: the patterns are defined precisely, apply to six correlated tests, and persist across 90.5% of Monte Carlo samples (seed 42, N = 2,000), across 338Canada's April 2026 polling input, and across a symmetric test-selection audit revealing additional minority-map patterns (Lethbridge and Red Deer 4-way splits, §5.6) not present in the majority.

Multiple-comparison posture. A hostile reviewer may legitimately observe this audit runs on the order of 20 distinct statistical tests against overlapping 2023 vote data (four partisan-bias metrics × two maps × three blending weights ≈ 24 tests, plus the non-partisan-bias dimensions), and ask whether a family-wise-error-rate (FWER) correction has been applied. The paper **explicitly does not apply a Bonferroni or Benjamini-Hochberg correction**, because the correction assumes the tests are independent significance claims — and this audit does not advance individual significance claims *as formal pre-registered hypothesis tests at* α = 0.05. (Reporting that individual metrics exceed a pre-specified ensemble percentile threshold is a descriptive screening claim, not a classical hypothesis test; the distinction is explained in §4.3.1.) Every per-metric per-map result is reported inside a Monte Carlo sensitivity interval already bracketing its own parametric uncertainty (§5.2.3, §5.2.2, §5.4), and the aggregate synthesis is framed as directional consistency across correlated dimensions, not as a count of independent rejections.

Applying Bonferroni to a consistency frame inflates false-negative rates on related measurements without correcting the actual inferential artefact (cross-dimensional agreement); the audit therefore uses the consistency-across-correlated-tests framing of Katz, King, and Rosenblatt (2020) and the ensemble-reporting discipline of Altman and McDonald (2011) in preference to a multiple-comparison correction. Readers who prefer an FWER-adjusted lens should note (a) post-Amendment-10: three of four individual partisan-bias metrics (MM p99.98, declination p98.79, seats@50/50 p99.99) exceed the uncorrected p95 threshold; under 4-comparison Bonferroni (threshold p98.75), MM and s50 clearly survive and declination sits at the boundary; (b) the audit's positive claim does not depend on individual-metric significance — the six-dimensional directional consistency and the Mahalanobis joint-outlier score ($p=1.40\times10^{-6}$) remain the core framing regardless of correction strategy. (Pre-Amendment-10 formulation said "the individual partisan-bias metrics do not cross their own uncorrected 95th-percentile sensitivity interval"; Amendment 10 corrected this 2026-06-12.)

Three qualifications inherited from the stress-test pass narrow this synthesis:

- Under the Chen-Rodden (2013) natural-packing framing (see §5.2.5), some portion of the minority-to-majority partisan-bias gap is not attributable to engineering — it reflects how any neutral map interacts with Alberta's urban-NDP / rural-UCP geography. This lowers the claim from "the minority was engineered against the NDP" to "the minority produces more UCP-favourable results under 2023 voter geography." The structural findings (§5.1 population equality, §5.1.2 Calgary zone gap, §5.8.2 visible anomalies, §5.8.4 community splits, §5.9 procedural) are not affected by the natural-packing caveat because they measure geographic and procedural properties the natural-packing argument does not address.
- Under RT3 (cross-election stability), the vote-based asymmetry flips sign when 2019 votes replace 2023 votes. The six-dimension synthesis rests mostly on the structural dimensions; the single vote-based dimension (§5.2) is qualified accordingly.
- Under the submission-archive verification (§5.9.4), the procedural concern rests primarily on the two configurations without documented public support (Airdrie 4-way, Nolan Hill-Cochrane) rather than on the chair's full five-item sweep.

Direction disagreement: three-layer reconciliation. A careful reader of §5.2 and §5.4 will notice the partisan-bias metrics do not all point the same way, and the direction can flip depending on which vote input is used. This paragraph names each layer explicitly so they are not mistaken for methodological inconsistency.

Layer 1 — metric agreement under 2023 votes (Amendment 10 sign correction applied). B2 (efficiency gap), B3 (mean-median), B4 (seats at 50/50), and B6 (declination) all show the minority 2026 map as more UCP-favourable than the majority under Phase 4C and 2023 vote input: the minority produces substantially more NDP vote waste (EG +3.96% vs +0.04%), a more NDP-packed mean-median distribution (+1.03 pp vs −3.64 pp, placing minority at p99.98 in the UCP-tail), gives NDP five fewer seats at a tied provincial election (43 vs 48), and shows a corrected Warrington declination of +0.077 (p98.79 UCP-tail). Earlier drafts of this paragraph said "B6 declination disagrees" / "minority at p2.17 NDP-tail" — that was an artefact of a swapped-operand bug at `mcmc_ensemble.py:215` corrected 2026-06-12 (Amendment 10 in `findings/pre_registration_amendment_log.md`). All four metrics now agree on the UCP-favoured direction; the asymmetric-packing reading at §6.2.2 is restated accordingly.

Layer 2 — vote-input dependence. Under 2023 votes the minority gives one more UCP seat than the majority at a tied election (minority 52 UCP / 37 NDP, majority 51 UCP / 38 NDP). 338Canada's April 2026 per-riding projection (§Track J), run through the same audit crosswalks, gives the minority one more NDP seat than the majority (minority 23 NDP, majority 22 NDP). Both differences are at the 1-seat noise floor and are consistent with two interpretations: the minority's marginal-seat calibration is efficient near the swing point the boundaries were designed for (roughly 2023 vote levels) but produces NDP-efficient results when the UCP advantage expands past the point; or both differences are within crosswalk estimation error and are not meaningfully distinguishable.

Layer 3 — magnitude threshold (Amendment 10 declination sign correction applied). Under 2023 vote input and the 1,010,000-plan canonical ensemble (§5.4.9), three individual partisan-bias metrics carry outlier flags on the minority map: mean-median at p99.98 (UCP-favoured tail), **declination at p98.79 (UCP-favoured tail)**, and seats@50/50 at p99.99 (individual flag reinstated; n_eff=1,495, ESS-adjusted lower bound ≈p98, above p95 threshold). Efficiency gap at p94.4 remains below the p95 threshold; the EG flag is withdrawn (directionally aligned with the other three, but not a tail flag). The majority is not flagged in the UCP-advantage direction on any metric (EG p15, declination p20.4, seats@50/50 p77.8); it sits at p0.92 on mean-median on the NDP-cracking tail (pre-registered Row 8, expected direction). The aggregate finding rests on the joint Mahalanobis result ($p=1.40\times10^{-6}$) and on directional consistency across the structural dimensions (§5.1, §5.8, §5.9).

Summary of the direction picture. Under structural tests, the direction is unambiguous and multi-confirmed. Under partisan-bias tests, all four standard metrics (EG, mean-median, seats@50/50, declination) agree that the minority is more UCP-favourable under 2023 votes (declination sign-convention corrected 2026-06-12 per Amendment 10; see Layer 1 above). The sole directional reversal is on the crosswalk-based seat count at April 2026 polling levels (§Track J: 338Canada April 2026 projection gives the minority one more NDP seat than the majority). A reviewer citing 338's April 2026 numbers as evidence the minority is the *fairer* map is making a legitimate point about one layer of the evidence; the audit does not suppress the layer and reports it in §Track J. The audit's overall position is the structural dimensions are the load-bearing evidence, and on those dimensions the direction is stable.

6.1 How to interpret these findings — NP-hardness, statistical improbability, and precision as armor

The results above are NOT findings of "the correct map was X and the Commission drew something different." The redistricting problem is **NP-hard**: the Alberta *Electoral Divisions Act* constraint set (±25 % population deviation + contiguity + compactness + community of interest + Indigenous effective representation under s.15(2) + public-hearing input) does not admit a unique optimum. No mathematical procedure produces *the* correct Alberta map; any map satisfying the constraints is one of an enormous family of legal maps, and the space of legal maps contains trade-offs — preserve COI at Airdrie at the cost of a slightly wider population dispersion, or tighten population equality at the cost of splitting Airdrie. These are not errors; they are **choices**. Fairness in this domain is not a discovery; it is a negotiation.

The audit therefore does not say **"this map is wrong."** It says "**this map is statistically improbable within the constraint set, and the improbability has a measurable direction**." Three concrete translations of the main results into the language:

- **Airdrie fragmentation (§5.3.2).** Of the large family of ReCom-legal Alberta maps satisfying population deviation, contiguity, and compactness, the overwhelming majority preserve Airdrie as a 2-ED community (Airdrie's population of approximately 85,805 at 2024 vintage sits at 1.56× provincial quota [City of Airdrie, 2024]; the statutory ±25 % band requires *at least* a 2-way split and permits a clean 2-way split, corrected 2026-06-12 per T1.7 R2 Ref #11 — earlier drafts erroneously said "a single-ED draw or a clean 2-way split"; single-ED is mathematically infeasible at the 2024 vintage). The minority map is one of a small minority of constraint-legal maps splitting Airdrie across four EDs. **The Commission could have kept Airdrie as a 2-ED community. They chose 4-way, among many legal alternatives.** The audit's question is not "why isn't Airdrie whole?" It is: "given the constraint set permitted a clean 2-way, why was a 4-way split selected?"
- **Population dispersion (§5.1).** The Median Absolute Deviation of ED population from the provincial quota is 3,180 on the majority and 4,707 on the minority — a 48 % wider dispersion. The constraint set permits both: the ±25 % population deviation band leaves room for either. **The Commission did not have to draw a map with 48 % wider dispersion; they chose to, among many constraint-legal alternatives.** (*The municipal-anchoring example previously listed in this position did not survive canonical recomputation — both maps fall within the Canadian comparator norm. See §5.8.5.*[4])
- **MCMC constraint-bound expectation (§5.4).** The ReCom ensemble places the minority map at the p99.98 tail on mean-median (canonical 1,010,000-plan run; §5.4.9). Interpretation: the constraint set admits 1,010,000 legal plans on the canonical ensemble; the minority map sits further from the mean-median median than ~99.98 % of them. **The Commission was not forced to draw a p99.98-tail map. They produced one, among hundreds of thousands of legal alternatives closer to the constraint-bound expectation.**

In every case, the audit's claim is the same shape: the Commission paid a measurable **fairness cost** to get the specific map they produced, where "fairness cost" means the number of statistical standard deviations (or the ReCom-ensemble percentile, or the anchoring ratio) by which the chosen map is displaced from the constraint-bound expectation. **This is the audit's only substantive claim.** It does not claim the minority is "wrong"; it claims the minority is improbable by a measurable amount, and the improbability has a direction.

6.2 Author's verdict — was either map a gerrymander?

This subsection departs from the audit's measurement-only voice and offers the author's stated opinion on the question motivating this work. It is presented as opinion, not as a peer-reviewed conclusion. The author's prior — UCP-disinclined Alberta voter — is disclosed in the Author Disclosure block in §1, along with the three findings retained in this paper running *against* the prior. Readers who weight the audit only on its measurable claims should stop at §6.1 and treat this subsection as ignorable.

A note on what "verdict" means here. The audit cannot judge intent; intent is not observable from public data, and the §1 author-disclosure block already commits the paper to this restraint. What the audit *can* do is observe and quantify *effects*, and report the pattern those effects form. If the pattern looks like a gerrymander signature in the academic literature, the audit says so. Whether the signature reflects deliberate engineering or unlucky choice among legal alternatives is a question for the reader to weigh; the audit's job is to make sure the evidence on which the weighing happens is on the table.

Definition of the pattern this section evaluates. A "gerrymander signature" is the constellation of measurable effects the academic redistricting literature (Stephanopoulos and McGhee 2015; DeFord et al. 2021; Warrington 2018; Cain et al. 2018; Katz, King and Rosenblatt 2020) treats as evidence of partisan engineering. Narrower than "any map with a partisan-asymmetry signal" (which would over-fire on natural-packing geography); broader than "any map whose author admits intent in writing" (essentially never observable). It maps onto the Canadian *Saskatchewan Reference* [1991] framework as: a map fails effective representation when its measured effects systematically dilute one community of voters' electoral weight in a way that **cannot be justified by any of the *Carter* factors** (geography, community history, community interests, minority representation, or better government). Earlier drafts used the formulation "beyond what geographic constraints require" — that is a *necessity* test *Carter* rejected. The corrected definition (per T1.7 Referee #8) tracks *Carter*'s actual holding: deviations need only be *justified* by a Carter factor, not strictly *required* by geographic constraints. The audit measures the effects; the constitutional question of whether those effects amount to a failure of effective representation is for counsel and a court to assess (Appendix F).

6.2.1 Lane 1 — Partisan-bias magnitude (against Alberta-calibrated thresholds)

The audit's §5.2.8 derives **four Alberta-calibrated EG thresholds**, of which the most empirically grounded is the MCMC-ensemble 95th percentile drawn from 1,010,000 ReCom plans on the official Elections Alberta shapefiles under EBCA constraints. None of these thresholds is borrowed from US federal judicial standards; each is calibrated to Alberta law or Alberta data. The full provenance of each threshold is in §5.2.8; the cross-tabulation against both 2026 maps follows.

The Lane 1 verdict uses **Phase 4C spatial attribution** as the sole canonical measurement (`packing_cracking_analysis.py` v0.3, 2026-05-18; exact VA-level centroid-in-polygon on official EA shapefiles). The two earlier attribution layers are retired: Reading A (crosswalk-blend, Layer 1) is superseded by Phase 4C (§5.2.2); Reading B (spatial against v0_8 DPG geometry, Layer 4) used geometries derived from commission PDF thumbnails — the DPG sunset clause is satisfied by Phase 4C on official shapefiles (§4.1.4). Phase 4C EG: majority +0.04%, minority +3.96% (S-M positive=UCP-favoured), consistent with canonical MCMC (majority +0.10%, minority +4.02%) to within 0.06 pp.

THRESHOLD	SOURCE	VALUE	PHASE 4C MAJORITY +0.04%	PHASE 4C MINORITY +3.96%
Stephanopoulos & McGhee (2015) reference	US historical calibration	7%	sub-threshold	sub-threshold
Option A — Assembly-size sensitivity	First-principles scaling: 2/89 seats	~2.2%	sub-threshold	**over threshold**
Option B — EBCA statutory-proportional	One-fifth of EBCA ±25%	5%	sub-threshold	sub-threshold
Option C — Alberta historical-swing	2015/2019/2023 swing analysis	1.01–9.71%	sub-threshold in all contexts	sub-threshold under 2023 (p94.4) and 2019 (p70.4) contexts; **over under 2015** (+10.54%, p99.45)
Option D — MCMC ensemble p95	**1,010,000-plan canonical (official EA shapefiles, seed 1432864451, ±25%)**	**4.10%**	sub-threshold (p15.5)	sub-threshold (p94.4; 0.14 pp below 4.10% threshold)

Lane 1 verdict. *Under Phase 4C canonical attribution, the majority is sub-threshold on all Alberta-calibrated options — a clean Lane 1 result (p15.5 on canonical ensemble). The minority exceeds the assembly-size Option A threshold (2.2%) and, under 2015 electoral-context votes, exceeds the Alberta historical-swing Option C ceiling (9.71%). Under 2023 votes — the operative electoral context for the 2026 boundaries — the minority falls just below the Option D canonical ensemble threshold (p94.4; 0.14 pp below the 4.10% threshold). The qualitative framing the §6.2 verdict ultimately rests on — the* ***directional engineering*** *distinguishing the two maps lives in the structural lane (§6.2.2), not the partisan-bias-magnitude lane — is robust to Phase 4C. The majority is clean on EG magnitude; the minority is near but below the most stringent Alberta-calibrated threshold under current electoral conditions.*

Effect Size Interpretation (updated 2026-05-10 for 1,010,000-plan canonical run; joint-statistic framing revised 2026-06-10; declination sign corrected 2026-06-12 per Amendment 10). Under the 1,010,000-plan canonical ensemble (official EA shapefiles, seed 1432864451, §5.4.9), the minority map places at p99.99 on seats@50/50 — approximately 66 of 1,010,000 neutral plans reach its canonical value of 51.69%. The dependence-robust Bonferroni upper bound for Ch1 (Mahalanobis, $p = 1.40\times10^{-6}$), with a ×2 conservative factor for the two channels examined, is $p \le 2.80\times10^{-6}$ (≈ 1 in 357,000) — **exploratory, not pre-registered; see §1.2**; see §4.3.2 for why this bound replaces the earlier Fisher combination of 6.87×10^{-8} (SZAT is retained as exploratory only — its standard-permutation result does not survive block-permutation null correction). The earlier DPG-era estimate (p=0.015, +2.21 pp vs neutral median, Cohen's d=0.31) is superseded. On the 1,010,000-plan canonical ensemble, **three partisan metrics carry tail flags (≥p95) simultaneously: MM at p99.98 (UCP-tail), declination at p98.79 (UCP-tail under the corrected Warrington convention; earlier drafts cited p1.21 NDP-tail under the swapped-operand δ_audit = −δ_Warrington convention now retired), seats@50/50 at p99.99**. EG at p94.4 falls *just below* p95 and **is not counted as an outlier flag** under the audit's pre-registered 4.10% / p95 Alberta-calibrated threshold — though EG is directionally aligned with the other three. The minority crosses three of four pre-registered partisan flags; the joint-tail bound above incorporates the three-flag tail pattern plus EG's directional alignment, not a four-tail reading. The three-tail-flag pattern is the asymmetric-packing signature: the minority over-concentrates NDP votes in high-share districts (MM and seats flags) while also producing unusually tight winning margins for NDP-won districts that translate into UCP-favoured declination geometry (declination flag). An unexpected finding from the 1,010,000-plan run: the majority map's mean-median places at p0.92 — also in the extreme NDP-tail (see §5.4.9 narrative). Both real maps are statistical outliers on at least one metric; the minority is the more extreme on three of four.

6.2.2 Lane 2 — Structural and procedural pattern

Lane 1 having returned a clean result for the majority on partisan-bias magnitude and a near-threshold result for the minority (p94.4; majority sub-threshold at p15.5), the question becomes whether the structural evidence reinforces the Lane 1 signal. The audit's other test families are vote-data-independent and detect engineering signatures that don't show up in EG magnitude alone.

TEST FAMILY	THRESHOLD (WHERE DEFINED)	MAJORITY 2026	MINORITY 2026
Mean-median percentile vs MCMC ensemble	p > 95 (UCP-tail) = outlier; p < 5 (NDP-tail) = opposite-direction outlier	**p0.924 (NDP-tail; opposite direction from UCP-gerrymander hypothesis; explained by commission-convention effect — §5.4.9; see §6.2.3)**	**p99.98 (UCP-tail outlier; conservative lower bound under ReCom-typical null — §5.4.9)**
Municipal anchoring	Canadian comparator norm 70–85 %	80.0 % (in norm)	72.0 % (in norm) — not a divergence signal on canonical geometry[4]
Population MAD	tighter is better	3,180	**4,707 (48 % wider)**
Calgary geographic-zone gap	smaller is more neutral	0.4 %	**12.2 %**
Chair-flagged geographic anomalies	count, lower is better	0	**3** (RMH–Banff Park, Nolan Hill–Cochrane, Olds → N Airdrie)
Airdrie community split	1- or 2-way is constraint-minimum	2-way	**4-way**
Rationale-failure pattern (symmetric audit; **qualitative, not pre-registered**)	0 of contested redraws fails	not applicable	**5 of 6 fail** (sixth — St. Albert-Sturgeon — is constraint-forced; the audit's earlier seventh — Lethbridge federal-boundary match — has been withdrawn as unsourced)
Public-submission support for contested configurations	every contested config has documented support	n/a	**2 configs (Airdrie 4-way, Nolan Hill) have no documented submissions** in the 1,140+ archive

TEST FAMILY	THRESHOLD (WHERE DEFINED)	MAJORITY 2026	MINORITY 2026
Pre-registered structural-irregularity count	≥ 4 of 5 = outlier per §5.5	0 of 5	**4 of 5** — meets threshold[4]

Lane 2 reading. *The minority crosses every structural threshold by a wide margin; the majority crosses none.* ***Note on anchoring:*** *the DPG-era 4.9× anchoring departure did not survive canonical recomputation (both maps fall within the 70–85 % norm at 72% vs 80%)*[4] *and is not carried forward as a standalone signal. The evidential core in Lane 2 is the MCMC ensemble (p = 1.40×10⁻⁶) supported by four geometry-independent structural dimensions: population dispersion (48 % wider), Calgary zone asymmetry (12.2 % vs 0.4 %), Airdrie fragmentation (4 vs 2 districts), and spatial anomalies (3 chair-flagged vs 0). The 5-of-6 rationale-failure pattern (with the 6th — St. Albert-Sturgeon — classified as constraint-forced because both the majority and minority maps independently arrive at the same two-district structure under the Act's constraints) is the cleanest rationale signal: a symmetric audit finding the minority's own published justifications fail on five of six configurations, while the alternatives satisfying the rationales are constraint-legal. (The audit had previously listed a seventh configuration — Lethbridge / Taber-Warner federal-boundary match — which has been withdrawn because the underlying minority claim cannot be located in the report; see* `analysis/methodology/lethbridge_federal_boundary_check.md`*.)*

6.2.3 Verdict on the majority 2026 proposal: not a gerrymander

Both lanes return a clean result. On partisan-bias magnitude (Lane 1), the majority crosses no UCP-tail Alberta-calibrated threshold. On structural dimensions (Lane 2), the majority crosses no UCP-tail threshold. The canonical 1,010,000-plan run (§5.4.9) finds the majority mean-median at p0.924 — an NDP-tail reading (opposite direction from the UCP-gerrymander hypothesis) — but §5.4.9 explains this as a commission-convention effect: the ReCom-typical neutral ensemble is free to dissolve rural districts a commission legally must preserve, pulling the neutral distribution's mean-median higher than commission constraints allow. Per the severity-audit disclosure at §5.4.9 (T1.17), this commission-convention correction applies symmetrically: it reduces the majority's apparent NDP-tail extreme AND makes the minority's UCP-tail readings conservative lower bounds (a constraint-enforcing ensemble would make the minority's UCP-tail position more extreme, not less). All other majority metrics are within null (EG p15.5, declination p79.6, seats@50/50 p77.8, MAD p15.8); Mahalanobis p=0.125 (50k pre-reg) / p=0.097 (1,010,000-plan canonical). Under canonical 1,010,000-plan geometry, the majority also retreats from the 2019 enacted baseline in joint-metric space (D^2=12.75 → 7.85; see §5.4.10), consistent with a normalizing commission process. The Chen-Rodden decomposition (§5.2.5–§5.2.6) attributes the majority's UCP-favourable point to dispersed rural margins rather than to drawing choices.

My verdict: the majority is the kind of map a competent independent commission would produce. It is not a gerrymander on any Alberta-calibrated metric.

6.2.4 Verdict on the minority 2026 proposal: exhibits a gerrymander signature on the structural lane; coverage-sensitive on the partisan-bias-magnitude lane

The honest reading of the two lanes:

1. **On partisan-bias magnitude (Lane 1) — Phase 4C canonical verdict (official EA shapefiles, 1,010,000-plan ensemble; authoritative per §6.2.1, supersedes Reading A and Reading B).** Three of four pre-registered partisan-bias metrics individually flag the minority map in the UCP-favoured tail: mean-median p99.98, declination p98.79 (Amendment 10 sign-correction applied), seats@50/50 p99.99. **Note on severity asymmetry (T1.17; §5.4.9 commission-convention correction applied symmetrically per T1.7 R2 S24):** the same commission-convention effect that explains the majority's NDP-tail MM outlier (§6.2.3) also means the minority's UCP-tail readings are *conservative lower bounds* — a constraint-enforcing ensemble, which the ReCom-typical ensemble is not, would produce fewer plans dissolving rural blocks, shifting the null distribution toward the minority and making these percentiles more extreme. The reported values understate how anomalous the minority is relative to a constraint-enforcing commission distribution. Efficiency gap at p94.4 is directionally aligned but sub-threshold (pre-registered Alberta-calibrated floor: p95 / 4.10%). The joint Mahalanobis D^2=32.67 places the minority at $p=1.40\times10^{-6}$. Under the retired DPG-era attribution layers: under crosswalk-blend (Reading A, retired per §6.2.1) the minority was sub-threshold on every Alberta-calibrated threshold; under v0_8 full-coverage spatial (Reading B, retired per §6.2.1, Run #4 in §5.4.4) the minority was over every threshold including the MCMC-derived 4.10 % at p100. Both readings — and Phase 4C canonical — agree the minority sits in the UCP-favoured tail; they differ on whether individual metrics cross their thresholds. Phase 4C canonical is the authoritative reading (3-of-4 individual flags + joint $p=1.40\times10^{-6}$).
2. **On structural and procedural patterns (Lane 2) the minority crosses every comparator threshold the audit applies, by a wide margin.** Four of five pre-registered structural-irregularity tests fire — meeting the ≥ 4 outlier threshold;[4] five of six contested redraws fail their stated rationales (the sixth — St. Albert-Sturgeon — is constraint-forced; an earlier seventh — Lethbridge federal-boundary match — has been withdrawn as unsourced); the commission chair flagged three geographic anomalies on this map and zero on the majority. Lane 2 is invariant across the coverage-attribution choice. Under canonical 1,010,000-plan geometry, the minority amplifies the 2019 enacted baseline's joint-space position (D^2=12.75 → 32.67), adding context to the Lane 2 reading without introducing an independent test (see §5.4.10).

Under Phase 4C canonical, the two lanes both point in the same direction. Lane 1 has three of four metrics individually above p95 and joint Mahalanobis $p=1.40\times10^{-6}$; Lane 2 is at maximum structural-irregularity count. A reader who limits Lane 1 to individual EG and counts only the sub-threshold p94.4 result may read the minority as not exhibiting a gerrymander signature on the EG dimension — but the three individually-flagged metrics, the joint D^2 result, and Lane 2 together point toward one. The literature is on the side of weighting the joint result: Katz, King and Rosenblatt (2020) argue no single magnitude metric is dispositive and that any valid measure must satisfy symmetry — a criterion this audit's ensemble approach satisfies; Cain et al. (2018) argue municipal-anchoring departure is itself a partisan-engineering enabler; the chair-flag, rationale-failure, and submission-archive patterns are not partisan-bias *measurements* but partisan-engineering *signatures*.

The verdict the audit's measurements support. On the cumulative structural-and-procedural pattern, the minority 2026 proposal exhibits the constellation of effects the academic literature treats as a gerrymander signature. On the partisan-bias-magnitude lane under Phase 4C canonical, three of four individual metrics and the joint Mahalanobis score are above their pre-registered thresholds; EG alone is directionally aligned but sub-threshold. (Pre-Amendment-10 DPG-era verdict used Reading A/B framing — Reading A sub-threshold on every metric, Reading B over every metric; both readings are retired per §6.2.1 in favour of Phase 4C canonical.) Whether the observed effects reflect deliberate engineering against the NDP, unlucky choice among legal alternatives, or some combination of the two is a question the audit cannot answer from public data alone. The author's restraint on intent is a deliberate methodological choice, not an evasion: intent is not observable from boundary geometry, vote totals, public submissions, or chair-flag counts. What is observable is the pattern of effects, and the pattern is reported above without hedging.

Where intent may suggest itself to the reader. Five of six contested minority redraws fail the *minority's own published rationales* under symmetric audit (the sixth, St. Albert-Sturgeon, is constraint-forced — both maps independently arrive at the same configuration), and two of those redraws have no documented public-submission support in the commission's 1,140+ archive. Each individual signal has a non-engineering explanation available. Whether the cumulative pattern's most parsimonious explanation is engineering or unlucky drafting is the inference the reader is invited to weigh — the audit does not make it on the reader's behalf.

A reasonable expert who weights partisan-bias magnitude over structural pattern would reach a more cautious verdict on the minority — closer to "no measurable gerrymander signature on the partisan-bias dimension; structural irregularities require explanation but do not by themselves establish anything about how the map was drawn." This disagreement is documented rather than hidden.

6.2.5 How close did the minority come?

A "how close" answer needs to be lane-specific because the two lanes give different answers.

Lane 1 (partisan-bias magnitude) — Phase 4C canonical (authoritative; supersedes Reading B below). Under canonical 1,010,000-plan ReCom on official EA shapefiles: mean-median p99.98 (UCP-tail), declination p98.79 (UCP-tail; Amendment 10 sign-correction applied), seats@50/50 p99.99 (~66 of 1,010,000 neutral plans reach this value). Efficiency gap +3.96 % at p94.4 — directionally aligned but 0.6 pp below the pre-registered p95 / 4.10 % Alberta-calibrated threshold. The Cannon et al. (2023) short-bursts test (§5.4.8) reaches 52.87 % UCP seats@50/50 in 40,000 hill-climbing steps, confirming the minority sits within the non-neutral procedure's reachable space. **Distance to Phase 4C thresholds: three individual metrics (MM, Decl, s50) above p95 and above the 4-comparison Bonferroni threshold for MM and s50; EG at p94.4 is 0.6 pp sub-threshold on the individual flag but directionally aligned.** DPG-era Lane 1: Under Reading B (retired, full-coverage spatial, Run #6 2M), the minority's EG was +9.21 % at p100 of the 2,000,000-map ensemble; on seats@50/50 (89-of-89 attribution, §5.4.7), 52.8 % sat above the 2M-ensemble converged ceiling of 51.72 %. These DPG-era figures are superseded by Phase 4C canonical and should not be cited independently.

Boundary-choice decomposition (SZAT, exploratory, §5.2.10). The SZAT bootstrap (p=0.0024 under standard i.i.d.-flip permutation, AsPredicted #289,469) is retained as exploratory context. Under a block-permutation null correcting for spatial autocorrelation across adjacent Voting Areas, SZAT returns p≈0.19 — not significant — and is therefore retired as a confirmatory channel. The regional decomposition of the SZAT score (+0.039211 total) is: Rest of Alberta +0.015 (dominant), Edmonton +0.008, Mountain-West / Canmore-RMH +0.006, Calgary −0.008 (partially offsetting). The signal is distributed across the province; Calgary swing zones run in the *opposite* direction to the headline score. This is inconsistent with a Calgary-centric explanation of the minority's UCP-favourable position. The dependence-robust Bonferroni upper bound for Ch1 (Mahalanobis, $p=1.40\times10^{-6}$), with a conservative ×2 factor for the two channels examined, is $p \le 2.80\times10^{-6}$ (~1 in 357,000) — exploratory, not pre-registered (§1.2); see §4.3.2.

Lane 2 (structural pattern). The minority crosses every threshold by a wide margin:

- Mean-median percentile p99.98 vs literature outlier line p95 — **~5 percentile points over** (canonical 1,010,000-plan run; 50k preliminary run had p99.99)
- Structural-irregularity count 4 of 5 vs pre-registered cut-off ≥ 4 — **meets threshold**[4]
- Chair-anomaly count 3 vs 0 — **3 over**
- Airdrie 4-way vs constraint-minimum 2-way — **2 splits over**
- Rationale-failure 5 of 6 vs symmetric-audit cut-off ≥ 1 — **4 over** (sixth — St. Albert-Sturgeon — is constraint-forced; an earlier seventh — Lethbridge federal-boundary match — has been withdrawn as unsourced)
- SZAT 2019→minority score = +0.016 (p=0.053, marginally outside null CI [−0.017, +0.006]); minority boundary choices moved EG in the UCP-favoring direction relative to 2019, opposite to what random boundary changes from 2019 produce — consistent with but not establishing a pattern of directed movement (exploratory, not pre-registered; see §5.4.10)

Distance to the nearest structural gerrymander threshold: 0 (already over). On the structural lane the minority is not "close" to a threshold — it crosses every threshold the audit applies.

The honest summary: *the minority maxes out the structural-irregularity scoring while staying inside the partisan-bias-magnitude band*. This is the observed effect. One drafting process producing this effect is structural engineering (off-reference boundaries, community splits, chair-flagged shapes) whose seat-count consequence is not measurable on current vote distributions — either because the structural choices did not translate to seats, or because the seat translation depends on a vote distribution not yet observed. The audit reports the effect; whether the effect reflects engineering or some other explanation is for the reader to weigh.

6.2.6 What would change the author's verdict

1. **An independent reviewer produces a constraint-legal Alberta map satisfying the minority's stated COI rationales (community-of-interest preservation in Airdrie, Cochrane, Nolan Hill, RMH–Banff Park) and *also* matches majority-comparable municipal anchoring (≥ 60 % CSD/DA edge alignment).** The minority's rationale-failure pattern would then be evidence of a constraint trade-off rather than engineering. (Issue #14 on the audit's GitHub repository invites this exact counter-map.) Strongest single falsifier.
2. **The 2019 cross-election direction reversal extends to 338Canada April 2026 polling.** Currently the polling supports the 2023 direction; if more recent polling shows the direction flipping under multiple 2020s-era voter distributions, the structural pattern would become harder to read as engineering.
3. **A pre-2026 internal commission document is published showing the minority's anchoring and split choices were a deliberate response to documented community submissions rather than drafting choices.** The §5.9.4 submission-archive verification could not find such submissions for two of the contested configurations; if they exist and were missed, the rationale-failure finding is wrong.
4. **The November 2026 Lunty-committee 91-seat map produces structural-irregularity counts in the same range as the minority (≥ 4 of 5).** This would suggest the minority's structural profile reflects Alberta-specific drawing difficulty rather than minority-specific engineering. Pre-registered held-out test — the cleanest single check.

The audit's pre-registration commits the author to publishing any of these retractions publicly within 30 days of receiving the falsifying evidence.

6.2.7 Verdict on the April 16 government pivot itself

Separate question, addressed in §5.9.2 as a procedural finding. The author's opinion: the substitution of a UCP-MLA-chaired committee for the independent commission's drafting process during the same redistricting cycle is the most government-controlled response among the three most commonly cited Canadian comparator cases (the other two being the federal 2012 redistribution challenge and Saskatchewan's 1989 court referral). It is not, on its own, evidence the resulting Lunty-committee map will be a gerrymander. It is evidence the procedural guardrails normally relied upon to make redistricting credible to the losing party have been weakened relative to comparator practice. The November 2026 Lunty-committee map will be evaluated under the same pre-registered scorecard applied to the two commission proposals; if it scores in the range of the minority on structural-irregularity count, the procedural opinion will be vindicated, and if it scores in the range of the majority, the procedural opinion will be tempered.

6.2.8 Bottom line

- **Majority 2026:** does not exhibit a gerrymander signature on either Alberta-calibrated lane.
- **Minority 2026 (Phase 4C canonical — authoritative):** three of four pre-registered partisan-bias metrics flag the minority individually in the UCP-favoured tail (mean-median p99.98, declination p98.79, seats@50/50 p99.99; ~66 of 1,010,000 neutral plans reach the minority's canonical s50 value); efficiency gap +3.96 % at p94.4 is directionally aligned but sub-threshold (0.6 pp below the pre-registered p95 / 4.10 % Alberta-calibrated floor); joint Mahalanobis $p=1.40\times10^{-6}$; and exhibits the academic literature's gerrymander signature on the cumulative structural-and-procedural lane by a wide margin on every structural threshold the audit applies. Cannon et al. (2023) short-bursts hill-climb reaches 52.87 % UCP seats@50/50 (§5.4.8), confirming the minority sits in the non-neutral procedure's reachable space. The DPG-era Reading B figures (EG +9.21 % at p100 of 2M; s50 = 52.8 % above 2M ceiling) are superseded per §6.2.1. The minority map is the kind of map a non-neutral procedure produces under EBCA constraints; it is not the kind of map a neutral procedure produces. The literature recommends the cumulative reading. The audit reports the effects; what those effects suggest about how the map was drawn is for the reader to weigh.

Neither summary is a legal finding. Both are an attempt to give the citizen-reader of this audit a plain answer to the question they probably came here with — calibrated to Alberta's own thresholds, not to US federal judicial standards. The audit's restraint on intent is deliberate: intent is not observable from public data, and the pattern of effects the audit *can* observe is reported plainly enough the reader does not need the author to draw the inference for them.

Why this level of methodological precision? The audit's apparatus — seven measurement layers, three-chain R-hat convergence, DPG tier-aware perturbation CIs, machine-readable dependency DAG, pre-committed pass criteria, per-finding retraction pathway — is deliberately over-engineered relative to the Commission's drawing process. The Commission drew with a broad brush (political negotiation, public-hearing notes, professional judgement). The audit looks through a microscope (adjacency chains, surplus-vote rates, constraint-bound percentiles). This asymmetry is intentional: **precision is armor**. If the audit presented a single number — "the minority's Efficiency Gap is 2.4σ from neutral" — the Commission would legitimately respond "Alberta's geography is unusual; your one number is too simple to capture it." The over-engineering forces the distinction between an accidental brush-stroke and a systematic pattern, and it is what allows the audit to answer the Commission's legitimate objections in advance.

Translation discipline. The math provides authority; the prose provides conviction. The table in §6 above is the audit's formal synthesis. The three bulleted translations above are the audit's human-readable reading of the synthesis. Both are included because neither alone is sufficient: math without translation is unactionable for a reader who is not a redistricting specialist, and translation without math is polemic. This section exists to bridge them explicitly.

What the audit does not claim. It does not claim intent (§4.5 is explicit). It does not claim the Commission acted in bad faith. It does not claim any specific individual commissioner is responsible for the minority's patterns. It claims only the minority map, relative to the majority map drawn by the same Commission under the same statutory constraints on the same voter geography, is displaced from the constraint-bound expectation in a measurably consistent direction across five structural dimensions. The explanation for the displacement is not within the audit's scope; the measurement is.

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7. Limitations and Falsifiability

7.0 Hypotheses tested during the audit and their disposition

The audit's central directional claim (*minority more UCP-favourable than majority across population, spatial, partisan-bias, and procedural dimensions*) is one of several working hypotheses running through the project. Three of those working hypotheses did not survive contact with the data and are recorded here as part of the audit's evidentiary discipline. The 2026-04-26 external code audit by Google Gemini 3.1 Pro (full memo at `analysis/methodology/external_code_audit_findings_gemini_2026-04-26.md`) and the falsification tests subsequently run against the audit's own pipeline produced this disposition.

#	HYPOTHESIS	FINAL DISPOSITION	WHAT KILLED IT
H1	The 2,000,000-step MCMC ensemble shows the minority map's `seats@50/50` is out of distribution (p100, no map in 2M reaches it)	**REJECTED**	The chunked-MCMC orchestration (`mcmc_ensemble_250k_v0_8.py:_run_chain_chunked()`) was silently re-seeding the chain from the 2019 baseline at every chunk boundary, so the "2M-step deep continuous chain" was structurally a stack of 100 independent 20,000-step short bursts. Bug fixed in commit `73544a3`; finalized 250k topological ensemble re-run; finalized percentile is p98.5 (3,750 of 250,000 reac or exceed)

#	HYPOTHESIS	FINAL DISPOSITION	WHAT KILLED IT
H2	The minority map's `seats@50/50` value is 52.8% (the published headline figure)	**REJECTED**	The v0_8 polygon reconstruction left 6 of the 89 minori districts effectively unscored under the centroid-in-polygon spatial join (`coverage_pct=100%` was misleading because covered VAs were unevenly distributed; `n_districts` in `seat_results` was 83, not 89). The Gemini-authored topological VA-dissolve (the v0_9 substrate, commit `7cf47a4`) recovered full 89/89 coverage. The corrected `seats@50/50` is **48.3%**, not 52.8
H3	Non-compact geometry (the chair-flagged lasso shapes, the urban-rural hybridizations, the Airdrie four-way split) is the load-bearing mechanism through which the minority commissioners reached their UCP `seats@50/50` advantage. Concretely: the SMC plans reaching the minority's 0.4831 value should have systematically lower mean Polsby-Popper compactness than the SMC plans that don't	**REJECTED (twice — falsification test, then independent v0_9 confirmation)**	Falsification test designed by the principal investigator before running, in keeping with pre-registration disciplin Among the 5,000-plan R-`redist` SMC ensemble, mean per-district PP for plans with `seats@50/50` ≥ 0.4831 (n=2,762) was **0.2391** vs **0.2339** for the rest (n=2,238) Welch t-test p = 7.7 × 10^{-234}, in the wrong direction. Independently, direct PP measurement on the v0_9 substrate (commit `c170f49`) found the chair's named lass themselves score in the *moderate* compactness band: Calgary-Nolan Hill-Cochrane PP = 0.402, Rocky Mountai House-Banff Park PP = 0.414. The minority did not brea Area/Perimeter ratio to build the firewall; the corridors are drawn thick enough to make PP look innocent. The empirical finding R-SMC reaches the minority's value mo often than Python-ReCom does is preserved; the *mechanistic* claim compactness is what makes the gap is rejected
H4	**[DPG-era hypothesis; precondition retracted on canonical geometry — see §5.8.5 and `[^anchoring_canonical]`]** The minority map's DPG-era municipal-anchoring deficit (DPG: 14.5% vs majority 71.0% on Statistics Canada CSDs) generalises to other anchoring substrates. If the minority simply substituted natural features (highways, rivers) for CSDs, all three maps should score similarly under a "natural anchoring" measurement	**MOOT on canonical geometry; the precondition (the 14.5% / 71.0% / 4.9× anchoring deficit) did not survive canonical recomputation, so the question of whether it generalises across substrates no longer applies. DPG-era result: REJECTED (informative).**	DPG-era direct measurement on highways + waterways substrate (commit `6a39960`) found 2019 = 40.2 %, majority = 38.4 %, minority = 40.1 % — all three maps cluster within 2 pp under natural anchoring. On canonic geometry the underlying CSD anchoring is majority 80.0 / minority 72.0%, both within the 70–85% Canadian comparator norm; the DPG-era "minority abandoned anchoring" framing is therefore retracted at the source. Hostile-witness counter ("commission anchored to highways instead") is moot because the DPG-era CSD-anchoring gap it was constructed to explain no longer exists on canonical geometry

#	HYPOTHESIS	FINAL DISPOSITION	WHAT KILLED IT
H5	The minority's `seats@50/50` percentile is robust to the uniform-swing assumption. Under empirical regional swing (Calgary swings 1.34× the provincial swing, Edmonton 0.50×, rural 0.95×), the minority should still place in the right tail of the ReCom ensemble	**REJECTED — signal substantially weakens; finding no longer relocates to majority under canonical geometry**	**DPG-era (v0_9 substrate):** minority collapses from p98. → p50.7; majority rises to p99.5 (opposite direction). **Canonical EA shapefiles — election-day votes (MCMC-consistent):** minority 0.5169 (p99.99) under uniform → 0.4607 under regional (≈ p65–70; above ensemble median 0.4483, below p95 0.4828); majority 0.4607 (p77.8) under uniform → 0.4270 under regional (near ensemble p5 0.4253; NDP-tail). **Canonical — full votes including advance ballots (SZAT-consistent):** minority 0.5056 → 0.4719; majority 0.4719 → 0.4494 (≈ ensemble median). Across both canonical substrates: the minority's outlier signal substantially weakens under regional swing but does not fully collapse to median; the majority no longer rises to p99.5 (rather, it shifts towarc the NDP-tail or median depending on substrate). The DP era "finding relocates to majority" result does not hold o canonical geometry. Canonical outputs: `data/outputs/regional_swing_canonical_ed.json` (election-day); `data/outputs/regional_swing_canonical_full.json` (f votes)
H6	The Lane-1 metrics computed under the original `seat_results()` (UCP share normalised by total votes including third-party; ReCom chain destroying s15(2) protected districts to enforce ±25%) are unbiased estimators of partisan effect under Alberta's actual electoral conditions	**DISPOSED — corrected 250k v0_9 ensemble retains the headline finding**	Two independent methodological flaws identified in `mcmc_ensemble.py`: (a) `seats@50/50` win threshold compared UCP share (out of total votes including third-party) against a 50% threshold, which is unreachable in districts with material third-party vote — fixed in commi `972b04a` to use two-party (UCP+NDP) totals throughout; (b) the s15(2) freeze attempt (commit `972b04a`) identified the legally protected districts correctly but the unfrozen 85-district subgraph was infeasible to seed-balance via `recursive_tree_part` — disabled in commit `3484351` wit a documented limitation (per second-pass Gemini analys destroying s15(2) protection in the baseline makes the minority's percentile MORE conservative). The corrected 250k ReCom ensemble was run on the v0_9 substrate (script `mcmc_ensemble_250k.py`, output `data/v0_1_mcmc_ensemble_*_250k_v0_9.*`). **Result: the minority's `seats@50/50` corrected percentile is p98.52** (versus the original v0_8/250k figure of p98.6 — Two-Party normalisation moved the value by <0.1pp because third-party share was negligible in 2023). The audit's Lane-1 percentile claim of "top 1.5% under uniform swin on the corrected ensemble" survives the correction withi rounding

#	HYPOTHESIS	FINAL DISPOSITION	WHAT KILLED IT
H7	The minority's `seats@50/50` outlier status is a pure-2023-election artefact. Under 2015 or 2019 vote distributions projected onto the v0_9 substrate, the minority should look indistinguishable from the majority	**REJECTED (in part)**	Cross-election recomputation on the v0_9 substrate usi cached 338Canada historical projections (`analysis/scripts/cross_election.py`, commit `e21994` direction of minority > majority on `seats@50/50` holds 3-of-3 elections (2015 tie at 0.652/0.652; 2019 minority 0.562 / majority 0.517; 2023 minority 0.472 / majorit 0.461). The 2015 tie is the honest cost — defends again the "pure-2023-swing" attack on direction; does not support a stronger "minority is a structural pro-UCP outlier under any vote distribution" claim. The 2023 percentile vs the 2023-trained ensemble is 92.99% (slightly below the original p98.6 because of the Two-Party normalisation)
H8	Excluding advance/mobile/special ballots (~47% of 2023 votes; reported only at ED level, not VA level) materially biases the audit's Lane-1 metrics	**REJECTED — empirical refutation**	Apportioning the missing advance ballots back to VAs proportionally to election-day two-party shares (`analysis/scripts/advance_vote_splat.py`, commit `72e3369`) shifts both v0_9 maps by **exactly +1.12 pp** in `seats@50/50` (majority 0.4607 → 0.4719, minority 0.4831 → 0.4944). The 2.25-pp inter-map gap is preserved to 4 decimal places. The advance-vote omissic is a real methodological caveat (every map gains 1.12pp UCP under the smear) but is **inter-map invariant** at the audit's percentile-comparison resolution. The hostile-witness attack ("you're missing half the electorate") is empirically refuted as a methodological objection. Side observation: the majority's efficiency gap flips sign (-0.0149) under the smear while the minority's barely moves (+0.0170 vs +0.0175), reinforcing the surgical-fortification framing
H9	The targeted hill-climbing procedure used to bound the minority's distance from neutral is asymmetric (UCP-direction works, NDP-direction does not, so the comparison with neutral median is rigged)	**REJECTED — symmetric**	NDP-maximizing short-bursts run on the v0_9 substrate (`analysis/scripts/targeted_gerrymander_burst_ndp.p` mirror of UCP-direction config; commit `72e3369`). UCP-targeted reaches 52.87% UCP `seats@50/50`; NDP-targete reaches 37.93%. Neutral 250k v0_9 ensemble bounds a 39.08%–50.57%. Both targeted procedures push beyond the neutral envelope (UCP +2.30pp above max, NDP −1.15pp below min). Asymmetry in degree (UCP has 8.04pp targeted headroom from neutral median; NDP ha 6.90pp) is a property of Alberta's political geography, no the test apparatus. The minority sits 2.3× closer to the UCP-targeted ceiling than to the NDP-targeted floor — t audit's existing framing holds. **Side correction:** `report_public.md:333` lists "Neutral 250k MCMC, min produced = 37.9%" — the value is in fact the NDP-burst result, not the neutral floor. Corrected neutral floor on t canonical v0_9 250k ensemble is 39.08%

#	HYPOTHESIS	FINAL DISPOSITION	WHAT KILLED IT
H10	The "Polsby-Popper says the chair's named lasso (Calgary-Nolan Hill-Cochrane) is moderately compact" finding (PP = 0.402) reflects the actual geometry of the lasso. If a different but equally well-known compactness metric (Reock — pre-registered alongside PP in the audit's protocol) also scores this district as moderate-or-better, then the lasso shape claim must rest entirely on chair judgement and not on any mathematical compactness metric	**REJECTED — Reock catches it**	Reock = $4\cdot\text{area}/(\pi\cdot D^2)$ where D is the diameter of the smallest enclosing circle. Computed on v0_9 in commit `e219943` (`analysis/scripts/reock.py`). Calgary-Nolan Hill-Cochrane scores Reock = **0.230** — *flagged* under the conventional Reock < 0.30 threshold. The two metrics measure different aspects of compactness: PP penalises perimeter wiggle, Reock penalises elongation. A lasso corridor is exactly the elongation pattern Reock catches and PP does not. The H3 conclusion — "compactness is not the mechanism for the *seats@50/50* gap" — still holc (the falsification test was on plan-level mean PP, and the v0_9 majority and minority have similar mean Reock too but the *individual lasso shape* IS quantitatively flagged by Reock. Both pre-registered metrics now in the record

State of the audit's central claim after H1–H10. *The audit's pre-shapefile Lane 2 finding — the minority municipal anchoring dropped from the 2019 baseline of 75.2 % to 14.5 % while the majority continued 2019 practice at 71.0 % — was the central published claim through April 2026 and survived every DPG-era methodological revision (sampler, swing, compactness, ReCom population tolerance). It did* ***not*** *survive canonical recomputation against the official 2026 shapefiles: minority 72.0 %, majority 80.0 %, both within the 70–85 % Canadian comparator norm.*[4] *H1–H10 are preserved here as the audit's historical stress-test record; the H4 row in particular references the DPG-era anchoring values not surviving canonical recomputation. H5's canonical regional-swing recomputation* ***substantially weakens but does not collapse*** *the minority's seats@50/50 finding: under canonical EA shapefiles and election-day votes (MCMC-consistent substrate), the minority moves from p99.99 under uniform swing to approximately p65–70 under empirical regional swing — the signal persists above the ensemble median but no longer reaches the extreme tail. The DPG-era "finding relocates to majority at p99.5" result does not replicate on canonical geometry; the majority instead drifts toward the NDP-tail or ensemble median. H6's corrected 1,010,000-plan canonical ensemble retains the headline at p99.98 on mean-median and p99.99 on seats@50/50. Lane 1 is therefore reported as a multi-scenario sensitivity table (uniform swing / regional swing × two vote substrates / cross-election direction holds 3-of-3). The audit's central published finding has reorganized: the MCMC ensemble (Mahalanobis p = 1.40×10^{-6} against the 1,010,000-plan canonical ensemble) is now the load-bearing Lane 1 result, supported by the four geometry-independent Lane 2 dimensions (population dispersion, Calgary zone, Airdrie split, chair anomalies).*

Empirical confirmation of the surgical-fortification thesis (corrected 250k v0_9 ensemble). Two findings from the H6 disposition stand out for the audit's framing:

1. **The 2019 enacted map and the 2026 majority sit at *identical* percentile on `seats@50/50`** (both p77.8 against the canonical 1,010,000-plan ensemble). The majority continues 2019 Alberta practice on the partisan-fairness axis. On canonical geometry the majority's municipal anchoring (80.0 %) also sits in the 2019 baseline neighbourhood (75.2 %); the minority's canonical anchoring (72.0 %) sits slightly below but within norm.[4] The partisan-fairness continuation framing — the majority continues 2019 Alberta practice, the minority diverges on the metrics surviving canonical recomputation — remains supported.
2. **The minority's Lane-1 metric pattern, against the canonical 1,010,000-plan ensemble, is three tail flags and one near-but-below — all four directionally UCP-favoured (Amendment 10 declination sign correction applied 2026-06-12).** Mean-median p99.98 (UCP-tail),

declination p98.79 (UCP-tail under the corrected Warrington 2018 convention; earlier drafts cited p1.21 NDP-tail under the swapped-operand implementation), and seats@50/50 p99.99 are the three flags; efficiency gap at p94.4 sits *near*, *but below*, the audit's pre-registered Alberta-calibrated p95 / 4.10 % threshold (directionally UCP-favoured but sub-threshold). (An earlier draft of this paragraph cited "near-median" Lane-1 placements based on the v0_9 ensemble; those numbers — EG p56.1, MM p53.2, declination p62.2 — are superseded by the canonical 1,010,000-plan placements.) The pattern of three tail flags plus a near-miss on EG is consistent with the "surgical fortification" reading the methodological-defenses appendix predicted (§3.2 "One-Metric Gerrymander"), and consistent with an asymmetric-packing signature: tight NDP winning margins (declination), over-concentration of NDP support in high-share districts (MM), and a seat ceiling reached well below 50 % UCP vote share (seats@50/50). EG measures average wasted-vote efficiency, which a fortification strategy distributed across a small number of marginal districts can leave only weakly perturbed; that is why the EG flag sits just below threshold while the structural tail metrics fire.

The DPG-substrate SMC cross-validation had a seed-placement defect: three runs with the same nominal `set.seed(88)` produced 5.6%, 28%, and 58% of weighted plans reaching the DPG minority's 0.4831 value, because `library(redistmetrics)` consumed RNG state before the sampler ran. The defect was diagnosed and corrected by placing `set.seed(852751799)` immediately before `redist_smc()`. The canonical re-run (2026-05-18; input: `va_2023_election_day_votes.gpkg`; 5,000 SMC plans, ESS 1,116) corroborates the Python ReCom finding: **0% of 5,000 SMC plans reached 0.5169** (ensemble maximum 0.4943, placing the canonical minority map above the 99.98th SMC percentile). Both standard samplers on official Elections Alberta geometry agree: the canonical minority map's `seats@50/50` exceeds the ensemble tail. The Python ReCom ensemble retains priority for percentile placement — it passes the Gelman-Rubin diagnostic ($\hat{R} \in [1.007, 1.017]$ across all four metrics, below the 1.05 publication threshold) and has 200× the sample size — but the canonical SMC result now provides algorithm-independence corroboration rather than a contradicting signal. (The H3 falsification test was run on the v0_9 DPG substrate against the 0.4831 threshold; on canonical geometry the threshold is never reached, making the PP comparison of high-vs-low plans moot. The direction of H3 — high-UCP-advantage plans more compact, not less — is preserved: the stronger statement now is no SMC plan reaches the canonical target at all.)

7.1 Missing evidence and scope limits

1. **2026 polygon geometry.** Phase 4A resolved 2026-05-06 (canonical EA shapefiles received; MCMC canonical ensemble complete). C1 (Polsby-Popper, 2026-05-18) and C2 (Reock, 2026-05-18) executed on canonical EA shapefiles — null findings; v0_9 direction reversed (documented in DOCUMENTED CORRECTIONS). B5 (MCMC seed variant) pending.

2. **Hybrid-ED blend vs. spatial measurement disagreement (Phase 4C vs. v0.2).** Phase 4C spatial assignment on **canonical EA shapefiles** (official Elections Alberta geometry, re-run 2026-05-17; `phase4c_canonical_attribution.py`) produces a **+3.92 pp** EG asymmetry (election-day votes; majority NDP@50/50 = 48, minority = 43). An earlier Phase 4C v2 run using derived shapefiles had produced +0.31 pp — a smaller figure attributable to derived boundary approximation of rural-fringe VAs. v0.2's 70/30 hybrid-ED blend on full-vote totals produces −1.41 pp (reversed sign). H8's empirical test shows advance-ballot addition shifts both maps by an identical +1.12 pp (inter-map invariant), confirming the method-vs-blend tension is not a vote-data gap. Full reconciliation: `findings/assignment_gerrymander_comparison.md`.
3. **Independent verification of the no-public-support claim (Section D).** Requires text-search of the commission's 1,140+ submission archive.
4. **Full-symmetry visual audit for majority.** Requires majority-proposal Alberta overview, Edmonton, and other-cities map images.
5. **2019-era population data.** Would permit A1/A2 symmetric analysis of the 2019 baseline alongside the two 2026 proposals.
6. **Multiple Testing (Family-Wise Error Rate).** The audit evaluates five structural tests (four partisan-bias dimensions plus one geometric compactness dimension). Under a formal Holm-Bonferroni correction for $m=5$ tests at a one-tailed $\alpha = 0.10$ threshold (standard for directional gerrymandering bounds), the strictest threshold is $0.10 / 5 = 0.02$. The minority map's `seats@50/50` result ($p = 0.0148$) satisfies this corrected threshold ($0.0148 \le 0.02$). Therefore, the outlier finding formally survives conservative FWER correction.
7. **Pre-Registration Authority.** While historical analyses rely on GitHub commit history for provenance (which lacks the strict immutability of third-party registries like OSF), the future evaluation of the November 91-seat map utilizes a cryptographically secure, third-party `drand` randomness beacon (Round 6062459, locked April 27, 2026) to establish true, immutable pre-registration before the data exists.

7.2 Falsifiability statement

The audit's directional claim — *minority more UCP-favorable than majority across population, spatial, partisan-bias, and procedural dimensions* — would be falsified by any of the following:

- An alternative Calgary classification producing near-null minority-majority asymmetry (≤1%) while A2's current rule produces >10%. Tested; both rules produce the same direction.
- Submission-archive evidence the five disputed minority configurations (Airdrie, Cochrane, Chestermere, Red Deer, St. Albert) did have substantial public support in the 1,140+ record. Refutes Section D claim.

- Visible spatial anomalies in the majority's rural or Edmonton districts of a severity comparable to the minority's three flagged ridings. Requires majority non-Calgary imagery.
- A comprehensive survey of Canadian provincial redistributions 1991–2025 finding comparable mid-cycle government-drafting-process replacements. Refutes Section D uniqueness framing.

7.3 Academic Limitations and Methodological Caveats

1. **Ecological Inference:** We use 2023 election results to simulate hypothetical 50/50 vote splits. Actual 2026 vote distributions at the block-level may differ structurally from the uniform-swing projections.
2. **Single Election Dependency:** The 2023 election was highly competitive (UCP 52.6%, NDP 44.0%). In a landslide environment, structural partisan metrics become noisy or uninformative.
3. **Geographic Constraints vs Intent:** Alberta's deep urban/rural divide creates severe natural packing. The "neutral" ReCom ensemble attempts to control for this, but residual natural packing may still be misattributed as intentional engineering (Chen and Rodden 2013).
4. **Model Dependence:** ReCom ensembles assume contiguous, population-balanced districts as the primary constraints. If the commission heavily weighted unmeasured criteria (e.g., historical communities of interest not aligned with municipal boundaries), the neutral baseline may be misspecified.
5. **Causal Identification:** The audit mathematically detects structural advantage, not human intent. The commission may have had legitimate, non-partisan reasons for their boundary choices happening to perfectly correlate with partisan advantage.

DOCUMENTED CORRECTIONS

Findings not surviving methodological review or data upgrade. Retained per the audit's pre-committed policy of never deleting failed findings. Full per-finding retraction conditions: `analysis/methodology/retraction`

C1 — SZAT SUBSTRATE DESCRIPTION (RETRACTED 2026-05-10).

An earlier version of §5.2.10's Post-registration VA file update paragraph described the file switch from `va_polygons_with_2023_votes.gpk` to `va_polygons_with_full_2023_vote` as a substrate change to "full advance-vote" data with NDP two-party share rising 42.60% → 44.66%. The description was false: both files carry election-day vote columns (`va_ndp`/`va_ucp`); the switch changed only 2 VA centroid assignments. The retracted paragraph incorrectly implied the SZAT was re-run on a different vote substrate.

C2 — V0_9 REOCK BETWEEN-MAP ASYMMETRY (RETRACTED 2026-05-18).

Analysis on the v0_9 topological substrate (VA-polygon dissolves, derived shapefiles) produced minority 34.8% of EDs below Reock 0.30 vs majority 13.5% — a 2.58× asymmetry in the direction of minority less compact. Under official Elections Alberta canonical shapefiles (received 2026-05-06), the direction reversed: minority 4/89 = 4.5% below threshold, majority 6/89 = 6.7%. The substrate effect: VA-polygon dissolves impose detailed perimeter geometry differentially penalising the minority's corridor-shaped EDs; clean administrative boundary lines are less sensitive to this pattern. The "minority less compact" between-map claim and 2.58× ratio do not survive canonical geometry. Primary: `findings/reock_verdict.md`.

C3 — V0_9 POLSBY-POPPER BETWEEN-MAP ASYMMETRY (SUPERSEDED 2026-05-18).

The v0_9 PP analysis produced minority 30.3% vs majority 22.5% below threshold (1.35× asymmetry).

Under canonical EA shapefiles, the direction reversed: minority 6/89 = 6.7%, majority 8/89 = 9.0%. Same substrate effect as C2. Primary: `findings/polsby_popper_verdict.`

C4 — PHASE 4C DERIVED-GEOMETRY EG ASYMMETRY (SUPERSEDED 2026-05-17).

An earlier Phase 4C v2 run using derived shapefiles produced a +0.31 pp EG asymmetry (majority − minority, election-day votes). The canonical Phase 4C re-run on official EA shapefiles (2026-05-17) produces +3.92 pp, with majority NDP@50/50 = 48 vs minority = 43. The discrepancy is attributable to derived boundary approximation affecting rural-fringe VA assignments. Primary: `findings/assignment_gerrymander`

C5 — V0.2 BLEND §5.2.1 TABLE VALUES (SUPERSEDED 2026-05-18).

The v0.2 urban/rural blend model (URBAN_WEIGHT_DEFAULT = 0.85) produced majority EG −0.40% / minority EG −1.81% (both UCP-favourable in negative=UCP convention), B3 majority −0.66 pp / minority −1.20 pp, B4 majority 45 / minority 46, and actual seats majority 38/51 / minority 38/51. Phase 4C (v0.3) exact VA-level spatial attribution gives: standard EG majority +0.04% / minority +3.96% (both UCP-favourable in S-M positive=UCP convention; majority approximately neutral at p15.5, minority substantially UCP-favourable at p94.4; +3.92 pp asymmetry), B3 majority −3.64 pp / minority +1.03 pp (B3 direction reverses: majority now more negative on MM than minority), B4 majority 48 / minority 43, actual seats majority 34/55 / minority 29/60. The blend model had the EG sign wrong: it showed minority as *more* UCP-favourable than majority (both negative, minority more negative); Phase 4C agrees on the direction (minority more UCP-favourable) but at much higher magnitude. Phase 4C and MCMC canonical agree to within 0.06 pp. Primary: `analysis/scripts/packing_cracki` v0.3; `findings/partisan_bias_summary.` §A1 (2026-05-18).

C6 — OPTION A THRESHOLD VERDICT CHANGE (UPDATED 2026-05-18).

Earlier blend values (majority −0.40%, minority −1.81%, negative=UCP convention) placed both maps below the 2.2% assembly-size threshold (Option A). Under Phase 4C (majority +0.04%, minority +3.96%), the minority exceeds the 2.2% threshold. The §5.2.8 table Option A row verdict changes from "Yes (both sub-threshold)" to "No — minority +3.96% exceeds threshold." The majority remains sub-threshold under all options. The audit headline updates accordingly: the majority is sub-threshold under all four Alberta-calibrated options; the minority exceeds Option A (2.2%) and approaches but does not cross Option D (4.10%). Primary: `analysis/methodology/threshold_` §B.2.1.

C7 — §5.2.3 CROSSWALK-STABILITY CLAIMS RE-SCOPED (2026-06-13).

An earlier version of the §5.2.3 "Direction of the 1-seat asymmetry" and "Implication" paragraphs stated: (a) "≤ 5 seats across all tested inputs" and (b) "the direction reverses" in UCP-landslide environments as if these were unconditional findings of the full §5.2.3 analysis. The claims derived exclusively from the per-riding crosswalk stability probe (`data/reference/polling_338_hist` The canonical uniform-swing probe (`data/outputs/338canada_uniform_` added 2026-05-12) uses Phase 4C canonical per-ED results without the hybrid crosswalk and finds: majority gives NDP more seats in 74/76 non-tied snapshots; mean advantage 5.7 seats; range 0–9 seats; no direction reversal at any polling level. Both probes are retained; the "≤ 5 seats" magnitude claim is falsified by the canonical probe's 0–9 range and has been removed; the direction-reversal claim is re-scoped to the crosswalk probe specifically and explained as a crosswalk-model artifact of hybrid ED treatment. The underlying finding — that the minority map is UCP-favorable relative to the majority at competitive polling levels — is confirmed by both probes. Primary: §5.2.3 reconciliation paragraphs (2026-06-13); `data/outputs/338canada_uniform_`

8. Conclusion

Three questions opened the paper. The answers are now on the table.

First, do the two commission proposals diverge in measurable, reproducible ways? Yes. Across eight sub-sections of §5 — population equality, partisan bias, signature detection, MCMC ensemble placement, the pre-registered checklist, symmetry-of-test-selection, geographic coherence, and procedural record — the minority 2026 proposal shows wider distribution, higher packing and cracking signals, more anomalies, and a more government-controlled procedural path than the majority. Every number is reproducible via `python3 analysis/<script>.py` against checked-in data anchored in `FROZEN_MANIFEST.md`.

Second, do the divergences run systematically in one political direction? Yes, with explicit qualifications. Five of the six structural dimensions point in the same direction without depending on vote data. The one vote-based partisan-bias dimension is directionally consistent under 2023 vote input and under April 2026 338Canada polling, but reverses under 2019 vote input. The audit reports the contingency as a property of the finding rather than a defect: the boundary effect is sensitive to which electorate is asked, and the direction holds at approximately 90% confidence under Monte Carlo over modelling choices across 2020s-era voter distribution.

Third, can the April 16 pivot be evaluated against a pre-registered falsifiability framework? Partially now, fully in November. The audit's test battery is pre-registered and the checklist is prepared for Open Science Framework submission with embargoed release scheduled for 2026-11-02. The OSF time-stamp converts the signature framework from an audit-voice discipline into a classical pre-registration against a future map. When the Lunty committee tables its 91-seat map by November 2, 2026, the same scripts producing §5 will be re-run against the new map; the pre-registered scorecard will flag signatures, outliers, and rationale failures at the same thresholds applied here.

Scope of the finding. The minority 2026 proposal shows a pattern of structural irregularities and rationale failures beyond what the majority exhibits, at magnitudes below the Stephanopoulos-McGhee 7% investigable-bias threshold (proposed in the academic literature; never judicially adopted) but above the noise floor of non-partisan redistricting variance as measured against the 1,010,000-plan canonical ReCom ensemble. The April 16 government action replaces the commission's drafting process rather than amending its output — the most government-controlled response among the three most commonly cited Canadian comparator cases. Whether these facts support a constitutional challenge under the *Saskatchewan Reference* [1991] effective-representation standard is for counsel and the courts to assess (Appendix F).

Next steps. Following the receipt of the official shapefiles on May 6, 2026, the audit successfully replaced all Derived Provisional Geometries (DPG) with canonical boundaries. On May 7, 2026, Elections Alberta's GIS team responded to the audit's technical methodology questions (personal communication, Raymond Mok, Geomatics Team Lead). Three specific points were clarified: (1) VA polygons for the 2026 proposed boundaries do not yet exist, because the government's Select Special Committee may redraw the boundaries before they take effect — confirming the audit's approach of spatially joining 2023 VA polygons to the 2026 ED boundaries is the correct and only available methodology; (2) the population estimation methodology was described and the EA contact noted it was reasonable for the study's purposes, without being able to confirm whether the underlying population data could be shared; (3) EA uses the current Alberta OpenData municipal boundary dataset for boundary work, and Communities of Interest determinations were made through EBC member knowledge and public hearings rather than a formal GIS protocol — confirming the audit's municipal anchoring analysis uses the same data source as the commission. The audit's remaining modelling approximations (election-day-only vote substrate; 2021 census-based population) reflect structural features of EA's published data, not gaps EA's internal data resolves. The November 2026 MLA-committee 91-seat map remains the held-out test closing the pre-registration residual (§5.5). In response to speculation regarding "dummymander" vulnerability, the November analysis will explicitly test for **microtargeting** and **demographic lockout**. By applying projected 10-year suburban population growth models to the canonical MCMC ensemble, the audit will test whether the minority's "Surgical Fortification" firewall is designed to systematically dilute opposition vote share as population grows, establishing a permanent structural lockout over the decennial cycle.

What stands in the evidentiary record. The audit's contribution is documenting two commission proposals diverge systematically on six measurable dimensions, the direction of divergence consistently favors the governing party under the available vote inputs, and the process being used to promote the more-favorable proposal departs from comparator Canadian practice in specific ways. These facts are reproducible from public data using checked-in code. They do not prove intent, and they do not by themselves establish a constitutional violation. What they do provide is the evidentiary substrate on which such judgments — by counsel, by courts, by voters, and by future commissioners — can be constructed.

8.1 Postscript — the audit as framework for the held-out test

In policy terms, the verdicts in §5 and §6.2 are now substantially moot. On April 16, 2026 the Alberta government set both commission proposals aside and assigned redistricting to a five-MLA committee chaired by Brandon Lunty (UCP, Leduc-Beaumont). The map this audit measured is no longer a candidate to become law. Whatever Alberta uses for the 2027 election will come from the Lunty committee, not from either commission proposal.

This does not make the work in §4–§7 unnecessary. The commission audit was a framework-building exercise. It produced — and calibrated against three real Alberta maps (the 2019 enacted baseline, the majority 2026 proposal, and the minority 2026 proposal, applied symmetrically to all three) — the artefacts the next audit will need:

1. The four Alberta-calibrated EG thresholds (§5.2.8), including the 4.10 % MCMC-derived 95th percentile (1,010,000-plan canonical ensemble on official EA shapefiles).
2. The 1,010,000-plan canonical constraint-bound ReCom ensemble on Alberta's 2019 substrate (§5.4), against which any 89- or 91-seat Alberta map can be percentile-placed for the four partisan-bias metrics.
3. The structural-irregularity scorecard (anchoring %, MAD, Calgary geographic-zone gap, Airdrie-class community-split count, chair-anomaly count), with majority-comparable benchmarks documented for every test.
4. The pre-registered rationale-failure framework classifying a contested redraw symmetrically across maps (§5.6).
5. The DPG construction pipeline (§4.1.5–§4.1.6), now extended through stage v0_8 with the perfecter and the nested-polygon ownership-inversion refinement, and the alignment-proof toolkit, all of which will accept the Lunty committee map without modification when its data is published.
6. The two-lane verdict structure (§6.2) distinguishing partisan-bias-magnitude readings from cumulative-structural-pattern readings, so the next audit can report both lanes honestly even if they disagree.

When the Lunty committee tables its 91-seat map on or before November 2, 2026, this audit's apparatus will be re-pointed at the new map and re-run as fulsomely and as faithfully as it was for the commission proposals. The pre-registered scorecard at `analysis/reports/pre_registration_draft.md` and its OSF time-stamped binding (§5.5) are the audit's commitment to the follow-through; the directional flag the §6.2 verdict places on the November map will be made on the same evidence, against the same thresholds, with the same restraint on intent.

Two contextual reminders for the November audit.

Alberta's natural geography will always give conservative voters an edge. The province's NDP supporters concentrate in city cores; UCP supporters spread across suburbs and rural ridings winning by efficient margins. Any neutrally-drawn Alberta map will produce a UCP-favourable efficiency gap as a structural starting point. The Chen-Rodden decomposition in §5.2.5 quantifies how much of any given map's partisan-bias signal is geography-derived versus drawing-derived; a future map cannot escape the geography component, only manage the drawing component. The audit's job is to separate the two clearly enough so legitimate criticism cannot be brushed off as "you're complaining about geography."

There are many ways to gerrymander, and not all of them register on the partisan-bias-magnitude lane. The minority 2026 proposal demonstrated this empirically: it sat sub-threshold on every Alberta-calibrated EG line while crossing every structural-pattern threshold by a wide margin. A drafting process wanting to engineer outcomes without leaving an EG fingerprint has the structural lane available — community splits, off-reference borders, anchoring departures, chair-flagged shapes — and the audit will be looking at both lanes when the Lunty committee map arrives. Sub-threshold on Lane 1 is necessary for a clean verdict, but not sufficient.

The standard the November audit will hold the Lunty committee map to is the same standard this audit held the commission proposals to: skew toward neutrality wherever the constraint set permits, document the choices when the constraints force a departure, and be prepared to defend any departure from comparator Canadian practice with evidence stronger than aesthetic preference. Whether the new map meets the standard is a question this audit cannot pre-judge. It is a question the same scripts will answer publicly, on the same timeline, against the same evidence base — because the held-out test is the entire reason the framework was built.

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Draft. Falsifiability gates, robustness checks, and APA citations documented throughout.

Appendix A — Reproducibility

A.1 Three-election direction-stability of minority-majority asymmetry

The minority map's partisan-bias asymmetry relative to the majority map is robust across three Alberta election cycles (2015, 2019, 2023). Minority-majority efficiency-gap asymmetry (percentage points):

ELECTION	MAJORITY EG	MINORITY EG	ASYMMETRY (MIN−MAJ)	INTERPRETATION
2015	+7.25%	+7.28%	+0.03 pp	Minority essentially equal to majority
2019	+0.16%	+0.90%	+0.75 pp	Minority more UCP-favorable than majority
2023	−0.85%	−1.36%	−0.51 pp	Minority less UCP-favorable than majority

Caveat. *The 2015 and 2019 asymmetries reverse sign relative to 2023, a cycle-contingent result explained in §5.2.3: the minority's hybrid districts (Springbank, Bearspaw, Cochrane) occupy the blend model's transition zone between urban and rural NDP performance. Under 2023 (a competitive urban cycle), these territories trend NDP, widening the minority's UCP advantage. Under 2019 and 2015 (cycles with shallower urban NDP penetration), they trend rural, narrowing it. The inter-map asymmetry direction thus depends on Alberta's election-cycle geography — not a measurement artifact or boundary instability. By contrast, both individual maps consistently show UCP advantage under all three elections; the direction of individual-map partisan bias is stable, only the inter-map gap direction changes. Full method and derivation at `findings/cross_election_2015.md`.*

All scripts run from repository root:

```
python3 analysis/scripts/packing_cracking_analysis.py # §5.2 symmetric three-m
python3 analysis/scripts/electoral_forensics_population.py # §5.1 with A2 robu
python3 analysis/scripts/poll_attribution_skeleton.py # §4 parse validation
```

Each script prints a gate PASS/FAIL line. Numbers in §2, 3 above must match the corresponding gate-passed output.

Reproducibility artifacts. A version-pinned environment manifest (`requirements.txt` at repo root) lists every Python package with exact version; an interpreter pin (`setup.md`) names the tested Python version; `FROZEN_MANIFEST.md` lists every external URL accessed during the audit with its access date. A third party running the pipeline 12+ months from today can (a) install the pinned environment, (b) check each URL's state against the frozen snapshot, and (c) reproduce every numeric finding to the tolerance stated in Gate G0. Reproducibility-artifact provenance follows the ICLR 2022 Reproducibility Checklist and Nosek et al. (2018) Open Science Framework pre-registration standards.

Appendix B — Supporting Analysis Documents

- Section A
- Section C
- Section D
- Section 4
- Bias audit — self-audit of this audit's own methodology
- Design critique — hostile stress-test pass
- Methodological defenses — pre-emptive adversarial review: attacks on substrate, MCMC baseline, and partisan metrics with per-attack empirical responses
- Test apparatus defense — per-test criticism and response; addresses "are you making up metrics to have metrics?"
- Fisher combination defense — statistical rationale for combining Ch1 and Ch2 via Fisher's method
- Fisher independence defense — argument the two channels are sufficiently independent for Fisher combination
- Warrington declination defense — declination metric applicability and its direction-disagreement with EG
- Urban weight defense — blend model urban-weight parameter justification
- Uncertainty analysis
- Academic literature review
- Submission search findings — §5.9.4 evidence base
- Chair's Recommendation 5 analysis — §5.9.2 evidence base
- Track C checklist baseline scoring — §5.5 full scorecard and comparison template for the November map
- MCMC ensemble baseline — §5.4 method, 10k-sample ReCom chain against 2019 baseline, per-metric percentile tables
- Plan B compliance + contested-config cross-check — note on population-data provenance evidence base

- Cycle-lag analysis — province-wide ED drift under mid-2025 populations
- Proposed Act §12 amendment — legislative reform proposal addressing the census / cycle-lag tension
- Calgary data-sources audit — 16 Calgary-specific sources catalogued; ward-level modelled A2 sensitivity is feasible from public data
- Adversarial stress-test passes and their fortifications are preserved in `historical/` for historical reference (see `historical/README.md`).
- Chen-Rodden Alberta validation — mechanism-level test of the natural-packing argument
- Canadian redistribution base-rate catalogue — C4 partial catalogue (quantitative acquisition flagged as future work)
- Alberta government database survey — Track N, composite-basis source recommendations for §12 reform
- Commission source provenance audit — Track O, verified 4,888,723 matches StatsCan Q2 2024 postcensal estimate
- Byelection data and assessment — Track S, 2022–2025 byelections evaluated and not incorporated into RT3
- A1 legal-baseline computation — Appendix C companion; 2019-map MAD on 2021 Census directly
- Threshold provenance compendium — every numeric threshold justified with source + sensitivity
- Canadian inter-map base-rate computation — comparative distribution across seven Canadian redistribution cycles
- External pre-registration draft and platform analysis / platform analysis — OSF submission package for the November signature-detection checklist
- Minority rationales validation — §5.8.4 and §5.9.2 evidence base (25 rationales inventoried, 3 contradicted)
- Minority rationales inventory — source quotes with citations
- Cochrane journey-to-work — §5.8.4 StatsCan 98-10-0459 pull
- CSD-level community splits — §5.8.4 robustness check
- 338Canada riding-level cross-validation — §5.2.3 independent cross-check
- Submission OCR log preserved at `historical/v0_1_submission_ocr_log.md` — §5.9.4 partial extension of the 88 non-text-layer submissions

Appendix C — 2021-census legal-baseline A1 for the 2019 map

Purpose. §5.1.1 reports A1 MAD on the commission's own tables, which derive from the July 2024 OSI population estimate. A reviewer committed to strict §12(3) statutory-basis discipline can argue the §5.1.1 numbers inherit the commission's data-source status. This appendix provides an independent 2021-Census-direct computation of A1 on the 87 existing 2019 EDs as the §12(3)-operative reference point. The 2026 proposals cannot receive the equivalent treatment because their ED shapefiles have not been publicly released.

Method. 2021 Census population at the dissemination-area level (6,203 Alberta DAs, `data/alberta_2021_da_populations.csv`) aggregated to the 87 2019 EDs via geopandas overlay on `data/alberta_2019_eds/EDS_ENACTED_BILL33_15DEC2017.shp`. Reproducible script at `analysis/scripts/a1_legal_baseline_2021_census.py`. Per-ED output at `data/a1_legal_baseline_2019eds_2021census.csv`.

Headline figures.

MAP & BASIS	QUOTA	MAD	SOURCE
2019 map on 2017-report basis (commission-quoted)	46,803	2,886	EBC 2017 Final Report pp. 60–61
2019 map on 2021 Census (this appendix)	48,996	**4,745**	This computation
2026 majority map on 2024 TBF estimate	54,929	3,180	Majority Report variance table
2026 minority map on 2024 TBF estimate	54,929	4,707	Minority Report variance table

Ordinal comparison. *2026 majority MAD (3,180) < 2019-on-2021-Census MAD (4,745) ≈ 2026 minority MAD (4,707). The minority 2026 proposal's distribution-tightness is effectively equal to the 2019 map's distribution-tightness at 2021-Census time; the majority 2026 proposal is meaningfully tighter than either benchmark.*

Seven 2019 EDs outside ±25 % under 2021 Census. Central Peace-Notley (−44.77 %), Lesser Slave Lake (−44.73 %) — both s.15(2)-protected — plus Edmonton-South (+40.89 %), Edmonton-Ellerslie (+38.60 %), Edmonton-South West (+33.71 %), Airdrie-Cochrane (+29.76 %), Calgary-North East (+25.55 %). The five positive outliers are urban-growth EDs already exceeding the +25 % ceiling by 2021-census time; Track L's mid-2025 analysis (`analysis/cycle_lag_analysis.md`) confirms all five remain out-of-band under 2025 populations.

Interpretation. The audit's §5.1.1 ordering (majority tighter than minority) is preserved under the §12(3)-operative basis. The minority proposal, drawn four years after the 2021 Census, reproduces the same population-distribution tightness the 2019 map exhibited against the 2021 Census; the majority proposal improves on the benchmark. A strict §12(5) reviewer's attack on §5.1.1's data basis does not change the direction of the A1 finding. Full discussion in the companion file `analysis/methodology/appendix_c_legal_baseline.md`.

Appendix D — Mathematical Formalism

D.1 Efficiency Gap (Stephanopoulos and McGhee 2015)

$$\text{EG} = \frac{W_A - W_B}{N}$$

```
EG = (W_A - W_B) / N
```

Wasted votes W_X for party X are defined as losing-district votes plus winning-district votes above the victory threshold $\lceil N_d/2 \rceil + 1$, summed across districts. The 7 % threshold is academic-literature authority only — never judicially adopted (see §5.2.8).

Sign-convention reconciliation. When party A is indexed first, Stephanopoulos-McGhee canonical EG uses a 2:1 slope baseline (positive EG = party A disadvantaged, i.e., party B has seat-advantage). This paper reports EG under the 1:1 proportional-seat baseline; in Alberta's rural-UCP-blowout context, this produces negative EG values whose magnitude correlates with UCP outcome advantage through seat count, despite UCP also being the more-wasteful party by the wasted-vote measure. The two conventions give the same ordinal ranking of Alberta's three maps (2019 / Majority 2026 / Minority 2026) and therefore the same minority-vs-majority direction. Full derivation and verification at `analysis/methodology/sign_convention_resolution.md`.

D.2 Mean-Median Gap (McDonald and Best 2015)

$$\text{MM} = \bar{v} - \tilde{v}$$

```
MM = mean(v) - median(v)
```

$\bar{v}$ is the mean NDP vote share across districts; $\tilde{v}$ is the median. Positive MM indicates mean > median, consistent with party voters packed into fewer high-share districts (cracking of the opposing party, or packing of own party).

D.3 Declination (Warrington 2018)

For an election with n total districts in which party R wins ℓ_R districts and party D wins ℓ_D districts ($\ell_R + \ell_D = n$), let $\bar{y}_R$ be the mean vote share for R across R's *won* districts and $\bar{y}_D$ the mean vote share for D across D's won districts. Define two angles measuring how far each party's mean winning share sits above the 50 % threshold, scaled by that party's seat fraction:

$$\theta_R = \mathrm{atan2}\!\left(\bar{y}_R - 0.5,\; \tfrac{\ell_R}{2n}\right), \qquad \theta_D = \mathrm{atan2}\!\left(0.5 - \bar{y}_D,\; \tfrac{\ell_D}{2n}\right)$$

(D's deviation from 0.5 is measured *downward* because party D's relevant vote-share axis from R's perspective is $1 - \bar{y}_D$; θ_D is constructed so that a perfectly symmetric election produces $\theta_R = \theta_D$ and $\delta = 0$.)

Warrington's declination is δ = (2/π)(θ_D − θ_R). Positive δ indicates the party indexed as R is favoured by the seat geometry (R's wins clustered tightly near 50 % — efficient — while D's wins are clustered with wider margins — wasted votes); mathematically equivalent to packing of party D. (Note: earlier drafts of this Appendix wrote the formula as (2/π)(θ_R − θ_D), which is *negated* relative to Warrington 2018 — that error was corrected 2026-06-12 as part of Amendment 10. See `findings/pre_registration_amendment_log.md` Amendment 10 for the verification.)

Convention used throughout this audit (corrected 2026-06-12 per Amendment 10; consistent with `analysis/scripts/mcmc_ensemble.py:215`): R = UCP and D = NDP, so **positive δ = UCP-favourable boundary geometry (NDP packed); negative δ = NDP-favourable boundary geometry (UCP packed).** This is Warrington's original convention applied with R = the dominant party in the modelled election (UCP in 2023 Alberta). It is now the convention in every chain CSV, every real-map JSON, and every reported percentile in this audit; earlier (pre-Amendment-10) chain CSVs, JSONs, and percentiles carried the swapped-operand sign and have been migrated in place via `analysis/scripts/amendment_10_declination_migration.py`.

```
# Discrete computation (chain code reference: mcmc_ensemble.py:215 post-Amendm
# For each party, sort districts by UCP vote share.
# y_bar_ucp_in_ucp_won  = mean UCP share in UCP-won districts
# y_bar_ucp_in_ndp_won  = mean UCP share in NDP-won districts (= 1 - mean NDP
# l_R = # UCP-won districts; l_D = # NDP-won; n = l_R + l_D
theta_R = math.atan2(y_bar_ucp_in_ucp_won - 0.5, l_R / (2 * n))
theta_D = math.atan2(0.5 - y_bar_ucp_in_ndp_won, l_D / (2 * n))
declination = (2 / math.pi) * (theta_D - theta_R)  # Warrington 2018
```

Verification (T1.7 R1 Ref #5; reproduced T1.7 R2 Ref #5): a textbook 10-seat UCP gerrymander (8 seats at 55 % UCP, 2 seats at 20 % UCP) returns EG = +0.34 (UCP-favoured) and δ = +0.716 (UCP-favoured). Under the pre-Amendment-10 (swapped-operand) formula the same gerrymander returned δ = −0.716, contradicting both EG's direction *and* Warrington's NC-2014 published analog of +0.54 for the same geometry — the discrepancy that surfaced the bug.

Alberta results (canonical shapefiles, 2023 votes; corrected 2026-06-12 per Amendment 10):

MAP	Δ (CORRECTED)	PERCENTILE VS 1,010,000-PLAN CANONICAL ENSEMBLE
2019 enacted	**+0.045**	p91.05 (mild UCP-favoured side)
Majority 2026	**−0.027**	p20.38 (mild NDP-favoured side of median, within null)
Minority 2026	**+0.077**	p98.79 (UCP-favoured tail)

Under the corrected sign, all four partisan-bias metrics agree on the UCP-favoured direction for the minority map (EG +3.96 %, MM +1.04 %, δ +0.077, s50 0.5169). Three carry tail flags at ≥p95; EG is directionally aligned but sub-threshold (p94.4). The pre-Amendment-10 "B6 disagrees by design" reading at earlier drafts of §5.2.4 was an artefact of the swapped operand, not a real cross-metric divergence; Warrington (2018) and Katz, King & Rosenblatt (2020) document EG↔δ tension on a non-trivial fraction of US-state maps, but the Alberta canonical minority on canonical substrate is not one of those cases once the sign is corrected. See §5.2.4 for the corrected reconciliation paragraph.

Substrate-iteration history (see also `findings/post_audit_recompute_deltas.md`). Under the swapped-operand convention used in earlier reports, the minority's canonical declination read −0.0666 (v0_8 DPG substrate) → +0.0105 (canonical first-pass, before the corrected Vote-Anywhere attribution landed) → −0.0770 (final canonical with corrected attribution). Under the Warrington-2018 convention restored by Amendment 10, the equivalent values are **+0.0666 → −0.0105 → +0.0770**; the v0_8 and final canonical agree on UCP-favoured direction once corrected, and the transient first-pass +0.0105 (old)/−0.0105 (Warrington) reflected an attribution refinement (~9.4 % of votes) that has since been superseded. The earlier §6.2 reading of "p62.2, near median" derives from the transient first-pass value and is superseded by the final canonical **p98.79** reading.

D.4 Polsby-Popper compactness

$$\text{PP} = \frac{4\pi A}{P^2}$$

```
PP = 4 * pi * A / P^2
```

A = polygon area, P = perimeter. Range $[0, 1]$; 1 = circle. Values near 0 indicate elongated or convoluted boundaries. Gate threshold: mean PP < 0.15 (extremely fragmented). See Polsby & Popper (1991); Niemi et al. (1990).

Results (CANONICAL EA shapefiles, EPSG:3401, full 89-ED coverage; script `analysis/scripts/compactness_metrics.py`):

> *⚠ **SUBSTRATE NOTE 2026-06-12 per T1.7 R2 Ref #10.** The table immediately below is the **v0_7 DPG historical** reading, preserved for trail-of-work. Canonical EA shapefile values supersede it (canonical reading reported in the §5.6 / §6.2.2 / §5.8.4 narrative; the canonical Edmonton-Enoch-Devon PP is 0.534, not the 0.0652 minimum in the v0_7 historical table). The v0_7 minimum 0.0652 (Edmonton-Enoch-Devon) was retracted at §5.8.4 as a DPG transcription artefact; the canonical reading reverses the inter-map ordering (canonical: minority mean ≈ 0.432 vs majority mean ≈ 0.437; canonical EDs < 0.25: 6 minority vs 8 majority). The v0_7 table below is **historical only** and should not be cited for geometric-consequence claims; use the canonical numbers in §5.6 instead.*

MAP (V0_7 HISTORICAL)	N	MEAN PP	MEDIAN PP	STD DEV	MIN	MAX	EDS < 0.30	EDS < 0.40	GATE
2019 enacted	87	0.4186	0.3994	0.1189	0.1717	0.7691	16	44	PASS
Majority 2026	89	0.3564	0.3591	0.1375	0.0902	0.6910	29	60	PASS

MAP (V0_7 HISTORICAL)	N	MEAN PP	MEDIAN PP	STD DEV	MIN	MAX	EDS < 0.30	EDS < 0.40	GATE
Minority 2026	89	0.3344	0.3374	0.1470	0.0652	0.6994	39	63	PASS

The minority 2026 map appeared to have the lowest mean Polsby-Popper (0.334) under v0_7. Canonical recomputation reverses this finding: the minority mean is ≈ 0.432 vs majority ≈ 0.437; below-0.25 counts run 6 minority vs 8 majority (i.e., the majority has more elongated districts on canonical substrate). Full per-ED CSV at `findings/compactness_metrics.csv` (v0_7 historical); canonical values at `findings/polsby_popper_verdict.md`. All three maps pass the gate (mean PP > 0.15) under either substrate.

D.5 Reock compactness

$$\text{R} = \frac{A_d}{A_c}$$

```
R = A_district / A_smallest_enclosing_circle
```

Range $[0, 1]$. R < 0.25 typically flagged. Results are included in `findings/compactness_metrics.csv` alongside Schwartzberg and convex-hull-ratio scores from the same v0_7 run.

Appendix E — Geometric Data Provenance

The subsections below document the historical geometric-data paths evaluated in the first phases of this audit prior to the release of official shapefiles. § E.1–E.5 describe the initial attempts (direct-shapefile, DA-dissolve, VA-polygon-attribution, OSM reconstruction, and QGIS paths). § E.6 is the technical data statement. § E.7 reports the approximate-geometry analysis (Tier A/B compactness) produced as a substitute for the not-yet-released 2026 shapefiles. All provisional geometry findings were officially superseded on May 6, 2026, when Elections Alberta provided the official canonical shapefiles. An additional "Note on population-data provenance and cycle-lag robustness" is preserved at the end of the appendix for completeness; its core findings are summarised in §3.3.

E.1 4A (direct shapefiles)

Resolved. Initially blocked on 2026-04-22 as 2026 proposal shapefiles were not published on the Elections Alberta maps portal. However, on May 6, 2026, Elections Alberta officially provided the 2026 shapefiles (`ea_majority_2026_eds.gpkg` and `ea_minority_2026_eds.gpkg`), triggering the sunset clause and transitioning the entire audit from Derived Provisional Geometries (DPG) directly to Canonical Geometry.

E.2 4B (DA dissolve)

Deprecated. This path would have attempted to reconstruct boundaries by dissolving 2021 Census DAs if the Commission's PDF report contained DAUID lists. It was formally deprecated on May 6, 2026, when the official shapefiles (E.1) provided the canonical boundaries directly, rendering DA-dissolve approximation obsolete.

E.3 4C (VA-polygon attribution)

Pipeline validated; full execution intentionally bypassed. Skeleton (`analysis/scripts/poll_attribution_skeleton.py`) correctly parses the 2023 Statement of Vote: 1,973 poll records matched to four-figure official totals. The full execution (geocoding 1,973 polls directly into the 2026 shapefiles) would require ~4–8 hours of compute. This path was evaluated and intentionally bypassed because it does not yield a measurable precision gain.

Reasoning: 47.5% of 2023 valid votes are "Vote Anywhere" Advance/Mobile ballots. Elections Alberta verified on May 7 — disaggregated VA-level mapping for these ballots does not exist. Even if the 53% Election Day polls were perfectly geocoded, the 47.5% Advance vote would still require a proportional area-weighted spread. Phase 4F (`va_attribution_area_weighted.py`) proved the residual geographic error on the Election Day straddle impacts exactly 1,396 votes (0.15% of the province) and shifts the minority map's `seats@50/50` metric by exactly +0.0000 pp. The current area-weighted substrate is therefore the mathematical ceiling of precision possible with public data; expending compute on this path would be mathematically redundant.

E.4 4D/4E (OSM Reconstruction / QGIS Manual)

Deprecated. 4D (OSM feature-class snapping) and 4E (manual QGIS painting) were alternative paths to reconstruct the missing boundaries visually. Both were permanently deprecated on May 6, 2026, upon receipt of the canonical shapefiles, which provide exact vertices and eliminate the need for visual transcription.

E.5 4F validation

Superseded. Validation of the approximate geometries against Commission population targets was superseded by the canonical runs on the official shapefiles.

E.6 Technical Data Statement

- **Source data for Sections A, B:** CSV files in `data/` (populations for both 2026 maps; per-ED 2023 and 2019 vote totals); raw Statement of Vote in `data/2023_results.xlsx`.
- **Source data for Section C:** JPG map images from the commission's final report (majority Calgary only; full coverage for minority).
- **Source data for Section D:** Electoral Boundaries Commission Act, commission report via prompt context, comparator-case general knowledge.
- **Geometric reconstruction:** Superseded. The audit operates entirely on the canonical shapefiles.
- **Coordinate system / resolution / aggregation:** EPSG:3401 (NAD83 / Alberta 10-TM).
- **Integrity metric:** Population checksum threshold (0.5% warn, 2% hard stop) cleanly passed on canonical shapes.

- **Geometric shift log:** None required. No manual geometric adjustments were applied to the canonical shapefiles.
- **Transformation log:** No CRS transformations applied.
- **Symmetry consistency:** B1–B4 use identical Phase 4C spatial attribution applied to both 2026 maps via the same `packing_cracking_analysis.py` v0.3 run (canonical EA shapefiles, exact VA-level centroid-in-polygon). The earlier 85/15 blending methodology is superseded by Phase 4C (§5.2.2). A1–A3 use identical variance computation against the same provincial average. A2 uses identical classification rule plus an alternative-rule robustness check (G4). Section C has a symmetry data gap (only majority Calgary imagery available) which is disclosed.

E.7 Approximate 2026 geometry — Tier A/B compactness (Archived)

Archived (Superseded May 6, 2026). Prior to Elections Alberta's release of official 2026 shapefiles, the audit constructed a provisional three-tier geometry framework (Tier A/B/C) for compactness testing. All findings have been superseded by the canonical measurements reported in §5.4.9 using the official shapefiles. Technical methodology and version history at `analysis/methodology/commission_reference_shapes.md` and `archive/provisional_geometries/approximate_shape_analysis.md`.

Appendix F — Legal Interpretive Note

This audit does not offer a legal conclusion. It provides the evidentiary basis on which a legal challenge under *Saskatchewan Reference* [1991] 2 SCR 158's "effective representation" standard could be constructed. The question whether the minority proposal, as potentially modified by the November 2, 2026 MLA-committee process, would satisfy the effective-representation requirement is for counsel and the courts to assess. The audit's core contribution is documenting :

1. The two commission proposals diverge systematically on six measurable dimensions.
2. The direction of divergence consistently favors the governing party.
3. The process being used to promote the more-favorable proposal departs from comparator Canadian practice in specific ways.

These facts are reproducible from public data using checked-in code. They do not prove intent, and they do not by themselves establish a constitutional violation.

Admissibility and conduct posture

Admissibility of any expert evidence built on this audit's data is for counsel and a court to assess against the *R v Mohan* [1994] 2 SCR 9 / *White Burgess Langille Inman v Abbott and Haliburton Co.* [2015] 2 SCR 182 framework. The audit does not pre-adjudicate that question. The author's posture is the following:

- **The data and code are public and reproducible.** Every number in this monograph is recoverable by an independent analyst running the publicly committed scripts against the publicly available Elections Alberta inputs. A skeptical analyst can assess methodological choices directly without depending on the author's interpretation.
- **Pre-registration of structural tests is documented at OSF** (registrations w2s8k, r3zm7, qsgy8, 6pt83; drand-pinned timestamps); the audit's own §4.3.3 disclosure distinguishes the prospectively pre-registered structural battery from the exploratory partisan-bias channels.
- **Conflict-of-interest disclosure is on file in the cover-page CoI block** and in `analysis/methodology/reference/conflict_of_interest.md`. The disclosure records the author's voting and donation history across parties, the absence of any consulting relationship with Elections Alberta, the commission, or any political party, and the self-funded nature of the research.
- **The author makes no representation regarding personal qualification as an expert witness.** If this audit's evidence becomes relevant to a constitutional proceeding, qualified counsel can engage independent experts in redistricting science, GIS, and electoral statistics to assess and present it.

Saskatchewan Reference framing. The "effective representation" standard established in *Reference re Provincial Electoral Boundaries* (*Saskatchewan*) [1991] 2 SCR 158 is permissive on deviation from population equality: McLachlin J (as she then was) wrote the guarantee of §3 is "the right to effective representation" (para. 26), with "relative parity of voting power" to be weighed against other factors "including geography, community history, community interests and minority representation" (para. 33). Importantly, the Court provided no bright-line variance ceiling; instead, it requires a holistic assessment in which deviations are defensible *if justified by the specified non-quantitative factors*.

The standard has been applied in *Raîche v. Canada* (*Attorney General*), 2004 FC 679, which examined the federal Electoral Boundaries Commission for New Brunswick's redrawing of Acadian boundaries. The Federal Court found **statutory** violations (ss. 15(1)(b) and 15(2) of the EBRA; Part VII of the Official Languages Act) and referred the boundaries back to the Commission. The Court did *not* find a Charter s.3 violation: applying the deference standard from *Reference re Prov. Electoral Boundaries* (*Sask.*), [1991] 2 SCR 158 (the "*Carter*" standard, named for counsel Roger Carter QC — style of cause corrected 2026-06-12 per T1.7 R2 Ref #8; round-1 of the Raîche correction styled this as "*Carter v. Saskatchewan* [1991] 2 SCR 158," which is not the case's actual reporter caption), the boundary transfer survived the constitutional challenge. Earlier drafts of this report cited Raîche as a "s.3 strike-down" attributed to "Heneghan J." — both attributions were corrected 2026-06-12 per T1.7 Referee #8. *Cassista v. Canada* (*Attorney General*), 2014 FC 398, was cited in earlier drafts as the companion "boundaries-sustained" case; that citation is flagged for verification at the same date (see §2 discretion-review discussion). Under the corrected Raîche reading: partisan asymmetry, standing alone, does not trigger either a Charter s.3

violation or an EBRA s.15(1)(b) statutory violation. Asymmetry becomes constitutionally or statutorily material only when paired with either (a) unjustified deviations from population parity unattributable to geography, community boundaries, or other statutory constraints, or (b) evidence permissible factors (geography, community of interest) have been distorted *in service of* partisan outcomes rather than applied in service of the statute's non-partisan purposes. This is a *narrower* and more defensible doctrinal foundation than the prior framing claimed; it strengthens commission deference under *Carter* and places the analytical burden squarely on the audit's two-lane evidence rather than on the constitutional precedent.

Under Pal's (2015) analysis, which refines the Saskatchewan Reference framework, commission outputs are reviewable under the post-*Vavilov* reasonableness standard (*Canada (Minister of Citizenship and Immigration) v. Vavilov*, 2019 SCC 65, collapsing the pre-Vavilov patently/manifestly-unreasonable distinction into a unified reasonableness standard — wording corrected 2026-06-12 per T1.7 R2 Ref #12; earlier drafts here used the pre-Vavilov "manifest unreasonableness" vocabulary). Commissioners operate within a broad "discretion space" including legitimate design preferences. The boundary between justified discretion and reviewable partiality turns on factual evidence about whether the stated non-partisan constraints genuinely drove the choices or were invoked post-hoc to rationalize a predetermined partisan outcome. The audit's findings — directional partisan asymmetry coupled with population-variance distortions not explained by geography or municipal boundaries, coupled with procedural departure from the pattern established by the majority under identical statutory constraints — are the kinds of evidence a court applying the effective-representation standard, refined through Pal's reasonableness lens, would weigh in assessing whether a commission's output exceeded justified margins of discretion. Whether the weighing produces a constitutional violation is for a court, not this audit, to determine.

Voter-impact translation of structural findings. The *Saskatchewan Reference* standard asks whether voters can exercise "effective representation," which requires districts enable rather than structurally impede the translation of votes into legislative representation. This paragraph translates the audit's structural findings into the specific voter-impact terms effective-representation analysis requires.

Population dispersion. The minority map's MAD of 4,707 persons (48% wider than the majority's 3,180) understates the individual voter effect. The minority's maximum positive deviation is +24.06%, corresponding to a district population of approximately 68,100 against the provincial mean of 54,929. A resident of the district commands 54,929 / 68,100 = 80.7% of the representative weight of a voter in a mean-population district — a dilution of roughly 19 percentage points. Three minority districts exceed +20% deviation (§5.1.1); none in the majority does. The majority's worst positive deviation is +14.28%, conferring 87.5% effective weight. Neither map breaches the ±25% statutory floor, and the *Saskatchewan Reference* standard explicitly permits deviation where justified by the enumerated factors. The question a court would weigh is whether the allocation of underweighted districts is spatially random (consistent with geographic necessity) or patterned (consistent with systematic placement). The audit's finding the minority's inflated districts concentrate in the Calgary Zone A geography (see below) is the evidence going to the patterning question.

Calgary zone asymmetry. Under the minority map, Zone A (NE / central Calgary — the NDP-competitive zone) averages 61,225 residents per district against Zone B (SW / S Calgary) at 54,569 — a 12.2% intra-city gap. A resident of a Zone A district has 54,569 / 61,225 = 89.1% of the representative weight of a Zone B resident. Across 17 Zone A districts, this dilution structurally affects approximately 1,040,000 Calgary residents concentrated in the northeastern and central city. The majority produces no comparable patterned dilution (Zone A–Zone B gap 0.4%, or 205 persons). The intra-city pattern is not explained by geographic factors — both zones are suburban and urban in character, served by comparable highway access, and draw from the same municipal services. The deviation is therefore facially inconsistent with the factors *Saskatchewan Reference* identifies as justifying population variance.

Airdrie community fragmentation. The minority map distributes the Airdrie region across four electoral districts. Airdrie (85,805 at the July 2024 municipal census per City of Airdrie 2024; 74,100 at the 2021 Census; 90,044 at the April 2025 municipal census — see §5.3.2 C1 footnote for full vintage reconciliation) is a single incorporated municipality with integrated planning jurisdiction, municipal services, and a distinct local economy. A 4-way split disperses Airdrie's legislative advocacy across four representatives, none of whom has Airdrie as a primary constituency. The majority's 2-way split concentrates Airdrie's representation in two districts. The 2022 federal redistribution sub-commission for Alberta applied a 2-way split, establishing comparator precedent consistent with the majority approach. Under *Saskatchewan Reference*, community of interest is a recognized justification for deviating from exact population parity; a configuration fragmenting a single community across four districts works against effective representation rather than in its service.

Conduct posture toward named-individual characterisations. This audit characterises the conduct of named public officials — Commission Chair Dallas K. Miller, Premier Danielle Smith, and the minority commissioners (Sela Clark, Nadia Samson, Sandra Evans, David Martin) — solely on the basis of those individuals' on-record statements in their public-official capacity, with each characterisation anchored to (a) a direct quotation of the public statement, (b) a primary-source citation, and (c) the audit's reasoning. Every characterisation appears alongside the underlying evidence so a reader can form an independent judgment. The audit does not investigate confidential deliberations, attribute motive without evidence, or characterise individuals in their private capacity. Whether any specific characterisation gives rise to a cause of action — under *Grant v. Torstar Corp.*, 2009 SCC 61, *WIC Radio Ltd. v. Simpson*, 2008 SCC 40, or otherwise — is for counsel and a court to assess on the actual record. The author does not pre-adjudicate that question and does not assert any defence in this monograph.

Appendix G — Mahalanobis D^2 Distribution: QQ-Diagnostic Against $\chi^2(4)$

The Channel 1 Mahalanobis p-value (1.40×10^{-6}) is computed as `scipy.stats.chi2.sf(D², df=4)`, which assumes the ensemble's four-metric joint distribution is multivariate Gaussian, making D^2 approximately $\chi^2(4)$-distributed. One metric (seats@50/50) is discrete, so multivariate normality is approximate. This appendix documents a post-hoc empirical check of that approximation.

Method. The same covariance matrix used by `joint_outlier_score_canonical.py` (sample mean, `np.cov(rowvar=False)`, `np.linalg.pinv`) was applied to all 1,010,000 canonical ensemble plans to compute a D^2 for each. A QQ-plot of the empirical D^2 quantile distribution against theoretical $\chi^2(4)$ quantiles is shown at `data/maps/mcmc/qqplot_mahalanobis_vs_chi2_4.svg` (generated by `analysis/scripts/mahalanobis_qqplot.py`). Quantiles were evaluated at 5,000 evenly spaced probability levels across [0.0001, 0.9999].

KS D-statistic. The Kolmogorov-Smirnov maximum CDF deviation between the empirical D^2 distribution and $\chi^2(4)$ is **D = 0.0103**. The KS p-value is not reported: with n = 1,010,000 observations it approaches zero for any trivial discrepancy regardless of practical fit quality, and the ensemble rows are not independent ($n_{eff} \approx 1{,}428$ per the convergence diagnostics). A D below approximately 0.01–0.02 indicates a practically excellent fit; D = 0.0103 is at the upper end of that range.

QQ-plot description. The bulk of the distribution — from $D^2 \approx 0$ through approximately the 99th percentile ($D^2 \approx 13.9$) — lies very close to the 45° reference line, indicating the $\chi^2(4)$ approximation is accurate throughout the mass of the distribution. At extreme percentiles the empirical tail is modestly heavier than $\chi^2(4)$ predicts: at the 99.0th percentile the empirical quantile is 13.93 against the theoretical 13.28; at the 99.9th it is 20.00 vs 18.47; at the 99.99th it is 24.45 vs 23.51. The departure is consistent with mild positive skew from the discrete seats@50/50 component.

Conservative or anti-conservative? A heavier empirical tail than $\chi^2(4)$ predicts means the empirical tail probability at any fixed D^2 threshold is larger than the parametric approximation returns. The parametric p-value therefore slightly understates the true tail probability — a mildly anti-conservative direction. However, the minority map's observed $D^2 = 32.67$ (Mahalanobis distance 5.716) lies entirely outside the empirical envelope: the largest D^2 among all 1,010,000 canonical plans is 27.74. No plan in the ensemble approaches the minority observation. The empirical tail count at D^2_{obs} is zero (0 / 1,010,000). This does not contradict the parametric p: under $\chi^2(4)$, the expected number of ensemble plans at or beyond $D^2 = 32.67$ is $1.40\times10^{-6} \times 1{,}010{,}000 \approx 1.4$ plans, so observing zero is fully expected (Poisson $P(0 \mid 1.4) \approx 24\%$). Because the observation lies beyond the empirical range, no direct empirical p-value is estimable and the parametric approximation is the only available figure.

Practical conclusion. The $\chi^2(4)$ model provides an excellent fit through the broad body of the ensemble distribution (KS D = 0.0103, just above the 0.01 threshold for an excellent fit; median 3.318 vs theoretical 3.357; the sample mean equals 4.000 by construction for D^2 and is not itself a fit diagnostic). At quantiles between the 99th and 99.99th percentiles the empirical tail is 4–8% heavier than the theoretical, suggesting the true p-value in that region is modestly larger (less significant) than the $\chi^2(4)$ formula returns. The minority map's D^2 = 32.67 lies beyond the empirical maximum of 27.74 across all 1,010,000 plans; the empirical tail count is 0. This is fully consistent with the parametric estimate: under $\chi^2(4)$, the expected number of ensemble plans at or beyond D^2 = 32.67 is $1.40\times10^{-6} \times 1{,}010{,}000 \approx 1.4$ plans, so observing zero is unsurprising (Poisson $P(0 \mid 1.4) \approx 24\%$). The mild anti-conservative tendency visible at the 99th–99.99th percentiles cannot be reliably extrapolated beyond the empirical maximum, so the quantitative effect on the headline p is unknown. The parametric assumption is defensible for this analysis: systematic departure at practical levels is small (< 10%), the qualitative conclusion that the minority observation is extraordinarily rare under the neutral null is not sensitive to it, and the Hotelling T^2 n_eff-corrected p uses an F-distribution adjustment that does not rely on the large-sample $\chi^2(4)$ limit.

Draft. Falsifiability gates, robustness checks, and APA *citations documented throughout.*

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1. The minority map proposes a 4-way split of the Airdrie region (Airdrie, Airdrie-Chestermere, Rocky Mountain House-Banff, Cochrane-Olds); the majority proposes a 2-way split (Airdrie, Cochrane). This structural divergence is documented and evaluated in §5.3.2 (packing/cracking-signature analysis) and §5.8 (municipal-boundary anchoring). A direct Canadian comparator is provided by the 2022 Alberta federal redistribution sub-commission, which applied a 2-way split to Airdrie, aligning with the provincial majority (see §5.9.3 for federal precedent discussion). ↩
2. Both MADs are computed against the majority's mean (54,929) rather than each map's own mean (majority mean: 54,929; minority mean: 54,930). Using a common reference enables direct apples-to-apples comparison across maps. The 1-person difference is negligible: re-computing the minority MAD against its own mean of 54,930 produces 4,707 (unchanged at this precision level). ↩
3. Monte Carlo median values vary by approximately ±0.3 pp between independent runs with different random seeds. The reported median (−1.40 pp) is from the canonical Run #4 (seed 42, 250,000 samples); an earlier 2,000-sample preliminary run reported median −1.44 pp. This variance is consistent with Monte Carlo sampling error and does not affect the directional claim (negative in 90.5% of draws) or the sensitivity interval bounds. (*This interval reflects parameter uncertainty in the vote-share model — urban weight, rural baseline, and per-hybrid jitter — not boundary position or spatial assignment uncertainty. It is not a frequentist confidence interval: the draws are not i.i.d. samples from a population, and under spatial autocorrelation the effective* N *is approximately* 20–30, *which would widen these bounds by a factor of* 1.5–2×.) We report this as a directional

observation at approximately 90% of draws and do not assert classical statistical significance. The magnitude claim (specifically 0.47–1.48 pp across the 0.70–0.80 weight range; 1.41 pp central) does not meet the 95% threshold. **Small-N noise floor.** In an 89-district assembly, the standard error of the efficiency gap under homogeneous swing is approximately 1–2 pp (Stephanopoulos and McGhee 2015, §III.B), placing the 1.41 pp central asymmetry near the lower bound of reliable EG detection. The metric is most informative in large assemblies (US Congressional, 435 seats); at 89 seats a 1–2 pp inter-map asymmetry is consistent with assembly-size rounding variation and does not independently constitute a structural anomaly. The inter-map EG comparison is therefore reported as directional evidence only; the magnitude is not meaningfully precise at this scale. The minority-vs-majority seat-count gap is 1 seat under both 2023 Statement-of-Vote data and April 2026 338Canada polling, but historical 338 stability testing shows the *direction* of the 1-seat gap is not invariant across vote inputs (see "338 historical stability" paragraph below). If measured attribution from Phase 4C produces an asymmetry of magnitude and direction inconsistent with 2023-vote attribution, the directional claim is falsified. ↩

4. **Canonical geometry recomputation, anchoring (2026-05-10) — full reconciliation note.** Pre-shapefile (DPG v0_10) result, computed April 2026: minority anchoring 14.5%, majority 71.0%, ratio 4.9× — reported during the audit as the structural lane's cleanest single signal. Canonical recomputation (official Elections Alberta shapefiles, received 2026-05-06): minority 72.0%, majority 80.0%, 2019 baseline 75.2% — both 2026 maps within the 70–85% Canadian comparator norm; ratio 1.11×. The 4.9× gap is not a property of the official maps. **Mechanism of the DPG error:** `score_anchoring.py` measures the percentage of ED perimeter falling within 500 m of a Statistics Canada CSD edge on a per-segment basis (without the ≥ 1 km run-length filter specified in the pre-registration — see §5.8.5 contiguity-filter disclosure; the implementation produces slightly more permissive values than the pre-registered definition). The DPG pipeline preserved ED *area* to within ~0.0004% but did not constrain ED *boundaries* to lie on CSD polygon edges — the minority map's DPG approximation in particular was sourced from non-CSD reference geometry (road centrelines, census-tract proxies) and therefore failed the perimeter-alignment criterion even where the underlying area was correct. Area fidelity does not imply perimeter fidelity; this is a structural distinction in the DPG pipeline, not a hidden property of the minority map. The minority swung 14.5 → 72.0% (≈ 58 pp) while the majority swung 71.0 → 80.0% (≈ 9 pp) because the majority's DPG already had an official 2019 reference geometry available during construction, while the minority's was novel and had no reference shapefile to seed from. **Disposition in this audit.** The DPG-era anchoring finding (4.9× departure, cited in earlier drafts as the load-bearing structural signal) did not survive canonical recomputation and is not carried forward as a standalone result. The draft audit was never published with the 4.9× claim — the figure appeared only in working drafts and was superseded once the canonical shapefiles arrived. Where the DPG figures previously appeared in prose, the canonical values now appear instead. This footnote is the single preserved record of the DPG-era

reading. The audit's structural case (§5.0.1, §6.2.2) now rests on the four geometry-independent dimensions: population dispersion, Calgary zone asymmetry, Airdrie fragmentation, and chair-flagged spatial anomalies, plus the MCMC ensemble (Mahalanobis $p = 1.40 \times 10^{-6}$). For forward analysis and replication, readers should use the canonical geometry (official Elections Alberta shapefiles, SHA-256 verified via `analysis/utils/canonical_manifest.py`) as the authoritative substrate. Parallel canonical recompute deltas for partisan metrics are documented in `findings/post_audit_recompute_deltas.md`. ↩↩↩↩↩↩↩↩↩↩↩↩↩↩↩↩